GENERAL THETA FOUNDATIONS I: CURRENT TECHNICAL SUPPLEMENT

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Abstract. This is the current technical supplement to General Theta Foundations I. It contains the complete inherited positive-realization, contrast, nonabelian multiplicity, mean–contrast, spectral and arithmetic theory, the return-polytope approximation and sharp physical-dictionary law, and all independent causal, minimax, dual, finite-order measurement, two-stage reset, communication, discrimination and geometric results. The main article proves memory-preserving regularization and uniform stability for general nonreset instruments with a fixed hidden model, using the inherited support-certificate theory developed here. It proves a matching square-root singularity over all stochastic two-label controllers and an arithmetic finite-saturation criterion for any finite number of imperfect search locations with arbitrary positive costs. Their complete proofs are in the main article. This supplement begins with the complete two-location proof and its sharper normalized approximation constants; the general argument does not depend on that special refinement. It also retains the complete reset cost frontier, its strict resource hierarchy and stability, together with the inherited shared-preparation, tied-action reset-memory and unrestricted-decoder certificates. The statements retained here keep their individual hypotheses and full proof bodies. Cross-document references and the theorem map identify their actual logical roles; historical preservation is not an additional mathematical premise.

References to the main article use ordinary section numbers; statements numbered S belong to this supplement. Native paired compilation resolves references in both directions. The current main chain is general support verification, same-label regularization, sharp primitive stability and the scoped finite-memory saturation classification. The first section here retains the two-location refinement. The following sections supply the exact group and Fourier premises used by the shared-device theorems. The inherited return geometry supplies the upper counterpart to the unrestricted-decoder lower. Subsequent sections retain the complete independent developments. Relocation does not change a statement's hypotheses or priority; two explicit output-visible-core clarifications from the parallel author branch are identified in the source-provenance and proof-audit records.

S1. Arithmetic obstructions to finite-memory saturation

The convergence in Corollary 4.4 need not terminate at any finite memory. We give an exact criterion for termination in a two-model family with rational examples. Its search ordering is classical: the index rule and the distinction between cyclic and acyclic search sequences belong to the search-game literature [95, 96, 97]. The issue here is the implementation of an optimal law by an autonomous stochastic controller with a counted pure start. A mixture of finitely many infinite search sequences need not have a finite positive-memory implementation. We prove that distinction, its arithmetic converse, and a quantitative finite-memory approximation below.

Fix $r, s \in (0,1)$ and write

$$q_a = 1 - r, \qquad q_b = 1 - s, \qquad u = -\log r, \qquad v = -\log s.$$

There are two scalar models, two commands a, b, one failure report, and compulsory stopping at success. In model 0, command a succeeds with probability q_a and b never succeeds. In model 1, command b succeeds with probability q_b and a never succeeds. The deterministic call costs are arbitrary $c_a, c_b > 0$, respectively, in either model. Put $\tau_a = c_a/q_a$ and $\tau_b = c_b/q_b$, the known-model

Date: October 2026.

optimal costs. The model is selected once and remains hidden. Thus the instrument is feasible, has positive bounded costs, and has the same observation and initialization conventions as Section 3. We use V_M and V_{hist} for its finite-label and unrestricted-history robust cost values.

S1.1. **Search words, the classical exchange rule, and robust duality.** A word $w = (w_0, w_1, \dots) \in \{a, b\}^{\mathbb{N}_0}$ specifies what to do on the hypothetical path on which every call fails. Let i_t, j_t be the numbers of a, b in its first t letters. The actual costs of that deterministic policy are

$$C_0(w) = \sum_{t\geq 0} c_{w_t} r^{i_t}, \qquad C_1(w) = \sum_{t\geq 0} c_{w_t} s^{j_t}. \tag{S1.1}$$

These nonnegative sums also define infinite costs without a convergence convention. A randomized causal policy induces a probability law on such words, by sampling its command kernels along the hypothetical all-failure path. Conversely every word law is implemented by its conditional next-letter probabilities. These probabilities do not require the hidden model: for a fixed command prefix the physical likelihood of all its failures is common to every private realization of that prefix and cancels from the conditional controller law. Tonelli's theorem therefore gives the cost vector as the expectation of (S1.1) under that word law.

The following search lemma, including interior least-favorable priors, equalized model costs and a mixture of at most two consistent tie rules, specializes the classical two-location search theory. In particular these conclusions are covered by Proposition 3 and Theorem 10 of the arXiv version of Clarkson–Lin–Glazebrook [96]; the index rule is attributed to Blackwell, as reported by Matula [95]. We reproduce a self-contained proof because the finite-memory obstruction needs its exact statement about all optimal word laws. These search and saddle conclusions are not claimed as new results.

Lemma S1.1 (Greedy words and an attained interior preparation saddle)**.** *For a prior $\pi \in (0, 1)$, a finite-cost word minimizes $\pi C_0 + (1-\pi)C_1$ if and only if, at every prefix, it chooses*

$$\begin{cases} a, & (\pi/\tau_a) r^{i_t} > ((1-\pi)/\tau_b) s^{j_t}, \\ b, & (\pi/\tau_a) r^{i_t} < ((1-\pi)/\tau_b) s^{j_t}, \end{cases} \tag{S1.2}$$

with either order permitted at equality. Every word obeying this rule has finite cost in both models. Its asymptotic frequency of a is

$$\lim_{t\to\infty} \frac{i_t}{t} = \frac{v}{u+v}. \tag{S1.3}$$

The robust history value is attained. There is an interior prior $\pi_* \in (0, 1)$ whose Bayes value is V_{hist}. At that prior, a robust optimum is a mixture of at most two greedy words: one places a first at every tied pair, and the other places b first at every tied pair. The resulting risk vector is $(V_{\mathrm{hist}}, V_{\mathrm{hist}})$.

Proof. First consider two consecutive different letters after a prefix with counts (i, j). Exchanging ab with ba leaves the counts after the pair, and hence every later contribution, unchanged. The two model-cost differences, with ab first in the subtraction, are exactly

$$(-c_b q_a r^i, c_a q_b s^j). \tag{S1.4}$$

The Bayes difference is therefore $c_a c_b(((1-\pi)/\tau_b) s^j - (\pi/\tau_a) r^i)$. The greedy rule merges the two strictly decreasing lists

$$\frac{\pi}{\tau_a}, \frac{\pi}{\tau_a} r, \frac{\pi}{\tau_a} r^2, \dots, \qquad \frac{1-\pi}{\tau_b}, \frac{1-\pi}{\tau_b} s, \frac{1-\pi}{\tau_b} s^2, \dots$$

in decreasing order. If a finite-cost competing word disagrees with a prescribed greedy letter, the desired letter occurs later: finite cost in both models requires infinitely many occurrences of both commands. Move its next occurrence to the desired position through the intervening copies of the other letter. Each adjacent exchange is nonincreasing in Bayes cost. Indeed the desired list item is

at least the first skipped item, and all subsequent skipped items are smaller. A strict violation of (S1.2) gives a strict improvement.

By finitely many such exchanges one can match any prescribed finite greedy prefix without increasing the competitor's full cost. The greedy prefix cost is nonnegative and no greater than that transformed full cost. Letting the prefix length increase proves optimality of every greedy word once its finite cost is verified. The strict-improvement observation proves necessity. An infinite-cost competitor cannot improve a finite Bayes value.

For finiteness and frequency put

$$z_0 = \log \frac{\pi\tau_b}{(1-\pi)\tau_a}, \qquad z_t = z_0 - ui_t + vj_t.$$

The rule subtracts u when $z_t > 0$, adds v when $z_t < 0$, and allows either operation at zero. Consequently, for all t,

$$\min(z_0, -u) \le z_t \le \max(z_0, v). \tag{S1.5}$$

Since $i_t + j_t = t$, the identity

$$i_t = \frac{z_0 + vt - z_t}{u+v}$$

proves (S1.3). More precisely, with $\kappa = \exp(-uv/(u+v)) < 1$,

$$r^{i_t} = \exp\left(\frac{u(z_t - z_0)}{u+v}\right)\kappa^t,$$
$$s^{j_t} = \exp\left(\frac{v(z_0 - z_t)}{u+v}\right)\kappa^t.$$

The strip bounds both factors uniformly, proving convergence of both cost sums. These bounds are uniform over all tie choices at the fixed prior.

We next establish the robust duality without presupposing a finite optimal word or an infinite-dimensional minimax exchange. Represent a word law by its cylinder probabilities. They lie in $[0,1]$, start with mass one, and satisfy the consistency equalities and identities for a binary probability tree. Every sequence of these laws has a subsequence on which all cylinder probabilities converge, by successive subsequence selection on the countable set of cylinders. The limit again defines a word law. Every cost truncated at a fixed number of calls is a continuous linear function of finitely many cylinders; the full cost is the supremum of these continuous functions.

Let $\mathcal{R} \subset \mathbb{R}^2$ be the upward risk set: $(x_0, x_1) \in \mathcal{R}$ when some word law has both costs finite and no greater than x_0, x_1. It is convex, since word laws can be mixed. It is closed: for a convergent sequence of such upper bounds, take a cylinderwise convergent subsequence of the associated laws. Every truncated cost of the limit is bounded by the limiting coordinate, and monotone convergence gives the same bound for the full cost. A fixed full-support command row makes $\mathcal{R}$ nonempty. Thus $(V_{\text{hist}}, V_{\text{hist}}) \in \mathcal{R}$.

Separate $\mathcal{R}$ from the disjoint open convex set $\{(x_0, x_1) : x_0 < V_{\text{hist}},\ x_1 < V_{\text{hist}}\}$. A separating linear functional has nonnegative coefficients: the open set is unbounded in each negative coordinate, while $\mathcal{R}$ is upward closed. Normalize the nonzero coefficients to $(\pi, 1-\pi)$. The separator and the displayed boundary point give

$$\inf_{(x_0,x_1)\in\mathcal{R}} \{\pi x_0 + (1-\pi)x_1\} = V_{\text{hist}}. \tag{S1.6}$$

This prior is interior. Under model 0, every proper finite-cost policy has expected number $1/q_a$ of a calls: summing their success probabilities gives total success probability one. Likewise model 1 requires expected number $1/q_b$ of b calls. The corresponding cost lower bounds are τ_a and τ_b. In fact both inequalities are strict for every law with finite cost in both models. Properness in model 1 requires positive probability of a b call at some finite commanded prefix. Every failure in such a prefix has strictly positive probability in model 0 as well. Thus model 0 incurs a strictly positive

additional expected cost for wrong-command b calls. The argument with the models exchanged proves strictness for model 1. Apply this to an attaining robust law: it gives $V_{\text{hist}} > \max(\tau_a, \tau_b)$.

At either endpoint prior the infimum over $\mathcal{R}$ is the corresponding τ_a or τ_b: a full-support memoryless row can approach the appropriate pure command, with finite cost in the other model at every stage. Thus an endpoint cannot satisfy (S1.6). Write the resulting interior prior as π_*. We record also a bound used later in the normalized case $c_a = q_a, c_b = q_b$. If the first-command probability of a is p, the two costs are at least $1 + q_b(1-p)$ and $1 + q_a p$. Consequently

$$\inf_{\mathcal{R}}\{\pi x_0 + (1-\pi)x_1\} \geq 1 + \min\{\pi q_b, (1-\pi)q_a\} \quad (c_a = q_a,\ c_b = q_b). \tag{S1.7}$$

At an interior prior the proved greedy optimum is finite in both models, so its Bayes value equals the infimum over $\mathcal{R}$. An attaining robust law must be supported on Bayes-optimal words: each word's Bayes cost is at least this value and its expectation equals the value. Both of its model costs equal V_{hist}, since both are at most that number and the two prior weights are positive.

For completeness, tied words reduce to two extreme risk vectors. The merged numerical lists have a fixed order between unequal items. Every tie consists of exactly one item from each list; its two letters are consecutive. Changing their order affects no later count. Formula (S1.4) lowers C_0 and raises C_1 when the order is ab. At every tie, $\pi_* r^i/\tau_a = (1-\pi_*)s^j/\tau_b$, so all these cost differences lie on the same Bayes-level line. The differences are absolutely summable, since tied indices are distinct and both geometric series converge. Consequently every greedy risk vector lies on the line segment joining the word with ab at every tie and the word with ba at every tie. The same is true of any randomized greedy law. The robust risk vector on that segment is obtained by mixing its two endpoints. This proves the last assertion. □

S1.2. **The finite-state obstruction.**

Lemma S1.2 (Deterministic tails of finite stochastic controllers)**.** *Suppose an autonomous controller with finitely many labels and fixed command and update rows, on one perpetually repeated nonterminal report, emits one prescribed infinite command suffix with probability one after some fixed finite time. That suffix is ultimately periodic. The conclusion allows arbitrary stochastic updates and command rows, and does not assume that the retained label is determined by the visible history.*

Proof. Let S_t be the nonempty set of labels of positive probability after t hypothetical failures. Once the command suffix is deterministic, every label in S_t must have the same pure command row, equal to its prescribed next symbol. Otherwise a label of positive probability would emit a different symbol with positive probability. Given that symbol, the next support is the union of the supports of the corresponding update rows from labels in S_t. Thus S_{t+1}, and its next prescribed symbol, are determined by S_t. There are only finitely many nonempty subsets of the finite label set. Once one support recurs after the suffix begins, its entire future support and symbol sequence recur. This proves ultimate periodicity. The argument uses supports, not the more restrictive assumption of deterministic internal transitions. □

Theorem S1.3 (Arithmetic dichotomy and exponentially accurate finite memory)**.** *For the two-model family above,*

$$V_1 = \tau_a + \tau_b, \qquad (\exists M < \infty :\ V_M = V_{\text{hist}}) \quad \Longleftrightarrow \quad \frac{\log r}{\log s} \in \mathbb{Q}.$$

If $\log r/\log s$ *is irrational, then every finite budget has an attained strict gap*

$$V_M > V_{\text{hist}}, \qquad M = 1, 2, \ldots, \qquad V_M \downarrow V_{\text{hist}}. \tag{S1.8}$$

This statement ranges over all randomized autonomous controllers with a prescribed pure initial label, not only deterministic cyclic command schedules.

For the normalized costs $c_a = q_a, c_b = q_b$, there is an explicit upper estimate for both arithmetic cases. Put

$$m = \min(q_a, q_b), \qquad \delta = \frac{m^2}{16}, \qquad L = \log\frac{1-\delta}{\delta},$$
$$A = \exp\left(\frac{uv + \max(u,v)L}{u+v}\right), \qquad \kappa = \exp\left(-\frac{uv}{u+v}\right).$$

Then for every $M \geq 2$,

$$0 \leq V_M - V_{\text{hist}} \leq 2A\kappa^{\lfloor M/2 \rfloor}. \tag{S1.9}$$

The upper rate is not claimed to be a matching lower rate or an exact formula for intermediate memory values. In the rational case a finite implementation is constructed below, without a claim that its label count is minimal.

Proof. For a one-label row with command-a probability $p \in (0,1)$ put $x = pq_a$ and $y = (1-p)q_b$. Its mean cost per call is $\tau_a x + \tau_b y$, and its model success probabilities are x and y. The costs are $\tau_a + \tau_b y/x$ and $\tau_b + \tau_a x/y$. If $x \geq y$ the second is at least $\tau_a + \tau_b$; if $x \leq y$ the first is. Equality holds at $x = y$, or $p = q_b/(q_a + q_b)$. Endpoint rows are improper in one model. This proves $V_1 = \tau_a + \tau_b$. In particular the normalized case has a common one-label fallback of robust cost two.

Take the interior prior π_* from Lemma S1.1. If u/v is irrational, there is at most one tied pair: two distinct equalities $\pi_* r^i/\tau_a = (1-\pi_*)s^j/\tau_b$ would imply $u(i-i') = v(j-j')$, contrary to irrationality. Thus there is one greedy word, or two which differ only in the order of a single adjacent pair and have an identical suffix. Every such suffix has the irrational frequency $v/(u+v)$ by (S1.3). It is therefore not ultimately periodic.

Suppose some finite-label controller attained the robust history value. Its planned-word law would be Bayes-optimal at π_*, and hence supported on those one or two words. After their possible single tied pair its command suffix would be prescribed with probability one. Lemma S1.2 would make that suffix ultimately periodic, a contradiction. All failure probabilities in this family are strictly positive, so the hypothetical-failure support argument introduces no physical zero-likelihood history. At each fixed M the optimum is attained by Theorem 3.4; therefore the absence of an optimal controller at that M gives the strict inequality in (S1.8), rather than only nonattainment. Its decreasing convergence follows either from Corollary 4.4 or from the explicit construction below.

Suppose instead that u/v is rational. Write $u = h\ell$, $v = k\ell$ with positive integers h, k and $\ell > 0$. For either fixed tie rule, the logarithmic state z_t belongs to the coset $z_0 + \ell\mathbb{Z}$ and to the bounded strip (S1.5). This intersection is finite. The fixed tie rule makes its next state a deterministic function of its present state, so each of the two extreme greedy words is ultimately periodic. An ultimately periodic word has a finite autonomous realization by one label for each position in its finite prefix and cycle.

The two-word robust mixture in Lemma S1.1 also has a finite realization with a counted pure start. Use disjoint finite realizations of the two tails and one initial label. If their mixing weights are λ_j and their first symbols are a_j, the initial command row assigns probability $p_a = \sum_{j:a_j=a} \lambda_j$ to symbol a. On its failure, select tail j with probability λ_j/p_a among those with $a_j = a$; an empty class is never selected. These are allowed command-dependent stochastic updates. Conditional on the sampled command, its failure probability is independent of which such tail will be selected, so this constructs exactly the desired mixture in both models. Its branch choice is stored in counted disjoint tail labels. No freely randomized initial label or persistent external coin is used. The finite implementation proves the rational direction.

For the quantitative approximation, now assume $c_a = q_a, c_b = q_b$, so $\tau_a = \tau_b = 1$. The constants will depend only on r, s. A memoryless row with $p/(1-p) = (q_b/q_a)\sqrt{\pi/(1-\pi)}$ has Bayes cost

$$1 + 2\sqrt{\pi(1-\pi)}.$$

Together with (S1.7) at $\pi = 1/2$ this gives, at every least-favorable interior prior,

$$1 + \frac{m}{2} \le V_{\mathrm{hist}} \le 1 + 2\sqrt{\pi_*(1-\pi_*)}.$$

Consequently $\pi_*(1-\pi_*) \ge m^2/16$ and $\delta \le \pi_* \le 1-\delta$, so $|z_0| \le L$. The strip and the exact survival expressions in the proof of Lemma S1.1 now give, simultaneously for both extreme words and both models,

$$\Pr(\text{first } T \text{ calls all fail}) \le A\kappa^T \qquad (T \ge 0). \tag{S1.10}$$

Follow the robust two-word mixture for at most $T \ge 1$ calls; after failure of its Tth call, switch to the one-label fallback of cost two. The truncated prefix costs at most V_{hist} in either model, and the added expected cost is at most $2A\kappa^T$ by (S1.10). This is an actual counted controller with at most $2T$ labels: one pure initial label, at most $2(T-1)$ disjoint labels for the remaining prefix positions of the two words, and one fallback label. The same command-conditioned initial update used above realizes the mixture. Depth and branch identity are stored in these labels. Thus $V_{2T} \le V_{\mathrm{hist}} + 2A\kappa^T$. Taking $T = \lfloor M/2 \rfloor$ and allowing unused labels proves (S1.9). □

Corollary S1.4 (An effective saturation test for rational primitives)**.** *Suppose $r, s \in \mathbb{Q} \cap (0,1)$. Factor their numerators and denominators into primes, and let $\nu(r), \nu(s)$ be their signed prime-exponent vectors on the union of those primes. The history optimum has a finite-label realization if and only if these vectors are positive rational multiples of one another. This is a finite exact test on the primitive data; it does not search unboundedly over label budgets.*

In the affirmative case one can compute a rational $t \in (0,1)$ and coprime positive integers h, k with $r = t^h$, $s = t^k$. For the normalized costs $c_a = q_a, c_b = q_b$, keep δ from Theorem S1.3, and choose any integer $J \ge 0$ with

$$t^{-J} \ge \frac{1-\delta}{\delta}.$$

Then the explicitly computable label bound

$$M_0 = 1 + 2\big(\max(J,h) + \max(J,k) + 1\big) \tag{S1.11}$$

satisfies $V_{M_0} = V_{\mathrm{hist}}$. The history value and an attaining finite controller are therefore algebraically computable by the fixed-budget procedure of Theorem 3.4. Neither M_0 nor the factorization procedure is asserted to be optimal in its resource use.

Proof. A rational logarithmic ratio is equivalent to $r^k = s^h$ for some positive integers h, k, and unique factorization makes this equivalent to $k\nu(r) = h\nu(s)$. The latter is tested by finitely many integer cross-products. Reducing h, k to coprime integers shows that every coordinate of $\nu(r)$ is divisible by h and every coordinate of $\nu(s)$ by k. Their common quotient vector defines the rational number t with $r = t^h$, $s = t^k$. It lies in $(0,1)$ because r does. Repeated rational multiplication finds an integer J satisfying the displayed inequality.

Put $\ell = -\log t$. The least-favorable logarithmic prior obeys $|z_0| \le L \le J\ell$. The entire orbit of either fixed tie rule lies in its coset $z_0 + \ell\mathbb{Z}$ and in

$$[-\max(J,h)\ell,\ \max(J,k)\ell].$$

There are at most $N = \max(J,h) + \max(J,k) + 1$ points of that coset in this interval. The deterministic orbit, including its transient and cycle, therefore uses at most N states. Each extreme word and each of its tails has a realization with at most N labels. The counted initial-mixture construction in the preceding proof uses at most $1 + 2N = M_0$ labels and attains the robust history value. The fixed-M_0 algebraic synthesis uses only the rational instrument and costs; it does not require an exact evaluation of the logarithmic prior. This proves the effective conclusion. □

Example S1.5 (A rational primitive with no finite saturation)**.** Take $r = 1/2$ and $s = 1/3$. The scalar nonterminal multipliers are $(1/2, 1)$ in model 0 and $(1, 1/3)$ in model 1, and the command costs are $1/2, 2/3$. All instrument and cost data are rational. The ratio $\log 2/\log 3$ is irrational: a

rational ratio would give $2^n = 3^m$ for positive integers m, n. Therefore $V_1 = 2$ and $V_M > V_{\mathrm{hist}}$ at every finite memory, although (S1.9) supplies exponentially accurate finite controllers. Each fixed-M value and an attaining controller are algebraically computable by Theorem 3.4. No transcendence assertion about V_{hist} is needed or made.

This is an exact saturation classification for the stated search-type family, not a classification of arbitrary quantum instruments or general partially observed control. The exchange ordering is inherited from classical search theory. The finite-support rigidity argument is what excludes every stochastic finite-label implementation of its irrational optimal law. The positive approximation bound and the negative exact realization statement address different resource questions; neither replaces the original squared-sign risk frontier or the independent geometric and Theta construction problems.

S2. A permutation normal form with an exact memory charge

Theorem 11.1 identifies when finite memory exists. We now determine the form of every possible finite realization, retaining the cost of its prescribed pure initial label. The distinction between a minimal linear representation and a positive realization remains essential. In particular, the permutation representation carried by the observable space does not by itself provide a decoder with values in the probability simplex.

Let b be a compactly reversible response admitting a finite realization. Write G for its finite observable group, $d = \dim V_b$, and N for its residual cardinality. We use the chronological convention $g(u)g(v) = g(uv)$: the right action on V_b is $f \cdot g(w) = L_w f$, where $L_w f(z) = f(wz)$. Thus G in this notation is the opposite of the group of operators with the usual convention for composition; inversion identifies the two groups. No group cardinality, faithfulness, or representation-theoretic statement depends on this convention. Every element has a representative in A^*, including its inverse, since a finite group generated by the command elements is already generated by their positive words. The value $b(g) = b(w)$ is consequently well defined.

A *probability factorization on a finite G-set* consists of a finite set X, row permutation matrices P_g with $P_g P_h = P_{gh}$, a probability row λ on X, and a row-stochastic matrix D with columns indexed by C, such that

$$b(g) = \lambda P_g D \qquad (g \in G). \tag{S2.1}$$

All rows of D are probability laws, including rows associated with states that are not distinguished by the immediate terminal measurement. Unused orbits may be deleted. Define

$$\kappa_G(b) = \min\{|X| : \text{(S2.1) holds}\}.$$

The residual realization shows that this minimum is finite and at most N. It permits a mixed λ without charging for its preparation; the stochastic memory $m(b)$ continues to charge every live label, beginning at the specified point mass as in (11.1).

Theorem S2.1 (Reversible stochastic normal form)**.** *Every M-label stochastic realization of b gives a faithful finite G-set X and a factorization* (S2.1) *satisfying*

$$|X| + \mathbf{1}_{\{\lambda \text{ is not a point mass}\}} \le M. \tag{S2.2}$$

Conversely, every such factorization has a realization with the displayed number of labels. Its transitions on X are permutations; if λ is mixed, its only additional label initializes the distribution on the first command. Thus arbitrary continuing stochastic dynamics confers no advantage over one initialization mixture followed by a permutation action.

Proof. Fix a realization $(e_1^{\mathsf{T}}, \{Q_a\}, D_0)$, and retain the reachable row space R and the surjection $\pi : R \to V_b$ from the proof of Theorem 11.1. Write

$$\mathcal{L} = \{w \in A^* : g(w) = 1\}.$$

These are identity words in the observable group; their matrices need not be identity matrices on hidden states. Choose finitely many words $\ell_1, \dots, \ell_s \in \mathcal{L}$ whose positive matrix entries cover all directed edges occurring in any Q_ℓ, $\ell \in \mathcal{L}$. There are at most M^2 distinct edges, so $s \le M^2$ is possible. If no nontrivial choice is needed, the empty word suffices. Set

$$T = \frac{I + Q_{\ell_1} + \cdots + Q_{\ell_s}}{s+1}.$$

Its closed communicating classes $C_1, \dots, C_k$ are closed for every Q_ℓ with $\ell \in \mathcal{L}$. The positive diagonal makes the restriction of T to each such class aperiodic. The elementary convergence argument in Lemma S17.1 gives a stochastic idempotent

$$E = \lim_{n\to\infty} T^n, \qquad \Delta_M E = \operatorname{conv}\{\mu_1, \dots, \mu_k\},$$

where μ_i is the unique stationary probability row of T on C_i. Their supports are disjoint, so this image is a simplex with exactly k vertices. Every identity-word matrix preserves each C_i; hence

$$EQ_\ell E = E \qquad (\ell \in \mathcal{L}). \tag{S2.3}$$

This is a sandwiched equality. We do not assert $Q_\ell E = E$, $EQ_\ell = E$, or common stationarity of the μ_i under all identity words.

Each T^n is a convex combination of identity-word matrices. Consequently $RE \subseteq R$ and $\pi(rE) = \pi(r)$ for $r \in R$. For a word u, put $F_u = EQ_uE$, considered as a stochastic map of $\Delta_M E$ to itself. Choose a word $\bar{u}$ representing $g(u)^{-1}$. Replacing the middle E by the convex combinations T^n and using $u\ell\bar{u} \in \mathcal{L}$ gives

$$F_u F_{\bar{u}} = E = F_{\bar{u}} F_u.$$

Thus F_u is an affine automorphism of that simplex. Such an automorphism permutes its vertices, because both it and its inverse preserve the simplex and therefore preserve extremality. If $g(u) = g(v)$, the same inverse word shows $F_u F_{\bar{u}} = F_v F_{\bar{u}} = E$, so $F_u = F_v$. Next, $EQ_uT^nQ_vE$ is a convex combination of $F_{u\ell v}$, all equal to F_{uv}. Passing to the limit proves $F_uF_v = F_{uv}$. We have obtained a permutation action P_g on the vertices $X = \{\mu_1, \dots, \mu_k\}$.

This action is faithful. If F_u is the identity on $\Delta_M E$, then $rEQ_uE = rE$ for all rows r. For $r \in R$, application of π shows $L_u\pi(r) = \pi(r)$. Surjectivity gives $L_u = I$ on V_b, and so $g(u) = 1$. Write $e_1^{\mathsf{T}} E = \sum_i \lambda_i \mu_i$ and set $D(i, \cdot) = \mu_i D_0$. Intertwining on R gives, for every word w,

$$\lambda P_{g(w)} D = e_1^{\mathsf{T}} EQ_w E D_0 = b(w),$$

including the empty word. If the original initial label lies in C_i, its row under E is exactly μ_i. A mixed λ therefore requires that the initial label lie outside the union of the closed classes. The k nonempty disjoint classes then use at most $M-1$ labels. This proves (S2.2); a pure λ only requires $k \le M$.

For the converse, when λ is a point mass its support point is the initial label in X. If it is mixed, adjoin a label $\star$ with decoder $D(\star, \cdot) = \lambda D$. On command a, its transition row into X is $\lambda P_{g(a)}$, and no transition returns to $\star$. The rows on X are those of $P_{g(a)}$. Induction gives (S2.1) for every nonempty word, and the decoder at $\star$ gives the empty word. The first input is applied during initialization, so it is not lost or encoded in an uncounted preparatory clock. □

The general structure of nonnegative idempotents and bounded groups of nonnegative matrices is classical; see Brown [60], Schwarz [61], and Flor [62]. The proof above supplies the additional steps required for an arbitrary semigroup realization: the identity-word projection is constructed before passing to the observable quotient, the sandwiched action is proved to factor through that quotient, and the prescribed pure start is charged. It does not assume that the original stochastic matrices themselves form a group.

Theorem S2.2 (Exact memory formula)**.** *With randomized terminal readout allowed and the initial label prescribed pure, the minimum stochastic memory is*

$$m(b) = \min_{(X,\lambda,D)} \Big(|X| + \mathbf{1}_{\{\lambda \text{ is not a point mass}\}} \Big) = \min\{N, 1 + \kappa_G(b)\}. \tag{S2.4}$$

The minimum ranges over precisely the probability factorizations (S2.1)*, not arbitrary linear factorizations.*

Proof. The first equality is both directions of Theorem S2.1. A point-mass factorization gives a deterministic-transition realization on X, so $|X| \geq N$ by Theorem 11.1. A mixed factorization costs at least $1 + \kappa_G(b)$. Conversely, the residual machine has N labels, and a factorization attaining $\kappa_G(b)$ costs at most $1 + \kappa_G(b)$. These inequalities give the second equality, also when a minimum factorization happens to have a pure start. □

Corollary S2.3 (Sharp group bounds)**.** *An M-label realization of a compactly reversible response satisfies*

$$|G| \leq M!, \qquad M < N \implies |G| \leq (M-1)!.$$

Both bounds are attained by families below. In particular, a stochastic memory saving over the residual count requires a faithful permutation core on at most $M-1$ points.

Proof. The core action in Theorem S2.1 embeds G into S_k. If $M < N$, its start cannot be pure, since that would produce a deterministic machine with at most M labels. Thus $k \leq M-1$ in this case. □

Let $\nu_G(V_b)$ be the least cardinality of a finite G-set whose real permutation module contains V_b, and let $\mu(G)$ be the least cardinality of a faithful finite G-set. All references to V_b here use the right-action convention already fixed. A probability factorization gives V_b as a subquotient of $\mathbb{R}^X$: take the cyclic span of λ, then its observable quotient. Averaging an inner product over finite G splits every invariant subspace orthogonally, so this subquotient embeds in $\mathbb{R}^X$. We obtain

$$d \leq \nu_G(V_b) \leq \kappa_G(b), \qquad m(b) \geq \max\{d, \mu(G), \min(N, 1 + \nu_G(V_b))\}. \tag{S2.5}$$

These are lower bounds, not an identification of a linear module with a bounded decoder. Proposition S2.6 makes the distinction strict.

Corollary S2.4 (Rank rigidity)**.** *For a finite compactly reversible response,*

$$m(b) = d \quad \iff \quad N = d.$$

In that case every realization with d labels has permutation transition matrices. If $N > d$, then $m(b) \geq d + 1$.

Proof. Every M-label realization has $d \leq M$. If $M = d$, the core in Theorem S2.1 has $k \geq d$ by its linear rank, hence $k = d$ and its start is pure. Thus $N \leq d$, whereas $N \geq d$ because the residuals span V_b. Conversely $N = d$ gives a deterministic machine with d labels. For the assertion about the original matrices, a d-label realization has $R = \mathbb{R}^d$ and the observable map π is an isomorphism. An inverse word for $g(a)$ therefore gives a stochastic inverse of Q_a on all of $\mathbb{R}^d$. A stochastic map with a stochastic inverse is an affine automorphism of the full simplex and hence a permutation. □

S2.1. **Two commands and a factorial deterministic gap.** Let $k \geq 3$, let P_g be the natural row permutation action of S_k, choose a nonconstant $\lambda \in \Delta_k$, and choose a nonconstant column $t \in [0,1]^k$. Two commands are the k-cycle $(1\,2\,\cdots\,k)$ and the transposition $(1\,2)$; these generate S_k. Supply the binary response

$$b(g) = (\lambda P_g t,\ 1 - \lambda P_g t). \tag{S2.6}$$

If the distinct coordinate values of λ have multiplicities $n_1, \ldots, n_j$, put $N_\lambda = k!/(n_1! \cdots n_j!)$.

Proposition S2.5 (Exact symmetric-group family)**.** *The response* (S2.6) *has observable group S_k, dimension k, residual cardinality N_λ, and*

$$\kappa_{S_k}(b) = k, \qquad m(b) = \min\{N_\lambda, k+1\}.$$

When the coordinates of λ are pairwise distinct, the least stochastic memory is $k+1$ and the least deterministic-transition memory is $k!$. For a point-mass λ, both are k. If either λ or t is constant, the response instead has one residual and one-label memory.

Proof. For any nonconstant vector, differences between it and its transposes yield a nonzero multiple of some $e_i - e_j$. Conjugating by all permutations shows that the span of its orbit differences is $\mathbf{1}^\perp$. The orbit of λ therefore spans $\mathbb{R}^k$, because every row has mass one. The orbit of t also spans $\mathbb{R}^k$: its differences span $\mathbf{1}^\perp$, and its coordinate sum is positive, since a nonconstant nonnegative vector is nonzero. Consequently the map

$$r \longmapsto \big(h \longmapsto (rP_h t,\, r\mathbf{1} - rP_h t)\big)$$

is injective on $\mathbb{R}^k$. It identifies the reachable row space with V_b, and identifies the observable action with the faithful natural permutation action. Its residuals are precisely the distinct permutations of λ, giving $d = k$ and $N = N_\lambda$. The displayed factorization and the rank lower bound give $\kappa = k$. The memory formula proves the remaining assertion; it also shows $N_\lambda \geq k$. The two constant cases are immediate from (S2.6). □

This is an exact prepared quantum experiment in every dimension k: take $\rho = \mathrm{diag}(\lambda)$, permutation unitaries for the commands, and terminal effect $E = \mathrm{diag}(t)$. The construction already lies in the diagonal, classical subalgebra. It demonstrates a stochastic versus deterministic memory separation, and does not assert a quantum advantage for this family. It also proves sharpness of Corollary S2.3: the point-mass family has $M = k$ and $|G| = k!$, while pairwise distinct coordinates have $M = k + 1 < N = k!$ for $k \geq 3$.

For a concrete rational example, take $k = 3$, $\lambda = (1/2, 1/3, 1/6)$, and $t = (0, 1/4, 1)^\mathsf{T}$. The six first-outcome values are

$$\left\{\frac{1}{4}, \frac{3}{8}, \frac{7}{24}, \frac{13}{24}, \frac{11}{24}, \frac{7}{12}\right\}.$$

It has $d = 3$, $N = 6$, $\kappa = 3$, and $m = 4$.

S2.2. **The decoder constraint can remove the entire saving.** The representation-theoretic lower bound in (S2.5) need not attain the positive factorization minimum, even for the same group, rank, and residual count as this example.

Proposition S2.6 (A strict bounded-decoder obstruction)**.** *Let $u = (1, -1, 0)$ and let S_3 act by its natural permutation matrices. The binary response with first coordinate*

$$\beta(g) = \frac{1}{2} + \frac{1}{4} u P_g u^\mathsf{T} \tag{S2.7}$$

has $d = 3$, $N = 6$, and $\nu_{S_3}(V_b) = 3$, but

$$\kappa_{S_3}(b) = m(b) = 6.$$

It is realized by a prepared qubit with

$$U_0 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad U_1 = \frac{1}{2}\begin{pmatrix} 1 & \sqrt{3} \\ \sqrt{3} & -1 \end{pmatrix}, \qquad \rho = E = |+\rangle\langle +|.$$

Thus the least stochastic memory can differ for responses having the same observable group, module, and residual cardinality.

Proof. The orbit of u consists of its six distinct permutations and spans the two-dimensional zero-sum subspace. Pairing with all permutations of u is nondegenerate on that subspace. Including the constant outcome sum gives dimension three and six distinct residuals. This representation is the natural permutation module $\mathbb{R}^3$, and the rank bound gives $\nu = 3$.

Write $s = (1\,2)$ and $r = (1\,2\,3)$. Directly from (S2.7),

$$\beta(1) = 1, \quad \beta(s) = 0, \quad \beta(r) = \beta(r^2) = \frac{1}{4}, \quad \beta(\tau) = \frac{3}{4} \quad \text{for the other two transpositions } \tau.$$

Suppose a probability factorization exists on at most five points. Its action is faithful, by the same residual argument as in Theorem S2.1. Transitive S_3-sets have sizes $1, 2, 3, 6$, since their stabilizers have orders $6, 3, 2, 1$; the subgroups of order two are conjugate and the subgroup of order three is unique. Thus our faithful action has one natural three-point orbit and, at most, one two-point sign orbit together with fixed points. Two three-point orbits already exceed the budget, and actions on only fixed and sign orbits annihilate the alternating subgroup.

Let a be any point of positive initial mass and write q for the first-coordinate decoder. The endpoint values force $q(a) = 1$ and $q(a \cdot s) = 0$. No fixed point can carry mass. Within the natural orbit, the point fixed by s cannot carry mass, and both of the points exchanged by s cannot carry positive mass simultaneously. Thus the initial mass in that orbit is concentrated at one moved point a; denote its total by A. If a sign orbit is present its mass is concentrated at a single point; write this total as B. We have $A + B = 1$. Every even permutation has decoder one on the mass in the sign orbit, and every odd permutation has decoder zero there.

The images $a \cdot r$ and $a \cdot r^2$ are the other two points in the natural orbit. One is $a \cdot s$, with decoder zero, and the other's decoder is some $t \in [0, 1]$. The two equal values $\beta(r) = \beta(r^2) = 1/4$ give $B = 1/4$ and $At = 0$. Hence $A = 3/4$ and $t = 0$. Of the two transpositions different from s, one fixes a and the other sends it to the third point. Their first-outcome values would therefore be A and zero, whereas both must be $3/4$. This contradiction rules out all factorizations on at most five points. The regular six-point residual machine gives $\kappa \le 6$, so $\kappa = 6$ and Theorem S2.2 gives $m = 6$.

For the physical realization, the adjoint actions of U_0 and U_1 on the real Bloch plane are two reflections whose product is a rotation by $2\pi/3$. The initial and terminal Bloch vector is the x-axis vector. The first reflection sends it to its negative, the other two reflections have inner product $1/2$ with it, and the two nonidentity rotations have inner product $-1/2$. The Born probability is one half of one plus that inner product. It has exactly the displayed six group values, so it is (S2.7) after the specified generator assignment. □

The cyclic partition formula of the preceding revision is recovered as a special case in Proposition S17.2. For the general finite group, however, both the start charge and the bounded-decoder constraint belong in the exact minimization problem. Neither can be removed by a rank calculation.

S3. Exact memory under terminal readout contrast

The normal form separates the cost of a permutation module from the positivity of its preparation and decoder. We now determine when that linear lower bound is attained, for every finite observable group, and describe the exact memory transitions as the terminal contrast varies. This is a change of terminal readout. It differs from the length-dependent background damping of Section S12.

Let b be a nonconstant compactly reversible response with finite observable group G, residual cardinality N, and real observable module V_b. Set

$$\pi = \frac{1}{|G|} \sum_{g \in G} b(g), \qquad b_t(g) = \pi + t\{b(g) - \pi\}, \qquad 0 \le t \le 1. \tag{S3.1}$$

If $\pi_c = 0$, nonnegativity implies $b(g, c) = 0$ for every g; such an output can be removed and later restored as a zero decoder column. Thus positivity of all remaining π_c imposes no additional

hypothesis. For prepared unitary commands, (S3.1) is realized by the terminal measurement $E_c(t) = tE_c + (1-t)\pi_c I$, with the preparation and commands unchanged.

Theorem S3.1 (Attainment of the module bound at small contrast)**.** *For every finite observable group G there is an explicitly positive t_0 such that*

$$\kappa_G(b_t) = \nu_G(V_b), \qquad m(b_t) = \min\{N, 1 + \nu_G(V_b)\} \qquad (0 < t \le t_0). \tag{S3.2}$$

For every $t > 0$, the observable group, module and residual cardinality agree with those of b. At $t = 0$ the minimum is one label. The assertion allows arbitrary finite groups and arbitrary finite terminal outcome sets.

Proof. Every $f \in V_b$ has a suffix-independent total mass $\ell(f) = \sum_c f(z, c)$, because this holds for its spanning residuals. Group averaging sends every residual to the same constant response $\bar{b}(z) = \pi$. Its image on V_b is therefore $\mathbb{R}\bar{b}$, and

$$V_b = \mathbb{R}\bar{b} \oplus W, \qquad W = \ker\left(\frac{1}{|G|}\sum_{g\in G} L_g\right).$$

The module W has no invariant vector, and $\ell(W) = 0$.

Choose a G-set X of minimum size $k = \nu_G(V_b)$ and an equivariant injection $j : W \to \mathbb{R}^X$. The permutation inner product is invariant. Since group averaging commutes with j, its image is orthogonal to the whole invariant subspace, including the uniform row $p = (1/k, \dots, 1/k)$. In particular its vectors have coordinate sum zero. Let $F : \mathbb{R}^X \to W$ be orthogonal projection onto $j(W)$ followed by j^{-1}. It is equivariant, $Fj = I_W$, and $F(p) = 0$.

Put $v = b - \bar{b}$, $a = j(v)$, and

$$T(x, c) = F(e_x)(\varnothing, c).$$

The rows of T sum to zero, $pT = 0$, and equivariance gives $aP_gT = b(g) - \pi$. Since $a \neq 0$, we may take

$$\epsilon = \frac{1}{2k\|a\|_\infty}, \qquad B = \max_{x,c}\frac{|T(x, c)|}{\pi_c} > 0, \qquad t_0 = \min\{1, \epsilon/B\}.$$

The row $\lambda = p + \epsilon a$ is a probability row with every entry at least $1/(2k)$. Since $(t/\epsilon)B \le 1$, for $0 \le t \le t_0$ the matrix

$$D_t(x, c) = \pi_c + (t/\epsilon)T(x, c)$$

is row-stochastic. Direct multiplication yields

$$\lambda P_g D_t = \pi + (t/\epsilon)\{pP_gT + \epsilon aP_gT\} = b_t(g).$$

This proves the upper $\kappa_G(b_t) \le k$ with a bounded decoder.

The linear map $f \mapsto tf + (1-t)\ell(f)\bar{b}$ is equivariant and invertible for $t > 0$: it is the identity on $\mathbb{R}\bar{b}$ and multiplication by t on W. It sends each residual of b to the corresponding residual of b_t. It preserves the group kernel and the equality pattern among residuals. The module lower bound (S2.5) now gives equality, and Theorem S2.2 charges the pure start. At zero contrast the decoder is constant. □

The construction is an equivariant positivity argument. Symmetric cone factorizations and coordinate-symmetric lifts have an established theory [67, 68]; ordinary nonnegative rank also has classical barycentric perturbation results [69]. The assertion here is the exact value $\nu_G(V_b)$ while preserving one common action for every command word and the prescribed-start charge. It is not a claim that positivity under mixing or equivariant factorization was previously unknown.

There is a useful distinction at an infinite compact observable group. The same invertible change of residuals preserves that group for every $t > 0$, so Theorem 11.1 still excludes every finite exact stochastic realization. Thus making the terminal probabilities strictly interior need not make exact

memory finite. Terminal readout contrast, repeated channel damping, and perturbations of an entire instrument are different operations.

Theorem S3.2 (Attained exact contrast thresholds)**.** *For each integer $k \geq 1$ there is $\tau_k \in [0,1]$ such that*

$$\kappa_G(b_t) \leq k \quad \Longleftrightarrow \quad 0 \leq t \leq \tau_k. \tag{S3.3}$$

These thresholds are nondecreasing in k, satisfy $\tau_N = 1$, and obey

$$k < \nu_G(V_b) \Longrightarrow \tau_k = 0, \qquad \tau_{\nu_G(V_b)} > 0.$$

The lower memory value is retained at every threshold itself. There are only finitely many positive-contrast transitions. If all $b(g,c)$ are real algebraic, all thresholds are effectively computable real algebraic numbers. When all subgroups of G are normal, they are obtained by finitely many linear programs; rational tables then have rational thresholds and rational optimal controllers.

Proof. For a fixed action on at most k points, the variables λ, D, t lie in a compact product of probability simplexes and $[0,1]$. The equations $\lambda P_g D = b_t(g)$ define a closed set, so its feasible contrasts form a compact set. If $s > 0$ is feasible, the same λ and the decoder $(t/s)D + (1 - t/s)\mathbf{1}\pi$ make every $0 \leq t \leq s$ feasible. There are only finitely many actions on at most k labelled points; their feasible sets therefore form a closed initial interval. A single fixed point realizes b_0, including when no positive contrast is feasible.

The residual machine gives $\kappa_G(b_t) \leq N$; the remaining inequalities follow from Theorem S3.1 and its module lower bound. The exact memory is recovered from $m(b_t) = \min\{N, 1 + \kappa_G(b_t)\}$ for $t > 0$. For algebraic data the equations for each action are finite polynomial constraints over the real algebraic numbers. Classical real quantifier elimination computes their projected interval and endpoint, as in Section S11. This is a proved finite procedure, not a reported general quantifier-elimination execution. The final assertion is the linear reduction and rationality proved in Theorems S5.2 and S5.3 below. □

S3.1. **A sharp transition for the retained prepared-qubit experiment.** The bounded-decoder obstruction of Proposition S2.6 has a complete contrast law. Let $u = (1, -1, 0)$ and

$$\beta_t(g) = \frac{1}{2} + \frac{t}{4} u P_g u^{\mathsf{T}}, \qquad g \in S_3,$$

and let $b_t = (\beta_t, 1 - \beta_t)$. This is the same prepared-qubit experiment with terminal effect $t|+\rangle\langle+| + (1-t)I/2$.

Proposition S3.3 (An exact qubit readout threshold)**.** *For $0 < t \leq 1$, this response has $d = 3$, $N = 6$, and*

$$\kappa_{S_3}(b_t) = \begin{cases} 3, & 0 < t \leq 2/3, \\ 6, & 2/3 < t \leq 1, \end{cases} \qquad m(b_t) = \begin{cases} 4, & 0 < t \leq 2/3, \\ 6, & 2/3 < t \leq 1. \end{cases}$$

At $t = 0$ its memory is one.

Proof. The group, rank and residual count are unchanged for $t > 0$ by Theorem S3.1. A faithful S_3-set of size at most five has exactly one natural three-point orbit and otherwise only fixed points or one two-point sign orbit, as in the proof of Proposition S2.6. Let A be the initial mass on its natural orbit. Write its unnormalized initial row and scalar decoder as

$$p = (A/3)\mathbf{1}^{\mathsf{T}} + r, \qquad D = q\mathbf{1} + s, \qquad r\mathbf{1} = 0, \quad \mathbf{1}^{\mathsf{T}} s = 0.$$

The contribution is $Aq + rP_g s$. All remaining orbits contribute a linear combination of the constant and sign functions. The functions $rP_g s$ and $uP_g u^{\mathsf{T}}$ are orthogonal to both: their group sums vanish because $|S_3|^{-1} \sum_g P_g = \mathbf{1}\mathbf{1}^{\mathsf{T}}/3$, and $\sum_g \operatorname{sgn}(g) P_g = 0$ entrywise. Consequently equality of the full responses implies

$$rP_g s = (t/4) u P_g u^{\mathsf{T}} \qquad (g \in S_3). \tag{S3.4}$$

Here no unproved matrix-coefficient injectivity is needed. On the row plane $\mathbf{1}^{\perp}$, use the basis $u, (1,1,-2)$. The matrices of the identity, (12), (23) and the chronological cycle (123) are respectively

$$I_2, \quad \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 3 & -1 \end{pmatrix}, \quad \frac{1}{2}\begin{pmatrix} -1 & 1 \\ -3 & -1 \end{pmatrix}.$$

They span all real two-by-two matrices: the first two span the diagonal ones, and the latter two have independent off-diagonal parts. Therefore (S3.4) holds with every endomorphism of this plane in place of P_g. Equality of the associated rank-one bilinear functionals, with $t > 0$, forces $r = au$, $s = bu^{\mathsf{T}}$ and $ab = t/4$. Nonnegativity of p gives $|a| \le A/3$, while the bounded decoder gives $|b| \le \min(q, 1-q) \le 1/2$. Hence

$$t/4 = |ab| \le A/6 \le 1/6.$$

No core of size at most five is possible above $t = 2/3$.

For $0 < t \le 2/3$ use the natural three-point action with $\lambda = (2/3, 0, 1/3)$ and scalar decoder $D = \frac{1}{2}\mathbf{1} + (3t/4)u^{\mathsf{T}}$. Its entries lie in $[0,1]$ and direct multiplication gives β_t; the rank bound proves $\kappa = 3$. Above the threshold the six-residual machine supplies the matching upper. The pure-start formula gives the stated minima and the constant case is immediate. □

S4. Orbit multiplicity, positive synthesis, and the convexity boundary

The quotient reduction of Section S5 has a precise boundary. A single transitive orbit admits convexification without enlarging the orbit exactly when its stabilizer is normal. For general finite groups, irreducible multiplicities measure the obstruction and determine the exact number of prescribed orbit copies at sufficiently small positive contrast. All representations below are finite dimensional. We use classical complete reducibility, matrix-coefficient orthogonality, and Frobenius reciprocity [75]; their representation-theoretic content is not claimed as new.

Let G be a finite group, not necessarily the faithful observable group of the particular table under discussion. For a nonempty finite right G-set X, let $\mathcal{R}_X(C)$ denote the probability tables

$$b(g) = \lambda P_g D, \qquad \lambda \in \Delta(X), \quad D(x,\cdot) \in \Delta(C),$$

where $P_g P_h = P_{gh}$. Thus the same preparation and decoder are used for every group element. Taking an observable quotient does not change the table; keeping G explicit lets us compare different allowed actions. For this ambient-group formulation, $\kappa_G(b)$ denotes the minimum $|X|$ over all such factorizations; an action need not be faithful unless G is the observable group. Write $\widehat{G}$ for the complex irreducible representations, choose a unitary representative ρ of degree d_ρ, and define

$$\widehat{b}_c(\rho) = \frac{1}{|G|}\sum_{g\in G} b(g,c)\rho(g^{-1}), \qquad A_\rho(b) = \begin{pmatrix} \widehat{b}_{c_1}(\rho) \\ \vdots \\ \widehat{b}_{c_{|C|}}(\rho) \end{pmatrix}, \qquad r_\rho(b) = \operatorname{rank}_{\mathbb{C}} A_\rho(b). \tag{S4.1}$$

Changing unitary bases does not change these ranks.

Lemma S4.1 (Fourier multiplicity and real descent)**.** *Let V_b be the real span of the translates $b^h(g) = b(hg)$. The multiplicity of the right irreducible ρ in $V_b \otimes_{\mathbb{R}} \mathbb{C}$ is $r_\rho(b)$. In particular $r_{\mathbf{1}}(b) = 1$ and $r_\rho(b) \le d_\rho$. If U and V are real G-modules, then U embeds equivariantly in V if and only if every complex irreducible multiplicity in $U \otimes \mathbb{C}$ is at most the corresponding one in $V \otimes \mathbb{C}$.*

Proof. The Fourier transform sends b^h to the blocks $A_\rho(b)\rho(h)$. Matrix-coefficient orthogonality identifies $\mathbb{C}[G]$ with $\bigoplus_{\rho\in\widehat{G}} M_{d_\rho}(\mathbb{C})$: the group algebra can independently prescribe an arbitrary matrix on each irreducible block. Consequently the ρ-part of the cyclic module generated by b identifies with

$$\{A_\rho(b)M : M \in M_{d_\rho}(\mathbb{C})\}.$$

If A_ρ has rank r, a choice of a basis of its column space identifies this right module with r independent rows of length d_ρ. It therefore consists of r copies of ρ. For the trivial representation the stacked block is the nonzero probability vector $\pi = |G|^{-1}\sum_g b(g)$, so its rank is one.

Necessity of the second assertion follows by complexification. For sufficiency, complete reducibility over $\mathbb{C}$ and the multiplicity inequalities give a complex equivariant injection $F : U \otimes \mathbb{C} \to V \otimes \mathbb{C}$. In real bases write $F = A + iB$. Both A and B are real intertwiners. A maximal nonzero minor of F is the value at $z = i$ of a real polynomial minor of $A + zB$. That polynomial is not identically zero, so some real z makes this minor nonzero. The real map $A + zB$ is the required injection. One may choose z among $0, 1, \ldots, \dim U$, since the nonzero polynomial has degree at most $\dim U$. This proves the assertion without excluding complex or quaternionic representation types. □

For $H \le G$ put

$$m_\rho(H) = \dim_{\mathbb{C}} \rho^H = \operatorname{rank}\left(\frac{1}{|H|}\sum_{h\in H}\rho(h)\right).$$

By Frobenius reciprocity these are the multiplicities of $\mathbb{C}[H\backslash G]$. Let $q_H(b)$ be the least positive integer q for which b belongs to $\mathcal{R}_{q(H\backslash G)}(C)$, where $q(H\backslash G)$ is the disjoint union of q copies; set $q_H(b) = \infty$ if no such factorization exists. This is an action-restricted resource. It is not $\kappa_G(b)/[G : H]$ in general, since an unrestricted optimal core may use other orbit types.

Theorem S4.2 (Exact orbit multiplicity at small contrast)**.** *Let $b : G \to \Delta(C)$, let $\pi = |G|^{-1}\sum_g b(g)$, and set $b_t = \pi + t(b - \pi)$ for $0 \le t \le 1$. For $H \le G$ define*

$$Q_H(b) = \max_{\rho\in\widehat{G}}\left\lceil \frac{r_\rho(b)}{m_\rho(H)}\right\rceil, \tag{S4.2}$$

with a positive numerator and zero denominator interpreted as ∞ and $0/0$ as zero. For every $t > 0$, $q_H(b_t) \ge Q_H(b)$. If $Q_H(b) < \infty$, an explicitly positive t_0 exists such that

$$q_H(b_t) = Q_H(b) \qquad (0 < t \le t_0).$$

If $Q_H(b) = \infty$, no positive contrast is realizable using any finite number of copies of this orbit. At zero contrast one copy suffices.

More generally, choose one representative of each subgroup conjugacy class. The unrestricted permutation-module bound has the exact finite integer formulation

$$\nu_G(V_b) = \min\left\{\sum_H q_H[G : H] :\ q_H \in \mathbb{Z}_{\ge 0},\quad \sum_H q_H m_\rho(H) \ge r_\rho(b)\quad (\rho \in \widehat{G})\right\}. \tag{S4.3}$$

Consequently this integer minimum equals $\kappa_G(b_t)$ for every sufficiently small positive contrast, with the pure-start charge given by Theorem S2.2 when G is the observable group.

Proof. A factorization through a finite G-set Y makes V_{b_t} a quotient of a submodule of $\mathbb{R}^Y$. Averaging an inner product over the finite group makes every invariant subspace have an invariant orthogonal complement; hence this subquotient embeds into $\mathbb{R}^Y$. The contrast change is the identity on the invariant line generated by π and multiplication by t on the zero-average part, so for $t > 0$ its module is isomorphic to V_b. On $q(H\backslash G)$ the available complex multiplicities are $qm_\rho(H)$, proving the lower bound and the assertion when an active type is absent.

For the matching construction, let Y be any nonempty finite G-set whose complex multiplicities dominate $r_\rho(b)$. The lemma gives a real equivariant injection of V_b into $\mathbb{R}^Y$. Here are the positivity details, to make clear that no complex factorization is being substituted for a stochastic one. Delete the output columns with $\pi_c = 0$; nonnegativity makes them identically zero, and they can be restored afterwards. Decompose

$$V_b = \mathbb{R}\bar{b} \oplus W, \qquad \bar{b}(g) = \pi,$$

where W is the kernel of group averaging. Every element of W has zero total output mass. Put $k = |Y|$, let p be its uniform probability row, and choose an equivariant injection $j : W \to \mathbb{R}^Y$. The image of j is orthogonal to all invariant vectors. Let $F : \mathbb{R}^Y \to W$ be orthogonal projection onto $j(W)$ followed by j^{-1}. Then $Fj = I_W$ and $F(p) = 0$. Set

$$a = j(b - \bar{b}), \qquad T(y, c) = F(e_y)(e, c).$$

Equivariance gives $aP_gT = b(g) - \pi$, while $pT = 0$ and each row of T sums to zero. If b is nonconstant, choose

$$\epsilon = \frac{1}{2k\|a\|_\infty}, \qquad B = \max_{y,c} \frac{|T(y, c)|}{\pi_c}, \qquad t_0 = \min\{1, \epsilon/B\}.$$

The probability row $\lambda = p + \epsilon a$ and the row-stochastic decoder $D_t(y, c) = \pi_c + (t/\epsilon)T(y, c)$ then satisfy $\lambda P_g D_t = b_t(g)$ for $0 \le t \le t_0$. For a constant table the constant decoder works at every contrast. This is precisely the retained positive construction, applied to the specified action rather than only to a globally minimum one. Taking $Y = Q_H(b)(H\backslash G)$ proves equality in the copy formula.

Every finite G-set is a disjoint union of coset orbits, and each such orbit contributes the multiplicities $m_\rho(H)$. The real descent lemma therefore makes the displayed integer constraints equivalent to real module containment. Conversely every feasible integer vector defines that disjoint-union action. This proves (S4.3); Theorem S3.1 or the preceding construction gives its positive realization. The regular orbit makes the minimum finite, since $m_\rho(\{e\}) = d_\rho \ge r_\rho(b)$. □

Theorem S4.3 (Normality is the exact single-orbit convexity boundary)**.** *For $H \le G$ and the transitive right action $X = H\backslash G$, the following conditions are equivalent:*

(1) *H is normal in G;*
(2) *$\mathcal{R}_X(\{0, 1\})$ is convex;*
(3) *$\mathcal{R}_X(C)$ is convex for every finite outcome set C.*

If H is normal, this response set is exactly the polytope of probability tables constant on the left cosets of H. If H is not normal, nonconvexity occurs in every neighborhood of the interior constant binary table $(1/d, 1 - 1/d)$, where $d = [G : H]$.

An explicit witness is supplied by the return tables

$$f_x(g) = \mathbf{1}_{\{xg=x\}}, \qquad x \in X, \qquad F_X(g) = \frac{1}{d} \sum_{x \in X} f_x(g) = \frac{\chi_X(g)}{d}.$$

Every $(f_x, 1 - f_x)$ lies in $\mathcal{R}_X(\{0, 1\})$, whereas $(F_X, 1 - F_X)$ does not when H is not normal. The same conclusion holds after replacing every scalar f by $d^{-1} + t(f - d^{-1})$, for every $0 < t \le 1$. More quantitatively, for any irreducible ρ with $0 < m_\rho(H) < d_\rho$, put $F_{X,t} = d^{-1} + t(F_X - d^{-1})$. Then

$$\inf_{(a,1-a) \in \mathcal{R}_X(\{0,1\})} \max_{g \in G} |F_{X,t}(g) - a(g)| \;\ge\; \frac{t\, m_\rho(H)}{d\, d_\rho} > 0. \tag{S4.4}$$

This is a certified separation margin; optimality of its constant is not asserted.

Proof. If H is normal, for $b(g, c) = \sum_{Hx} \lambda(Hx) D(Hxg, c)$ one has $b(hg, c) = b(g, c)$ for every $h \in H$: normality gives $Hxh = Hx$. Conversely every probability table with this left invariance is realized by the point preparation H and decoder $D(Hg, c) = b(g, c)$. This proves the stated polytope identity and (i)⇒(iii)⇒(ii).

Suppose H is not normal. The group-algebra element $e_H = |H|^{-1} \sum_{h \in H} h$ is an orthogonal projection in every unitary irreducible representation, of rank $m_\rho(H)$. If every one of these ranks were either zero or d_ρ, each matrix $\rho(e_H)$ would be zero or the identity. The faithful decomposition of the complex group algebra into its irreducible matrix blocks would then make e_H central. Its coefficients are constant under conjugation exactly when H is normal. Hence some irreducible ρ has

$$0 < m_\rho(H) < d_\rho.$$

This irreducible is nontrivial.

The return table at x is realized by initial point x and the indicator decoder of x. Since $\chi_X = \sum_\sigma m_\sigma(H)\chi_\sigma$, orthogonality gives

$$\widehat{F}_X(\rho) = \frac{m_\rho(H)}{d\, d_\rho} I_{d_\rho}.$$

Its rank is $d_\rho > m_\rho(H)$. Lemma S4.1 and the module lower bound therefore exclude a single-orbit factorization. This average of members of the response set proves nonconvexity. For positive contrast, the nontrivial Fourier block is multiplied by t and keeps its rank. Each contrasted return table is still realized on its original orbit by replacing its decoder with $d^{-1} + t(D - d^{-1})$. Letting t decrease to zero proves the neighborhood assertion. Thus (ii) implies (i).

For the quantitative assertion let $A = \widehat{a}(\rho)$ for any single-orbit response. It has rank at most $m_\rho(H) < d_\rho$, so there is a unit vector v with $Av = 0$. The Fourier block of $F_{X,t}$ is αI, where $\alpha = t\, m_\rho(H)/(d\, d_\rho)$. Therefore $\|\widehat{F}_{X,t}(\rho) - A\|_{\mathrm{op}} \geq \alpha$. Because every $\rho(g)$ is unitary, the definition of the normalized Fourier transform also gives

$$\|\widehat{F}_{X,t}(\rho) - \widehat{a}(\rho)\|_{\mathrm{op}} \leq \frac{1}{|G|} \sum_{g \in G} |F_{X,t}(g) - a(g)| \leq \max_{g \in G} |F_{X,t}(g) - a(g)|.$$

Combining these inequalities proves the margin for every allowed probability preparation and bounded decoder. □

For this witness the exact small-contrast number of copies is

$$Q_H(F_X, 1 - F_X) = \max_{\rho : m_\rho(H) > 0} \left\lceil \frac{d_\rho}{m_\rho(H)} \right\rceil. \tag{S4.5}$$

Indeed every active Fourier block of the character is full rank, including the trivial block. Thus nonnormality entails a strict copy cost at arbitrarily small positive contrast, with a matching positive realization. This obstruction is not caused by a decoder touching the boundary of its simplex.

Lemma S4.4 (Full-contrast return-law rigidity)**.** *Let X be a transitive d-point G-set. If a scalar response is a convex combination of responses on copies of X, has value one at e, and has group average $1/d$, every positive-weight component uses a point preparation and its point indicator decoder. In particular every component is a return law f_x.*

Proof. For a component write $\lambda \in \Delta(X)$ and $D \in [0,1]^X$. The total value one at e forces $\lambda D = 1$ for every component of positive weight, hence $D(x) = 1$ on the nonempty support of λ. Transitivity gives its group average $d^{-1} \sum_x D(x) \geq 1/d$. Equality of the weighted mean with $1/d$ forces equality for each component. Consequently $\sum_x D(x) = 1$, so the support of λ consists of exactly one point and D is its indicator. □

These statements separate three matters: the classical multiplicities of a permutation representation, the positivity of its preparation and decoder, and the cost of an externally prescribed pure start. The integer formula is exact at small positive contrast; it does not replace the general full-contrast bilinear synthesis problem by a linear program. The convexity theorem pinpoints why the retained normal-stabilizer absorption cannot simply be extended to arbitrary subgroups.

S4.1. **An unbounded nonabelian family with a strict positivity cost.** We apply the multiplicity theory to all symmetric groups S_n with $n \geq 5$. The natural permutation character $\chi_n(g)$ counts the fixed points of g on $[n]$. Consider the binary response whose first coordinate is

$$\beta_t(g) = \frac{1 - t}{n} + \frac{t}{n}\chi_n(g), \qquad b_t = (\beta_t, 1 - \beta_t), \qquad 0 \leq t \leq 1. \tag{S4.6}$$

Its group average is $1/n$, and at the identity it equals $[1 + (n-1)t]/n$. This supplies a family with growing group, observable rank, and exact memory, rather than a fixed-dimensional illustration.

The positivity threshold is controlled by the following matrix extremum. No symmetry or positive-semidefiniteness hypothesis is imposed on its coefficient matrix.

Lemma S4.5 (A sharp singular-coefficient bound and its equality case)**.** *Let $n \geq 5$, $P = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/n$, and $t > 0$. If an entrywise nonnegative real $n \times n$ matrix A satisfies*

$$\operatorname{rank} A < n, \qquad \sum_{i,j} A_{ij} = 1, \qquad PAP = \frac{t}{n}P,$$

then

$$t \leq \tau_n := \frac{\sqrt{n}-1}{\sqrt{n}+1}. \tag{S4.7}$$

Equality holds if and only if, after the same permutation of rows and columns,

$$A = A_* := \frac{1}{n-1}\sum_{i=1}^{n-1} \ell_i^{\mathsf{T}}\ell_i, \qquad \ell_i = \left(1 - \frac{1}{\sqrt{n}}\right)e_i + \frac{1}{\sqrt{n}}e_n. \tag{S4.8}$$

Here e_i and ℓ_i are row vectors. In particular symmetry of the extremal matrix is a conclusion.

Proof. Put $\alpha = t/n$. The centered identity implies $A = \alpha I + x\mathbf{1}^{\mathsf{T}} + \mathbf{1}y^{\mathsf{T}}$: indeed $P(A - \alpha I)P = 0$ decomposes the difference into a matrix constant on each row and one constant on each column. Let v be a nonzero left null vector of A. If $v^{\mathsf{T}}\mathbf{1} = 0$, then $v^{\mathsf{T}}AP = v^{\mathsf{T}}PAP = \alpha v^{\mathsf{T}}$, a contradiction. Normalize $\sum_i v_i = 1$. Substitution of $v^{\mathsf{T}}A = 0$ in the preceding decomposition gives a vector c such that

$$A = \alpha\{I + c\mathbf{1}^{\mathsf{T}} - \mathbf{1}v^{\mathsf{T}}\}, \qquad v \cdot c = 0, \qquad 1 = t\sum_i c_i. \tag{S4.9}$$

We bound $\sum_i c_i$. Relabel so that $v_1 = M$ is maximal and $v_2 = L$ is second largest. Since $\sum_i v_i = 1$, $M > 0$. Entrywise nonnegativity gives

$$c_1 \geq \max\{L, M-1\}, \qquad c_i \geq M \quad (i > 1).$$

If v is nonnegative, then $M \leq 1$ and $L \geq 0$. Thus $v \cdot c = 0$ forces $v_i = 0$ for every $i > 1$, $v_1 = 1$, and $c_1 = 0$. Consequently $\sum_i c_i \geq n - 1$.

Otherwise put $m = -\min_i v_i > 0$ and $b_0 = \max\{L, M-1\}$. Write $c = (b_0, M, \ldots, M) + u$ with $u \geq 0$. The dot product of v with the displayed base vector is $M(b_0 + 1 - M) \geq 0$. Since $v \cdot u \geq -m\sum_i u_i$, the condition $v \cdot c = 0$ therefore yields

$$\sum_i c_i \geq b_0 + (n-1)M + \frac{M(b_0 + 1 - M)}{m}. \tag{S4.10}$$

The normalization of v also implies

$$M \geq M_- := \frac{1+m}{n-1}, \qquad L \geq \frac{1 - M + m}{n-2}.$$

The latter follows by removing one maximum and one minimum from the sum of the coordinates and bounding each remaining coordinate by L. The bounds $M - 1$ and $(1 - M + m)/(n-2)$ coincide at $M_+ := 1 + m/(n-1)$.

If $M \geq M_+$, use $b_0 \geq M - 1$ in (S4.10) to obtain $\sum_i c_i \geq nM - 1 \geq n - 1$. If $M_- \leq M \leq M_+$, use $b_0 \geq (1 - M + m)/(n-2)$. The right side of (S4.10) is increasing in b_0, and becomes

$$F(M) = \frac{1+m}{n-2} + (n-1)M + \frac{n-1}{m(n-2)}(M - M^2).$$

This is a strictly concave quadratic. Its minimum on the stated interval occurs at an endpoint, where

$$F(M_-) = \frac{1+m}{n-1}\left(n + \frac{1}{m}\right), \qquad F(M_+) = nM_+ - 1 \geq n - 1.$$

The arithmetic–geometric mean inequality gives

$$\frac{1+m}{n-1}\left(n+\frac{1}{m}\right)=\frac{n+1+nm+1/m}{n-1}\geq\frac{(\sqrt{n}+1)^2}{n-1}.$$

For $n>4$ this last constant is strictly less than $n-1$. The two cases together prove $\sum_i c_i\geq(\sqrt{n}+1)/(\sqrt{n}-1)$; (S4.9) proves (S4.7).

Suppose equality holds, and write $s=\sqrt{n}$. The case $v\geq 0$ and the branch $M\geq M_+$ give strictly larger bounds. Equality in the remaining chain forces $m=1/s$ and $M=M_-$: strict concavity excludes the interior, and the other endpoint has value at least $n-1$. Thus

$$M=\frac{1}{s(s-1)},\qquad v=(M,\dots,M,-1/s)$$

after a coordinate permutation. Indeed the sum of the $n-1$ coordinates other than one minimum is $(n-1)M$, forcing each to equal its maximum. Now $b_0=M$. Equality in $v\cdot u\geq -m\sum_i u_i$ places all of u at the unique negative coordinate, and its value there is M/m. Hence $c=(M,\dots,M,M+M/m)$. Substituting in (S4.9) gives precisely (S4.8).

Conversely the rows ℓ_i in that formula are probability vectors and linearly independent. If $p=\mathbf{1}^{\mathsf{T}}/n$ is regarded as a row, their centered rows satisfy

$$(\ell_i-p)\cdot(\ell_j-p)=\left(1-\frac{1}{\sqrt{n}}\right)^2\delta_{ij}.$$

They are therefore an orthogonal basis of the zero-sum subspace. Consequently A_* is nonnegative, has total sum one and rank $n-1$, and

$$PA_*P=\frac{(1-1/\sqrt{n})^2}{n-1}P=\frac{\tau_n}{n}P.$$

This proves sufficiency and completes the equality characterization. □

Theorem S4.6 (The exact two-level nonabelian memory law)**.** *For $n\geq 5$ and $t>0$, the response* (S4.6) *has observable group S_n, residual cardinality $n!$, and observable dimension $1+(n-1)^2$. Its permutation-module bound is*

$$\nu_{S_n}(V_{b_t})=n(n-1).$$

At the sharp threshold $\tau_n=(\sqrt{n}-1)/(\sqrt{n}+1)$, the unrestricted positive core and pure-start memory satisfy

contrast	$\kappa_{S_n}(b_t)$	$m(b_t)$
$0<t\leq\tau_n$	$n(n-1)$	$n(n-1)+1$
$\tau_n<t\leq 1$	n^2	$n^2+1.$

The smaller memory is attained at the threshold itself. At $t=0$ both minima are one.

Every minimum permutation core at $t=\tau_n$ is the disjoint union of $n-1$ natural orbits. Its probability preparation and binary decoder, after relabeling the orbit copies and simultaneously relabeling their natural coordinates, have exactly the form

$$\lambda_i=\ell_i,\qquad D_i=\frac{\ell_i}{(n-1)w_i},\qquad \sum_{i=1}^{n-1}w_i=1,\qquad w_i\geq\frac{1-1/\sqrt{n}}{n-1},\tag{S4.11}$$

where ℓ_i is defined in (S4.8)*, w_i is the preparation mass on orbit i, and D_i is its first-output decoder written as a row. The second-output decoder is $1-D_i$. Conversely every displayed weight vector gives an optimal factorization. This is a classification of minimum permutation-core factorizations under the inherited normal form, not a classification of all stochastic transition realizations before that reduction.*

Proof. The real natural permutation module is the direct sum of the invariant line and the $(n-1)$-dimensional standard representation Std_n. Its complexification is irreducible on that zero-sum subspace. One direct proof uses the transpositions: from a nonconstant vector in an invariant subspace, a vector $v-(ij)v$ produces a nonzero multiple of e_i-e_j, whose orbit spans the whole zero-sum space. The standard representation is faithful, because a permutation fixing every difference e_i-e_j fixes all points. Matrix-coefficient orthogonality gives the nontrivial block

$$\widehat{\beta}_t(\mathrm{Std}_n)=\frac{t}{n(n-1)}I_{n-1},$$

and all other nontrivial irreducible blocks vanish. Therefore the observable module is $\mathbf{1}\oplus\mathrm{Std}_n^{\oplus(n-1)}$, of dimension $1+(n-1)^2$. Equality of two residuals forces equality of their standard representation matrices because this Fourier block is invertible; faithfulness then makes the group elements equal. This proves the claims about the group and residual count.

We next record the subgroup index estimate needed for an unrestricted memory converse. For $K\le S_n$, let r be the number of its orbits on $[n]$. The multiplicity of Std_n in $\mathbb{C}[K\backslash S_n]$ is

$$m(K)=\dim\mathrm{Std}_n^K=r-1.$$

If $m(K)>0$, then

$$[S_n:K]\ge n\,m(K). \tag{S4.12}$$

More sharply, if $m(K)>0$ and K is not a full point stabilizer,

$$[S_n:K]\ge n\{m(K)+1\}. \tag{S4.13}$$

To prove these, let the natural orbit sizes of K be $a_1,\dots,a_r$. Since K is contained in the product of the symmetric groups on those orbits,

$$[S_n:K]\ge\frac{n!}{a_1!\cdots a_r!}.$$

For $r\ge3$, this is at least $n!/(n-r+1)!$: transferring one unit from the smaller of two nonunit parts to the larger increases their factorial product until only one nonunit part remains. The last lower bound is at least nr for $n\ge5$. Indeed at $r=3$ it is $n(n-1)\ge3n$, and increasing r multiplies the bound by at least two while r increases by one. If $r=2$ and one part is a singleton, K lies in a point stabilizer; it is that full stabilizer of index n, or has index at least $2n$. If both parts have size at least two, the index is at least $\binom{n}{2}\ge2n$. These cases prove both inequalities. The argument uses the natural K-orbits and requires no classification of low-index subgroups. In particular, an exceptional subgroup of S_6 that is transitive in the natural action has $m(K)=0$ and does not supply a standard carrier.

A G-set containing V_{b_t} must have total standard multiplicity at least $n-1$. Summing (S4.12) over its standard-carrying orbits gives at least $n(n-1)$ points. Exactly $n-1$ natural orbits supply this multiplicity, so the module minimum is $n(n-1)$.

For sharp positive attainment, use the preparations ℓ_i of (S4.8) on $n-1$ natural orbits, each of mass $1/(n-1)$, and use $D_i=\ell_i$ as first-output decoders. The coefficient matrix is A_*. For any permutation matrix P_g, the sum of its entrywise products with a matrix A will be denoted by $\langle A,P_g\rangle$; in particular $\lambda P_gD^{\mathsf{T}}=\langle\lambda^{\mathsf{T}}D,P_g\rangle$ when the scalar decoder is written as a row. The identities $PA_*P=(\tau_n/n)P$ and $\sum_{ij}(A_*)_{ij}=1$ imply

$$\langle A_*,P_g\rangle=\frac{1}{n}+\frac{\tau_n}{n}\{\chi_n(g)-1\}=\beta_{\tau_n}(g).$$

Indeed the centered part contributes $(\tau_n/n)(\chi_n(g)-1)$; the row-constant and column-constant parts contribute their common group mean $1/n$. Mixing each scalar decoder with $1/n$ by the factor t/τ_n realizes every $0\le t\le\tau_n$ on these same points. The module bound proves optimality for positive t in this interval.

A second, rational construction on the subinterval $0 \le t \le 1/(n-1)$ is useful independently. Use $n-1$ natural orbits indexed by $i = 1, \dots, n-1$, each with initial mass $1/(n-1)$ concentrated at its point i. Give orbit i the scalar decoder

$$D_i(j) = \frac{1}{n} + \frac{(n-1)t}{n}\{\mathbf{1}_{\{j=i\}} - \mathbf{1}_{\{j=n\}}\}. \tag{S4.14}$$

For $0 \le t \le 1/(n-1)$ these numbers lie in $[0,1]$. Since a permutation has exactly one preimage of n,

$$\frac{1}{n-1}\sum_{i=1}^{n-1} D_i(g(i)) = \frac{1}{n} + \frac{t}{n}\sum_{i=1}^{n-1}\{\mathbf{1}_{\{g(i)=i\}} - \mathbf{1}_{\{g(i)=n\}}\} = \frac{1}{n} + \frac{t}{n}\{\chi_n(g) - 1\} = \beta_t(g).$$

This retains a realization using only rational probabilities whenever t is rational. Its endpoint is not the sharp threshold when $n \ge 5$.

At full contrast, n natural orbits realize β_1: give each mass $1/n$, start orbit i at i, and use the indicator decoder of i. Hence $\kappa_{S_n}(b_1) \le n^2$. Suppose, to the contrary, that a factorization exists on fewer than n^2 points. The total standard multiplicity is at least $n-1$. If any standard-carrying orbit were not natural, (S4.13) and the ordinary estimates on the others would force at least $n(n-1) + n = n^2$ points. Thus all standard-carrying orbits are natural. There must be exactly $n-1$ of them: each supplies multiplicity one and costs n points. Every other orbit has total size less than n.

Write the factorization as a mixture over its positive-mass transitive orbits. Since $\beta_1(e) = 1$, each component's decoder equals one on its initial support. On a transitive orbit of degree d its mean is therefore at least $1/d$. The natural components have mean at least $1/n$; any additional orbit here has $d < n$ and would have mean strictly greater than $1/n$. The mean of β_1 is exactly $1/n$, so all additional orbits have zero initial mass and every active natural component has mean $1/n$. Lemma S4.4 now makes each component a deterministic return law. Let w_i be the aggregate mass of the components starting at natural point i. Evaluating at a transposition (ij) gives

$$1 - w_i - w_j = \frac{n-2}{n}, \qquad w_i + w_j = \frac{2}{n}.$$

For $n \ge 3$ these equations imply $w_i = 1/n$ for every i. Thus at least n positive components are needed, contradicting the availability of only $n-1$ natural orbits. This proves $\kappa_{S_n}(b_1) = n^2$.

We next exclude every intermediate positive-core value. Fix $t > 0$ and suppose a factorization has fewer than n^2 points. The same index argument gives exactly $n-1$ natural standard-carrying orbits; the remaining orbits have standard multiplicity zero. Write $b_t = B_N + B_E$ for their unnormalized response contributions. The Fourier support of B_N lies in the trivial and standard types, because these are the only types in a natural permutation module. The target has the same support. On the other hand, B_E has zero standard block. Subtracting B_N from b_t shows that all nontrivial blocks of B_E vanish, so $B_E(g) = v_E$ is a constant nonnegative vector. Its total mass $w_E = \sum_c v_E(c)$ equals the preparation mass on the extra orbits. Let $w_N = 1 - w_E$. Since $t > 0$, the nonzero standard block forces $w_N > 0$. Restrict the original preparation and decoder to the natural orbits, calling them λ_N and D_N. On those same $n(n-1)$ points use

$$\lambda'_N = \lambda_N / w_N, \qquad D'_N = w_N D_N + \mathbf{1} v_E.$$

Every row of D'_N is nonnegative and sums to $w_N + w_E = 1$, and

$$\lambda'_N P_g D'_N = B_N(g) + v_E = b_t(g).$$

Thus any positive core below n^2 compresses to the module minimum $n(n-1)$. No intermediate value is possible.

The full-contrast construction realizes every b_t by mixing its decoder with the constant $1/n$, so $\kappa \le n^2$ everywhere. It remains to exclude the smaller core for every $t > \tau_n$. By the absorption

argument, any core below n^2 would give a factorization on exactly $n-1$ natural orbits. Write its orbit weights, probability rows, and scalar decoder rows as w_i, λ_i, and D_i. The matrix

$$A = \sum_{i=1}^{n-1} w_i \lambda_i^{\mathsf{T}} D_i$$

is entrywise nonnegative and has rank at most $n-1$. Its response is $\langle A, P_g \rangle = \beta_t(g)$. The group mean is $(1/n)\sum_{ij} A_{ij} = 1/n$, so $\sum_{ij} A_{ij} = 1$. Moreover $PAP = (t/n)P$. Here is an elementary way to see the last implication without a choice of Fourier bases. Put $B = A - (t/n)I$. All permutation sums $\langle B, P_g \rangle$ are equal. Comparing two permutations that differ only by interchanging the assigned columns of two distinct rows gives

$$B_{ij} + B_{k\ell} = B_{i\ell} + B_{kj} \quad (i \neq k,\ j \neq \ell).$$

Thus $B_{ij} = B_{i1} + B_{1j} - B_{11}$, so $PBP = 0$. Lemma S4.5 now forces $t \leq \tau_n$. Together with the matching construction and the absence of intermediate values, this proves the exact two-level law. Theorem S2.2, namely $m = \min\{n!, 1+\kappa\}$, gives the displayed pure-start values since $n^2 + 1 < n!$ for $n \geq 5$. The explicit quadratic threshold is also consistent with the retained algebraic-threshold theorem; no quantifier-elimination calculation is needed here.

Finally consider a minimum core at $t = \tau_n$. The index bounds force exactly $n-1$ natural orbits and no others, and the equality part of Lemma S4.5 forces $A = A_*$ after simultaneous coordinate relabeling. Set $a = 1 - 1/\sqrt{n}$ and $b = 1/\sqrt{n}$, so $a > b > 0$ for $n \geq 5$. The leading $(n-1) \times (n-1)$ block of A_* is the positive diagonal matrix $a^2 I/(n-1)$. Each nonnegative rank-one summand $w_i \lambda_i^{\mathsf{T}} D_i$ can cover at most one of these diagonal positions: covering two would give a positive off-diagonal entry. All $n-1$ positions need coverage and there are exactly $n-1$ summands. Therefore each is covered by a different positive-weight summand. Relabel that summand by its position i. The zero entries in row and column i then force both λ_i and D_i to be supported on $\{i, n\}$. Only this summand contributes to the entries (i,i), (i,n), and (n,i). Their ratios in A_* are

$$\frac{(A_*)_{in}}{(A_*)_{ii}} = \frac{(A_*)_{ni}}{(A_*)_{ii}} = \frac{b}{a}.$$

Normalization of λ_i and the diagonal value $(A_*)_{ii} = a^2/(n-1)$ give $\lambda_i = a e_i + b e_n = \ell_i$ and $D_i = \ell_i / \{(n-1) w_i\}$. The condition $D_i \in [0,1]^n$ is exactly $w_i \geq a/(n-1)$ because $a > b$. Conversely these weights make every decoder valid and give $w_i \lambda_i^{\mathsf{T}} D_i = \ell_i^{\mathsf{T}} \ell_i / (n-1)$. This proves precisely the claimed classification. □

Prepared quantum realization. The family has a finite-dimensional prepared unitary realization, so it remains inside the experiment framework. On $\mathbb{C}^n \otimes \mathbb{C}^n$ take

$$\rho = \frac{1}{n} \sum_{i=1}^{n} |ii\rangle\langle ii|, \qquad U_g = P_g^{\mathsf{T}} \otimes I, \qquad E = \sum_{i=1}^{n} |ii\rangle\langle ii|.$$

Here P_g is the row-transition permutation matrix, so $U_g |x, y\rangle = |xg, y\rangle$, consistently with chronological command multiplication. Then $\mathrm{tr}(E U_g \rho U_g^*) = \chi_n(g)/n$. The binary effect $E_t = tE + (1-t)I/n$ produces (S4.6). Two command types suffice uniformly in n: an n-cycle and a transposition of adjacent points generate S_n, since conjugation by the cycle supplies the adjacent transpositions. They give one common experiment valid for every word in that fixed-size command alphabet, with a binary terminal outcome set. The pure maximally entangled preparation $n^{-1/2} \sum_i |ii\rangle$ gives the same terminal probabilities with this effect, if a pure preparation is desired. The statement is a physical realization of the response, not a claim that this Hilbert-space dimension is minimal or that it establishes a quantum-versus-classical separation under a common resource metric.

The gap n between the positive core at full contrast and its module minimum is unbounded. The relative ratio $n/(n-1)$ tends to one; no growing relative separation is asserted. The exact transition tends to one as n grows. Its proof combines a general orbit-multiplicity converse, a sharp positivity

inequality with equality rigidity, and explicit optimal implementation; none of these counts requires the completion of the wider Theta pipeline.

S5. Quotient decompositions, linear duality and sharp transitions

For groups whose subgroups are normal, the probability optimization in the normal form has an exact linear reduction. This includes every finite abelian group and allows arbitrary probability tables and terminal outcome sets. Neither a special phase choice nor a positive-amplitude trigonometric ansatz is imposed. The reduction uses normal stabilizers; it is not an unrestricted claim that equivariant nonnegative factorization is linear programming.

Lemma S5.1 (Absorption of within-orbit preparation)**.** *Suppose the stabilizers in a finite G-set are normal. Every probability factorization on it can be replaced, without increasing its size, by one having at most one orbit $H\backslash G$ for each normal subgroup H and whose initial row is concentrated at the identity coset within each orbit. Only the mixture between different orbits is required.*

Proof. On an orbit of positive initial mass, normalize its local preparation and write its response as

$$q(g,c)=\sum_{Hx\in H\backslash G}\lambda(Hx)D(Hxg,c).$$

For $h\in H$, normality gives $Hxhg=Hxg$, so $q(hg,c)=q(g,c)$. Thus q itself is a probability decoder on $H\backslash G$. Replacing the preparation by the identity coset and the decoder by q leaves every command-word response unchanged. Orbits with the same H can be merged by averaging their decoders with their initial masses as weights. Delete zero-mass orbits. These operations weakly decrease the size. □

Assume henceforth, until the Fourier specialization, that every subgroup of G is normal. Such groups are called Dedekind groups. For a nonempty set $\mathcal{H}$ of distinct subgroups, put $c(\mathcal{H})=\sum_{H\in\mathcal{H}}[G:H]$.

Theorem S5.2 (Exact quotient linear programs)**.** *The invariant $\kappa_G(b)$ is the minimum $c(\mathcal{H})$ for which there are $p_H\ge 0$ and functions $f_H: H\backslash G\times C\to\mathbb{R}_{\ge 0}$ satisfying*

$$\sum_c f_H(q,c)=p_H\quad(q\in H\backslash G),\qquad \sum_H p_H=1,\qquad \sum_H f_H(Hg,c)=b(g,c)\quad(g\in G,c\in C). \tag{S5.1}$$

The residual cardinality is $N=|G|$, and $m(b)=\min\{|G|,1+\kappa_G(b)\}$. For a rational response table a minimum controller can be chosen rational. More generally, if the table lies in a specified real algebraic number field, the controller can be chosen in that same field.

Proof. Apply Lemma S5.1 to a minimum probability factorization. Its orbit contributions give (S5.1). Conversely, if $p_H>0$, put $D_H=f_H/p_H$, place mass p_H at its identity coset, and use the right quotient action. If $p_H=0$, all its entries vanish and the orbit is discarded. This constructs the whole response with at most $c(\mathcal{H})$ points. The regular orbit $H=\{1\}$ is always available, and there are finitely many subgroup families.

The stabilizer H_0 of the initial residual is normal. If $h\in H_0$ and $g\in G$, moving h past g using normality shows that h fixes the residual after g. It therefore fixes their span and is the identity in the minimal observable group. Thus $H_0=\{1\}$ and orbit-stabilizer gives $N=|G|$. The pure-start assertion follows from Theorem S2.2.

For each family the displayed feasible set is a bounded closed polyhedron: $0\le f_H(q,c)\le p_H\le 1$. A feasible vertex is obtained by solving a linear system over the coefficient field. It is rational for rational data and remains in the given real algebraic field otherwise. The construction uses only these entries and division by positive p_H. If the residual machine is smaller than the initializer construction, its decoder is the given table and has the same arithmetic property. □

This theorem optimizes the common permutation action supplied by the inherited normal form. The relation between symmetry and positive factorizations is classical [67, 68], and positivity can cost more than a factorization inside the target linear space [70]. The additional step here is the absorption of preparation on normal quotient orbits, which removes the bilinear unknowns without changing the counted resource.

S5.1. **An explicit payoff dual.** Let $\mathcal{P}_H$ be the polytope of all probability tables on $H\backslash G$, pulled back to G. For a real table y on $G \times C$, use the unnormalized pairing $\langle y, b\rangle = \sum_{g,c} y(g,c)b(g,c)$ and set

$$h_H(y) = \sum_{q \in H\backslash G} \max_{c \in C} \sum_{g \in q} y(g,c), \qquad h_{\mathcal{H}}(y) = \max_{H \in \mathcal{H}} h_H(y). \tag{S5.2}$$

Theorem S5.3 (Payoff certificates and exact contrast endpoints)**.** *The family $\mathcal{H}$ is feasible in* (S5.1) *if and only if*

$$\langle y, b\rangle \le h_{\mathcal{H}}(y) \quad \text{for every real table } y. \tag{S5.3}$$

If infeasible, it has a strict rational witness y. A witness of margin $\Delta = \langle y, b\rangle - h_{\mathcal{H}}(y) > 0$ still excludes that family for any table $\widetilde{b}$ with $\|\widetilde{b} - b\|_\infty < \Delta/\|y\|_1$.

For the contrast line (S3.1)*, put $v = b - \pi$. Its endpoint for this family is*

$$\tau_{\mathcal{H}} = \min\left\{1, \min_{\langle y, v\rangle = 1} \big(h_{\mathcal{H}}(y) - \langle y, \pi\rangle\big)\right\}. \tag{S5.4}$$

Both minima are attained. Equivalently, maximize t in $[0,1]$ subject to (S5.1) *with right side $\pi + tv$. The endpoint for budget k is the maximum of these linear-program values over $c(\mathcal{H}) \le k$. Rational tables give rational endpoints and optimal controllers; no polynomial bound on the number of families is claimed.*

Proof. The feasible response set for a fixed family is exactly $\mathcal{P}_{\mathcal{H}} = \operatorname{conv} \bigcup_H \mathcal{P}_H$. Each $\mathcal{P}_H$ is a product of probability simplexes. Maximizing a payoff independently on each coset gives its support function h_H; taking a convex hull of the union gives $h_{\mathcal{H}}$. Membership proves (S5.3). If b is outside this compact convex set, let q be a closest point. The first variation along the segment from q to any point of the set gives $\langle b - q, x - q\rangle \le 0$. With $y = b - q$ this yields a strictly positive gap of at least $\|b - q\|_2^2$. Approximate y by a rational table sufficiently closely to preserve that gap; the support function is a finite maximum of linear functions and is continuous. The perturbation assertion is the inequality $|\langle y, \widetilde{b} - b\rangle| \le \|y\|_1 \|\widetilde{b} - b\|_\infty$.

The constant table π belongs to every $\mathcal{P}_H$. Along its ray $\pi + tv$, a support inequality with $\langle y, v\rangle \le 0$ imposes no restriction for $t \ge 0$; every other inequality can be normalized by $\langle y, v\rangle = 1$. This gives (S5.4). Attainment requires no strict-feasibility assumption. If v lies outside the linear space parallel to the affine hull of $\mathcal{P}_{\mathcal{H}}$, an affine-hull equation gives an attained inner minimum zero. Otherwise the bounded polytope has a finite facet description in that affine hull; the first boundary reached along the ray activates a facet with positive directional coefficient. Its normalized supporting functional attains the inner minimum. The polytope is bounded and $v \ne 0$, so the ray endpoint is finite.

One explicit linear formulation of the inner dual introduces z_{Hq} and s with

$$z_{Hq} \ge \sum_{g \in q} y(g,c) \quad (H, q, c), \qquad s \ge \sum_q z_{Hq} \quad (H \in \mathcal{H}), \qquad \langle y, v\rangle = 1,$$

and minimizes $s - \langle y, \pi\rangle$. The primal formulation is obtained by replacing the right side of (S5.1) by $\pi + tv$. It is bounded and contains $t = 0$. Vertex arithmetic proves the endpoint and controller assertions, and the finite maximum over admissible families gives the budget endpoint. These are exact linear certificates in the specified finite problem, not a formal proof-assistant verification. □

S5.2. **Fourier support and a sharp positivity transition.** For an abelian G, let S be the complex characters whose vector Fourier coefficient in b is nonzero. It contains the trivial character and is closed under conjugation.

Corollary S5.4 (The abelian module cover)**.** *The number $\nu_G(V_b)$ is the minimum $\sum_{H\in\mathcal{H}}[G:H]$ over distinct subgroup families such that each character in S is trivial on at least one $H \in \mathcal{H}$. This lower bound is the exact value of $\kappa_G(b_t)$ for all sufficiently small positive contrasts. It need not be attained at full contrast.*

Proof. Character averaging projects the cyclic vector b onto each of its nonzero Fourier modes. Thus its complex cyclic module contains each active character once; conjugate pairs give the corresponding real two-dimensional summands. The permutation module on $H\backslash G$ contains exactly the characters trivial on H, each once. Containment of V_b is therefore exactly the stated cover. Repeated subgroups add no new characters and may be deleted. Apply Theorem S3.1. □

The following example computes a complete rational threshold and an explicit dual obstruction. Let $G = (\mathbb{F}_2)^4$, with the four coordinate flips as commands, and let $\chi_1, \dots, \chi_4$ be its coordinate sign characters. Put $\chi_5 = \chi_1\chi_2\chi_3\chi_4$ and

$$\beta_t(g) = \frac{3}{8} + \frac{t}{8}\sum_{i=1}^{5}\chi_i(g), \qquad b_t(g) = (\beta_t(g), 1-\beta_t(g)), \qquad 0 \le t \le 1. \tag{S5.5}$$

Proposition S5.5 (A sharp transition invisible to the observable module)**.** *For every $t > 0$,* (S5.5) *has observable group $(\mathbb{F}_2)^4$, dimension six, residual cardinality sixteen, and $\nu_G(V_b) = 10$. Its exact positive core and pure-start memory are*

$$\kappa_G(b_t) = \begin{cases} 10, & 0 < t \le 3/5, \\ 12, & 3/5 < t \le 1, \end{cases} \qquad m(b_t) = \begin{cases} 11, & 0 < t \le 3/5, \\ 13, & 3/5 < t \le 1. \end{cases}$$

At zero contrast the minimum is one. In particular a twelve-label minimum does not occur in this family.

Proof. The five signs have product one, so their sum is $5, 1$, or -3. Consequently β_1 takes the values $1, 1/2, 0$, and every β_t is a valid convex combination with the constant $3/8$. Any four of the five characters are independent over $\mathbb{F}_2$. For $t > 0$ all five character coefficients and the constant mode are nonzero, giving dimension six and trivial joint kernel. Thus G is the observable group and $N = 16$.

For a quotient orbit write $W = H^\perp$ and $r = \dim W$; its size is 2^r. A proper W contains at most r of the five active nontrivial characters, and $2^r \ge 2r$ for $r = 1, 2, 3$. An orbit with $r = 4$ costs sixteen. Covering the five modes therefore costs at least ten, and the five two-point quotients attain this module bound.

Suppose that a positive core has at most eleven points. A size-eight orbit covers at most three active characters; the remaining budget at most three covers at most one more. Size sixteen is unavailable. The nontrivial orbits thus have sizes two or four. Their total cost must be ten and their active-character blocks partition the five modes into singletons or pairs: each dimension must carry an active character and no active character can be covered twice. At most one fixed point can remain. Products belonging to different pairs are distinct and inactive, because equality would give a nontrivial relation involving at most four of the five circuit characters.

For this family of blocks let $\mathcal{B}_2$ denote its pairs and define

$$z(g) = 1 - \sum_{i=1}^{5}\chi_i(g) + \sum_{\{i,j\}\in\mathcal{B}_2}\chi_i(g)\chi_j(g).$$

Averaging z over the stabilizer of a singleton orbit gives $1-\chi_i$; over a pair orbit it gives $(1-\chi_i)(1-\chi_j)$; over a fixed orbit it gives 1. The absence of any relation of length at most four ensures that no

other term survives these averages. Each resulting function is nonnegative. Hence any nonnegative quotient contributions satisfy

$$\frac{1}{16}\sum_{g} z(g)\beta(g) = \sum_{H} \frac{1}{16}\sum_{g} z(g) f_H(Hg, 1) \geq 0.$$

For the target (S5.5), character orthogonality gives the value $(3 - 5t)/8$. Thus every core of size at most eleven requires $t \leq 3/5$. This is also an explicit instance of Theorem S5.3: take $y(g, 1) = -z(g)/16$, $y(g, 0) = 0$. Its family support value is zero and its strict margin above the threshold is $(5t - 3)/8$.

For $0 < t \leq 3/5$, use five two-point quotient orbits, putting initial mass $1/5$ at the plus point of each. The first-coordinate decoder on orbit i is

$$D_i(g, 1) = \frac{3}{8} + \frac{5t}{8}\chi_i(g).$$

Its values lie in $[0, 1]$, and the weighted sum is β_t. This attains the ten-point core.

For $t = 1$, use the four-point quotient on (χ_1, χ_2) and the eight-point quotient on (χ_3, χ_4, χ_5), with initial mass $1/2$ at the all-plus point of each. Their decoders are respectively the indicator of two plus signs and the indicator of at least two plus signs. For signs a, b, c, the latter is $1/2+(a+b+c-abc)/4$. The weighted sum of the two decoders is

$$\frac{1 + \chi_1 + \chi_2 + \chi_1\chi_2}{8} + \frac{1}{4} + \frac{\chi_3 + \chi_4 + \chi_5 - \chi_3\chi_4\chi_5}{8} = \beta_1,$$

because $\chi_1\chi_2 = \chi_3\chi_4\chi_5$. Replace each decoder D by $(1 - t)3/8 + tD$ for other contrasts. This proves the twelve-point upper everywhere, matching the strict lower above $3/5$. The pure-start formula now gives eleven and thirteen, since $N = 16$. □

Corollary S5.6 (A robust finite-table obstruction)**.** *Fix $3/5 < t \leq 1$ and a binary probability table $\widetilde{\beta}$ on the same group $G = (\mathbb{F}_2)^4$. Supply it under the same four generator commands, so the whole response is $w \mapsto \widetilde{\beta}(g(w))$. If*

$$\max_{g \in G} |\widetilde{\beta}(g) - \beta_t(g)| < \frac{5t - 3}{18}, \tag{S5.6}$$

its exact positive core has at least twelve points and its pure-start memory is at least thirteen. This is a sufficient lower certificate on finite group tables. It does not assert a sharp neighbourhood radius or a matching thirteen-label upper for every perturbed table.

Proof. Write η for the left side of (S5.6). Every normalized Fourier coefficient changes by at most η. Since $(5t - 3)/18 < t/8$ for $t \leq 1$, all five prescribed coefficients remain nonzero. Their joint kernel is trivial, so the perturbed table still has observable group G and sixteen residuals. An action of size at most eleven must therefore cover those same five modes, and the preceding proof gives its singleton-and-pair partition even if additional Fourier coefficients are present.

There are zero, one, or two pairs in such a partition. Every permutation of the five circuit characters extends to a linear automorphism of the dual group, because any four are a basis and the fifth is their sum. Thus the absolute sum of its witness z depends only on the number of pairs. Direct evaluation on the sixteen sign patterns gives the following values and multiplicities:

pairs	value : multiplicity of z	$16^{-1}\sum_g \lvert z(g)\rvert$
0	$-4:1,\ 0:10,\ 4:5$	$3/2$
1	$-3:1,\ -1:6,\ 1:4,\ 3:2,\ 5:3$	$17/8$
2	$-2:5,\ 0:4,\ 2:2,\ 4:4,\ 6:1$	$9/4$

The normalized pairing with $\widetilde{\beta}$ is at most

$$\frac{3 - 5t}{8} + \frac{9}{4}\eta < 0.$$

It cannot have the required nonnegative quotient decomposition. Hence $\kappa_G(\widetilde{b}) \geq 12$, and $m(\widetilde{b}) = \min\{16, 1 + \kappa_G(\widetilde{b})\} \geq 13$. The upper thirteen on the exact contrast line was supplied by a specific construction and is not inferred for arbitrary perturbations. □

For an exact physical realization one may use the sixteen-dimensional diagonal model: prepare the identity basis state, use regular permutation unitaries for the four generators, and take the diagonal terminal effect with entries $\beta_t(g)$. This is already classical; no quantum advantage is asserted for this example. The general low-contrast theorem, the normal quotient linear programs, and this exact transition have different scopes.

S6. The retained instrument realization theory and its boundary

The group results concern terminal responses. For an outcome at every call, one must instead preserve each command–outcome history and every subsequent decision made from that visible history. This stronger interface is already covered, under a sufficient uniform posterior condition, by the earlier instrument theory of this paper. We recall its exact scope before proving the new conditional-memory classification.

For fixed known completely positive branches $\mathcal{I}_{a,y}$ whose command sums are trace preserving, the hypothesis is

$$\mathcal{I}_{a,y}(X) \succeq \frac{\eta}{h} \operatorname{tr}(\mathcal{I}_{a,y}(X)) I \quad (X \succeq 0), \qquad \eta > 0.$$

It is a condition on every recorded branch, including the zero-branch convention, and follows when depolarizing noise of weight η is applied after each branch. In every finite dimension $h \geq 2$, and for k prescribed preparations, Theorem S23.2 gives at most

$$k + \left\lceil (4h/\eta)^{h-1} \right\rceil$$

hidden labels, with at most h^2 successors per branch row. All commands use the same physical dictionary. The complete proof in Section S23 gives the operator identity for every history; it preserves all adaptive stopped laws, including nontermination, and nonnegative costs. The hidden label is not exposed to the observer. Corollary S23.3 consequently preserves the risk of every common report-based observer, including constrained boundary budgets; it does not identify hidden simulator size with optimal observer memory.

The same retained theory includes a terminating algebraic construction (Proposition S23.5), interiority after a finite block with the clock and intermediate reports counted (Theorem S23.6), and matching powers of η^{-1} for universal state-positive representation (Theorem S23.4). The latter lower bound is for generators which are actual density matrices. It is not imported as a lower bound against arbitrary nonnegative hidden-state representations. The family in Section S15 will have a lower bound against that unrestricted class.

Invariant positive cones and their stochastic realizations are classical [48, 47]. Okay's quantum simulation framework explicitly includes invariant-polytope updates and adaptive composition [71]. Pure-state nets and approximation of the density-matrix body have an established convex-geometric theory [72]; extension complexity is a different resource [73]. These ingredients, and the paper's already proved interior-instrument theorem, are not asserted as new discoveries of this revision. Their role is to fix the sufficient condition and the resource convention against which the new sharp examples are compared. The next common-contraction construction gives an explicit certificate in supplied linear-response coordinates; it is also an application of positive-realization geometry, not a replacement for the retained instrument theorem.

S7. Sharp conditional memory with damping and adaptive commands

The one-dependent family of Section S15 also admits a sharp positive theory under conditional damping. The extension below realizes the complete observed instrument, with one physical

dictionary shared by every command. The classical sector inequality and the scalar thresholds of Section S20 are inherited ingredients, not new spectral-region results. Earlier noise-dependent process lower bounds, including an inverse-square-root worst-phase dependence, are due to Fanizza, Lumbreras and Winter [50, Theorem 2 and Section 4.2]. Here the lower and upper meet for an explicit uniformly positive, exactly depolarizing instrument, and the upper works simultaneously under adaptive rotation commands.

For a qubit set

$$U_\theta = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \qquad E = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}, \qquad \tau = |0\rangle\langle 0|, \qquad \mathcal{D}(X) = \operatorname{tr}(X) I/2.$$

For $0 \le \alpha \le 1$ define

$$\begin{aligned} \Psi_0^{\alpha,\theta}(X) &= \tfrac{1}{8}[\alpha U_\theta X U_\theta^* + (1-\alpha)\mathcal{D}(X)], \\ \Psi_1^{\alpha,\theta}(X) &= \tfrac{1}{4}\operatorname{tr}(EX)\tau, \\ \Psi_2^{\alpha,\theta} &= \mathcal{D} - \Psi_0^{\alpha,\theta} - \Psi_1^{\alpha,\theta}. \end{aligned} \tag{S7.1}$$

The initial state is $I/2$. An allowed command specifies θ; the report is $s \in \{0,1,2\}$. A command may be selected from the visible history. A finite known command alphabet is enough for every finite lower witness below. The same construction also gives Borel transition tables for the entire angle circle; this last formulation counts labels, not the encoding or exact sampling cost of a real-valued command.

This family does not satisfy the retained uniform-interior hypothesis on every posterior, even when $\alpha < 1$: report 1 always prepares the pure state τ. Repeated reports 1 keep that pure state, so passing to a fixed block length does not restore that hypothesis. The finite construction below instead uses the common geometry of the damped rotation, the reset vertex and the remaining branch.

Lemma S7.1 (Physical margins and predictive rank)**.** *For every $\alpha \in [0,1]$ and every angle, the maps* (S7.1) *form a quantum instrument, their sum is exactly $\mathcal{D}$, and their report probabilities lie in $\{1/8\}$, $[1/16, 3/16]$ and $[11/16, 13/16]$, respectively. The prepared response is predictively stationary under each unobserved command. At a fixed angle its output process is stationary and one-dependent. When $\alpha > 0$ and $\sin\theta \neq 0$, its word Hankel rank is three, and*

$$p(1\,0^n\,1) = \frac{8^{-n}}{64}\left(1 + \frac{\alpha^n}{2}\cos(n\theta)\right) \qquad (n \ge 0). \tag{S7.2}$$

At $\alpha = 0$ the complete command response has rank two and is independent of the angles. For all α, it has no one-label realization.

Proof. The first two Choi matrices are positive. Their operator norms are at most $1/4$ and $3/16$, while $J(\mathcal{D}) = I_4/2$. Hence $J(\Psi_2^{\alpha,\theta}) \succeq I_4/16$. The asserted trace bounds are unchanged by damping. Since $\mathcal{D}(I/2) = I/2$, averaging any unobserved command fixes the prepared predictive response, including arbitrary subsequent commands and terminal POVMs. At a fixed angle, summing an intervening report applies $\mathcal{D}$ and therefore factorizes the probabilities of the blocks on either side, exactly as in Theorem S15.1.

Report 1 has initial probability $1/8$ and prepares τ. Conditional on n zero reports its Bloch vector is $\alpha^n(\cos(n\theta), \sin(n\theta))$, giving (S7.2). All reached subnormalized states are real symmetric, so the Hankel rank is at most three. For $\alpha > 0$ and nonreal phase, the three modes $1/8$, $\alpha e^{i\theta}/8$ and $\alpha e^{-i\theta}/8$ are distinct with nonzero coefficients. The 3×3 submatrix with entries $p(1\,0^{i+j}\,1)$ is therefore a product of two nonsingular Vandermonde matrices and a nonsingular diagonal matrix. This proves rank three. At $\alpha = 0$ every reached state is diagonal, giving rank at most two; the 2×2 cylinder minor has determinant $2^{-19} > 0$. Finally $p(1) = 1/8$ and $p(11) = 3/128 \neq p(1)^2$ for every parameter, excluding a one-label realization. □

Write $N_{\mathrm{mix}}(\alpha,\theta)$ for the least label count reproducing every report word at one fixed angle, and define N_{pure} with a counted pure start. The same conclusions below hold if every terminal qubit POVM is included: the uppers realize the subnormalized quantum states themselves and the lowers use report words alone. Write $\mathcal{K}_M$ for the set of complex eigenvalues of $M \times M$ row-stochastic matrices, as in Section S20.

Lemma S7.2 (Relative cylinder obstruction and polygon realization)**.** *For $\alpha > 0$ and $\sin\theta \neq 0$, every M-label mixed-start realization satisfies*

$$\alpha e^{i\theta} \in \mathcal{K}_M. \tag{S7.3}$$

Conversely, let P_q be the regular q-gon with vertices $e^{2\pi ij/q}$, $q \geq 3$, one of which represents τ. If $\alpha e^{i\theta} \in P_q$, its q physical pure states form one dictionary invariant under all three normalized branches, and give a q-label mixed-start realization of the complete instrument. A single dictionary works for any collection of angles satisfying this condition.

Proof. For a candidate HMM put $v = \mu T_1$ and $d = T_1\mathbf{1}$, so that the cylinder in (S7.2) is $vT_0^n d$. Restrict T_0 to labels reachable from v and from which d is eventually positive, obtaining S. For every pair i, j in this productive set, fixed positive entry and exit paths give constants $c_{ij} > 0$ and integers $a_i, b_j \geq 0$ with

$$c_{ij}(S^n)_{ij} \leq p(1\,0^{a_i+n+b_j}\,1) \leq C_{ij}8^{-n}.$$

Thus $\rho(S) \leq 1/8$. Cayley–Hamilton and the three distinct modes in (S7.2) force $\alpha e^{i\theta}/8$ to be an eigenvalue of S. An irreducible diagonal block B carrying it has spectral radius $0 < R \leq 1/8$. The Perron normalization proved before Proposition S21.2 gives a stochastic matrix Q with eigenvalue $\alpha e^{i\theta}/(8R)$ and eigenvector u. For a stationary row π of Q, $\pi u = 0$ because this eigenvalue is nonreal. Consequently

$$Q' = 8RQ + (1 - 8R)\mathbf{1}\pi$$

is stochastic and has eigenvalue $\alpha e^{i\theta}$. Its size is at most M; padding by self-loop labels proves (S7.3). This normalizes only the irreducible block carrying the mode, and does not assume a positive Perron vector for a reducible productive core.

Identify the real Bloch plane with $\mathbb{C}$, taking τ to 1. Multiplication by $\alpha e^{i\theta}$ maps P_q into itself whenever that point is in P_q, by its invariance under multiplication by a qth root. This handles branch zero. Branch one resets to the vertex 1. If $r \in P_q$ and $h = \mathrm{tr}(\Psi_1(\sigma_r)) \in [1/16, 3/16]$, the normalized third-branch Bloch vector is

$$\frac{-\alpha R_\theta r/8 - h(1,0)}{7/8 - h}.$$

Its norm is at most $5/11$, strictly below the inradius $\cos(\pi/q) \geq 1/2$. Thus it too lies in P_q. Choose barycentric coefficients for each of these normalized images and multiply them by the branch probability. The resulting nonnegative matrices T_s have stochastic sum and satisfy, for every dictionary state,

$$\Psi_s(\sigma_i) = \sum_j T_s(i,j)\sigma_j.$$

Uniform initial mixture represents $I/2$. Induction proves this identity for every subnormalized word state; terminal effects and common adaptive policies therefore have the same laws. Triangulation of the polygon provides a piecewise continuous, hence Borel, choice when the angle varies. □

Theorem S7.3 (Sharp noisy-instrument memory)**.** *For $M \geq 3$, $0 < \theta \leq 2\pi/M$ and $0 < \alpha \leq 1$,*

$$N_{\mathrm{mix}}(\alpha,\theta) \leq M \quad \Longleftrightarrow \quad \alpha \leq \frac{1}{\cos\theta + \sin\theta\tan(\pi/M)}. \tag{S7.4}$$

At equality the mixed minimum is exactly M, and the pure minimum is $M+1$. In every M-label mixed realization at this boundary, its zero-report matrix is diagonally similar to $\frac{1}{8}[(1-t)I + tC_M]$

for some $0 < t \le 1$, up to cyclic relabelling and orientation. No analogous claim is made for its other report matrices.

For $0 < \alpha < 1$ put

$$K(\alpha) = \max\left\{3, \left\lceil \frac{\pi}{\arccos \alpha} \right\rceil \right\}. \tag{S7.5}$$

One dictionary of $K(\alpha)$ labels realizes simultaneously every rotation command in (S7.1)*, with a mixed start. This count is minimal for the full angle command family, and the corresponding pure minimum is $K(\alpha) + 1$. In particular*

$$\max_{\sin\theta \neq 0} N_{\mathrm{mix}}(\alpha, \theta) = K(\alpha), \qquad \max_{\sin\theta \neq 0} N_{\mathrm{pure}}(\alpha, \theta) = K(\alpha) + 1.$$

The same common upper holds for each finite known angle alphabet, with matching lower whenever that alphabet contains a phase attaining the displayed maximum.

At $\alpha = 0$, the complete command response has exactly two labels with a mixed start and three with a pure start. At $\alpha = 1$, the full angle family has no finite realization; for one nonreal angle its finite-order and infinite-order alternatives are those of Theorem S15.1. Whenever a relevant minimum is finite, pure initialization adds exactly one label.

Proof. The lower in (S7.4) follows from (S7.3) and Lemma S20.2. For the upper, the radial boundary of the edge between 1 and $e^{2\pi i/M}$ is the right side of (S7.4); use the polygon realization. At that boundary strict monotonicity of $\cot(\pi/M)$ excludes every smaller size at least three; the rank-three lower excludes sizes one and two. For any realizing size M, the irreducible block in the preceding proof must use all M labels. Applying the sector inequality before the final radial contraction gives

$$\frac{\alpha}{8R}[\cos\theta + \sin\theta \tan(\pi/M)] \le 1,$$

so $R \ge 1/8$, and hence $R = 1/8$. Equality in the inherited sector lemma therefore fixes its Perron normalization to the lazy directed cycle. The stationary initialization theorem supplies the pure count without requiring the original diagonal similarity to be scalar.

The disk of radius α lies in the regular $K(\alpha)$-gon, including the equality case $\alpha = \cos(\pi/K(\alpha))$. The preceding lemma therefore constructs one physical dictionary for every angle. If $K = 3$, any nonreal phase has rank three and requires three labels. If $K > 3$, take $\theta = \pi/(K-1)$. The definition of the ceiling gives $\alpha > \cos(\pi/(K-1))$, so the first-sector criterion excludes $K - 1$ labels for this single phase. It also excludes every smaller size for the full command family. The pure upper appends one initializer, and the single critical phase, together with Theorem 15.1, gives the matching lower.

At $\alpha = 0$, use the two diagonal density matrices with Bloch coordinates 1 and $-3/11$. Their initial probabilities are $3/14$ and $11/14$. In that order the report matrices are

$$T_0 = \begin{pmatrix} 3/112 & 11/112 \\ 3/112 & 11/112 \end{pmatrix}, \qquad T_1 = \begin{pmatrix} 3/16 & 0 \\ 19/176 & 0 \end{pmatrix}, \qquad T_2 = \begin{pmatrix} 0 & 11/16 \\ 7/88 & 121/176 \end{pmatrix}. \tag{S7.6}$$

Their sum has both rows $(3/14, 11/14)$, and direct substitution verifies the subnormalized state identities. The rank-two lower and the stationary initialization law give the asserted exact counts. Finally the undamped full family includes a non-root phase, so its infinite lower follows from the retained theorem. For every finite angle alphabet the prepared predictive state is fixed by every unobserved command, and $p(11) \neq p(1)^2$; thus the exact initialization law applies at all parameters, not just at spectral boundaries. □

For a fixed phase the lower in Lemma S7.2 and its regular-polygon upper also transfer the already proved arithmetic envelope of Theorem S19.1, with the same constants, to the complete observed instrument. Consequently its two logarithmic exponents and its badly-approximable criterion transfer as well. This is a change from a scalar response to an entire instrument, not a

new number-theoretic assertion. No equality between the full-instrument minimum and the scalar spectral minimum is asserted outside the proved sectors and boundary cases.

S8. The complete mean–contrast law and optimal-factor rigidity

The mean of a response and the size of its nonconstant part impose distinct positivity constraints. The return family of Theorem S4.6 fixes the mean at $1/n$. We now allow every feasible mean and both signs of contrast, for every $n \geq 5$. A second positivity inequality changes the sharp boundary: simply rescaling the old threshold gives the wrong answer on an interval of means. The proof yields the complete boundary factorization on that interval. Throughout, the mean is the average terminal probability over the group; it is not a prior on an unknown parameter.

Bounded simplex-structured factorization and stacking complementary nonnegative slacks with a common factor are established techniques [77, Section IV-C]. Threshold attainment and boundary identifiability also occur in the geometric theory of nonnegative rank [78, Section 4.3]. Our additional conclusion is the sharp simultaneous coefficient extremum below, its three-piece parameter region, and the resulting unrestricted group-memory and boundary-factor laws.

The following elementary extremum treats two nonnegative coefficient matrices sharing a left null vector. Both outcomes must be represented through the same preparation; this common-null-vector requirement is stronger than separate rank bounds on the two outcome matrices.

Lemma S8.1 (Sharp simultaneous positivity inequality)**.** *Let $n \geq 5$ and $v, c, d \in \mathbb{R}^n$ satisfy*

$$\sum_i v_i = 1, \qquad v \cdot c = v \cdot d = 0, \qquad c_i \geq v_j - \delta_{ij}, \quad d_i \geq \delta_{ij} - v_j \quad (1 \leq i, j \leq n).$$

Put $C = \sum_i c_i$, $D = \sum_i d_i$, and $\gamma_n = 2(n-2)/(n-4)$. For every real $\gamma \geq \gamma_n$,

$$\gamma C + D \;\geq\; L_n(\gamma) := \frac{\gamma(n+1) + n^2 - 2n - 1 + 2\sqrt{n(\gamma-1)(\gamma+n-2)}}{n-1}. \tag{S8.1}$$

Equality holds exactly for the following triples, up to a simultaneous coordinate permutation. Set

$$m = \sqrt{\frac{\gamma+n-2}{n(\gamma-1)}}, \qquad M = \frac{1+m}{n-1}.$$

Then

$$\begin{aligned} v &= (M, \ldots, M, -m), \\ c &= (M, \ldots, M, M + M/m), \\ d &= (1-M, \ldots, 1-M, (n-1)M(1-M)/m). \end{aligned} \tag{S8.2}$$

In particular $1/\sqrt{n} < m \leq (n-2)/n$. No sign assumption on v or symmetry assumption on the resulting coefficient matrices is made.

Proof. The diagonal inequalities give $D \geq n-1$. If $v \geq 0$, the argument in Lemma S4.5 applies verbatim to c: a largest coordinate $M > 0$ forces all other positive coordinates to vanish, hence $C \geq n-1$. We verify below that $L_n(\gamma) < \gamma(n-1) - 1/(n-1)$ in the stated parameter range; therefore this case is strict.

Otherwise write $m = -\min_i v_i > 0$, relabel so that $v_1 = M$ is maximal, let L be the second largest coordinate, and put $b_0 = \max\{L, M-1\}$. The inequalities for c give

$$c = (b_0, M, \ldots, M) + u, \quad u \geq 0, \qquad v \cdot (b_0, M, \ldots, M) = M(b_0 + 1 - M) \geq 0.$$

Using $v \cdot u \geq -m \sum_i u_i$ and $v \cdot c = 0$ gives

$$C \geq b_0 + (n-1)M + \frac{M(b_0 + 1 - M)}{m}. \tag{S8.3}$$

This is the first-outcome estimate from the retained sharp singular-matrix proof. For the second outcome, $d = \mathbf{1} - v + z$ with $z \geq 0$ and $v \cdot z = \|v\|_2^2 - 1$. Consequently

$$D \geq n - 1 + \frac{1 - \|v\|_2^2}{m}.$$

The bound $-m \leq v_i \leq M$ implies $v_i^2 \leq (M - m)v_i + Mm$; summing it yields

$$D \geq n + \frac{1 - M}{m} - nM. \tag{S8.4}$$

This bound is valid even when its right side is below $n - 1$.

As in the retained proof,

$$M \geq M_- := \frac{1 + m}{n - 1}, \qquad L \geq \frac{1 - M + m}{n - 2}, \qquad M_+ := 1 + \frac{m}{n - 1}.$$

If $M \geq M_+$, (S8.3) with $b_0 \geq M - 1$ gives $C \geq nM - 1 \geq n - 1$. Together with $D \geq n - 1$ this case is again strict. If $M_- \leq M \leq M_+$, substitute $b_0 \geq (1 - M + m)/(n - 2)$ in (S8.3); its right side increases with b_0. This gives

$$C \geq F(M) := \frac{1 + m}{n - 2} + (n - 1)M + \frac{n - 1}{m(n - 2)}(M - M^2).$$

Therefore

$$\gamma C + D \geq G(M) := \gamma F(M) + n + \frac{1 - M}{m} - nM.$$

The quadratic G is strictly concave in M, so its minimum on this interval is at an endpoint. At the left endpoint its value is

$$G(M_-) = \frac{\gamma(n + 1) + n^2 - 2n - 1 + n(\gamma - 1)m + (\gamma + n - 2)/m}{n - 1} \geq L_n(\gamma), \tag{S8.5}$$

by the arithmetic–geometric mean inequality. At the right endpoint,

$$G(M_+) = (\gamma - 1)nM_+ + n - \gamma - \frac{1}{n - 1} \geq E_n(\gamma) := \gamma(n - 1) - \frac{1}{n - 1}.$$

For completeness, $E_n(\gamma) > L_n(\gamma)$ throughout the asserted range. After subtracting the rational terms and multiplying by $n - 1$, this is equivalent to

$$n\{(n - 3)\gamma - n + 2\} > 2\sqrt{n(\gamma - 1)(\gamma + n - 2)}.$$

The left side is positive. On squaring and dividing by n, the difference is

$$Q_n(\gamma) = n\{(n - 3)\gamma - n + 2\}^2 - 4(\gamma - 1)(\gamma + n - 2).$$

For $n \geq 6$ its quadratic coefficient is positive, $Q_n(2) = n(n - 6)(n - 2) \geq 0$, and $Q_n'(2) = 2n(n - 3)(n - 4) - 4n - 4 > 0$. Hence it is strictly positive at $\gamma \geq \gamma_n > 2$. For $n = 5$, $\gamma_n = 6$ and $Q_5(\gamma) = 16\gamma^2 - 68\gamma + 57$ is positive and increasing for $\gamma \geq 6$. This proves the strict endpoint comparison and (S8.1).

Equality forces $M = M_-$ and equality in (S8.5); thus m has the stated value. The sum of the coordinates other than one minimum is $(n - 1)M$, so each of them is M. Equality in (S8.3) places every excess coordinate of c at the unique negative coordinate and gives $c_n = M + M/m$. Equality in the bound for D similarly places every excess over $\mathbf{1} - v$ at that coordinate, giving the stated d_n. The range $\gamma \geq \gamma_n$ is precisely $1/\sqrt{n} < m \leq (n - 2)/n$, so $M + m \leq 1$. Thus $d_i = 1 - M \geq m$ for $i < n$, and $d_n \geq 1 + m$; all the required inequalities hold. The displayed c also satisfies its inequalities and $v \cdot c = 0$. Direct substitution verifies $v \cdot d = 0$ and equality in the bound. □

Let $\chi_n(g)$ count fixed points in the natural action of S_n. For a mean $p \in [0,1]$ and real contrast a, define

$$\beta_{p,a}(g) = p + a\{\chi_n(g) - 1\}, \qquad b_{p,a} = (\beta_{p,a}, 1 - \beta_{p,a}). \tag{S8.6}$$

This is a probability table if and only if

$$0 \le p - a \le 1, \qquad 0 \le p + (n-1)a \le 1. \tag{S8.7}$$

Both endpoint values occur, at a derangement and at the identity. Put

$$p_0 = \frac{1}{n - \sqrt{n}}, \qquad p_1 = \frac{2}{n}, \qquad F_n(p) = \begin{cases} p\,\dfrac{\sqrt{n}-1}{\sqrt{n}+1}, & 0 \le p \le p_0, \\ \dfrac{(n-1)p - 1}{(n-1)(np-1)}, & p_0 \le p \le p_1, \\ \dfrac{1-p}{n-1}, & p_1 \le p \le 1. \end{cases} \tag{S8.8}$$

The three formulas agree at the two meeting points.

Theorem S8.2 (Sharp synthesis for bounded centered coefficients)**.** *Let $n \ge 5$, $P = I - J/n$, $0 < p < 1$ and $a > 0$. The following are equivalent:*

(1) *There are a probability column u and a real matrix A such that*

$$0 \le A \le u\mathbf{1}^{\mathsf{T}}, \qquad \sum_{i,j} A_{ij} = np, \qquad PAP = aP, \qquad \operatorname{rank}[A \;\; u] < n.$$

(2) *There exists such a pair admitting a common nonnegative factorization*

$$A = \sum_{i=1}^{n-1} w_i \lambda_i^{\mathsf{T}} D_i, \qquad u = \sum_{i=1}^{n-1} w_i \lambda_i^{\mathsf{T}}, \qquad w \in \Delta_{n-1}, \quad \lambda_i \in \Delta_n, \quad D_i \in [0,1]^n.$$

(3) $a \le F_n(p)$.

Thus joined ordinary rank deficiency and a positive factor of width $n-1$ have the same parameter-feasibility region in this centered coefficient problem. This does not assert that every feasible pair away from the boundary has that nonnegative factorization.

For $0 < p \le p_0$ and $a = F_n(p)$, all width-$n-1$ factorizations in (ii), up to factor and simultaneous coordinate relabeling, are

$$\lambda_i = \ell_i, \qquad D_i = \frac{np}{(n-1)w_i}\ell_i, \qquad \sum_{i=1}^{n-1} w_i = 1, \qquad w_i \ge \frac{np(1 - 1/\sqrt{n})}{n-1}, \tag{S8.9}$$

where $\ell_i = (1 - 1/\sqrt{n})e_i + e_n/\sqrt{n}$ and D_i is the first-output decoder. For $p_0 \le p \le p_1$ and $a = F_n(p)$, all width-$n-1$ factorizations in (ii) have exactly the form

$$w_i = \frac{1}{n-1}, \qquad \lambda_i = (1-b)e_i + be_n, \qquad D_i = e_i + (np-1)e_n, \qquad b = \frac{p}{np-1}, \quad 1 \le i < n. \tag{S8.10}$$

At $p = p_0$ these classifications agree. In the middle interval the coefficient pair (A, u) itself is unique up to simultaneous coordinate permutation; the proof gives it explicitly.

Proof. Start with any matrix pair in (i); no positive factorization is assumed in the converse. Put $B = u\mathbf{1}^{\mathsf{T}} - A \ge 0$. A nonzero common left null vector v of A and u also annihilates $B = u\mathbf{1}^{\mathsf{T}} - A$. If its coordinate sum were zero, $v^{\mathsf{T}}AP = av^{\mathsf{T}}$ would contradict $v^{\mathsf{T}}A = 0$. Normalize that sum to one. Exactly as in the retained null-vector decomposition,

$$A = a\{I + c\mathbf{1}^{\mathsf{T}} - \mathbf{1}v^{\mathsf{T}}\}, \quad B = a\{-I + d\mathbf{1}^{\mathsf{T}} + \mathbf{1}v^{\mathsf{T}}\}, \quad v \cdot c = v \cdot d = 0.$$

Nonnegativity gives all the inequalities of Lemma S8.1, and

$$C = \frac{p}{a}, \qquad D = \frac{1-p}{a}. \tag{S8.11}$$

For $p \le p_0$, apply Lemma S4.5 to $A/(np)$: its centered coefficient is $a/(np)$, so its parameter there is a/p, giving $a \le p\tau_n = F_n(p)$. For $p \ge p_1$, the diagonal inequalities for d give $D \ge n-1$, hence $a \le (1-p)/(n-1) = F_n(p)$.

For $p_0 < p \le p_1$, set

$$m_0 = (n-1)p - 1, \qquad \gamma = \frac{nm_0^2 + n - 2}{nm_0^2 - 1}.$$

Then $1/\sqrt{n} < m_0 \le (n-2)/n$, so $\gamma \ge \gamma_n$, and m_0 is the minimizing value in (S8.5). The equality triple (S8.2) with $M = p$ has sums

$$C_0 = p(n + 1/m_0), \qquad D_0 = (1-p)(n + 1/m_0).$$

Thus $L_n(\gamma) = (\gamma p + 1 - p)(n + 1/m_0)$. Substituting (S8.11) into the lemma and cancelling the positive factor $\gamma p + 1 - p$ gives

$$a \le \frac{1}{n + 1/m_0} = \frac{(n-1)p - 1}{(n-1)(np-1)} = F_n(p).$$

This proves the necessary boundary everywhere.

We give matching constructions at that boundary. In each case, smaller $a \ge 0$ is obtained on the same core by mixing the displayed decoder with the constant decoder p, with mixing weight $a/F_n(p)$. For $0 < p \le p_0$, use $n-1$ factors with equal weights, preparation rows ℓ_i and decoders $np\ell_i$. The largest decoder entry is $np(1 - 1/\sqrt{n}) \le 1$, precisely because $p \le p_0$. The coefficient matrix is npA_*, where A_* is the retained sharp singular extremizer. Its mean is p and its centered part is $p\tau_n P = F_n(p)P$, as required.

For $p_0 \le p \le p_1$, use (S8.10). Here $0 < b < 1$ and $0 < np - 1 \le 1$. Put $r = np - 1$ and $a_* = (1-b)/(n-1) = F_n(p)$. The resulting coefficient matrix has

$$A_{ii} = a_* \ (i < n), \quad A_{ij} = 0 \ (i \ne j < n), \quad A_{in} = a_* r, \quad A_{ni} = \frac{b}{n-1}, \quad A_{nn} = br = p.$$

The identity $r(nb - 1) = 1$ shows that all cross differences of $A - a_* I$ vanish. Thus $PAP = a_* P$ and the group mean is p; the response is β_{p,a_*}.

For $p_1 \le p < 1$, put $b = 2/n$, use the same preparations $(1-b)e_i + be_n$ and equal orbit weights, and let

$$d_0 = \frac{np - 2}{n - 2}, \qquad D_i(j) = d_0 + (1 - d_0)\mathbf{1}_{\{j \in \{i,n\}\}}.$$

These are valid decoders. At $p = p_1$ they agree with the preceding construction; mixing those decoders with the constant one shows that the response is $p + [(1-p)/(n-1)](\chi_n - 1)$ for every $p \ge p_1$. Equivalently this follows by direct permutation summation. This proves attainment at $F_n(p)$. The same constructions and decoder mixing give (iii)⇒(ii), while the displayed factorization immediately gives (ii)⇒(i). The converse already proved (i)⇒(iii), so the matrix synthesis equivalence is complete.

For the first rigidity statement, the equality case of the retained singular-coefficient lemma forces $A = npA_*$. The positive diagonal leading block and its zero off-diagonal entries force precisely one nonnegative rank-one summand at each of its $n-1$ diagonal positions. The adjacent last-row and last-column ratios then give (S8.9), exactly as in the retained proof. Its decoder bound is $w_i \ge np(1 - 1/\sqrt{n})/(n-1)$. At $p = p_0$ summing these lower bounds forces equal weights. Pair uniqueness at this endpoint also follows without presupposing a positive factorization. The singular lemma fixes $A = np_0 A_*$, whose normalized left null vector is $(p_0, \dots, p_0, -1/\sqrt{n})$. Thus $v \cdot u = 0$ and $\sum_i u_i = 1$ force $u_n = 1/\sqrt{n}$. The bounds $u_i \ge A_{ii} = 1/(n + \sqrt{n})$ for $i < n$ then sum to $1 - 1/\sqrt{n}$, forcing equality in every one of them.

If $p_0 < p \le p_1$ and $a = F_n(p)$, equality holds in the weighted lemma with the chosen γ. It forces $v = (p, \dots, p, -m_0)$ and the unique c, d in (S8.2). The coefficient matrix is therefore exactly the one displayed in the middle construction, while $u = a(c + d)$ has $u_i = a$ for $i < n$ and $u_n = b$. Its leading block again forces one rank-one summand at each diagonal position, with both factors

supported on $\{i, n\}$. The last-row ratios force $\lambda_i = (1-b)e_i + be_n$. The diagonal entries force $D_i(i) = a/\{w_i(1-b)\} = 1/\{(n-1)w_i\}$. The upper bound $D_i(i) \le 1$ implies $w_i \ge 1/(n-1)$; summing the weights makes every one equal to $1/(n-1)$. The last-column ratios then give $D_i(n) = np - 1$. This proves the complete middle-boundary classification. □

Corollary S8.3 (Exact memory throughout the mean–contrast plane)**.** *Let $n \ge 5$ and let (p, a) satisfy* (S8.7)*. At $a = 0$, both mixed-core and pure-start minima are one. If $a \ne 0$, put $q = p$ when $a > 0$ and $q = 1 - p$ when $a < 0$. Then*

$$\kappa_{S_n}(b_{p,a}) = \begin{cases} n(n-1), & 0 < |a| \le F_n(q), \\ n^2, & |a| > F_n(q), \end{cases} \qquad m(b_{p,a}) = 1 + \kappa_{S_n}(b_{p,a}). \tag{S8.12}$$

Every nonconstant table has observable group S_n, residual cardinality $n!$, and observable dimension $1 + (n-1)^2$. For $a > 0$ and $p \ge 2/n$, the smaller memory realizes every feasible contrast. Thus positivity causes no additional core cost anywhere in that part of the parameter plane.

At the nontrivial boundary $a = F_n(p)$ with $0 < p \le 2/n$, all minimum permutation-core factorizations are exactly those in (S8.9) *and* (S8.10)*. For negative contrast, exchange the two output labels. This classifies minimum permutation cores, without asserting uniqueness of every unreduced stochastic machine.*

Proof. It suffices to treat $a > 0$, because exchanging the two outputs replaces (p, a) by $(1-p, -a)$. The trivial block of the Fourier transform is $(p, 1-p)$, and the nontrivial standard block is $aI_{n-1}/(n-1)$ in the first coordinate. The faithful-standard-module argument in Theorem S4.6 therefore gives the claimed observable group, residual count and dimension for every $a > 0$. That theorem's subgroup-index estimate depends only on this module. It implies the lower bound $n(n-1)$ and proves that every core of size less than n^2 compresses to exactly $n-1$ natural orbits: extra orbits have no standard component, so their nonnegative constant contribution is absorbed into the natural-orbit decoders. This absorption preserves both output probabilities and does not use the value $p = 1/n$.

On n natural orbits, put weight $1/n$ on each, prepare point i on orbit i, and use

$$D_i(j) = p - a + na\,\mathbf{1}_{\{i=j\}}.$$

These decoders lie in $[0, 1]$ exactly under (S8.7). Their response is (S8.6). Thus the core is always at most n^2, and there are no possible intermediate values.

Suppose a realization on $n-1$ natural orbits exists. Its preparation rows λ_i, weights w_i and decoder rows D_i give

$$A = \sum_{i<n} w_i \lambda_i^{\mathsf{T}} D_i, \qquad u = \sum_{i<n} w_i \lambda_i^{\mathsf{T}}, \qquad B = u\mathbf{1}^{\mathsf{T}} - A \ge 0.$$

Here $A \ge 0$, u is a probability column, and both coefficient matrices have the same preparation factor. Their permutation sums give the two outcomes. The permutation-switch argument in the retained proof implies $PAP = aP$, and averaging gives $\sum_{ij} A_{ij} = np$. The preparation matrix has at most $n-1$ rows, so $\operatorname{rank}[A \;\; u] < n$. The matrix criterion in Theorem S8.2 is therefore necessary. Conversely its width-$n-1$ positive factorizations give exactly $n-1$ natural-orbit preparations and valid binary decoders. Their coefficient identities imply the required response for every permutation. Hence a core below n^2 exists exactly when $a \le F_n(p)$, and then the module lower bound makes its size $n(n-1)$. The retained pure-start formula gives $m = \min\{n!, 1+\kappa\}$, which equals $1+\kappa$ because $n^2 + 1 < n!$ for $n \ge 5$. At $a = 0$ the constant decoder uses one state. Every minimum core on the nontrivial boundary consists of $n-1$ natural orbits, so the factor classification follows from the matrix theorem. □

Physical realization and scope. Use the prepared unitary experiment of Theorem S4.6, with $\rho = n^{-1}\sum_i |ii\rangle\langle ii|$ and $E = \sum_i |ii\rangle\langle ii|$. Replacing the terminal effect by

$$E_{p,a} = (p-a)I + naE$$

produces (S8.6); its two eigenvalues are $p-a$ and $p+(n-1)a$, so the exact validity range is (S8.7), including negative contrast. The same two command generators realize every word. The theorem solves the whole displayed two-parameter family, including its constant boundary and its minimum-core factorization on the nontrivial transition curve. It is not an unrestricted high-contrast formula for every finite group, an identification result for an unknown model, or a solution of the original arbitrary-prior decision-risk problem.

S9. Approximation by truncating rare conditional excursions

Exact memory can be infinite although one finite simulator approximates every stopped experiment with a prescribed expected cost. The damping construction of Theorem S10.1 gives such an upper bound for the entire angle circle. For a fixed finite command alphabet we now obtain a smaller bound by preserving a long conditional excursion exactly and modifying only its overflow. The general mechanism depends on the growth of the continuation semigroup, and applies beyond the qubit family.

The ingredients have classical antecedents. Polyhedral positive realizations are part of the theory of Heller and of Monràs and Winter [48, 47, 76]. Martingale calculations for pattern waiting times are classical [80]. The use made here is a uniform approximation of complete adaptive stopped laws, with an explicit memory bound from a finite return polytope and the number of distinct continuation states. It is separate from the exact noisy realization problem of [50].

S9.1. **A sharp cost bound for a long conditional excursion.** Consider sequential calls, each producing a report declared either *continuing* or *returning*. Conditional on the entire past, the chosen command and any private randomization, suppose the probability of a continuing report is at most r, where $0 < r < 1$. Let η_L be the first call completing L consecutive continuing reports, with $L \geq 1$. A policy can stop initially or after any call; its number of calls is τ.

Lemma S9.1 (Sharp expected-cost excursion bound)**.** *For every such policy with $\mathbb{E}\tau \leq B < \infty$,*

$$\mathbb{P}\{\eta_L \leq \tau\} \leq \min\left\{1, \frac{(1-r)B}{r^{-L}-1}\right\}. \tag{S9.1}$$

The coefficient is optimal under the stated assumptions. For independent continuing indicators of probability r, the supremum over randomized stopping policies of expected cost at most B is exactly the right side.

Proof. Before the first completed run, let $K_t \in \{0, \ldots, L-1\}$ be the current run length. It increases by one on a continuing report and returns to zero otherwise. Absorb it at L after the first completed run, and put

$$u(k) = \frac{r^{-k}-1}{1-r} \quad (0 \leq k \leq L).$$

For $k < L$ and any conditional continuation probability $p \leq r$,

$$pu(k+1) + (1-p)u(0) - u(k) \leq ru(k+1) - u(k) = 1.$$

Stop this calculation at $\sigma_H = \min\{\tau, \eta_L, H\}$. Summing the conditional inequalities, including the indicator that the next call is actually made, gives

$$u(L)\mathbb{P}\{\eta_L \leq \tau,\ \eta_L \leq H\} \leq \mathbb{E}u(K_{\sigma_H}) \leq \mathbb{E}\sigma_H \leq B.$$

Let $H \to \infty$. Increasing events give (S9.1); no assumption about the expected duration of another process is used.

For sharpness, take independent Bernoulli indicators of probability r. The first completed run has finite expectation: disjoint blocks of length L succeed with probability r^L, so its expectation is at most L/r^L. In this case the displayed drift equals one. Stopping at $\eta_L \wedge H$ and then using bounded convergence for $u(K)$ and monotone convergence for the duration proves $\mathbb{E}\eta_L = u(L)$. With probability $\min\{1, B/u(L)\}$, selected before making any call, run this experiment until η_L; otherwise stop immediately. Its expected cost is at most B and its success probability equals the right side of (S9.1). □

Sharpness in this lemma concerns the probability of the exceptional run. It will not imply that every simulator must incur the approximation error estimated by that probability.

S9.2. **A general finite return polytope construction.** Let $\mathcal{H}$ be finite dimensional and let $\mathcal{C}$ be a closed convex set of density matrices invariant under every positive-probability normalized branch below. The full state space is allowed. For each command in a nonempty finite set A, let $\{\mathcal{I}_{a,y} : y \in Y_a\}$ be a quantum instrument, with finite Y_a. Select a subset $R_a \subseteq Y_a$ of continuing reports. Suppose that there are density matrices $\sigma_1, \dots, \sigma_k$, with $P = \operatorname{conv}\{\sigma_1, \dots, \sigma_k\}$, such that

$$\begin{aligned} &\rho_0 \in P, \qquad \sum_{y \in R_a} \operatorname{tr} \mathcal{I}_{a,y}(\rho) \le r < 1, \\ &\frac{\mathcal{I}_{a,y}(\rho)}{\operatorname{tr} \mathcal{I}_{a,y}(\rho)} \in P \quad (y \notin R_a,\ \operatorname{tr} \mathcal{I}_{a,y}(\rho) > 0) \end{aligned} \tag{S9.2}$$

for every command a and every $\rho \in \mathcal{C}$, with the seeds in $\mathcal{C}$. On the full state space, the continuation bound is equivalently the finite family of operator inequalities $\sum_{y \in R_a} \mathcal{I}_{a,y}^*(I) \preceq rI$. The return condition is a specified geometric hypothesis; it is not inferred merely from mixing of the unobserved channel. A prescribed preparation in $\mathcal{C}$ but outside P can be included by adding it to the seed list.

Write $\mathcal{E} = \{(a, y) : a \in A,\ y \in R_a\}$ and $b = |\mathcal{E}|$. For a word w over $\mathcal{E}$, let $\mathcal{I}_w$ be its chronological composition. For $L \ge 1$ define the finite set

$$\mathcal{D}_L = \left\{ \frac{\mathcal{I}_w(\sigma_j)}{\operatorname{tr} \mathcal{I}_w(\sigma_j)} : 1 \le j \le k,\ |w| < L,\ \operatorname{tr} \mathcal{I}_w(\sigma_j) > 0 \right\}, \qquad G(L) = |\mathcal{D}_L|. \tag{S9.3}$$

The empty word is included. Repeated states may be identified.

Theorem S9.2 (Finite-memory approximation from return geometry)**.** *Assume* (S9.2) *with* $0 < r < 1$. *For each* $L \ge 1$, *there is a single* $G(L)$*-label mixed-start classical model, chosen independently of the policy and of* B, *such that*

$$\sup_{\pi:\, \mathbb{E}^\pi_{\text{target}} \tau \le B} \| \mathcal{L}^\pi_{\text{target}} - \mathcal{L}^\pi_L \|_{\text{TV}} \le \min\left\{ 1, \frac{(1-r)B}{r^{-L} - 1} \right\}. \tag{S9.4}$$

Here policies can choose commands, stopping and arbitrary terminal POVMs from the visible record and their own randomness. The complete stopped record includes all commands and reports, the stopping time, the terminal outcome and nontermination. The bound also holds in trace distance for the classical–quantum terminal record, with nontermination represented by its scalar mass. There is no additional entangled quantum register.

A counted pure start needs at most $G(L) + 1$ *labels. In general*

$$G(L) \le k \sum_{j=0}^{L-1} b^j. \tag{S9.5}$$

If the $b \ge 1$ *continuation maps commute pairwise, then*

$$G(L) \le k \binom{L + b - 1}{b}. \tag{S9.6}$$

More generally any independently proved upper bound for the number of distinct normalized states in (S9.3) *can be used in place of these bounds. If that number is* $O(L^d)$*, the construction uses* $O((\log(B/\delta))^d)$ *labels for fixed* r *and* $B/\delta \to \infty$*.*

Proof. Use the states of $\mathcal{D}_L$ as hidden labels. At a label ρ, the probability of report y on command a is $p_{a,y}(\rho) = \operatorname{tr} \mathcal{I}_{a,y}(\rho)$. Zero probabilities require no successor choice. For a returning report, choose any convex decomposition of its normalized output in the seed states. For a continuing report whose normalized output belongs to $\mathcal{D}_L$, use that output label deterministically. In the remaining case, send the state to any one fixed seed. Multiply each successor distribution by $p_{a,y}(\rho)$ to obtain the report transition row. All rows are nonnegative and the sums over the reports are stochastic. Use a seed decomposition of ρ_0 for the initial distribution. At label ρ, the terminal decoder for any POVM $(E_z)_z$ is $\operatorname{tr}(E_z\rho)$.

To verify the error, first use all finite continuation words, without the length restriction in (S9.3). This countable refinement reproduces the target exactly. Every returning branch is expanded in the seeds; every continuing branch uses its exact normalized image. On each individual hidden label the weighted state identity is exact, so induction gives the target subnormalized quantum state for every command–report history. Initial mixing, terminal POVMs and common randomized policies preserve these identities. This is only a proof device, not an uncounted finite implementation resource.

Couple this exact refinement to the finite model by identical initial seed choices and by identical reports and successor choices while the states agree. The coupling first fails only when a continuing image leaves $\mathcal{D}_L$. After a return the exact label is a seed, so a product of fewer than L continuing branches is in $\mathcal{D}_L$ by definition. The same is true initially. Thus a failure requires a run of at least L consecutive continuing reports. Identifying duplicate states does not change this implication: their next normalized image depends only on the state itself, and every shorter word image is still present.

Before a failure the two visible histories, quantum states and policy choices can be identical. After a failure use arbitrary valid tails. For a fixed finite horizon, the stopped classical–quantum states have a common positive contribution from all paths with no failure. If the remaining mass is e, their trace distance is at most e, by positivity and the triangle inequality. The same assertion applies to the complete record after any terminal POVM. The target's probability of a failure before stopping is at most $\mathbb{P}^{\pi}_{\text{target}}\{\eta_L \le \tau\}$, which is bounded by Lemma S9.1. Its conditional continuation bound follows from (S9.2), also after conditioning on the virtual seed choices.

For unbounded stopping, retain every stopped history up to horizon H and combine the unfinished histories into one symbol. The sum of trace norms on the finite stopped blocks increases to its full sum; the two unfinished masses decrease to their respective nontermination masses. Their absolute difference therefore converges to the nontermination term. The uniform finite-horizon bound passes to the limit, giving (S9.4). The budget is always evaluated under the target. The argument does not require the approximate model to have expected duration at most B.

For a pure start append one initializer. On the first actual command it uses the initial mixture times the corresponding report matrix and enters the core. Its zero-call terminal decoder is that of ρ_0. It is never entered again. This proves the upper $G(L) + 1$ without an uncounted preliminary randomization. If a separate finite policy uses s labels, the same construction gives at most $sG(L)+1$ labels for the closed product implementation with a prescribed initial label.

There are b^j words of length j, proving (S9.5). For commuting maps the product depends only on a multi-index $m \in \mathbb{N}^b$ with $|m| < L$; there are $\binom{L+b-1}{b}$ such multi-indices. Normalization and deletion of zero images cannot increase this number. Finally it suffices to take

$$L \ge \max\left\{1, \left\lceil \frac{\log(1 + (1-r)B/\delta)}{\log(1/r)} \right\rceil \right\} \quad (B > 0,\ \delta > 0).$$

The asserted consequence of polynomial orbit growth follows. □

The construction preserves every finite history containing no run of L continuing reports, including its unnormalized terminal state. It does not claim that its approximate response is stationary or that the extra pure-start label is necessary. When all continuation maps have finite normalized orbit on the seeds, the full finite orbit instead gives an exact model. These are upper constructions, not a converse classification of general positive realizations.

S9.3. **A polylogarithmic simulator for finitely many commanded angles.** Apply the theorem to (S7.1) with $\alpha = 1$. Let the finite known angle alphabet be $\Theta = \{\theta_1, \dots, \theta_a\}$, where $a \geq 1$. The continuation report is 0, of probability $1/8$. In the real Bloch plane put

$$v_0 = (1, 0), \qquad v_1 = (-1/2, \sqrt{3}/2), \qquad v_2 = (-1/2, -\sqrt{3}/2).$$

Let σ_j be the corresponding pure states. Their triangle contains the disk of radius $1/2$, and $I/2$ is their equal mixture. Report 1 returns to $\sigma_0 = \tau$. By Lemma S7.2, every normalized report-2 image has Bloch norm at most $5/11 < 1/2$, so it returns to this same triangle, at every angle and from every real-plane state. The real-plane density matrices form an invariant closed convex set: the unitaries are real, and the other two branches preserve real symmetric matrices. Thus they supply the set $\mathcal{C}$ in the general theorem.

Corollary S9.3 (Uniform finite-alphabet approximation)**.** *For the undamped instrument at any fixed finite angle alphabet Θ and initial state $I/2$, there is, for every integer $L \geq 1$, one classical model with at most*

$$3\binom{L+a-1}{a} \quad \textit{mixed-start labels, or} \quad 1 + 3\binom{L+a-1}{a} \quad \textit{counted pure-start labels,} \tag{S9.7}$$

whose complete stopped-law error satisfies

$$\sup_{\pi:\, \mathbb{E}_1^\pi \tau \leq B} \|\mathcal{L}_1^\pi - \mathcal{L}_L^\pi\|_{\mathrm{TV}} \leq \min\left\{1, \frac{7B}{8(8^L - 1)}\right\}. \tag{S9.8}$$

The stronger stopped classical–quantum distance has the same bound. The statements include adaptive choices among all supplied angles and arbitrary terminal qubit POVMs.

For $B > 0$ and $\delta > 0$ it is enough to choose

$$L = \max\left\{1, \left\lceil \log_8\left(1 + \frac{7B}{8\delta}\right)\right\rceil\right\}. \tag{S9.9}$$

For a single commanded angle this uses at most $3L + 1$ pure-start labels, uniformly in that angle. For each fixed finite alphabet it uses $O_a((\log(B/\delta))^a)$ labels as $\delta/B \downarrow 0$. When the sine and cosine of each commanded angle are algebraic, every state and transition coefficient of the displayed construction is algebraic and is obtained by finitely many exact field operations.

Proof. The continuation maps are scalar multiples of commuting rotations, so their normalized orbit dictionary consists of

$$r_{j,m} = R_{m_1\theta_1 + \cdots + m_a\theta_a} v_j, \qquad 0 \leq j < 3, \quad m \in \mathbb{N}^a, \quad |m| < L. \tag{S9.10}$$

We may keep all these indices, even if some represent the same state. Initialize uniformly on $(0, 0), (1, 0), (2, 0)$. From a label with Bloch vector r, and command θ_i, set

$$h = \tfrac{1}{8} + \tfrac{1}{16} r_1, \qquad p_2 = \tfrac{7}{8} - h, \qquad u_2 = -\tfrac{1}{8} R_{\theta_i} r - h v_0.$$

Report 1 has a row of mass h into seed $(0, 0)$. Report 2 has rows into the three seeds of masses

$$t_{2,j} = \frac{p_2 + 2 v_j \cdot u_2}{3}, \qquad j = 0, 1, 2. \tag{S9.11}$$

Indeed, $\sum_j v_j = 0$ and $\sum_j v_j v_j^{\mathsf{T}} = \frac{3}{2} I_2$, so these masses sum to p_2 and their weighted Bloch vector is u_2. Moreover, $|u_2|/p_2 \le 5/11$ and $p_2 \ge 11/16$ give

$$t_{2,j} \ge \frac{p_2}{3}(1 - 10/11) \ge \frac{1}{48} > 0.$$

Thus the rows give the exact report-2 subnormalized state, with an explicit positive margin. On report 0, whose mass is $1/8$, move from (j, m) to $(j, m + e_i)$ if $|m| < L - 1$. At $|m| = L - 1$ send that mass to the three seeds uniformly. All report probabilities are unchanged on each represented input state, and their sum is one. Only the state following an overflow zero report is modified.

The resulting table uses words with the same multi-index identified at each depth and the same common-prefix proof as the theorem. It keeps the depth indices and may truncate at depth L even when a periodic phase makes that next physical state coincide with an earlier one. The first possible modified state still occurs on the Lth consecutive zero. Its state count is (S9.7), and substitution of $r = 1/8$ in (S9.4) gives (S9.8). Formula (S9.9) makes this bound at most δ. The label count and all error bounds are uniform in the numerical values of the finite supplied angles, although the transition table itself uses those values. The explicit rotations and (S9.11) prove the algebraic assertion. □

If $B = 0$, a single initial decoder suffices, with no call and no error. The damping upper in Theorem S10.1 remains available and can be smaller for particular B, δ, a; one may choose the smaller of the two proved upper counts. Its simultaneous all-angle statement also remains valid. In contrast, the finite dictionary in (S9.10) is for a *fixed finite* angle alphabet; the bound is not asserted for a continuum of possible commands.

For every non-root phase, exact finite classical realization is still impossible by Theorem S15.1. At that same fixed physical instrument the present bound is logarithmic in B/δ for a single angle. Thus an inverse-square-root exact noise law is not a lower bound on approximation of bounded-cost stopped experiments. Section S9.4 proves a matching logarithmic law within the explicitly restricted class of physical dictionaries; a matching lower for arbitrary classical terminal decoder tables remains open. The stopped-law bound also does not control the approximate model's expected duration: unbounded costs require a further tail or uniform-integrability hypothesis. External policy storage, table bits, arithmetic precision and exact sampling remain separate resources.

S9.4. **A matching logarithmic law for physical dictionaries.** The preceding uppers use genuine density matrices as hidden dictionary states. Within that representation class a matching lower, including the leading logarithmic constant, is possible. This restriction is substantive: an arbitrary classical hidden-state model can have terminal decoder tables that are not Born probabilities of fixed density matrices. The conclusion below does not impose that restriction on such a model without proof.

Fix one commanded angle $\theta = 2\pi\omega$ in the undamped instrument. A *physical-dictionary simulator* has finitely many labels with fixed qubit density matrices σ_i, arbitrary stochastic report transition tables, and terminal decoder $d_E(i) = \mathrm{tr}(E\sigma_i)$ for every requested effect E. The causal transition tables are fixed independently of any later terminal-measurement request. The transitions themselves need not implement the quantum instrument on each dictionary state. Denote by $N^{\mathrm{m}}_{\mathrm{phys}}(\delta; B)$ and $N^{\mathrm{p}}_{\mathrm{phys}}(\delta; B)$ the least mixed- and counted-pure-start label counts that approximate, in total variation to error at most δ, every common visible stopping and terminal-POVM policy whose target expected calls are at most B.

Theorem S9.4 (Sharp leading order for a fixed badly approximable phase)**.** *Suppose ω is irrational and has a supplied constant $c > 0$ such that*

$$\|q\omega\|_{\mathbb{R}/\mathbb{Z}} \ge c/q \qquad (q \in \mathbb{Z}_{\ge 1}). \tag{S9.12}$$

Necessarily $c \le 1/2$. Fix $B > 0$ and put $\zeta = \min\{1, 7B/8\}$. For every integer $J \ge 2$, if

$$0 < \delta \le \frac{\zeta}{8^J(1 + J^2/c^2)}, \tag{S9.13}$$

then

$$N^{\mathrm{m}}_{\mathrm{phys}}(\delta; B) \ge J, \qquad N^{\mathrm{p}}_{\mathrm{phys}}(\delta; B) \ge J + 1. \tag{S9.14}$$

There is also a finite integer K_ω, independent of B, δ, such that, with L from (S9.9),

$$N^{\mathrm{m}}_{\mathrm{phys}}(\delta; B) \le K_\omega + L - 1, \qquad N^{\mathrm{p}}_{\mathrm{phys}}(\delta; B) \le K_\omega + L. \tag{S9.15}$$

Consequently, for either start convention and every fixed $B > 0$,

$$\lim_{\delta \downarrow 0} \frac{N_{\mathrm{phys}}(\delta; B)}{\log_8(1/\delta)} = 1. \tag{S9.16}$$

For $\omega = (\sqrt{5} - 1)/2$, one may take $c = 1/3$ and $K_\omega = 5$. All approximation lower bounds in this theorem are for physical dictionaries with the stated Born terminal decoders.

Proof. For $n \ge 0$, consider the report word $1\,0^n$ and the pure target state ρ_n it prepares. Its Bloch vector is $v_n = (\cos(2\pi n\omega), \sin(2\pi n\omega), 0)$, with the coordinates ordered along the real Bloch plane. The word has probability $8^{-(n+1)}$. Use the following policy. With probability ζ begin the test; otherwise stop before any call. During the test stop as soon as a report differs from the specified prefix, or at its completion. Apply the terminal POVM $(\rho_n, I - \rho_n)$ on successful completion. The target expected number of calls is

$$\zeta \sum_{j=0}^{n} 8^{-j} \le \zeta\, 8/7 \le B.$$

Thus this is one of the policies in the definition of the approximation error, for every n, without a horizon-dependent loss of the budget.

In the simulator let $A_n = \sum_i a_{n,i}\sigma_i$, $a_{n,i} \ge 0$, be the subnormalized terminal state on the event that this test is begun and its complete prefix is observed. Put $p_n = \zeta 8^{-(n+1)}$. Two events in the full stopped record give

$$|\operatorname{tr} A_n - p_n| \le \delta, \qquad \operatorname{tr}[(I - \rho_n)A_n] \le \delta. \tag{S9.17}$$

The second event has zero target probability. If $p_n > \delta$, these inequalities imply that some dictionary state with positive weight obeys

$$\operatorname{tr}(\rho_n\sigma_i) \ge 1 - \frac{\delta}{p_n - \delta}.$$

This conclusion is derived from joint events. No error bound for a rare conditional distribution has been assumed.

For $0 \le n < J$, condition (S9.13) gives

$$\frac{\delta}{p_n - \delta} \le c^2/J^2.$$

If $r_i \in \mathbb{R}^3$, $|r_i| \le 1$, is the Bloch vector of the corresponding physical dictionary state, then

$$|r_i - v_n|^2 \le 2 - 2r_i \cdot v_n = 4[1 - \operatorname{tr}(\rho_n\sigma_i)] \le 4c^2/J^2.$$

On the other hand, for $0 \le m < n < J$, the elementary inequality $\sin(\pi x) \ge 2x$ on $[0, 1/2]$ and (S9.12) give

$$|v_n - v_m| = 2|\sin(\pi(n - m)\omega)| \ge 4\|(n - m)\omega\|_{\mathbb{R}/\mathbb{Z}} \ge \frac{4c}{n - m} > \frac{4c}{J}.$$

One dictionary vector cannot be within distance $2c/J$ of two such points. At least J distinct labels are therefore necessary.

For a pure start, the initial label has its own physical density matrix σ_o. Stopping before any call and choosing any binary terminal POVM is allowed at zero cost. The error condition therefore implies $\frac{1}{2}\|\sigma_o - I/2\|_1 \le \delta$, or $|r_o| \le 2\delta$. Its fidelity with any pure state is at most $1/2 + \delta$. Under (S9.13), $J \ge 2$ and $c \le 1/2$ give $\delta < 1/4$ and $c^2/J^2 \le 1/16$. Thus the initial label cannot be one of the J labels whose fidelity is at least $15/16$. This proves both inequalities in (S9.14).

For the improved upper, choose $K = K_\omega$ such that

$$P_K = \operatorname{conv}\{v_0, \dots, v_{K-1}\} \supseteq \{x \in \mathbb{R}^2 : |x| \le 5/11\}.$$

Such a finite K exists for every irrational ω. Here is a direct argument sufficient for this use. Pigeonholing finitely many fractional multiples of ω gives an integer $q > 0$ with $0 < d = \|q\omega\| < 1/4$. The points with indices $0, q, 2q, \dots, \lfloor 1/d \rfloor q$ have successive circular gaps at most d, in one orientation or the other. Taking a prefix containing all these indices, its inscribed polygon has every central gap less than $\pi/2$, and hence inradius greater than $1/\sqrt{2} > 5/11$. The origin is in this polygon. It is a valid seed polytope: report 1 returns to v_0, and every report-2 output is in its contained disk.

For one angle, all continuation images of these K seeds through depth $L - 1$ belong to $\{v_0, \dots, v_{K+L-2}\}$, a set of $K + L - 1$ states. Returning branches are expanded in the original K seeds. No overflow can occur before L successive zero reports, even when a return chooses the last seed. Theorem S9.2 therefore proves (S9.15). The finite barycentric coefficients can be chosen from any triangulation of P_K. All labels, including the optional initializer representing $I/2$, are physical states.

For fixed B, the upper has $L = \log_8(1/\delta) + O_B(1)$. Conversely, (S9.13) holds for an integer

$$J = \log_8(1/\delta) - 2\log_8\log_8(1/\delta) + O_{B,c}(1)$$

chosen sufficiently small, as follows by taking logarithms of that inequality. Dividing the two bounds by $\log_8(1/\delta)$ proves (S9.16).

Finally put $\omega = (\sqrt{5}-1)/2$, and let p be a nearest integer to $q\omega$. Its conjugate is $\omega' = -(\sqrt{5}+1)/2$. The product $(q\omega - p)(q\omega' - p) = p^2 + pq - q^2$ is a nonzero integer. Therefore

$$|q\omega - p| \ge \frac{1}{|q\omega' - p|} \ge \frac{1}{\sqrt{5}q + 1/2} > \frac{1}{3q},$$

proving (S9.12) with $c = 1/3$. The first five fractional orbit angles, in circular order, are

$$0, \quad 2\omega - 1, \quad 4\omega - 2, \quad \omega, \quad 3\omega - 1.$$

Their successive gaps are $2\omega - 1$ or $2 - 3\omega$. Since $3/5 < \omega < 5/8$, their maximum is $2\omega - 1 < 1/4$. This proves the required polygon inclusion with $K_\omega = 5$. □

The final limit fixes $B > 0$; it is not a uniform two-parameter asymptotic as B also varies. For arbitrary classical decoder tables, the general finite-alphabet upper remains valid, while the present packing lower need not apply. The badly approximable phase condition has an explicit quadratic-irrational instance above. It is not asserted for the separate algebraic rotation $(3 + 4i)/5$ studied earlier in the paper.

The improved upper does apply to that original algebraic rotation, with an explicit constant. Write $\theta = \arg((3 + 4i)/5) \in (0, \pi/2)$. The inequalities $\pi/6 < \theta < \pi/3$ put the first seven orbit points $0, \theta, \dots, 6\theta$ in their circular order with every gap less than π. The six consecutive edges have distance $\cos(\theta/2) = 2/\sqrt{5}$ from the origin. The closing edge has distance $-\cos(3\theta) = 117/125$. Both exceed $5/11$, so $K = 7$ is valid. This gives $L + 6$ mixed labels or $L + 7$ pure labels at the error in (S9.8). All those orbit coordinates are rational. Triangulating their rational polygon gives rational initial and returning barycentric rows, so the resulting finite transition tables can be chosen rational. This is an explicit upper for that algebraic instrument, not a claim about its unresolved Diophantine lower certificate.

S10. Finite-memory approximation uniform over bounded-cost policies

Exact finite memory and approximation of an experiment at bounded expected cost have different quantifiers. The noisy construction gives an explicit approximation of the undamped conditional process, using one finite model for an entire class of adaptive stopping policies. The following statement is an upper bound; it does not claim an optimal approximation exponent among arbitrary classical models.

Theorem S10.1 (Stopped-law approximation from conditional damping)**.** *Fix a finite angle command alphabet, initial qubit $I/2$, and the undamped instruments* (S7.1) *with $\alpha = 1$. Let $\mathcal{P}_B$ be any class of common randomized policies for the next command, stopping and a terminal qubit POVM, such that the number τ of physical calls satisfies $\mathbb{E}_1^\pi \tau \le B < \infty$ for every $\pi \in \mathcal{P}_B$. The policy has access to the visible record and its own classical randomness; there is no additional quantum register entangled with the qubit.*

For every $0 < \epsilon < 1$, replace the instrument by the one with $\alpha = 1 - \epsilon$ and use its common finite realization from Theorem S7.3. This one model, independently of the chosen policy, obeys

$$\sup_{\pi \in \mathcal{P}_B} \left\| \mathcal{L}_1^\pi - \mathcal{L}_{1-\epsilon}^\pi \right\|_{\mathrm{TV}} \le \epsilon B/8. \tag{S10.1}$$

Here the complete stopped record includes the stopping time, commands, reports, terminal measurement result and a nontermination symbol. The stronger bound holds for the block-diagonal classical–quantum terminal states before applying the terminal POVM; its nontermination block is one-dimensional and records only the nontermination mass. The simulator uses at most $K(1-\epsilon)$ labels with a free mixed start, or $K(1-\epsilon)+1$ with a counted pure start.

Thus for $0 < \delta < B/8$, one may take $\epsilon = 8\delta/B$ and use

$$1 + \max\left\{3, \left\lceil \frac{\pi}{\arccos(1 - 8\delta/B)} \right\rceil \right\} = \frac{\pi}{4}\sqrt{B/\delta} + O(1) \tag{S10.2}$$

pure-start labels as $\delta/B \downarrow 0$. For $\delta \ge B/8$ the $\alpha = 0$ model uses three pure-start labels and satisfies the same error requirement. Policy memory is separate: a policy with L counted labels and a prescribed initial label can be composed with the mixed core using at most $LK(1-\epsilon)+1$ labels, with its first actual call processed during entry into the product core.

Proof. For every positive qubit operator X and every angle,

$$\|\mathcal{D}(X) - U_\theta X U_\theta^*\|_1 \le \operatorname{tr} X.$$

Indeed, after division by $\operatorname{tr} X > 0$, the difference from $I/2$ has eigenvalues $\pm|r|/2$ for a Bloch vector of length at most one; the zero-trace case is immediate. The perturbation of branch zero is $\epsilon[\mathcal{D} - \mathrm{Ad}_{U_\theta}]/8$, that of branch one is zero, and that of branch two is the negative of branch zero. Hence the one-call flagged output satisfies

$$\frac{1}{2}\sum_{s=0}^{2} \|[\Psi_s^{1-\epsilon,\theta} - \Psi_s^{1,\theta}](X)\|_1 \le \frac{\epsilon}{8} \operatorname{tr} X. \tag{S10.3}$$

This bound is uniform in the command. Averaging it over a randomized command choice preserves it.

First cap the experiment at a deterministic horizon H. Retain the classical history and the subnormalized qubit state in every live or stopped branch. A subsequent common policy step, including a terminal measurement, is a positive trace-preserving map on this direct sum, and therefore contracts the trace norm of Hermitian differences. To see this directly, write such a difference as $Z_+ - Z_-$, with $Z_\pm \ge 0$. Positivity, trace preservation and the triangle inequality give $\|\Phi(Z)\|_1 \le \operatorname{tr} Z_+ + \operatorname{tr} Z_- = \|Z\|_1$.

Telescope between the undamped and damped H-step experiments, replacing one physical call at a time. In the term for call $t+1$, keep the first t calls undamped and the remaining tail damped.

The input to the changed call is then the target positive live state of total trace $\mathbb{P}_1^\pi\{\tau > t\}$. Apply (S10.3) there and contract the difference through its common tail. The resulting terminal trace distance is at most

$$\frac{\epsilon}{8}\sum_{t=0}^{H-1}\mathbb{P}_1^\pi\{\tau > t\} = \frac{\epsilon}{8}\mathbb{E}_1^\pi \min\{\tau, H\}. \tag{S10.4}$$

The reference distribution in this bound is the undamped target, not an uncontrolled distribution of a hybrid or approximate process.

To pass to the uncapped experiment, first retain separately every stopped history of length at most H and merge the unfinished branches into one symbol. Coarse graining cannot increase the distance. As H increases, the sum of the trace norms of the differences on the finite stopped blocks increases to its full sum, while the unfinished probabilities decrease to the respective nontermination probabilities. Thus the left side converges to the stated stopped classical–quantum distance, including the absolute difference at the nontermination symbol. The right side increases to $\epsilon\mathbb{E}_1^\pi\tau/8$. A terminal POVM and further classical processing contract this distance, proving (S10.1). The argument permits all the policy randomization because that randomness is simply part of the common classical transition at each step.

The noisy dictionary reproduces these complete subnormalized word states exactly, so replacing the noisy quantum system by its classical realization introduces no further error. The count follows from the preceding theorem. Substitution of $\epsilon = 8\delta/B$ proves (S10.2); use $\arccos(1-x) = \sqrt{2x} + O(x^{3/2})$ as $x \downarrow 0$. The same perturbation bound is valid at $\epsilon = 1$, where the exact two-state core was given explicitly. Finally a product with the L-label policy has LK core labels. Its separate initial label has the initial terminal decoder and, on the first call, the appropriate policy-weighted transition into that product core. This proves the claimed count without omitting the first physical call. □

The bound is uniform over a class defined by the target's expected number of calls. It does not imply that the approximate experiment has expected cost at most B: total variation controls bounded losses, whereas an unbounded stopping time requires a separate uniform integrability or tail hypothesis. In particular the possible nontermination mass of the approximate process is bounded by $\epsilon B/8$ but need not vanish. If a deterministic horizon is part of the policy class, both processes terminate within that horizon and (S10.4) applies directly. No extra quantum register, quantum input encoding, precision budget, or external policy memory is included in the stated simulator label count. If $B = 0$, the target makes no call almost surely. The same policy then also makes no call in the simulator, and a single label with the initial terminal decoder suffices with zero error.

S10.1. **Status of the inherited arithmetic consequences.** For completeness, the claim after Theorem S7.3 uses only the two implications already proved in Lemma S7.2. With $\alpha = 1 - \epsilon$, $x = \theta/(2\pi)$ and

$$\mathfrak{A}_x(\epsilon) = \min_{q\ge 1}\max\{3, q, \|qx\|/(q\epsilon)\},$$

the lower proof of Theorem S19.1 applies because $\alpha e^{i\theta} \in \mathcal{K}_M$ is necessary for a full-instrument M-state realization. Its upper proof constructs a regular polygon containing $\alpha e^{i\theta}$, so it also constructs the complete instrument, rather than just its scalar decoder. Therefore

$$\tfrac{1}{2}\mathfrak{A}_x(\epsilon) \le N_{\mathrm{mix}}(1-\epsilon, \theta) \le (1 + 4\pi^2)\mathfrak{A}_x(\epsilon).$$

The exact $+1$ pure-start increment does not change logarithmic exponents, divergence, or the constant-factor cubic-root criterion. All subsequent number-theoretic reasoning is inherited from Section S19; no new irrationality exponent is assigned to a particular algebraic rotation. In particular the full Karpelevič region alone has not been shown sufficient for this complete instrument away from the sectors treated above.

S11. Exact synthesis from algebraic observable data

The preceding normal form concerns existence over the real numbers. We now give a terminating exact procedure on algebraic input, including a certificate that no finite stochastic realization exists. The input is a supplied finite-dimensional probability response, rather than a finite sample from an unknown process. All algebraic numbers are specified by defining polynomials and isolating data, with a common real number field and its chosen embedding. The algorithms below concern exact label count and exact controller tables. They make no polynomial running-time or bounded-bit sampling claim.

Finite matrix-group recognition is classical, and substantially faster algorithms than the elementary enumeration below are known [64, 63]. Reduction of probabilistic-automaton minimization to the existential theory of the reals is also established [65]. Together with the inherited reversible criterion, these ingredients already yield decidability and bounded-size minimization on the input class considered here. The new normal form supplies the more structured conclusion: a smallest controller can be returned as a finite permutation core with a precisely counted initializer and a bounded multi-output decoder. We give a complete terminating construction, including finite certificates of the negative answer. Neither finite-group recognition nor real-algebraic automaton minimization is asserted to be a new algorithm.

S11.1. **Observable reduction and a finite stopping bound.** Write the supplied representation in chronological row convention as

$$b(w) = \beta T_w B, \qquad T_{uv} = T_u T_v, \tag{S11.1}$$

where the matrices, row and decoder have entries in a real number field K of degree e over $\mathbb{Q}$. The displayed representation is linear; its matrices and intermediate coordinates need not be stochastic or nonnegative. Probability validity means that the supplied $b(w)$ belongs to $\Delta(C)$ for every word. Compact reversibility is imposed on its *minimal observable* response space, as in Section 11.

For clarity, minimal reduction itself is finite. Starting with β, close its row span under right multiplication by each T_a. Starting with the columns of B, close their span under left multiplication by the T_a. Either iteration stabilizes once its dimension fails to increase, and neither dimension exceeds the supplied dimension. A reachable row is invisible precisely when it annihilates the second span. Quotienting the reachable span by this kernel gives the minimal representation, with dimension $d = \dim V_b$ and coefficients still in K. This is the familiar reachable/observable linear reduction; it does not identify arbitrary hidden stochastic rows with observable states. Since the response is normalized and nonempty, $d \geq 1$.

Set

$$h = ed, \qquad B_h = |\operatorname{GL}_h(\mathbb{F}_3)| = \prod_{j=0}^{h-1} (3^h - 3^j). \tag{S11.2}$$

This deliberately conservative bound has the advantage of a short complete proof. Restriction of scalars may use a larger field than necessary; doing so changes the computational bound, not the resulting exact label minimum.

Lemma S11.1 (Classical finite-group bound)**.** *A finite subgroup of* $\operatorname{GL}_d(K)$ *has order at most* B_{ed}*. Consequently a monoid generated by invertible* d*-dimensional matrices over* K *is finite if and only if breadth-first enumeration of its distinct positive-word products closes before producing* $B_{ed} + 1$ *elements.*

Proof. Restriction of scalars embeds the group faithfully in $\operatorname{GL}_{ed}(\mathbb{Q})$. For a finite subgroup H of $\operatorname{GL}_h(\mathbb{Q})$, let $\Lambda = \sum_{g \in H} g\mathbb{Z}^h$. There is a common denominator q with $\mathbb{Z}^h \subseteq \Lambda \subseteq q^{-1}\mathbb{Z}^h$. Thus Λ is a free abelian group of rank h, and it is invariant under every g and g^{-1}. In a lattice basis, H is a subgroup of $\operatorname{GL}_h(\mathbb{Z})$.

The kernel of reduction modulo 3 has no nonidentity element of finite order. Indeed, such an element would have a power A of prime order ℓ. Write $A = I + 3^s Z$, with $s \geq 1$ and with an integer matrix Z not zero modulo 3. The binomial expansion of $A^\ell = I$, divided by 3^s, gives $\ell Z = 0$ modulo 3, so $\ell = 3$. The expansion for this prime, divided by 3^{s+1}, instead gives

$$Z + 3^s Z^2 + 3^{2s-1} Z^3 = 0,$$

again contradicting $Z \not\equiv 0 \pmod 3$. Reduction is therefore injective on H. Counting independent ordered columns over $\mathbb{F}_3$ proves the bound in (S11.2). This is the standard Minkowski argument [63]; the proof is included to make the termination certificate explicit.

A finite monoid of invertible matrices is a group: for every element, two positive powers agree, and invertibility yields a positive power equal to the identity and another equal to its inverse. A finite generated monoid therefore has at most B_{ed} elements. In a breadth-first search, process each newly encountered product and multiply it by all generators. If the queue empties, the finite set found is closed under all generators and is the entire monoid. If the monoid is infinite, the search eventually finds $B_{ed} + 1$ distinct products. These are the two asserted stopping events. All equality tests are exact tests in K. □

Theorem S11.2 (Decidability and minimal algebraic synthesis)**.** *For a supplied compactly reversible probability response with algebraic linear representation* (S11.1)*, there is a terminating algorithm that decides whether an exact finite stochastic realization exists. If the answer is positive, the algorithm returns its least number $m(b)$ of live labels and an exact realization with algebraic entries, a prescribed initial label and a row-stochastic terminal decoder. It can return a permutation core with at most one transient initializer, as in Theorem S2.1.*

A negative answer has a finite certificate consisting of $B_{ed} + 1$ distinct observable command products. A positive answer has an exact controller, the finite observable-group multiplication table, and finite real-algebraic feasibility or infeasibility certificates for every smaller candidate size. The negative certificate concerns the promised compactly reversible input class; it is not a nonrealizability test for arbitrary dissipative responses.

Proof. Perform the observable reduction and apply Lemma S11.1 to the minimal command matrices. If enumeration exceeds its bound, the positive-word monoid and its compact group closure are infinite. Theorem 11.1 rules out every finite stochastic realization. The list of distinct products, each supplied with its generating word, is a finite independently checkable certificate of this conclusion.

Otherwise enumeration closes to a finite group G. Compute the distinct reachable rows βT_g in the minimal representation; their count n is the residual cardinality. The deterministic residual machine proves $m(b) \leq n$. Theorem S2.2 expresses the answer as a minimum over finite permutation actions, a probability row λ, and a stochastic decoder D. Only the following finite possibilities can improve or attain the upper n: a core of size $k \leq n$ with a point initial row, or a core of size $k \leq n - 1$ with a possibly mixed initial row and one additional initializer.

For each such k, enumerate assignments of permutations to the elements of G and retain those satisfying $P_1 = I$ and $P_g P_h = P_{gh}$. This is a finite search through at most $(k!)^{|G|}$ assignments, and it may of course be reduced using generators and relations. For a retained action impose

$$\lambda_i \geq 0, \quad \sum_i \lambda_i = 1, \qquad D_{ic} \geq 0, \quad \sum_c D_{ic} = 1, \qquad \lambda P_g D = b(g) \quad (g \in G). \tag{S11.3}$$

For the point case impose $\lambda = e_i^{\mathsf{T}}$ and enumerate i. For the other case it is harmless to permit point rows as well: that candidate then pays a redundant initializer, and the separately enumerated point case can only improve its cost. Equations (S11.3) have algebraic coefficients and degree at most two in their real unknowns. Exact real-algebraic elimination decides feasibility, and a nonempty such set has an algebraic sample point. One may see the latter directly from a sign-invariant algebraic cell decomposition: choose isolating algebraic sample coordinates successively in its nonempty sections

and sectors. Algebraic elimination is the classical decision theorem used here [15]; no numerical optimizer is being substituted for it.

Search in increasing total label cost and stop at the first feasible candidate. Termination follows from the deterministic upper. Theorem S2.2 proves that no stochastic competitor outside the enumerated normal forms can be smaller. For a point initial row use the permutation matrices directly. For a mixed initial row add the counted initial state, give it decoder $b(\varnothing) = \lambda D$, and on command a send it to λP_{g_a} in the core. All later transitions are permutations. Every returned entry is algebraic, and all response identities follow from the group action and (S11.3).

The finite word criterion of Theorem S12.1 supplies an alternative bounded-size formulation using arbitrary stochastic matrices, independently of the normal form. This is useful for checking a returned implementation against the original linear representation. Its existence does not give an efficient algorithm for minimizing probabilistic automata, a problem with established hardness results [65]. □

Corollary S11.3 (Algebraic prepared unitary experiments)**.** *Let the prepared density matrix, a finite family of unitary commands and a finite terminal positive-operator-valued measurement have algebraic real and imaginary entries. Whether their Born probabilities for all command words admit an exact finite classical stochastic realization is decidable. In the positive case a smallest exact classical realization and its bounded decoder can be synthesized with algebraic entries. For a protected Hilbert space of dimension d_{phys}, the observable linear dimension used in the algorithm is at most d_{phys}^2.*

Proof. Hermitian matrices have a real coordinate basis of size d_{phys}^2 consisting of diagonal matrix units and real and imaginary symmetric matrix-unit pairs. In this basis conjugation by each supplied unitary, preparation, and terminal evaluation have real algebraic coefficients. This constructs (S11.1). Positivity and normalization of Born probabilities follow from the declared density matrix and measurement, not from a finite test. The observable reduction is finite linear algebra over a common real number field. Corollary 11.2 proves compact reversibility of this quotient, so Theorem S11.2 applies. □

This corollary allows noncommuting commands, nonfaithful physical actions, mixed preparations and arbitrary finite terminal measurements. In particular, a physically infinite group with constant observable response is correctly reduced to the one-label case before group enumeration. The result concerns exact simulation of the supplied experiment. Choosing an optimal feedback experiment, deciding validity of an arbitrary algebraic series, approximate program size, exact sampling and workspace lower bounds are different questions. The complete retained causal and boundary theory keeps its own quantifiers and resource accounts.

S12. Finite certificates and stable exact memory

The normal form gives exact realizations at reversible responses. We next prove that their memory obstructions admit finite tests and persist under a specified class of physical perturbations. Finite-word equivalence for linear machines is classical [22]; the finite bound below is its direct difference-space argument. The additional conclusions concern the positive separation from *all* smaller stochastic machines, its effective algebraic certificate, and a matching realization under damping toward a common background response.

S12.1. **Finite tests against a fixed memory budget.** Suppose a response $b : A^* \to \Delta(C)$ has a supplied real linear representation

$$b(w) = \beta T_w B, \qquad \beta \in \mathbb{R}^{1\times d}, \quad T_a \in \mathbb{R}^{d\times d}, \quad B \in \mathbb{R}^{d\times |C|}.$$

Minimality of this representation is not required. For $M \geq 1$ let $\mathcal{P}_M$ be the compact product of the row-stochastic matrix simplexes for $(Q_a)_{a\in A}$ and $D \in \mathbb{R}^{M\times|C|}$. Its initial row is the prescribed pure

row e_1^{T}, as before. Set

$$L = d + M - 1, \qquad \delta_M(b) = \min_{(Q,D)\in\mathcal{P}_M} \max_{\substack{|w|\le L\\ c\in C}} \left| e_1^{\mathsf{T}} Q_w D_c - b(w,c) \right|. \tag{S12.1}$$

Theorem S12.1 (Finite obstruction and effective separation)**.** *An M-label stochastic machine realizes b on every word if and only if it does so on the words of length at most $d+M-1$. Consequently $\delta_M(b) > 0$ if and only if no exact M-label realization exists. If another supplied response $\widetilde{b}$ satisfies*

$$\max_{|w|\le L,\ c\in C} |\widetilde{b}(w,c) - b(w,c)| \le \eta,$$

then every M-label machine has error at least $\delta_M(b) - \eta$ on these same probes. This last assertion imposes no finite-rank hypothesis on $\widetilde{b}$.

If the given entries of β, T_a, B are real algebraic numbers, $\delta_M(b)$ is an effectively computable real algebraic number, and a minimizing machine with real algebraic entries can be computed. When it is positive, a positive rational number strictly below it and a real-algebraic quantifier-elimination certificate of that strict lower bound can be computed. No polynomial-time or practical-size bound is asserted.

Proof. For a fixed candidate machine use the block-diagonal matrices $S_a = \mathrm{diag}(T_a, Q_a)$, the row $u = (\beta, -e_1^{\mathsf{T}})$, and the output columns $v_c = (B_c, D_c)^{\mathsf{T}}$. The difference on w is $uS_w v_c$. Let R_j be the span of uS_w for $|w| \le j$. These spaces increase; if $R_j = R_{j+1}$, the common space is invariant under every S_a and all later spaces coincide. Since $u \neq 0$, the sequence starts in dimension one inside a space of dimension $d + M$ and must have stabilized by $j = d + M - 1$. Vanishing on the indicated finite set therefore implies vanishing on all words. This is the same linear-equivalence principle as the inherited finite-realization test.

The maximum in (S12.1) is a continuous function on the compact set $\mathcal{P}_M$, so its minimum is attained. It is zero precisely when one machine passes every finite probe, hence, by the first assertion, realizes every word. The perturbation claim follows pointwise from the triangle inequality and then from the minimum defining $\delta_M(b)$.

For algebraic input, every probe value is algebraic. The constraints defining row-stochasticity and the inequalities between the finitely many polynomial errors and a variable t are polynomial constraints over the real algebraic numbers. Real quantifier elimination decides their feasibility and computes the least feasible t together with an algebraic witness. A singleton semialgebraic set over the real algebraic numbers consists of an algebraic number. If $\delta_M(b) > 0$, effective isolation supplies a positive rational $\sigma < \delta_M(b)$; infeasibility with all errors at most σ is a finite real-algebraic certificate. This uses the classical decision procedure [15]; it does not assert that this procedure was executed by the finite diagnostics accompanying this article. □

For the prepared quantum response, these finite probes have an explicit physical perturbation bound. Suppose two density matrices differ in trace norm by at most h_ρ, corresponding channels differ in diamond norm by at most h_Φ, and corresponding terminal effects differ in operator norm by at most h_E. All channels are trace preserving and completely positive, and both lists of effects are valid measurements. Contractivity and telescoping give $\|\widetilde{\Phi}_w(\widetilde{\rho}) - \Phi_w(\rho)\|_1 \le h_\rho + |w|h_\Phi$. The state difference has trace zero; subtracting $I/2$ from a terminal effect, whose spectrum lies in $[0,1]$, gives

$$\max_{|w|\le L,c} |\widetilde{b}(w,c) - b(w,c)| \le h_E + \tfrac{1}{2}(h_\rho + Lh_\Phi). \tag{S12.2}$$

Thus a strict smaller-memory obstruction survives a specified open neighbourhood of physical primitives. This supplies a lower bound there; it does not assert that an arbitrary perturbation has a realization of the same size.

S12.2. **A matching upper under a common background.** Let b be compactly reversible, nonconstant, and have finite stochastic memory $m = m(b)$. Let G be its finite observable group, with the chronological convention of the normal-form theorem, and put

$$\overline{b} = |G|^{-1} \sum_{g \in G} b(g), \qquad C_b = \max_{g \in G, c \in C} |b(g, c) - \overline{b}(c)| > 0.$$

Choose a damping factor $\alpha_a \in [0, 1]$ for each command, put $\beta_w = \prod_{j=1}^{|w|} \alpha_{w_j}$, with $\beta_\varnothing = 1$, and define

$$b_{\boldsymbol{\alpha}}(w) = \beta_w b(w) + (1 - \beta_w)\overline{b}. \tag{S12.3}$$

Theorem S12.2 (Stable exact memory under background damping)**.** *The response* (S12.3) *has an exact realization with m labels for every choice of the damping factors. Put $L = d + m - 2$, where d is any supplied representation dimension of b, and let $\epsilon = \max_a(1 - \alpha_a)$. Then*

$$\epsilon < \min\left\{1, \frac{\delta_{m-1}(b)}{LC_b}\right\} \quad \Longrightarrow \quad m(b_{\boldsymbol{\alpha}}) = m. \tag{S12.4}$$

For algebraic input satisfying the hypotheses, a positive algebraic cutoff, or a smaller positive rational cutoff, is effectively computable. For a constant response every damping leaves that response unchanged and its minimum is one label; the expression involving $m - 1$ is not used.

Proof. Take a minimal realization in the normal form proved above. On its recurrent labels each Q_a is a permutation, and an extra initializer is present only when the initial distribution on these labels is mixed. Average that distribution over G, and call the resulting row π, extended by zero on the initializer when present. It obeys

$$\pi Q_a = \pi, \qquad \pi D = \overline{b}.$$

Indeed, the group average is invariant under each permutation. The initializer has no incoming transition, and its first-input row is the original initial distribution followed by the commanded permutation, so the same identities hold on the entire machine. Set $J = \mathbf{1}\pi$. Then $J^2 = J$, $Q_a J = J Q_a = J$, and

$$Q_{a,\boldsymbol{\alpha}} = \alpha_a Q_a + (1 - \alpha_a) J$$

is row-stochastic on the same labels. Multiplication gives, for every nonempty word and also trivially at the empty word,

$$Q_{w,\boldsymbol{\alpha}} = \beta_w Q_w + (1 - \beta_w) J.$$

Reading with the same decoder proves the upper.

By minimality $\delta_{m-1}(b) > 0$. For $|w| \le L$, the elementary product bound $1 - \beta_w \le |w|\epsilon$ gives an error from b of at most $LC_b\epsilon$. Theorem S12.1 rules out every $(m - 1)$-label realization when this is smaller than $\delta_{m-1}(b)$. Machines with fewer labels can be padded by unreachable rows, so they too are excluded. This and the upper prove equality. A nonconstant response has $m \ge 2$ and $d \ge 2$, hence $L > 0$. The effective assertion follows from the finite group enumeration and algebraic synthesis theorem, together with Theorem S12.1. Finite maxima and averages over the computed group preserve algebraicity. □

There is a physical realization of the common background for prepared unitary commands. Let $\mathcal{K}$ be the compact closure of their physical conjugation group. The compact convex hull of $\{\Phi(\rho) : \Phi \in \mathcal{K}\}$ has a unique element ρ_* of least Hilbert–Schmidt norm: strict convexity gives uniqueness. Every conjugation in $\mathcal{K}$ preserves that set and norm, so ρ_* is invariant. Its observable image belongs to the convex hull of the finite orbit of b and is invariant. Averaging any element of that orbit over G gives $\overline{b}$; applying this averaging projection to an invariant element in its convex hull shows that its observable image is exactly $\overline{b}$. Therefore the physical channels

$$\widetilde{\Phi}_a(X) = \alpha_a U_a X U_a^* + (1 - \alpha_a)\,\mathrm{tr}(X)\rho_*$$

produce (S12.3) under every command word. In general ρ_* need not be maximally mixed. The theorem concerns damping toward this common invariant background, not every depolarizing or noisy channel.

S12.3. **Periodic plateaux and an explicit rational certificate.** For the periodic mixture of Section S18, let the distinct nonreal frequencies have positive amplitudes γ_j, total $\Gamma = \sum_j \gamma_j$, and let M_* be its exact partition-and-anchor minimum. Its dimension is $d = 2r + 1$ and $C_b \le \Gamma/2$. Hence a common damping $\alpha = 1 - \epsilon$ retains exactly M_* labels whenever

$$0 \le \epsilon < \min\left\{1, \frac{2\delta_{M_*-1}(b)}{\Gamma(2r + M_* - 1)}\right\}. \tag{S12.5}$$

This is an effective finite certificate for each algebraic periodic input. It closes the previously existential plateau cutoff for this class, without claiming a sharp transition point or a fixed-memory risk asymptotic.

For clarity, one certificate can be calculated without a general quantifier-elimination run. Consider the unary response

$$b_\alpha(n) = \tfrac{1}{2}\{1 + \alpha^n \cos(\pi n/2)\}, \qquad n \ge 0. \tag{S12.6}$$

Proposition S12.3 (A certified four-label interval)**.** *For the undamped sequence $b = b_1$ and the dimension $d = 3$, $\delta_3(b) > 1/36$. In particular the exact stochastic memory of* (S12.6) *is four for $89/90 \le \alpha \le 1$. This interval is a certified sufficient interval, not a claimed sharp one.*

Proof. Let Q be any row-stochastic 3×3 matrix, put $t = \operatorname{tr} Q$ and $h = \det Q$, and write $y_n = e_1^{\mathsf{T}} Q^n D$ for any scalar decoder. Since $Q\mathbf{1} = \mathbf{1}$, its characteristic polynomial is

$$z^3 - tz^2 + (t + h - 1)z - h.$$

Thus $y_{n+3} - ty_{n+2} + (t+h-1)y_{n+1} - hy_n = 0$. We have $0 \le t \le 3$, $|h| \le 1$, and $|t+h-1| \le 3$. The determinant bound follows directly by bounding its absolute permutation sum by the product of the row sums. If the first five values of y are within η of $(1, 1/2, 0, 1/2, 1)$, substitution at $n = 0, 1$ yields

$$|t - h| \le 16\eta, \qquad |2 - t - h| \le 16\eta, \qquad |t - 1|, |h - 1| \le 16\eta.$$

Nonnegativity and the row sums imply a further exact identity. For $a_i = 1 - Q_{ii}$ it is

$$t^2 - 3t + 3 - 3h = \tfrac{1}{2}\sum_{i<j}(a_i - a_j)^2 + 3\sum_{i<j} Q_{ij}Q_{ji} \ge 0. \tag{S12.7}$$

To verify it, the second characteristic coefficient is both $t + h - 1$ and $\sum_{i<j}(Q_{ii}Q_{jj} - Q_{ij}Q_{ji})$; substitute $Q_{ii} = 1 - a_i$ and simplify. On the other hand, putting $u = t - 1$ and $v = h - 1$ gives

$$t^2 - 3t + 3 - 3h = -2 - u + u^2 - 3v \le -2 + 64\eta + 256\eta^2.$$

For $\eta \le 1/36$ the right side is at most $-2/81 < 0$, a contradiction. The compact minimum on the first five probes is therefore strictly greater than $1/36$, and so is (S12.1), which here uses the six probes $0 \le n \le 5$.

The undamped response has dimension three and a four-cycle realization with decoder $(1, 1/2, 0, 1/2)$. The same polynomial certificate excludes three labels, so its minimum is four. Damping that cycle toward the uniform stationary row realizes (S12.6) with four labels. Its difference from the undamped sequence on $0 \le n \le 5$ is at most $\frac{5}{2}(1 - \alpha)$. For $\alpha \ge 89/90$ this is at most $1/36$, strictly below $\delta_3(b)$. The finite obstruction proves the lower and completes the proof. □

The certificate uses a stochastic-matrix inequality rather than a sample search over three-label machines. Its accompanying rational diagnostics check the displayed polynomial identity, the constructed all-word damping identities on explicit finite models, and exact finite recurrence consequences. They do not replace the proofs, compute all algebraic gaps, or give an independent human or editorial assessment.

S13. Relation to the retained exact resource laws

The finite-group normal form has a cyclic specialization, but it does not replace the independent spectral and arithmetic arguments for responses outside the reversible class. We record the precise relation.

For the probability-valid periodic mixture of Theorem S18.2, with distinct nontrivial frequency orders $q_1, \dots, q_r$, let

$$K = \min_{\mathcal{P}} \sum_{B \in \mathcal{P}} \operatorname{lcm}\{q_j : j \in B\},$$

where $\mathcal{P}$ ranges over partitions of the frequency indices. The exact pure-start count is the minimum of the full period $\operatorname{lcm}(q_1, \dots, q_r)$ and $K + 1$, with the constant and single-frequency endpoints as stated in the retained theorem. A cyclic permutation representation is a disjoint union of cycles; a frequency of order q occurs on a cycle precisely when q divides its length. Assigning required modes to cycles therefore gives the same partition expression. The bounded decoders and positive-amplitude preparation in the retained theorem attain that lower bound. This agreement identifies a model-specific closed form inside the general factorization problem. Arbitrary bounded decoders do not have this same closed formula, as both the sharp S_3 and the five-character contrast transitions show. The quotient theorem applies to arbitrary finite abelian tables; it does not assume the positive-amplitude mixture of the periodic closed form.

The leakage theorem, Theorem S18.5, adds its separately proved inactive live label. Its lower bound uses a decay-normalized observable core and the no-critical-chain argument; the common-background plateau of Section S12 is a different perturbation, as is terminal readout contrast. The exact four-period damping threshold was already $1/\sqrt{3}$ by Theorem S20.3; the current supplement makes this inherited closure explicit. The exact spectral construction of Theorem S20.1, the relative-spectrum obstruction and the arbitrary-phase envelope of Theorem S19.1 remain proved with their own hypotheses. In particular a strictly damped irrational mode need not have compactly reversible observable dynamics.

The following actual-history transfer theorem remains a general statement about causal experiment comparison. It compares complete procedures, charges the common paid acquisition construction, and forms its lower witness on actual competing histories. None of its hypotheses is inferred from a finite observable group. The complete paired supplement retains the associated performed-law minimax and dual theory, finite-width transfer, reset control, optimal discrimination, predictive quotients and geometric results. The original ordered-measurement and two-stage-reset realizations remain independently identifiable by source commit and path.

S14. Joint contraction and bounded positive realizations

The reversible classification has a complementary dissipative regime. The condition below is imposed on all command products through one supplied norm, and the construction gives one machine for all words. A separate realization for each command would not establish that conclusion. Invariant-polytope positive realization is classical [48, 47]; the point of the statement is an explicit sufficient certificate with bounded probability decoders, a prescribed pure start, and a stated number of live labels.

Let A and C be nonempty finite sets, $a = |A|$, and let

$$b(w, c) = p_c + \ell_c(x_w), \qquad x_{wa} = F_a x_w, \qquad x_\varnothing = x_0 \in \mathbb{R}^n, \tag{S14.1}$$

where F_a are linear, $p \in \Delta(C)$, $\sum_c \ell_c = 0$, and the displayed values are probabilities for every word. For a norm $\|\cdot\|$ write $\|\ell\|_*$ for its dual norm. Suppose we are given $0 < \rho < 1$ and $r > 0$ such that

$$\|F_a x\| \le \rho \|x\| \quad (a \in A), \qquad r\|\ell_c\|_* \le p_c \quad (c \in C). \tag{S14.2}$$

In particular, strictly positive p_c give such an r for every finite collection of output functionals. Zero components are allowed only when the corresponding functional vanishes. The second condition certifies that the entire ball rB, not only the actual orbit, has legal decoder rows.

Theorem S14.1 (A common contracting realization)**.** *Under* (S14.2)*, the response* (S14.1) *has an exact stochastic realization from a prescribed pure label, valid for every command word, with at most*

$$\sum_{j=0}^{T-1} a^j + \left\lceil \left(1 + \frac{2}{1-\rho}\right)^n \right\rceil \quad \textit{live labels}, \tag{S14.3}$$

where $T \geq 1$ may be any integer satisfying $\rho^{T-1}\|x_0\| \leq r$. For $x_0 \neq 0$ one may take

$$T = 1 + \max\left\{0, \left\lceil \frac{\log(\|x_0\|/r)}{\log(1/\rho)} \right\rceil \right\};$$

for $x_0 = 0$, the response is constant and one label suffices. No minimality or computational-efficiency assertion is made for this upper bound.

Proof. Put $\varepsilon = 1 - \rho$. Choose a maximal ε-separated finite set V in the unit ball B, where distinct selected points have distance greater than ε. The open balls of radius $\varepsilon/2$ about these points are pairwise disjoint and lie in $(1 + \varepsilon/2)B$. Lebesgue volume, which scales by the nth power, gives

$$|V| \leq (1 + 2/\varepsilon)^n.$$

This bound also proves that the greedy selection stops; maximality says that V is an ε-net of B. If $P = \operatorname{conv}(V)$, then $\rho B \subseteq P \subseteq B$. Indeed, for every real linear functional f, choose a maximizer $x \in B$ and a point $v \in V$ at distance at most ε from x. Symmetry of the unit ball gives

$$\max_{v \in V} f(v) \geq f(x) - \varepsilon\|f\|_* = (1 - \varepsilon)\|f\|_*.$$

The separating-hyperplane characterization of convex containment proves $\rho B \subseteq P$. Consequently

$$F_a(rP) \subseteq \rho r B \subseteq rP \quad (a \in A).$$

Use one recurrent label for each $v \in V$, representing the vector rv. For every such label and command choose nonnegative barycentric coefficients of $F_a(rv)$ in rV; they sum to one and define a row of Q_a. Give that label decoder $p_c + \ell_c(rv)$. The decoder is nonnegative by the second condition in (S14.2), and its entries sum to one because p has mass one and $\sum_c \ell_c = 0$.

In addition use one transient label for each word w with $|w| < T$, representing its actual vector x_w. Its decoder is the actual probability row $b(w)$, legal by hypothesis. If $|w| < T - 1$, command a sends it deterministically to the label wa. If $|w| = T - 1$, then

$$\|F_a x_w\| \leq \rho^T \|x_0\| \leq \rho r,$$

so $F_a x_w$ is in rP. Choose its barycentric row in the recurrent labels. Start at the single empty-word label. There are exactly $\sum_{j=0}^{T-1} a^j$ such transient labels before any duplicate states are merged. Induction on the word shows that the probability-weighted represented vector is always x_w: the transition identity holds on each individual label, and linearity preserves it under a mixture. Pairing with all ℓ_c and adding p gives the entire response. This proves the count, including the pure initial label and every first command. □

The assumption may also be supplied as a uniform product bound. If a known norm $\|\cdot\|_0$ and constants $K \geq 1$, $0 < q < 1$ satisfy

$$\|F_w\|_0 \leq K q^{|w|} \qquad (w \in A^*),$$

choose any $q < \rho < 1$ and set

$$\|x\|' = \sup_{w \in A^*} \rho^{-|w|} \|F_w x\|_0.$$

This is a norm with $\|x\|_0 \le \|x\|' \le K\|x\|_0$ and $\|F_a x\|' \le \rho\|x\|'$. Hence Theorem S14.1 applies with $\|x_0\|' \le K\|x_0\|_0$ and any r satisfying $r\|\ell_c\|_{0,*} \le p_c$. This is a consequence of a supplied uniform certificate, not a decision procedure for joint spectral radius. A common Euclidean contraction certificate, for example, is the finite collection $F_a^{\mathsf{T}} H F_a \preceq \rho^2 H$ for a supplied positive definite H. The actual net and barycentric rows provide finite implementation data; their exact algebraic or sampling encodings remain distinct resources.

S15. Average mixing does not control conditional response memory

Positive realization via a common invariant polytope, including composition of adaptive instruments, is established machinery [47, 71]. A hypothesis forcing every normalized branch into an interior region is stronger than fast convergence of the channel obtained by ignoring its outcomes. The following exact example separates these statements even when every outcome has a probability bounded away from zero. The infinite-memory conclusion is a response-level consequence of the inherited relative spectral obstruction; it is not a claim that ordinary mixing theory has failed.

The phase obstruction itself has classical predecessors. The FRDN processes discussed in [50, Section 2.3] already exhibit irrational-phase obstructions to finite classical realization and phase-dependent finite minima in commensurate cases. Our assertion concerns the simultaneous qubit, rank-three, uniform-positivity and exact depolarization properties below, and the sharp additional charge for a counted pure start; it does not claim priority for the phase mechanism.

Fix $\sin\theta \neq 0$. For a qubit put

$$U_\theta = \begin{pmatrix} \cos(\theta/2) & -\sin(\theta/2) \\ \sin(\theta/2) & \cos(\theta/2) \end{pmatrix}, \qquad E = \begin{pmatrix} 3/4 & 0 \\ 0 & 1/4 \end{pmatrix}, \qquad \tau = |0\rangle\langle 0|, \qquad \mathcal{D}(X) = \operatorname{tr}(X) I/2.$$

Define three linear maps

$$\Psi_0(X) = \tfrac{1}{8} U_\theta X U_\theta^*, \quad \Psi_1(X) = \tfrac{1}{4}\operatorname{tr}(EX)\tau, \quad \Psi_2 = \mathcal{D} - \Psi_0 - \Psi_1. \tag{S15.1}$$

Theorem S15.1 (Exact memory in a uniformly positive one-dependent family)**.** *The maps* (S15.1) *form a quantum instrument. Its unconditioned channel is the completely depolarizing map $\mathcal{D}$. For every density matrix, its three outcome probabilities belong respectively to*

$$\{1/8\}, \qquad [1/16, 3/16], \qquad [11/16, 13/16].$$

Started at $I/2$, its stationary outcome process is one-dependent: two finite blocks separated by at least one unobserved outcome are independent. Its word-probability Hankel matrix has rank exactly three. Write $z = e^{i\theta}$. If z has finite order q, the least classical hidden-state count reproducing every word probability is exactly q when a mixed initial row is free, and exactly $q+1$ when the initial label must be pure and counted. If z is not a root of unity, no finite classical model exists, even with a freely mixed start and arbitrary real stochastic entries. All primitive superoperators are algebraic whenever $\cos\theta$ and $\sin\theta$ are algebraic.

Proof. Use the Choi convention $J(\Phi) = \sum_{i,j=0}^{1} |i\rangle\langle j| \otimes \Phi(|i\rangle\langle j|)$. Then

$$J(\mathcal{D}) = \tfrac{1}{2} I_4, \qquad \|J(\Psi_0)\|_{\mathrm{op}} = \tfrac{1}{4}, \qquad \|J(\Psi_1)\|_{\mathrm{op}} = \tfrac{3}{16}.$$

The first two branch Choi matrices are positive semidefinite. Subtraction therefore gives $J(\Psi_2) \succeq I_4/16$, proving complete positivity. The sum is trace preserving by construction. The probability bounds follow from $\operatorname{tr}\Psi_0(\sigma) = 1/8$ and $\operatorname{tr}\Psi_1(\sigma) = \operatorname{tr}(E\sigma)/4 \in [1/16, 3/16]$. Conjugation by U_θ depends algebraically on $\cos\theta$ and $\sin\theta$, which proves the last assertion.

The initial state $\sigma_* = I/2$ is fixed by $\mathcal{D}$, giving stationarity. To check one-dependence directly, write Ψ_u for the chronological branch product of a block u. Summing over one intervening outcome gives

$$\sum_{s=0}^{2} \operatorname{tr}\big(\Psi_v \Psi_s \Psi_u(\sigma_*)\big) = \operatorname{tr}\big(\Psi_v \mathcal{D} \Psi_u(\sigma_*)\big) = \operatorname{tr}\Psi_u(\sigma_*)\ \operatorname{tr}\Psi_v(\sigma_*).$$

Further unobserved outcomes leave $\mathcal{D}$ unchanged. Summing over specified block events proves the claimed independence.

After outcome 1 the normalized state is τ, and that first outcome has probability $1/8$. Repeating outcome 0 rotates this state with the factor 8^{-n}. The probability of the full word $1\,0^n\,1$ is

$$u_n = \frac{1}{64}\,8^{-n}\left(1+\frac{1}{2}\cos(n\theta)\right). \tag{S15.2}$$

All reachable subnormalized states are real symmetric matrices, a space of real dimension three, so the word-probability Hankel rank is at most three. For the reverse inequality take prefixes $1\,0^i$ and suffixes $0^j\,1$, $0 \le i, j \le 2$. Their Hankel submatrix is (u_{i+j}). Equation (S15.2) expresses it as the product of a three-point Vandermonde matrix, a diagonal matrix with three nonzero coefficients, and the transpose Vandermonde matrix. It is therefore nonsingular, proving rank three.

Suppose a finite classical model exists. Write its nonnegative outcome matrices as T_0, T_1, T_2, with $\sum_s T_s$ row-stochastic, and its initial probability row as μ. Then

$$u_n = vT_0^n d, \qquad v = \mu T_1 \ge 0, \quad d = T_1\mathbf{1} \ge 0.$$

Here T_0 is row-substochastic. Delete states unreachable from the support of v and states from which no positive terminal value of d is reachable. The remaining principal submatrix S still realizes u_n. For every remaining i, j, choose a positive path from v to i, of length a_i and weight $p_i > 0$, and a path from j to a positive terminal value, of length b_j and weight $r_j > 0$. Nonnegativity gives

$$p_i r_j (S^n)_{ij} \le u_{a_i+n+b_j} \le C_{ij} 8^{-n}.$$

Thus $(8S)^n$ is entrywise bounded, and every eigenvalue of S has modulus at most $1/8$.

The three exponential modes of (S15.2) are $1/8$, $z/8$, and $\overline{z}/8$, all distinct and with nonzero coefficients. Cayley–Hamilton and the Vandermonde independence of these sequences force $z/8$ to be an eigenvalue of S. In an upper block triangular decomposition of S, it belongs to an irreducible nonnegative diagonal block B. Its spectral radius is exactly $1/8$. Perron normalization by a strictly positive eigenvector therefore makes $8B$ similar to a stochastic matrix with eigenvalue z. This is the elementary normalization proved before Proposition S21.2. At a largest-modulus coordinate of a corresponding complex eigenvector, equality in the triangle inequality forces each positive successor to have the same modulus and value multiplied by z. Irreducibility propagates this equality to every coordinate, so the relation holds on every positive edge and every directed cycle. Following positive successors reaches a directed cycle; its length $l \ge 1$ gives $z^l = 1$. This identity proves that z must be a root of unity. If its order is q, every such directed cycle has length divisible by q, so every realization has at least q labels.

For the matching upper, suppose z has order $q \ge 3$. In the real Bloch plane let P be the regular q-gon with vertices

$$r_j = (\cos(2\pi j/q), \sin(2\pi j/q)), \qquad 0 \le j < q,$$

and let σ_j be the corresponding pure qubit states. The polygon contains the disk of radius $\cos(\pi/q) \ge 1/2$. The normalized branch 0 permutes its vertices and branch 1 resets to $\sigma_0 = \tau$. For an arbitrary state with Bloch vector r in the unit disk, write $h = \operatorname{tr}\Psi_1(\sigma) \in [1/16, 3/16]$. The normalized Bloch vector after branch 2 is

$$\frac{-R_\theta r/8 - h r_0}{1 - 1/8 - h},$$

whose norm is at most $(1/8 + 3/16)/(11/16) = 5/11 < 1/2$. It therefore lies in P. For each input vertex and outcome, expand the normalized output in the same vertices and multiply the barycentric coefficients by its outcome probability. These nonnegative rows form outcome matrices T_s with $\sum_s T_s$ row-stochastic. They reproduce each subnormalized branch state on each label. Induction proves equality of every word probability. The uniform mixture of the vertices represents $I/2$, giving the q-label upper with a free mixed start. A stationary hidden initialization can also be used: with

$T = \sum_s T_s$, every row μT^n represents $\mathcal{D}^n(I/2) = I/2$, where μ is the uniform row. A convergent subsequence of their Cesaro averages gives $\pi T = \pi$, and π still represents $I/2$, so it gives the same word probabilities on q labels. If a pure start is required, add one label representing $I/2$ and use its branch decompositions on the first outcome; the first outcome is generated during this counted transition. No transition returns to that label. This gives the $q + 1$ upper.

It remains to exclude a pure-start realization on exactly q labels. The preceding lower then forces the irreducible block carrying $z/8$ to use every label. Perron normalization makes $8T_0$ similar to a stochastic matrix with primitive qth-root eigenvalue and exactly q states. The equality argument above supplies a directed q-cycle using every state. An additional positive edge would close a directed cycle of length less than q, which is impossible. Thus, before normalization, T_0 is a weighted directed q-cycle. Its successive row weights are positive. From the prescribed starting point,

$$p(0^n) = 8^{-n} \qquad (n \geq 0).$$

For $n = 1, \ldots, q$, the left side is the product of the first n row weights encountered on this cycle. Successive ratios force every weight to be $1/8$, so $T_0 = C_q/8$ after relabelling. Next, $p(0^n 1) = 8^{-n}/8$ shows that every coordinate of $d = T_1\mathbf{1}$ is $1/8$. This would give

$$p(1\,0^n\,1) = e_1^{\mathsf{T}} T_1 (C_q/8)^n d = 8^{-n}/64,$$

contradicting (S15.2), already at $n = 0$. The pure-start minimum is therefore $q + 1$. □

In particular, the fully algebraic choice

$$U_\theta = 5^{-1/2} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \qquad e^{i\theta} = (3 + 4i)/5,$$

has no finite classical realization. Its superoperator coefficients are rational. The phase is not a root of unity, since otherwise $z + z^{-1} = 6/5$ would be a rational algebraic integer and hence an integer. Every finite-order choice, on the other hand, gives the exact counts q and $q + 1$ in a fixed physical qubit with the same uniform outcome probability bounds. The size of an exact hidden-state realization cannot be inferred from predictive rank, unconditional mixing time, or finite dependence range.

Stationary one-dependent processes without a finite hidden Markov model were established earlier, notably by the finitely dependent colorings of Holroyd and Liggett [74]. We make no first-existence claim for that phenomenon. Here the simultaneous properties are an explicit qubit instrument, rank-three linear prediction, uniformly positive outcome probabilities, and an exactly depolarizing unconditioned channel, together with a sharp finite-order memory law and a separate pure-initialization charge.

This example satisfies a stronger average-mixing property than exponential convergence: one unobserved step erases all prior state information. The infinite obstruction survives in a specified observed word, whose conditional maps retain an irrational phase. It therefore does not contradict a finite-realization theorem requiring uniform mixing of each normalized branch. It also explains why positive single-outcome probabilities cannot replace that hypothesis. The example supplies a precise boundary for the constructive theorem.

S16. An inherited sharp four-period transition

The four-period certificate in Proposition S12.3 provides an explicit finite-probe separation. Its sufficient interval was not the sharp exact-memory interval. The latter is already a consequence of the inherited first-sector theorem, and is recorded here to correct the research-status distinction rather than as a new spectral result.

Corollary S16.1 (Inherited complete four-period count)**.** *For $b_\alpha(n) = \frac{1}{2}[1 + \alpha^n \cos(\pi n/2)]$, $n \geq 0$, with a prescribed pure initial label and randomized bounded decoder, the exact live memory is*

$$m(b_\alpha) = \begin{cases} 2, & \alpha = 0, \\ 3, & 0 < \alpha \leq 1/\sqrt{3}, \\ 4, & 1/\sqrt{3} < \alpha \leq 1. \end{cases}$$

Proof. For $\alpha > 0$ the three distinct modes $1, i\alpha, -i\alpha$ give the rank lower bound three. Apply Theorem S20.3 with $M = 3$ and $\vartheta = \pi/2 \leq 2\pi/3$; its exact threshold is $1/(\cos(\pi/2)+\sin(\pi/2)\tan(\pi/3)) = 1/\sqrt{3}$. Above this value three labels are impossible, while the damped four-cycle construction of Proposition S12.3 works throughout $0 \leq \alpha \leq 1$. At $\alpha = 0$ the initial probability is one and all subsequent probabilities are $1/2$, requiring and admitting two labels. □

The unresolved general damping problem concerns other responses and later transition regions not covered by an existing exact theorem. In particular, the first transition for this four-period family is closed by the earlier first-sector result. Its explicit rational finite-gap certificate remains a different and valid conclusion.

S17. The recurrent simplex and the cyclic specialization

Lemma S17.1 (A lazy finite-chain projection)**.** *Let T be a finite row-stochastic matrix with strictly positive diagonal. Let $C_1, \ldots, C_k$ be its closed communicating classes. There is a stochastic idempotent $E = \lim_{n\to\infty} T^n$ whose image on the probability simplex is the simplex spanned by the stationary rows μ_i supported on the respective C_i. If a row-stochastic matrix Q preserves every C_i, then $EQE = E$. If T is a convex combination of matrices from a multiplicative semigroup $\mathcal{S}$ that contains the identity, then E belongs to the closed convex hull of $\mathcal{S}$.*

Proof. Consider first one closed class C. Irreducibility and the positive diagonal imply that some power K of the restricted matrix has every entry positive: choose a path between each ordered pair, and pad the finitely many path lengths to their maximum using diagonal self-loops. Put $\eta = \min_{i,j} K_{ij} > 0$ and $n_C = |C|$. Then

$$K = n_C \eta J + (1 - n_C \eta) K_0,$$

where all rows of J are uniform and K_0 is row-stochastic, with the second term omitted if $n_C\eta = 1$. For probability rows p, q,

$$\|pK - qK\|_1 \leq (1 - n_C\eta)\|p - q\|_1.$$

A stationary row for $T|_C$ exists by taking an accumulation point of the Cesàro averages $n^{-1}\sum_{j=0}^{n-1} pT^j$ in the compact simplex: their difference from their right translates by T tends to zero. Call it μ_C. The contraction shows it is unique and that all rows converge geometrically to μ_C along the chosen power. The remaining finitely many residues of the exponent converge to the same row because $\mu_C T = \mu_C$. This proves convergence on each closed class.

Let H be the complement of their union. If H is empty, the preceding closed-class argument already gives the asserted convergence. Otherwise, every vertex in H has a directed path to a closed class. Since the graph is finite, there are an integer q and a number $\zeta > 0$ such that, starting from any vertex in H, the probability of leaving H by time q is at least ζ. Paths can again be padded using the positive diagonal. In a permutation of coordinates, write

$$T = \begin{pmatrix} A & B \\ 0 & C \end{pmatrix},$$

where A is indexed by H and C is block diagonal over the closed classes. Then $\|A^n\|_\infty \le (1-\zeta)^{\lfloor n/q \rfloor}$, while $C^n \to E_C$, the block projection with all rows in C_i equal to μ_i. The upper-right block of T^n is

$$\sum_{j=0}^{n-1} A^j B C^{n-1-j}.$$

This converges to $\sum_{j=0}^{\infty} A^j B E_C$: for a fixed cutoff the finitely many factors C^{n-1-j} converge, and the tail is uniformly bounded by a convergent geometric series. Thus

$$E = \begin{pmatrix} 0 & \sum_{j=0}^{\infty} A^j B E_C \\ 0 & E_C \end{pmatrix}$$

exists. As a limit of stochastic matrices it is stochastic. Every row is a convex combination of the μ_i, each of which occurs as a row of E on its own class. Their disjoint supports give affine independence, and $\mu_i E = \mu_i$ proves $E^2 = E$.

If Q preserves C_i, the row $\mu_i Q$ is a probability law on C_i, so $(\mu_i Q)E = \mu_i$. Extend over the convex combinations constituting the rows of E to obtain $EQE = E$. This does not require $\mu_i Q = \mu_i$. Finally, every T^n is a convex combination of products from $\mathcal{S}$, which remain in $\mathcal{S}$. Taking the limit gives the last assertion. □

S17.1. **Cyclic modules and the retained positive-amplitude model.** The next statement identifies exactly where the representation-theoretic bound is sufficient for the periodic family; it is not asserted for every response on a cyclic group.

Let $G = C_Q$ and let the distinct nontrivial character pairs occurring in a real G-module V have orders $q_1, \dots, q_r$ dividing Q; equal orders are allowed for different pairs. Assume each complex-conjugate pair occurs once, the trivial character occurs once, and no other characters occur. For this assertion a real character of order two, if present, is counted once. For a partition $\mathcal{P}$ of $\{1, \dots, r\}$, put

$$\Psi(\mathcal{P}) = \sum_{B \in \mathcal{P}} \operatorname{lcm}(q_j : j \in B).$$

Proposition S17.2 (Cyclic permutation-module minimum)**.** *If $r \ge 1$, the least size of a permutation module containing V is*

$$\nu_{C_Q}(V) = \min_{\mathcal{P}} \Psi(\mathcal{P}). \tag{S17.1}$$

For the nondegenerate positive-amplitude periodic response of Section S18, the cycle-mixture construction there attains this bound with a probability decoder. Consequently $\kappa_G(b) = \nu_G(V_b)$ for that family, and (S2.4) *gives its retained partition minimum plus the pure-start alternative. A constant response has $\nu = \kappa = m = 1$.*

Proof. A transitive C_Q-set is a cycle of length ℓ dividing Q. Its real permutation module contains exactly the characters whose orders divide ℓ, once each after the conjugate pairs are combined over $\mathbb{R}$. This can be verified directly with the vectors $j \mapsto \exp(2\pi i m j/\ell)$, $0 \le m < \ell$, which are a basis and diagonalize cyclic translation. Given any collection of cycles whose module contains V, assign each required nontrivial character to one cycle that contains it, and discard cycles to which no character was assigned. For the nonempty index block B assigned to a cycle, its length is a multiple of every q_j, $j \in B$, and therefore at least their least common multiple. The total number of points is at least $\Psi(\mathcal{P})$ for the induced partition. Conversely, cycles of the lengths in any $\Psi(\mathcal{P})$ contain every required character and at least one trivial character. This proves (S17.1).

For the indicated positive-amplitude response, the construction in Section S18 uses one such cycle for each block, chooses its initial block weight W_B/Γ, where $W_B = \sum_{j \in B} w_j$ and $\Gamma = \sum_j w_j \le 1$,

and uses the binary decoder

$$d_{B,n} = \frac{1}{2}\left[1 + \frac{\Gamma}{W_B}\sum_{j\in B} w_j \cos(2\pi n a_j/q_j + \varphi_j)\right].$$

Each cosine lies in $[-1,1]$ and the coefficients have total $\Gamma \le 1$, so every decoder lies in $[0,1]$. It realizes the response with exactly the cycle count in (S17.1). The opposite inequality is (S2.5). The separate initial label is charged exactly as in Theorem S2.2; when the residual orbit itself is cheaper, its pure-start machine is used. □

The cyclic conclusion uses the amplitudes, frequencies, and normalization actually specified in the retained theorem. Opposite frequencies must be combined, zero amplitudes removed, and order-two modes treated as real one-dimensional modes before applying the module statement. It does not replace the original hypotheses by a mere list of eigenvalue orders.

S18. Exact periodic memory, initialization and leakage

At zero internal damping a finite mixture of rational frequencies is periodic. Its period, linear response rank and least stochastic memory are different quantities. We determine the last one exactly, with a prescribed initial label, and give a matching sparse implementation. This completes the rational multi-frequency endpoint left as bounds by Corollary S19.5. The spectral cycle method is classical Perron–Frobenius theory; periodic stochastic orbits and least-common-multiple constraints are established subjects [59, 54]. We prove the precise partition and initialization rule needed for the present bounded probability response rather than claim those ingredients as new.

Choose distinct nonreal roots $\zeta_j = e^{2\pi i a_j/q_j}$, $1 \le j \le r$, no two conjugate, with $\gcd(a_j, q_j) = 1$ and $q_j \ge 3$. Let $w_j > 0$, $\Gamma = \sum_j w_j \le 1$, and $\varphi_j \in \mathbb{R}$. The target is

$$b_n = \tfrac{1}{2}\left[1 + \sum_{j=1}^{r} w_j \cos(2\pi n a_j/q_j + \varphi_j)\right]. \tag{S18.1}$$

All listed complex modes have nonzero coefficients. For a nonempty index set B, set $\ell(B) = \operatorname{lcm}\{q_j : j \in B\}$. If $\mathcal{P}$ is a partition of $\{1,\dots,r\}$, define

$$\mathcal{M}(\mathcal{P}) = \sum_{B\in\mathcal{P}} \ell(B) + \mathbf{1}_{\{|\mathcal{P}|\ge 2\}}, \qquad M_* = \min_{\mathcal{P}} \mathcal{M}(\mathcal{P}). \tag{S18.2}$$

Equal orders are allowed when their nonconjugate primitive modes differ. They impose compatible divisibility constraints, not distinct mandatory recurrent classes.

Lemma S18.1 (Unit-circle modes of a reachable stochastic matrix)**.** *Every nonzero coefficient mode ζ in a finite exponential response $e_1^{\mathsf{T}} Q^n d$ is an eigenvalue of the restriction of Q to its reachable states. If $|\zeta| = 1$, it occurs in a reachable closed irreducible class. If that class contains primitive root modes of orders in a set J, its number of states is at least* $\operatorname{lcm} J$.

Proof. Cayley–Hamilton gives a characteristic recurrence for the response. Distinct exponential sequences are linearly independent by the Vandermonde determinant. Substituting the recurrence therefore forces its characteristic polynomial to vanish at each mode with nonzero coefficient.

Order the strongly connected components of the reachable transition graph to make Q block triangular. An eigenvalue belongs to an irreducible diagonal block B, whose row sums are at most one. For $Bv = \zeta v$, choose a coordinate with maximal nonzero modulus. If $|\zeta| = 1$, equality in the row-sum and triangle inequalities forces that row sum to be one and every positive successor to have the same modulus, with value ζ times the preceding value. Irreducibility propagates this to every coordinate and every row. Thus the block is closed in Q. Along each directed cycle of length h, one has $\zeta^h = 1$. Applying this to every required mode in the block shows that every directed cycle length is a multiple of their least common multiple. A simple directed cycle exists and has

length no greater than the number of states, proving the lower. This argument allows reducible Q and arbitrary real stochastic entries. □

Theorem S18.2 (Exact partition law for periodic stochastic memory)**.** *The least exact stochastic live-label count for* (S18.1)*, with prescribed initial label and decoder in* $[0,1]$*, is exactly* M_**. It is independent of the nonzero weights and preparation phases within the displayed probability-valid family.*

The least count when transitions must be deterministic but the terminal decoder may still be randomized is

$$P = \operatorname{lcm}(q_1, \dots, q_r).$$

For pairwise coprime $q_j \geq 3$ *with* $r \geq 2$*, the stochastic count is* $1 + \sum_j q_j$*, whereas the deterministic-transition count is* $\prod_j q_j$*. An invariant finite residual orbit therefore need not have minimal stochastic cardinality.*

Proof. Consider an exact realizing machine after removing unreachable labels. Lemma S18.1 assigns every required root to a reachable closed irreducible class. Assign each index to one such class and omit classes assigned no index. These assignments partition the indices, and their class sizes total at least $\sum_{B\in\mathcal{P}} \ell(B)$. If at least two assigned closed classes are reachable, the prescribed initial state cannot belong to any one of them: a closed class cannot reach another. At least one additional transient initial label is therefore necessary. This proves the lower M_*. It would be incorrect to omit this label by allowing a free mixed initial distribution.

For the upper fix a partition. With one block, a deterministic cycle of length P and decoder $d_k = b_k$ is sufficient. With $g \geq 2$ blocks, take disjoint deterministic cycles of lengths $\ell(B)$ and one new initial label. Put $W_B = \sum_{j\in B} w_j$, $\lambda_B = W_B/\Gamma$, and on phase k of block B set

$$d_{B,k} = \tfrac{1}{2}\left[1 + \frac{\Gamma}{W_B}\sum_{j\in B} w_j \cos(2\pi k a_j/q_j + \varphi_j)\right].$$

The total absolute oscillatory amplitude is at most $\Gamma \leq 1$, so all decoder values lie in $[0,1]$. Give the initial label decoder b_0. Its first transition selects block B with probability λ_B and enters phase one modulo $\ell(B)$. All later transitions simply advance that cycle. Thus at time $n \geq 1$ the output is $\sum_B \lambda_B d_{B,n} = b_n$, and equality at zero holds separately. There is no one-step phase error or uncounted block-selection state. This constructs exactly (S18.2) labels; minimizing proves the stochastic formula.

The exact period of b is P. Indeed any period annihilates every nonzero exponential mode under shift minus identity, hence is divisible by each q_j. A deterministic finite machine has an eventual cycle. Its length must be an eventual period of b, and for a sequence already periodic from zero an eventual period is also a period from zero: shift any given identity far enough by multiples of P. That cycle length is therefore at least P. The P-cycle gives the matching upper. Finally, for pairwise coprime orders at least three, merging two or more singleton blocks replaces their sum by their product, strictly increasing cost even if a single initialization label is saved. Repeated splitting proves the stated special formula. □

For example, orders $3, 5, 7$ give exact stochastic memory 16 and deterministic-transition memory 105. Orders $4, 6$ give $\min\{12, 4+6+1\} = 11$. Divisibility alone, e.g. orders $3, 6$, gives 6 rather than 10. These contrasting cases are covered by the same all-real-stochastic lower, not a search restricted to proposed cycles. If a listed coefficient vanishes or conjugate modes cancel, one must first remove that mode; the formula is not continuous across deletion of spectral support.

Corollary S18.3 (Constructive minimal synthesis and a zero-noise plateau)**.** *The optimum partition is computable by at most* $O(3^r)$ *subset comparisons and least-common-multiple calculations on integers of bit length at most* $\sum_j \lceil \log_2 q_j \rceil$*. After its partition is found, a minimal transition table*

has only deterministic cycle rows and at most one randomized initialization row. Producing all decoder entries requires at most $O(rM_*)$ evaluations and arithmetic operations in their specified exact number representation. This is an output-sensitive count, not a uniform polynomial bit bound for arbitrary real phases or exact sampling.

Let m_α be the least count for the internally damped response

$$b_n^{(\alpha)} = \tfrac{1}{2}\left[1 + \alpha^n \sum_j w_j \cos(2\pi n a_j/q_j + \varphi_j)\right], \quad 0 < \alpha \le 1.$$

Then $m_\alpha \le M_*$ for every α, and $m_\alpha = M_*$ for all α sufficiently close to one. This is an exact multi-frequency endpoint plateau, not merely a limiting lower bound for each separate pole.

Proof. For a subset S put $D(\varnothing) = 0$ and $D(S) = \min_{\varnothing \ne B \subseteq S}\{\ell(B) + D(S \setminus B)\}$. Compute these in increasing subset size, using a canonical distinguished index in B to remove duplicate partitions if desired. The total number of subset pairs is at most 3^r. For $r \ge 2$ compare $\ell(\{1,\dots,r\})$ with $1 + \min_{\varnothing \ne B \subsetneq \{1,\dots,r\}}[\ell(B) + D(B^c)]$. This separately charges the one-block and multiblock cases. The construction in the theorem proves the output counts. Real-algebraic weights and $e^{i\varphi_j}$ give algebraic controller entries, since the rational-angle roots are algebraic; arbitrary real phases retain their separate description requirement.

In that construction every cycle has mean decoder $1/2$, because each nonconstant root averages to zero around its cycle. Choose a stationary row π uniform on any one cycle, with zero mass at the transient initial label when present. Then $\pi d = 1/2$. Replacing Q by $Q_\alpha = \alpha Q + (1-\alpha)\mathbf{1}\pi$ preserves stochasticity on the same labels and gives

$$Q_\alpha^n = \alpha^n Q^n + (1 - \alpha^n)\mathbf{1}\pi.$$

This is the damped response, proving the upper. If sizes below M_* occurred along $\alpha \uparrow 1$, pad them to $M_* - 1$ labels and take a convergent subsequence of their compact stochastic and decoder rows. Every fixed response coefficient passes to the limit, producing an undamped realization smaller than M_*. This contradicts the theorem and proves the plateau. No numerical cutoff for its first plateau point is claimed. □

S18.1. **The exact leakage increment for the entire periodic mixture.** The preceding count also survives decay-normalization. This requires more than applying the single-pole bound to each frequency separately, since those bounds would lose both the recurrent-class partition and its initialization charge.

Lemma S18.4 (No chains of critical classes in a power-bounded core)**.** *Let $A \ge 0$ have bounded powers. No directed path connects two distinct irreducible diagonal blocks of spectral radius one.*

Proof. Suppose such blocks B, C are connected. Choose a positive-weight path from a vertex of B to a vertex of C whose interior avoids both blocks, and let W be the matrix with that path weight in the corresponding entry and zero elsewhere. Let its length be h. Perron vectors give a strictly positive left row $uB = u$ and a strictly positive column $Cv = v$. For each $0 \le i, j \le n$, concatenate a length-i path inside B, the fixed connecting path, and a length-j path inside C. Their first exit from B distinguishes different i, so they are distinct contributions to the powers. Therefore the B, C block of $\sum_{k=0}^{2n+h} A^k$ dominates $\sum_{i,j=0}^{n} B^i W C^j$. Pairing with u, v gives the lower $(n+1)^2 uWv > 0$. Bounded powers give an upper growing only linearly in n, a contradiction. The elementary Perron normalization used here is proved in Section S21. □

Theorem S18.5 (Periodic-mixture leakage law)**.** *For every fixed $0 < \lambda < 1$, the least stochastic live memory of $u_n = \lambda^n b_n$, with b as in* (S18.1)*, is exactly*

$$s(u) = M_* + 1.$$

The extra label is live until the actual commitment event. In the fixed squared-task embedding, these are exact optimal-observer counts at the original least cost $1/p$*, with event-stopped and state-stopped total-label upper counts* $\max\{M_* + 1, 2\}$ *and* $M_* + 3$*.*

Proof. The construction of Lemma S21.1 applied to a minimal realization of b gives the upper $M_* + 1$. For the reverse, take an arbitrary stochastic realization of u with M reachable labels. Its output-visible core S has at most $M - 1$ labels, includes the prescribed start, and realizes u. The path bound (S21.2), together with $0 \le u_n \le \lambda^n$, shows entrywise that $(S/\lambda)^n$ is bounded. Its required modes are the distinct roots $1, \zeta_j, \overline{\zeta}_j$ after scaling by λ, each with nonzero coefficient. Every required nonreal root lies in an irreducible block of radius one. Perron scaling of that block makes it stochastic without changing its graph or eigenvalues. Lemma S18.1 thus imposes the same least-common-multiple size for every group of assigned modes.

By Lemma S18.4, distinct assigned critical blocks cannot reach one another. If at least two are reachable from the prescribed initial label, that label lies outside all of them and adds a further core state. Hence the core has at least $\sum_B \ell(B) + \mathbf{1}_{\{|\mathcal{P}|\ge 2\}} \ge M_*$ labels. Including the necessarily inactive label gives $M \ge M_* + 1$. The argument does not impose a nonexistent globally positive Perron vector on a reducible core. The squared-task and stopping interpretation follows from the unchanged physical embedding and established role packing, not from interpreting a leakage event as an observed halt. □

A protected target for (S18.1) is obtained by the block-qubit preparation and terminal effect of Theorem S20.8; leakage appends its actual dark physical state. These are sufficient physical realizations, not minimal physical-dimension claims. The standard square-loss flag embedding is distinct from a probability decoder supplied as an oracle. For a prescribed smaller observer size, the finite rational-resolvent tests of Theorem S21.6 can also witness the new exact counts, using response order $2r + 1$. Their number of models grows with the tested memory. The partition law does not solve the different general-M squared-sign risk curve, nor declare arbitrary controlled boundary stopping classified.

S19. An arithmetic envelope for exact memory at every phase

The spectral criterion of Theorem S20.1 determines exact stochastic memory by membership in the classical stochastic eigenvalue regions. A worst-phase bound and a matching estimate at a badly approximable fixed phase do not describe all fixed phases. In particular, excellent rational approximations can produce both unusually small memory and long intervening ranges of larger memory. We determine a single arithmetic envelope that controls the entire memory function, and derive its exact lower and upper logarithmic growth exponents for every irrational phase. The arithmetic is classical continued-fraction theory; the conclusions here concern its conversion into the least exact stochastic observer memory. We retain the distinction from the noise-dependent classical process bounds in [50] and from the classical irrationality-exponent identity in [58].

Throughout this section, $0 < \epsilon < 1$, $\alpha = 1 - \epsilon$, and

$$b_n = \tfrac{1}{2}[1 + \gamma\alpha^n \cos(2\pi n x + \varphi)], \qquad 0 < \gamma \le 1, \quad \sin(2\pi x) \ne 0.$$

The phase and nonzero contrast describe one prepared target, not information revealed to its observer. Write $m_x(\epsilon) = m(1 - \epsilon, 2\pi x)$ for its least live-label count, with a prescribed initial label, stochastic transitions and a decoder in $[0, 1]$. Theorem S20.1 makes this independent of γ, φ. The actual squared-loss embedding of Theorem S25.1 forces exactly this probability response at the least physical cost $1/p$. Thus the conclusions below are about exact optimal implementation, not approximation of the target sequence or a new definition of loss.

For a positive integer q, put

$$e_x(q) = \frac{\|qx\|_{\mathbb{R}/\mathbb{Z}}}{q} = \min_{a\in\mathbb{Z}} |x - a/q|, \qquad \mathfrak{A}_x(\epsilon) = \min_{q\ge 1} \max\{3, q, e_x(q)/\epsilon\}.$$

The minimum is attained: its summand is at least q, while $q = 1$ gives a finite upper. No global Diophantine lower hypothesis is imposed.

Theorem S19.1 (Universal arithmetic envelope)**.** *For every nonreal phase and every $0 < \epsilon < 1$,*

$$\tfrac{1}{2}\mathfrak{A}_x(\epsilon) \le m_x(\epsilon) \le C_0\mathfrak{A}_x(\epsilon), \qquad C_0 = 1 + 4\pi^2. \tag{S19.1}$$

The constants are independent of the phase, contrast, preparation phase and damping. Each upper bound comes from an explicit stochastic circulant realization with a prescribed initial label and bounded decoder. The lower holds over all stochastic matrices and decoders, not only those circulants. The estimate is a constant-factor law for the full arithmetic envelope; it is not an assertion of an exact leading constant or a new classification of the stochastic spectral regions.

Proof. We use the polygon construction and the power-sector inequality in Lemma S20.4, whose positive-matrix proofs are included in Section S20. Let $M = m_x(\epsilon) \ge 3$. Among the $2M + 1$ fractional parts of $0, x, \dots, 2Mx$, two lie in the same one of $2M$ intervals of length $1/(2M)$. Their difference supplies $1 \le k \le 2M$ with $\|kx\| \le 1/(2M)$. If this norm is zero, using $q = k$ gives $\mathfrak{A}_x(\epsilon) \le 2M$ directly.

Otherwise put $\delta = 2\pi\|kx\|$ and $h = \pi/M$. Then $0 < \delta \le h \le \pi/3$. For $F = \cos\delta + \sin\delta\tan h$, one has $F \le \sec h \le 2$ and

$$F - 1 \ge \tfrac{3}{4}h\delta - \tfrac{1}{2}\delta^2 \ge h\delta/4.$$

Here $\sin\delta \ge 3\delta/4$, $\tan h \ge h$ and $1 - \cos\delta \le \delta^2/2$ on the stated interval. The eigenvalue of the stochastic power Q^k gives $(1-\epsilon)^k F \le 1$. Bernoulli's inequality therefore yields

$$k\epsilon \ge 1 - (1-\epsilon)^k \ge 1 - 1/F \ge h\delta/8.$$

Consequently $e_x(k)/\epsilon = \delta/(2\pi k\epsilon) \le 4M/\pi^2$. Since also $k \le 2M$, the envelope is at most $2M$. This proves the lower.

For the upper fix any denominator q, choose its closest integer numerator a, and choose the smallest positive multiple l of q satisfying

$$l \ge 3, \qquad l \ge 4\pi^2 e_x(q)/\epsilon.$$

Then $l < q + \max\{3, 4\pi^2 e_x(q)/\epsilon\} \le C_0 \max\{3, q, e_x(q)/\epsilon\}$. The rational angle $2\pi a/q$ is an lth-root angle. Thus the distance δ_l from $2\pi x$ to the nearest such root is at most $2\pi e_x(q)$. The directional polygon bound gives

$$1 - R_l(2\pi x) \le 2(\pi/l)\delta_l \le 4\pi^2 e_x(q)/l \le \epsilon.$$

Hence $\alpha e^{2\pi i x}$ lies in the regular l-gon. The positive combination of its two adjacent roots and zero gives the stochastic matrix described in Lemma S20.4; its normalized eigenvector gives the bounded decoder from Theorem S20.1. The original initial label suffices. Minimize over q to conclude. All constructions act on the observer's private memory and do not add a physical observation or a call. □

For irrational x, $\mathfrak{A}_x(\epsilon) \to \infty$ as $\epsilon \downarrow 0$: a bounded envelope would restrict q to finitely many integers whose errors have a positive minimum. For a rational nonreal phase of reduced denominator q, the exact memory is eventually q as $\epsilon \downarrow 0$. Indeed $\alpha C_q + (1-\alpha)J$ gives the upper q, while a bounded sequence of smaller stochastic realizations would converge, by compactness, to one with the forbidden unit eigenvalue of order q. This rational boundary is different from assigning an irrationality exponent to a rational number. Zero contrast and real phases remain governed by their separately stated degeneracies.

S19.1. **The arithmetic input, with proof.** For irrational x, changing it by an integer does not change any quantity above. Write its fractional part as $[0; a_1, a_2, \ldots]$, with convergents p_n/q_n. We use the standard initial values $p_{-1} = 1$, $p_0 = 0$, $q_{-1} = 0$, $q_0 = 1$ and the recursion $p_{n+1} = a_{n+1}p_n + p_{n-1}$, $q_{n+1} = a_{n+1}q_n + q_{n-1}$. The possible initial repeated denominator is irrelevant below; all asymptotic statements use the eventually strictly increasing sequence.

Lemma S19.2 (Convergent errors and the irrationality exponent)**.** *Set $E_n = q_n x - p_n$, with x reduced modulo one. Then*

$$\frac{1}{q_{n+1} + q_n} < |E_n| < \frac{1}{q_{n+1}}, \qquad |qx - a| \geq |E_n| \quad (0 < q < q_{n+1},\ a \in \mathbb{Z}). \tag{S19.2}$$

For all sufficiently large n, $e_x(q_n) = |E_n|/q_n$. Define

$$\mu(x) = \sup\{t :\ e_x(q) < q^{-t} \text{ for infinitely many } q\}.$$

Then

$$\mu(x) = 1 + \limsup_{n\to\infty} \frac{\log q_{n+1}}{\log q_n} \in [2, \infty]. \tag{S19.3}$$

Moreover x is badly approximable, in the sense that $e_x(q) \geq c/q^2$ for some $c > 0$ and all q, if and only if the ratios q_{n+1}/q_n are bounded.

Proof. The recursions give $p_n q_{n-1} - p_{n-1}q_n = (-1)^{n-1}$ by induction. If $t_{n+1} = [a_{n+1}; a_{n+2}, \ldots]$, repeated substitution in the continued fraction gives

$$x = \frac{p_n t_{n+1} + p_{n-1}}{q_n t_{n+1} + q_{n-1}}, \quad |E_n| = \frac{1}{q_n t_{n+1} + q_{n-1}}, \quad E_{n-1} = -t_{n+1}E_n.$$

Since $a_{n+1} < t_{n+1} < a_{n+1} + 1$, the denominator lies strictly between q_{n+1} and $q_{n+1} + q_n$, proving the first pair of bounds. These errors eventually have modulus below $1/2$, so the numerator is the nearest integer.

For the second assertion, suppose $|qx - a| < |E_n|$. The determinant identity makes the two consecutive convergent vectors an integral basis. Write $(a, q) = A(p_n, q_n) + B(p_{n-1}, q_{n-1})$ with integers A, B. Then $|A - Bt_{n+1}| < 1$. If $B = 0$, this forces $A = 0$ and $q = 0$. If $B \geq 1$, it forces $A \geq Ba_{n+1}$ and hence $q \geq Bq_{n+1} \geq q_{n+1}$. If $B = -b \leq -1$, it forces $A \leq -ba_{n+1}$ and hence $q \leq -bq_{n+1} < 0$. These exhaust the possibilities and prove the asserted best-error inequality. No best-approximation assertion for arbitrary fitted rational data has been used.

Put $\Lambda = \limsup \log q_{n+1}/\log q_n \geq 1$. The convergent upper $e_x(q_n) < 1/(q_n q_{n+1})$ shows $\mu \geq 1 + \Lambda$, including the infinite case. If $s > \Lambda$ is finite, eventually $q_{n+1} \leq q_n^s$. For $q_n \leq q < q_{n+1}$, the second-kind inequality gives

$$e_x(q) > \frac{1}{q(q_{n+1} + q_n)} \geq \frac{1}{2q^{s+1}}.$$

For every positive η, this excludes infinitely many errors below $q^{-(s+1+\eta)}$. Let $s \downarrow \Lambda$ and $\eta \downarrow 0$ to prove (S19.3).

If $q_{n+1}/q_n \leq B$ eventually, the same interval argument gives $e_x(q) > 1/[(B+1)q^2]$ there; reducing the positive constant handles the finitely many earlier denominators. Conversely $e_x(q_n) < 1/(q_n q_{n+1})$ is incompatible with one positive lower constant when the ratios are unbounded. This proves the last equivalence. The identity (S19.3) is classical; it is also stated in the cited arithmetic literature [58], and is proved here to make the memory argument independent of that reference. □

S19.2. **The complete logarithmic growth spectrum.** The envelope law retains more information than a single exponent. Its two logarithmic exponents nevertheless have a closed form for every irrational phase, including phases not known to satisfy a uniform approximation bound.

Theorem S19.3 (Exact lower and upper memory exponents)**.** *For every irrational x, the exact stochastic memory obeys*

$$\liminf_{\epsilon\downarrow 0} \frac{\log m_x(\epsilon)}{\log(1/\epsilon)} = \frac{1}{\mu(x)+1}, \tag{S19.4}$$

$$\limsup_{\epsilon\downarrow 0} \frac{\log m_x(\epsilon)}{\log(1/\epsilon)} = \frac{\mu(x)-1}{2\mu(x)-1}. \tag{S19.5}$$

For $\mu(x)=\infty$, the right sides mean 0 and 1/2. In particular the logarithmic exponent exists exactly when $\mu(x)=2$, and then it equals 1/3. This does not imply a constant-factor cubic-root law for every such phase. The assertions describe one fixed phase while the noise tends to zero, not a worst-phase choice depending on the noise.

Proof. By Theorem S19.1, multiplicative constants do not affect either logarithmic limit, so work with $\mathfrak{A}_x$.

First suppose $\mu<\infty$ and choose $t>\mu$. For all sufficiently large q, $e_x(q)\ge q^{-t}$. Therefore

$$\max\{q, e_x(q)/\epsilon\} \ge \max\{q, q^{-t}/\epsilon\} \ge \epsilon^{-1/(t+1)}.$$

Every one of the finitely many remaining denominators has positive error, and its error term grows as ϵ^{-1}, so those denominators do not invalidate this lower for sufficiently small ϵ. This gives a lower limit at least $1/(\mu+1)$. Conversely, for $1<t<\mu$, infinitely many q satisfy $e_x(q)<q^{-t}$. At $\epsilon=q^{-(t+1)}$, the envelope is at most q once $q\ge 3$. Taking $t\uparrow\mu$ proves (S19.4); for $\mu=\infty$ the same argument with arbitrarily large t, together with nonnegativity, gives zero.

For the upper limit with finite μ, again choose $t>\mu$. Pigeonhole approximation at denominator bound Q gives $1\le k\le Q$ with $e_x(k)\le 1/(kQ)$. Such k tend to infinity along $Q\to\infty$, because the phase is irrational. The eventual lower $e_x(k)\ge k^{-t}$ therefore forces $k\ge Q^{1/(t-1)}$. Hence

$$\mathfrak{A}_x(\epsilon) \le \max\{3, Q, \epsilon^{-1}Q^{-t/(t-1)}\}.$$

Choose $Q=\lceil \epsilon^{-(t-1)/(2t-1)}\rceil$ and then let $t\downarrow\mu$. This bounds the upper limit by the right side of (S19.5). If $\mu=\infty$, use only $k\ge 1$ and $Q=\lceil\epsilon^{-1/2}\rceil$ in the same pigeonhole argument to obtain the universal upper limit 1/2.

To prove the reverse upper-limit bound, take consecutive large convergent denominators $Q=q_n$, $Q'=q_{n+1}$ and set

$$\epsilon_n = \frac{1}{Q(Q')^2}. \tag{S19.6}$$

For every rational a/q with $0<q<Q'/2$, either $a/q=p_n/Q$, in which case its error is greater than $1/[Q(Q'+Q)]\ge 1/(2QQ')$, or the determinant of the two integer pairs is nonzero and

$$|x-a/q| \ge \frac{1}{qQ} - |x-p_n/Q| > \frac{1}{qQ} - \frac{1}{QQ'} > \frac{1}{QQ'}.$$

Thus all these denominators have $e_x(q)/\epsilon_n \ge Q'/2$. All other denominators have $q\ge Q'/2$. It follows that $\mathfrak{A}_x(\epsilon_n)\ge Q'/2$. On the other hand, denominator Q' has $e_x(Q')<1/(Q'q_{n+2})$, so $e_x(Q')/\epsilon_n < QQ'/q_{n+2}\le Q\le Q'$. Consequently

$$Q'/2 \le \mathfrak{A}_x(\epsilon_n) \le Q'. \tag{S19.7}$$

The corresponding logarithmic ratio is, up to a term tending to zero,

$$\frac{\log Q'}{\log Q + 2\log Q'} = \frac{\log Q'/\log Q}{1+2\log Q'/\log Q}.$$

The function on the right is increasing, continuous and tends to 1/2 at infinity. Lemma S19.2 proves precisely the remaining lower bound in (S19.5). Finally the two displayed exponents are equal exactly when $\mu=2$: solving the equality gives $\mu(\mu-2)=0$, and $\mu\ge 2$. □

Theorem S19.4 (Rigidity of the uniform cubic-root law)**.** *For an irrational fixed phase, the uniform estimate*

$$m_x(\epsilon) = \Theta(\epsilon^{-1/3}) \qquad (\epsilon \downarrow 0)$$

holds if and only if x is badly approximable. If it is not, the stronger oscillation statements hold:

$$\liminf_{\epsilon\downarrow 0} \epsilon^{1/3} m_x(\epsilon) = 0, \qquad \limsup_{\epsilon\downarrow 0} \epsilon^{1/3} m_x(\epsilon) = +\infty. \tag{S19.8}$$

Thus even an existing logarithmic exponent $1/3$ need not give uniform cubic-root constants.

Proof. For badly approximable x, Theorem S20.6 already proves both uniform bounds, and that result is inherited. Otherwise Lemma S19.2 supplies consecutive denominators Q, Q' with $Q'/Q \to \infty$ along a subsequence. At $\epsilon'_n = e_x(Q)/Q$, the envelope is at most Q, and the upper in (S19.1) gives

$$(\epsilon'_n)^{1/3} m_x(\epsilon'_n) \le C_0 (Q^2 e_x(Q))^{1/3} < C_0 (Q/Q')^{1/3} \longrightarrow 0.$$

At the crossover noise (S19.6), the lower in (S19.1) and (S19.7) instead give

$$\epsilon_n^{1/3} m_x(\epsilon_n) \ge \tfrac{1}{4} (Q'/Q)^{1/3} \longrightarrow +\infty.$$

Both noise sequences tend to zero. Nonnegativity proves the first equality in (S19.8), and the second rules out a uniform upper constant. No convergence of sampled continued-fraction coefficients has been substituted for a global arithmetic property. □

These conclusions distinguish three questions: the complete envelope at a given noise, the logarithmic exponents determined by μ, and uniform power-law constants governed by bounded convergent ratios. A quadratic irrational phase has bounded partial quotients and retains the inherited cubic-root law. For an explicit phase with exponent two but without uniform cubic constants, take $x = [0; 1, 2, 3, 4, \ldots]$. Its denominator ratios are unbounded, while the denominator recursion implies at least Fibonacci growth and $\log(q_{n+1}/q_n)/\log q_n \le \log(n+2)/\log q_n \to 0$. Hence $\mu = 2$ but (S19.8) holds. For $\mu = \infty$, the two exponents are 0 and $1/2$. The stronger universal theorem does not determine the numerical value of μ for the distinct algebraic unit phase $(3+4i)/5$; no cubic-root claim for that phase is introduced.

S19.3. **Several visible frequencies and leakage.** Let $r \ge 1$, let $x_j \in (0, 1/2)$ be distinct, and let $w_j > 0$ with $\sum_j w_j \le 1$. For prescribed phases φ_j, consider

$$b_n^{\mathrm{mix}} = \tfrac{1}{2} \left[1 + (1-\epsilon)^n \sum_{j=1}^{r} w_j \cos(2\pi n x_j + \varphi_j) \right].$$

Let m_{mix} be its least exact live memory. For $0 < \lambda < 1$, let s_{mix} be the least memory of the decaying response $\lambda^n b_n^{\mathrm{mix}}$.

Corollary S19.5 (Multi-frequency arithmetic envelopes)**.** *For every $0 < \epsilon < 1$,*

$$\max\{2r+1, \tfrac{1}{2} \max_j \mathfrak{A}_{x_j}(\epsilon)\} \le m_{\mathrm{mix}}(\epsilon) \le 2 + C_0 \sum_j \mathfrak{A}_{x_j}(\epsilon), \tag{S19.9}$$

$$\max\{2r+2, 1 + \tfrac{1}{2} \max_j \mathfrak{A}_{x_j}(\epsilon)\} \le s_{\mathrm{mix}}(\epsilon) \le 3 + C_0 \sum_j \mathfrak{A}_{x_j}(\epsilon). \tag{S19.10}$$

In particular each count is comparable, with constants depending only on r, to the largest of the component envelopes. If at least one phase is irrational, their upper logarithmic exponent is

$$\max_{j:\, x_j \notin \mathbb{Q}} \frac{\mu(x_j) - 1}{2\mu(x_j) - 1},$$

with the same infinite-exponent convention. We do not replace the lower limit of a maximum by the maximum of lower limits; good approximation scales for different phases need not coincide. For a

single leaking frequency, the exact identity $s = 1 + m_x$ transfers both exponents and the uniform-law rigidity unchanged, independently of λ.

These are actual boundary-optimal squared-task observer counts under the retained embedding. The protected target can be realized on dimension at most $2r + 1$ before leakage and at most $2r + 2$ after leakage; the flag embedding doubles these sufficient physical dimensions. No optimal physical-dimension claim or uniform controller for unknown target phases is made.

Proof. The distinct modes $1, (1 - \epsilon)e^{\pm 2\pi i x_j}$ have nonzero coefficients. Their rank and individual spectral obstructions, together with the counted block-mixture construction, are exactly those of Theorem S20.8. Apply (S19.1) to each component to obtain (S19.9).

For leakage, the $2r + 1$ decaying scalar modes gain the independent constant mode in the complementary output, giving rank $2r + 2$. The relative-pole obstruction of Proposition S21.2 requires $1 + m(1 - \epsilon, 2\pi x_j)$ labels for each j. Conversely append one live zero-output state to the entire unattenuated stochastic realization and scale its continuing matrix by λ. This realizes the decaying mixture from the same initial label. These arguments prove (S19.10).

For a finite list, the upper logarithmic limit of its maximum is the maximum of its upper logarithmic limits. Rational components have bounded exact size and do not change this maximum when an irrational component is present. Theorem S19.3 proves the formula. The single-frequency assertion follows from the exact leakage increment of Theorem S21.3. Finally the block-diagonal protected realization in Theorem S20.8, with one additional leakage state when needed, supplies the stated dimensions; the actual squared-loss and stopping-cost assertions follow from the same declared flag embedding. A private randomized block choice is stored in the counted labels and never reveals the protected block. □

The conclusions do not supply a sharp leading constant, a general exact intermediate-memory risk curve, or a minimal controller for an unspecified noncomputable phase. For a supplied computable phase and positive noise, interval bounds on $e_x(q)$ and the explicit polygon construction give finite certified upper constructions; exact minimality still uses the separate spectral-membership problem. Ideal finite labels, descriptions of transition probabilities, exact sampling and working storage remain distinct resources. The complete arithmetic envelope closes the fixed-phase constant-factor question for this response family, not the general multi-command foundations programme.

S20. Spectral and arithmetic laws of exact response memory

The instrument realization theorem preserves every terminal effect, whereas a prescribed task can require substantially less memory. We now determine the exact memory problem for a whole family of single-frequency protected responses, not just one damping factor or the previously solved three-label case. The criterion reduces the problem to classical stochastic eigenvalue regions. A self-contained sector argument and elementary rational approximation then give an exact worst-phase law and matching growth at every fixed badly approximable phase. None of the spectral-region, positive-realization, or sector machinery is claimed as newly invented [51, 52, 53]. The contributions considered here are their exact response-level equivalence with a prescribed initial label, their consequences for the physical squared task, and the phase-sensitive matched resource law.

S20.1. **A physical family and the exact spectral criterion.** Let $0 < \alpha \le 1$, $0 < \gamma \le 1$, and $\vartheta, \varphi \in \mathbb{R}$, with $\sin \vartheta \neq 0$. Consider the binary response

$$b_n = \frac{1}{2}\big[1 + \gamma \alpha^n \cos(n\vartheta + \varphi)\big], \qquad n \ge 0. \tag{S20.1}$$

The rotation, contrast and preparation phase specify one prepared instance; independence of the minimum size does not assert one fixed decoder for unknown phases or contrasts. It is physically

prepared on a qubit: take the Bloch vector $\gamma(\sin\varphi, 0, \cos\varphi)$, the channel

$$\Phi(X) = \alpha U_\vartheta X U_\vartheta^* + (1-\alpha)\,\mathrm{tr}(X) I/2, \qquad U_\vartheta = \begin{pmatrix} \cos(\vartheta/2) & -\sin(\vartheta/2) \\ \sin(\vartheta/2) & \cos(\vartheta/2) \end{pmatrix},$$

and terminal effect $E = (I + \sigma_z)/2$. Its Born probabilities are exactly (S20.1). Theorem S25.1 embeds this response into the same three-outcome squared-loss experiment, with protected dimension four, fixed decoder dictionary $\{-1, 1\}$, and least expected cost $1/p$. It forces the probability of output $+1$ after n failures to equal b_n. Thus the mathematical realization problem below is the exact optimal-observer problem of an actual bounded squared task, not a sequence supplied by a loss oracle.

Write $m(\alpha, \vartheta; \gamma, \varphi)$ for the least M for which

$$b_n = e_1^{\mathsf{T}} Q^n d \quad (n \geq 0), \qquad Q \geq 0, \quad Q\mathbf{1} = \mathbf{1}, \quad d \in [0,1]^M.$$

The initial label is prescribed and counted. Let

$$\mathcal{K}_M = \{z \in \mathbb{C} : z \text{ is an eigenvalue of some } M \times M \text{ row-stochastic matrix}\}.$$

These are the classical Karpelevič regions; their full boundary classification is prior work [52]. We will not use that classification as an unproved step in any bound below.

Theorem S20.1 (Exact spectral realization, with no extra initial label)**.** *For the nondegenerate family* (S20.1)*,*

$$m(\alpha, \vartheta; \gamma, \varphi) = \min\{M : \alpha e^{i\vartheta} \in \mathcal{K}_M\}. \tag{S20.2}$$

The value is independent of every nonzero contrast γ and of the preparation phase φ; denote it by $m(\alpha, \vartheta)$. The criterion is necessary and sufficient over all stochastic matrices, including reducible matrices and arbitrary randomized decoders. For $0 < \alpha < 1$ it is finite. For $\alpha = 1$ and $\vartheta/(2\pi) = a/q$ in lowest terms it is exactly q; for irrational $\vartheta/(2\pi)$ it is infinite.

In the squared-task embedding, these are exact live-label counts at cost $1/p$. At each fixed smaller memory bound, the attained risk is strictly above the full-history value 2. Event-stopped and state-stopped total-label upper counts remain respectively $\max\{M, 2\}$ and $M + 2$.

Proof. Put $z = \alpha e^{i\vartheta}$. The response has the three distinct exponential modes $1, z, \bar{z}$, with nonzero coefficients. Cayley–Hamilton for any realizing Q makes its characteristic polynomial an annihilating shift polynomial of b_n. Substitution and Vandermonde independence force that polynomial to vanish at z. This proves necessity without assuming a minimal linear representation has positive coordinates.

Conversely let $Qv = zv$ with $v \neq 0$. Normalize a largest-modulus coordinate to one by multiplying v by one complex scalar, and permute that coordinate into position one. Then $v_1 = 1$ and $|v_j| \leq 1$. The column

$$d_j = \tfrac{1}{2}[1 + \gamma\,\mathrm{Re}(e^{i\varphi} v_j)]$$

lies in $[0,1]^M$. Row normalization and $Q^n v = z^n v$ give $e_1^{\mathsf{T}} Q^n d = b_n$ for every n. The same M labels suffice: there is no preliminary random initialization or extra anchor. Unreachable labels can be pruned. Finiteness for $\alpha < 1$ follows from the regular-polygon construction proved below, or from the inherited polygon upper in Theorem S28.6.

At $|z| = 1$, equality in the triangle inequality at a largest-modulus eigenvector coordinate forces each positive successor to have the same modulus and value z times the current value. Following positive successors reaches a directed cycle of length $l \leq M$, so $z^l = 1$. An irrational phase is therefore impossible. If z has order q, every such cycle has length divisible by q, giving $M \geq q$; a q-cycle realizes it. Finally the squared embedding forces the whole response. Compact attainment at fixed M was proved in Proposition S24.8; equality of its risk with 2 would give the excluded exact response. The existing packing lemma gives the terminal counts. □

The nondegeneracy assumptions cannot be suppressed. Zero contrast gives one constant label. At $\alpha = 0$, $b_0 = (1 + \gamma\cos\varphi)/2$ and $b_n = 1/2$ for $n \geq 1$; one label suffices if $\gamma\cos\varphi = 0$, and otherwise two are necessary and sufficient. Real phases and constant cases are classified in the supplement. In particular a vanishing oscillation amplitude is not the limit of an exact-memory lower valid for nonzero amplitude.

For algebraic $x = \operatorname{Re} z$, $y = \operatorname{Im} z \neq 0$, spectral membership has the compact polynomial certificate

$$\begin{gathered} Q \geq 0,\ Q\mathbf{1} = \mathbf{1}, \quad Qa = xa - yc, \quad Qc = ya + xc, \\ a_1 = 1,\ c_1 = 0, \quad a_j^2 + c_j^2 \leq 1 \quad (1 \leq j \leq M). \end{gathered} \tag{S20.3}$$

It has $M^2 + 2M$ real variables and polynomial degree at most two. Normalization of an eigenvector proves equivalence in both directions. Real-algebraic elimination decides it at each fixed M, and the finite polygon upper terminates a minimal-size search for $|z| < 1$. This improves the form of the certificate, not the general complexity of real-algebraic elimination. A chosen algebraic preparation and contrast then yield algebraic decoder entries. State labels, encodings of those entries, exact-sampling resources and runtime remain separate accounts.

S20.2. **The sector bound, its equality case, and exact finite phases.** We include the classical stochastic sector argument to expose every load-bearing inequality. The equality structure is consistent with the Type-0 classification of Kirkland and Šmigoc [53]; it is not a new classification of stochastic spectral arcs.

Lemma S20.2 (Stochastic sector inequality and rigidity)**.** *If $z = x + iy$, $y \neq 0$, is an eigenvalue of an $M \times M$ stochastic matrix, then $M \geq 3$ and*

$$|y| \leq (1 - x)\cot(\pi/M). \tag{S20.4}$$

If equality holds, then, up to simultaneous relabelling and reversing the orientation when needed, that matrix is

$$Q = (1 - t)I + tC_M, \qquad 0 < t \leq 1, \tag{S20.5}$$

where C_M is the directed M-cycle. Its eigenvalue on the equality ray is $1 - t + te^{2\pi i/M}$, or its conjugate.

Proof. Conjugate the eigenvector when necessary so $y > 0$. Let P be the convex hull of its complex coordinates. The matrix $e^{s(Q-I)} = e^{-s}\sum_{n\geq 0} s^n Q^n/n!$ is stochastic for $s \geq 0$, so multiplication by $e^{s(z-1)}$ maps P into itself. Since z is nonreal, $1 - x > 0$, and the spiral of any nonzero coordinate visits every angle, a finite set of points of this spiral surrounds zero. Hence zero lies in the interior of P.

List its $l \leq M$ vertices counterclockwise as $r_j e^{i\theta_j}$, with $r_j > 0$. Their consecutive angular gaps Δ_j belong to $(0, \pi)$ and sum to 2π. Put $u = 1 - x > 0$ and $v = y > 0$. The initial velocity $(-u + iv)r_j e^{i\theta_j}$ must point into the half-plane to the left of the outgoing edge. Taking its signed two-dimensional cross product with that edge gives

$$r_j/r_{j+1} \leq \cos\Delta_j + (u/v)\sin\Delta_j.$$

All right sides are positive. Their product is at least one. Write $\phi = \arctan(u/v) \in (0, \pi/2)$. The function $\log(\cos\Delta + (u/v)\sin\Delta) = \log\sqrt{1 + (u/v)^2} + \log\cos(\Delta - \phi)$ is strictly concave on the interval of positive factors in $(0, \pi)$. Jensen's inequality therefore gives

$$1 \leq \prod_j [\cos\Delta_j + (u/v)\sin\Delta_j] \leq [\cos(2\pi/l) + (u/v)\sin(2\pi/l)]^l.$$

As $l \geq 3$, this implies $u/v \geq \tan(\pi/l) \geq \tan(\pi/M)$, proving (S20.4).

At equality, $l = M$, every gap is $2\pi/M$, and all radius ratios equal one: the polygon is regular and every coordinate is one distinct vertex. The velocity at a vertex is along its outgoing edge. But $(z - 1)v_j = \sum_k Q_{jk}(v_k - v_j)$ is a nonnegative combination of chords. Strict convexity of this polygon

makes all chords other than the self-chord and outgoing edge point strictly to the interior side of that edge. Their coefficients must therefore vanish. The remaining coefficient $t = Q_{j,j+1}$ satisfies $z - 1 = t(e^{2\pi i/M} - 1)$ and is independent of j. Row normalization gives (S20.5) and $t \le 1$. □

Theorem S20.3 (Exact first-sector memory threshold and optimizer)**.** *For $M \ge 3$ and $0 < \vartheta \le 2\pi/M$, a response realization with at most M live labels exists exactly when*

$$0 < \alpha \le r_M(\vartheta) := \frac{1}{\cos\vartheta + \sin\vartheta \tan(\pi/M)}. \tag{S20.6}$$

The threshold is attained. At $\alpha = r_M(\vartheta)$ the least memory is exactly M, the transition matrix is the lazy directed cycle in (S20.5)*, and, in cyclic coordinates with prescribed initial label zero, its decoder is uniquely*

$$d_j = \tfrac{1}{2}[1 + \gamma\cos(\varphi + 2\pi j/M)].$$

This is global over arbitrary stochastic transition and decoder rows, not an optimization restricted in advance to circulants.

Proof. The spectral criterion and (S20.4) prove necessity. For sufficiency put $\omega = e^{2\pi i/M}$ and $J = \mathbf{1}\mathbf{1}^{\mathsf{T}}/M$. The nonnegative coefficients

$$b = \frac{\alpha\sin\vartheta}{\sin(2\pi/M)}, \qquad a = \alpha\cos\vartheta - b\cos(2\pi/M), \qquad c = 1 - a - b$$

sum to one under (S20.6). The inequalities $a, b \ge 0$ follow from $0 < \vartheta \le 2\pi/M$; $c \ge 0$ is exactly the stated threshold. Then $Q = aI + bC_M + cJ$ has eigenvector $(1, \omega, \dots, \omega^{M-1})$ at eigenvalue $\alpha e^{i\vartheta}$ and the displayed decoder realizes the response.

At the threshold, strict monotonicity of $\cot(\pi/M)$ excludes every smaller size. Lemma S20.2 fixes Q. Here $b = t > 0$, so $C_M = (Q - (1-t)I)/t$. The rows $e_0^{\mathsf{T}}Q^n$, $0 \le n < M$, span all coordinate rows, because the rows $e_0^{\mathsf{T}}C_M^j$ do and each is a polynomial combination of the former. Thus all response coefficients determine the decoder uniquely. This proves the rigidity assertion including the endpoint $t = 1$. □

For the inherited algebraic angle $\cos\vartheta = 3/5$, $\sin\vartheta = 4/5$, let

$$a_j = \frac{5}{3 + 4\tan(\pi/j)}, \qquad j = 3, 4, 5, 6.$$

One has $0 < \vartheta < 2\pi/6$. Consequently exact memory is three on $(0, a_3]$, four on $(a_3, a_4]$, five on $(a_4, a_5]$, and six on $(a_5, a_6]$. In particular $a_4 = 5/7$. Above a_6 at least seven labels are needed; no assertion that a regular polygon is always globally optimal beyond the first sector is made. The old three-label threshold is recovered rather than reproved as a new independent result.

S20.3. **Exact worst-phase size and arithmetic improvements.** For an integer $q \ge 3$ define the distance to the nearest qth root angle by

$$\delta_q(\vartheta) = \min_{j\in\mathbb{Z}} |\vartheta - 2\pi j/q| \quad \text{modulo } 2\pi, \qquad 0 \le \delta_q \le \pi/q,$$

and define the radial boundary of the regular polygon by

$$R_q(\vartheta) = \frac{\cos(\pi/q)}{\cos(\pi/q - \delta_q(\vartheta))}. \tag{S20.7}$$

Lemma S20.4 (Directional polygon construction and power obstruction)**.** *If $0 < \alpha \le R_q(\vartheta)$, a q-label exact response realization exists. Conversely, any M-label realization satisfies, for every positive integer k,*

$$\alpha^k\big[\cos(k\vartheta) + |\sin(k\vartheta)|\tan(\pi/M)\big] \le 1. \tag{S20.8}$$

For $h = \pi/q$ and $0 \le \delta = \delta_q \le h$,

$$1 - R_q(\vartheta) \le 2h\delta. \tag{S20.9}$$

Proof. The intersection of the ray at angle ϑ with the chord joining its two adjacent unit roots has radius (S20.7). Thus $\alpha e^{i\vartheta}$ is in their triangle with zero. Write it as $a\omega^j + b\omega^{j+1} + c0$, with $a, b, c \geq 0$ and sum one. The stochastic matrix $aC_q^j + bC_q^{j+1} + cJ$ has that eigenvalue, giving the first assertion by Theorem S20.1. The matrix Q^k of any realization is still stochastic and has eigenvalue $\alpha^k e^{ik\vartheta}$. Apply the sector lemma; for a real eigenvalue the displayed inequality is automatic. Finally $\cos(h-\delta) \geq \cos h \geq 1/2$ and $\cos(h-\delta) - \cos h = 2\sin(h-\delta/2)\sin(\delta/2) \leq h\delta$. Division proves (S20.9). □

Theorem S20.5 (Exact worst-phase memory)**.** *For every* $0 < \alpha < 1$*,*

$$\max_{\sin\vartheta\neq 0} m(\alpha,\vartheta) = \max\left\{3, \left\lceil \frac{\pi}{\arccos\alpha} \right\rceil\right\}. \tag{S20.10}$$

In particular the worst-phase exact label count is asymptotic to $\pi/\sqrt{2(1-\alpha)}$ *as* $\alpha \uparrow 1$*. The phase may depend on* α*; this is not a lower rate for each fixed phase.*

Proof. Let the displayed integer be K. The regular K-gon contains the disk of radius $\cos(\pi/K) \geq \alpha$, so the polygon lemma gives the upper simultaneously for all phases. If $K = 3$, three distinct nonzero modes force at least three states. Otherwise put $M = K - 1 \geq 3$ and choose $\vartheta = \pi/M$. The definition of K gives $\alpha > \cos(\pi/M)$; the exact first-sector threshold at this angle is $\cos(\pi/M)$. No M-label realization exists, proving the lower and attainment of the maximum. The asymptotic follows from $1 - \cos t \sim t^2/2$. □

A number x is called badly approximable here with constant $c > 0$ when

$$\|kx\|_{\mathbb{R}/\mathbb{Z}} \geq c/k \qquad (k \geq 1). \tag{S20.11}$$

The constant is part of the hypothesis, not something inferred from a finite list of continued-fraction coefficients. The next theorem holds for every such phase and proves a different, matching fixed-phase rate.

Theorem S20.6 (Sharp cubic-root memory at a fixed Diophantine phase)**.** *Let* $x = \vartheta/(2\pi)$ *satisfy* (S20.11)*. Then, for* $0 < \alpha < 1$ *and* $\epsilon = 1 - \alpha$*,*

$$\left(\frac{\pi^2 c}{16\epsilon}\right)^{1/3} \leq m(\alpha,\vartheta) \leq \max\left\{4, \left\lceil \frac{3}{c} \right\rceil, \left\lceil \left(\frac{4\pi^2}{c^2\epsilon}\right)^{1/3} \right\rceil\right\}. \tag{S20.12}$$

Thus $m(\alpha,\vartheta) = \Theta_c((1-\alpha)^{-1/3})$*. The exact live-state bits are* $\frac{1}{3}\log_2(1/(1-\alpha)) + O_c(1)$ *in the ideal finite-label model. These are exact stochastic realizations at the original least expected-cost budget, not approximation of the response or additional physical observations. The statement does not supply matching program-bit or exact-sampling complexity bounds.*

Proof. We use only the following elementary approximation fact. For any irrational x and integer $Q \geq 1$, among the $Q+1$ fractional parts $0, x, \ldots, Qx$ two fall in the same one of Q equal intervals. Subtracting them gives $1 \leq k \leq Q$ with $0 < \|kx\| \leq 1/Q$.

For the lower, take a realizing size $M \geq 3$ and apply that fact with $Q = 2M$. Set $\delta = 2\pi\|kx\|$ and $h = \pi/M$. Then

$$\pi c/M \leq \delta \leq h \leq \pi/3.$$

Writing $F = \cos\delta + \sin\delta\tan h$, we have $F \leq \sec h \leq 2$. Also $\sin\delta \geq 3\delta/4$ on $[0, \pi/3]$, $\tan h \geq h$ and $1 - \cos\delta \leq \delta^2/2$, so

$$F - 1 \geq 3h\delta/4 - \delta^2/2 \geq h\delta/4.$$

By (S20.8), $\alpha^k F \leq 1$. Hence

$$2M\epsilon \geq k\epsilon \geq 1 - \alpha^k \geq 1 - 1/F \geq h\delta/8 \geq \pi^2 c/(8M^2).$$

This proves the lower constant.

For the upper, let Q be the integer on the right of (S20.12). The approximation fact provides integers $1 \le q \le Q$ and a with $|qx - a| \le 1/Q$. The lower bound (S20.11) forces $q \ge cQ \ge 3$. The distance from ϑ to a qth-root angle is at most $2\pi/(qQ)$. The polygon-error bound gives

$$1 - R_q(\vartheta) \le \frac{4\pi^2}{q^2 Q} \le \frac{4\pi^2}{c^2 Q^3} \le \epsilon.$$

Thus $\alpha \le R_q(\vartheta)$ and the explicit circulant construction realizes the response with $q \le Q$ labels. Both bounds apply for the same fixed phase as $\epsilon \downarrow 0$. Taking base-two logarithms and rounding the label bits proves the last assertion. □

The hypothesis is nonempty and has elementary physical examples. For $x = (3-\sqrt{5})/2$, choose the closest integer a to kx. The integer $a^2 - 3ak + k^2$ is nonzero. Since the other root is $x' = (3+\sqrt{5})/2$ and $|a/k - x'| \le \sqrt{5} + 1/2 < 3$, factorization gives $|a/k - x| > 1/(3k^2)$. Thus (S20.11) holds with $c = 1/3$. This provides a fixed qubit rotation with the matched exponent. Its trigonometric primitive entries are not asserted to be algebraic. In particular the Diophantine condition has not been proved here for the different algebraic rotation $(3 + 4i)/5$.

Corollary S20.7 (A stronger fixed-algebraic-angle exclusion)**.** *For the inherited angle $e^{i\vartheta} = (3+4i)/5$ and $0 < \alpha < 1$, every exact M-label stochastic realization satisfies*

$$1 - \alpha \ge \frac{3}{16M^2\, 25^M}. \tag{S20.13}$$

Consequently its exact memory is at least of order $\log(1/(1-\alpha))$ as $\alpha \uparrow 1$. This improves the earlier $\log / \log\log$ lower, without assigning the cubic-root law to this angle or claiming a matching upper for it.

Proof. Again choose $1 \le k \le 2M$ with $\delta = 2\pi\|k\vartheta/(2\pi)\| \le \pi/M$. The Gaussian integer $(3+4i)^k - 5^k$ is nonzero: otherwise the unit complex number would be a root of unity, forcing $6/5$ to be a rational algebraic integer. Its modulus is at least one. Therefore $\delta \ge |1 - e^{ik\vartheta}| \ge 5^{-k} \ge 25^{-M}$. Repeat the lower proof of Theorem S20.6 up to $2M(1-\alpha) \ge h\delta/8$, with $h = \pi/M$. This gives the stronger constant $\pi/(16M^2 25^M)$ and hence (S20.13). Taking logarithms gives $\log(1/(1-\alpha)) \le \log(16/3) + 2\log M + M\log 25$, proving the order assertion. □

S20.4. **Finite spectral mixtures and higher-dimensional protected systems.** The matching exponent is not confined to a two-dimensional physical carrier. Let $r \ge 1$, let $\vartheta_1, \dots, \vartheta_r \in (0, \pi)$ be distinct, and let $w_j > 0$ with $\sum_j w_j \le 1$. For fixed phases φ_j, consider

$$b_n^{\mathrm{mix}} = \tfrac{1}{2}\left[1 + \alpha^n \sum_{j=1}^{r} w_j \cos(n\vartheta_j + \varphi_j)\right]. \tag{S20.14}$$

Write $m_{\mathrm{mix}}(\alpha)$ for its least pure-start stochastic memory.

Theorem S20.8 (Spectral mixtures preserve the matched arithmetic exponent)**.** *For $0 < \alpha < 1$, the exact memory of* (S20.14) *satisfies*

$$\max\{2r+1, \max_j m(\alpha, \vartheta_j)\} \le m_{\mathrm{mix}}(\alpha) \le 1 + \mathbf{1}_{\{\sum_j w_j < 1\}} + \sum_{j=1}^{r} m(\alpha, \vartheta_j). \tag{S20.15}$$

All models are realized with the counted prescribed initial label. If every fixed $\vartheta_j/(2\pi)$ has a global badly-approximable certificate, then $m_{\mathrm{mix}}(\alpha) = \Theta((1-\alpha)^{-1/3})$, with constants depending on the finite phase list. The same conclusion holds when some phases are rational and at least one is certified badly approximable. This probability response is physically prepared on dimension at most $2r+1$, and its exact memory is the boundary-optimal observer memory of the same squared task after the standard two-dimensional herald-flag embedding.

Proof. The $2r+1$ modes $1, \alpha e^{\pm i\vartheta_j}$ are pairwise distinct and have nonzero coefficients. Their Hankel rank is $2r+1$ by Vandermonde independence. Every realizing characteristic polynomial vanishes at every nonconstant mode, giving both lower bounds through Theorem S20.1.

For each component choose a minimal stochastic realization (Q_j, d_j) starting in its first label. Keep their state sets disjoint and add one initial label o with decoder b_0^{mix}. On its first failure, its row enters block j with subprobability row $w_j e_1^{\mathsf{T}} Q_j$. If $w_0 = 1 - \sum_j w_j > 0$, its remaining mass enters one absorbing label with decoder $1/2$. Within each block use Q_j. Thus after $n \geq 1$ failures the output probability is $w_0/2 + \sum_j w_j e_1^{\mathsf{T}} Q_j^n d_j = b_n^{\mathrm{mix}}$, while the initial decoder handles $n = 0$. No physical call is added and no persistent choice is stored outside the counted labels. This proves the upper. Theorem S20.6 supplies matched exponents for the finite certified phase list. For a rational phase of order q, the matrix $\alpha C_q + (1-\alpha)J$ gives a uniform q-label upper, so adding those components does not change the exponent.

For the physical realization, take a direct sum of r qubits, prepare the phase-rotated pure state in block j with weight w_j, and use the damped rotation channel in that block. Drop off-diagonal block coherences. If $w_0 > 0$, add one scalar block of mass w_0 with effect $1/2$ and identity evolution. Each diagonal branch is completely positive and their direct sum is trace preserving. The terminal effect is the direct sum of the qubit Z effects and the scalar half effect. Its probability is exactly (S20.14). Theorem S25.1 therefore gives the same fixed squared loss and cost $1/p$, and forces this response at the optimum. The observer's independently randomized block selection does not reveal or copy the protected block. □

The three memory statements have different quantifiers: the exact worst-phase law permits the phase to vary with the noise, the cubic-root law fixes any phase with the displayed arithmetic certificate, and the last corollary fixes the older algebraic rotation without such a certificate. Noise-dependent classical memory bounds for quantum processes were already established by Fanizza, Lumbreras and Winter [50]; their worst-phase process bound is not a fixed-phase matched law for this response task. The stochastic regions and their Type-0 rigidity are also classical. The present proofs make the task-level reduction and the new arithmetic comparison explicit, rather than claiming earliest priority from a literature search. No statement in this section solves the different squared-sign prediction frontier or the phase-type timing frontier.

S21. Transient output faces and leakage-sensitive memory

Uniform decay of a physical state and decay of its observable probabilities do not measure the same obstruction to exact stochastic realization. In this section leakage removes population from a protected subspace, whereas internal depolarization changes the spectrum relative to that surviving population. We prove that the first operation costs exactly one additional live label in the nondegenerate single-frequency family, independently of how rapid the leakage is. We then obtain finite-model witnesses on a fixed six-dimensional squared-prediction experiment, and an exact finite-realization result when a positive background is added to the terminal effect.

Nonnegative realizations, dominant-pole obstructions and transient-state reductions have classical antecedents [54, 56, 57, 47]. We do not claim to originate these methods. The statements below specify the constrained stochastic decoder, prescribed initial label, sharp label increment, physical leakage channel and finite squared-task tests. All necessary positive-matrix arguments are included.

S21.1. **The output-visible transient core.** For a nonzero sequence $u = (u_n)_{n\geq 0}$ in $[0,1]$ tending to zero, let $s(u)$ be its least stochastic realization size:

$$u_n = e_1^{\mathsf{T}} Q^n d, \qquad Q \geq 0, \quad Q\mathbf{1} = \mathbf{1}, \quad 0 \leq d \leq \mathbf{1}.$$

The initial label is fixed, not a freely randomized initial distribution. Define $t(u)$ by the same formula with a nonnegative row-substochastic matrix S, $S\mathbf{1} \leq \mathbf{1}$, whose spectral radius is strictly below one. Both minima are infinite if no finite realization exists.

Lemma S21.1 (Exact transient-core reduction)**.** *For every such sequence,*

$$s(u) = t(u) + 1. \tag{S21.1}$$

More precisely, any M-label stochastic realization has a nonempty set of at most $M-1$ reachable labels that can eventually give positive output. Its restriction S realizes u, is transient and uses the same initial label. If also $u_n \le C\lambda^n$ for $0 < \lambda < 1$, then the spectral radius of this restriction is at most λ.

Proof. Remove labels not reachable from label one. Call a remaining label active if a positive-probability path from it reaches some label j with $d_j > 0$. The initial label is active because u is nonzero: there is some n with $e_1^{\mathsf{T}} Q^n d > 0$, so a positive-weight path from label one reaches a label with positive decoder output. This remains true when $u_0 = 0$. A path that leaves the active set cannot return: a return would make every intervening reachable label active. Thus, with S the active principal submatrix and d_A its decoder, $u_n = e_1^{\mathsf{T}} S^n d_A$.

For each active i, choose a path of length a_i from the start to i of weight $p_i > 0$. For each active j, choose a path of length b_j from j with positive terminal expected output $r_j > 0$. Positivity gives, for every n,

$$p_i r_j (S^n)_{ij} \le u_{a_i+n+b_j}. \tag{S21.2}$$

Consequently every entry of S^n tends to zero. In finite dimension this forces every eigenvalue to have modulus less than one. In the exponential case it gives $(S^n)_{ij} \le C_{ij}\lambda^n$, and applying this bound to an eigenvector proves the stronger spectral-radius bound. There must be a reachable inactive label: otherwise S would be stochastic and have eigenvalue one. Hence the active size is at most $M-1$, proving $t(u)+1 \le s(u)$.

Conversely a k-label transient realization extends to the stochastic matrix and decoder

$$\widehat{Q} = \begin{pmatrix} S & \mathbf{1} - S\mathbf{1} \\ 0 & 1 \end{pmatrix}, \qquad \widehat{d} = \begin{pmatrix} d_A \\ 0 \end{pmatrix}. \tag{S21.3}$$

They realize the same sequence from the same initial label with $k+1$ labels. This proves the reverse inequality, including the infinite case. The added label is a live zero-output label until the actual herald; it is not permission to stop at an unobserved physical leakage event. □

We recall a small positive-matrix fact with its proof. If $B \ge 0$ is irreducible and has a nonzero eigenvalue, it has a vector $h > 0$ with $Bh = Rh$, where R is its spectral radius. Indeed, for $B + \epsilon \mathbf{1}\mathbf{1}^{\mathsf{T}}$ the map sending a simplex vector to its normalized image has a fixed point by the finite-dimensional simplex fixed-point theorem. The fixed point is strictly positive. Along $\epsilon \downarrow 0$ a convergent subsequence gives $Bh = Rh$, $h \ge 0$, $\sum_i h_i = 1$. If a coordinate vanished, the eigenvector equation and irreducibility would propagate that zero to every coordinate. Thus $h > 0$, and $R > 0$. The similarity $R^{-1}\,\mathrm{diag}(h)^{-1} B\,\mathrm{diag}(h)$ is stochastic, so its maximum-row-sum bound shows that R is indeed the spectral radius. This is the Perron normalization used below, not a new Perron theorem.

Proposition S21.2 (A relative spectral obstruction for decaying outputs)**.** *Suppose an M-label stochastic response satisfies $0 \le u_n \le C\lambda^n$ with $0 < \lambda < 1$. If its rational generating function has a nonreal pole at ζ^{-1}, then*

$$\zeta/\lambda \in \mathcal{K}_{M-1}. \tag{S21.4}$$

Here $\mathcal{K}_j$ is the stochastic eigenvalue region defined in Section S20. It is the decay-normalized mode, not merely the absolute mode ζ, that constrains the active state count.

Proof. The core in Lemma S21.1 gives $\sum_n u_n z^n = e_1^{\mathsf{T}}(I - zS)^{-1} d_A$. Hence ζ is an eigenvalue of S. Order its strongly connected components to obtain a block triangular matrix. One irreducible diagonal block B has that eigenvalue. Its spectral radius R obeys $|\zeta| \le R \le \lambda$. Perron normalization gives a stochastic matrix T of size $l \le M-1$ with eigenvalue ζ/R and eigenvector v.

A stationary probability row π for T exists: take a convergent subsequence of the averages of any probability row and its first $n-1$ iterates. Since $\zeta/R \neq 1$, one has $\pi v = 0$. Put $a = R/\lambda \in (0,1]$. The stochastic matrix $T' = aT + (1-a)\mathbf{1}\pi$ satisfies $T'v = (\zeta/\lambda)v$. Padding by unreachable self-loop labels, if needed, proves membership in $\mathcal{K}_{M-1}$. No global Perron scaling of the reducible core has been assumed. □

The last qualification matters. A reducible matrix such as $S = \left(\begin{smallmatrix} 1/2 & 1/4 \\ 0 & 1/4 \end{smallmatrix}\right)$ has spectral radius $1/2$, but no strictly positive h satisfies $Sh \leq h/2$. The proof normalizes the irreducible block carrying the observed pole, rather than silently imposing that false global inequality.

S21.2. **An exact leakage law for every damping and phase.** Fix $0 < \lambda < 1$, $0 < \alpha \leq 1$, $0 < \gamma \leq 1$, and $\sin\vartheta \neq 0$. Let

$$u_n = \frac{\lambda^n}{2}[1 + \gamma\alpha^n \cos(n\vartheta + \varphi)]. \tag{S21.5}$$

The scalar λ is survival against leakage; α is coherence retention within the surviving qubit. These parameters are known properties of the prepared target channel, not extra observations available to the observer.

Theorem S21.3 (Exact leakage increment)**.** *For the nondegenerate family* (S21.5)*,*

$$s(u) = 1 + m(\alpha, \vartheta) = 1 + \min\{k : \alpha e^{i\vartheta} \in \mathcal{K}_k\}. \tag{S21.6}$$

This holds over all stochastic matrices and randomized decoder rows, with a prescribed initial label. In particular the value is independent of every $0 < \lambda < 1$, every nonzero contrast and every preparation phase. For $\alpha < 1$ it is finite. At $\alpha = 1$ it is $h+1$ if $e^{i\vartheta}$ has finite order h, and it is infinite otherwise.

The full two-output Hankel response has rank four, regardless of this possibly much larger or infinite exact memory. In the fixed squared-task embedding it is the least live memory at the original cost $1/p$, with event-stopped and state-stopped total-label upper counts $\max\{s(u), 2\}$ and $s(u) + 2$ when finite.

Proof. The three modes λ, $\lambda\alpha e^{i\vartheta}$ and $\lambda\alpha e^{-i\vartheta}$ in u are distinct and have nonzero coefficients because $\lambda > 0$, $\alpha > 0$, $\gamma > 0$ and $\sin\vartheta \neq 0$. The phase factor $e^{i\varphi}$ never vanishes, including at initial-output zeros. Thus its generating function has a pole at $(\lambda\alpha e^{i\vartheta})^{-1}$. Proposition S21.2 makes every finite realizing size M satisfy $\alpha e^{i\vartheta} \in \mathcal{K}_{M-1}$. Theorem S20.1 proves the required lower and the infinite case.

For the upper take a minimal realization (Q, d) of the unattenuated response $b_n = [1+\gamma\alpha^n \cos(n\vartheta + \varphi)]/2$. Replace its continuing block by λQ and append a live zero-output label, as in (S21.3). Its output is exactly $\lambda^n b_n$ from time zero. This proves equality without an extra initialization operation. The unit-circle alternatives follow from the already proved root-of-unity characterization.

The scalar response spans its three exponential modes. Its complement adds the independent constant mode one, because $\lambda < 1$. Vandermonde independence therefore gives full response rank four. Theorem S25.1 forces this entire probability response in the actual squared task; the role-packing statement is then inherited without identifying protected physical states with live observer labels. □

Corollary S21.4 (Noise laws depend on relative damping)**.** *For fixed $0 < \lambda < 1$ and $0 < \alpha < 1$, the worst-phase exact count is*

$$\max_{\sin\vartheta \neq 0} s(u) = 1 + \max\{3, \lceil \pi/\arccos\alpha \rceil\}.$$

For every fixed phase with the global badly-approximable certificate of Theorem S20.6, $s(u) - 1$ has the same matching order $(1-\alpha)^{-1/3}$. Neither rate depends on leakage λ, even when leakage is allowed to vary simultaneously with α. At the original algebraic phase, the first nontrivial threshold

is exactly four live labels for $0 < \alpha \le 5/(3+4\sqrt{3})$*, and at least five beyond it. No badly-approximable claim is made for that algebraic phase.*

Proof. Substitute (S21.6) into the exact worst-phase, fixed-phase and three-label results. All subtractions concern live labels, including the necessary zero-output label; no terminal label has been subtracted from an event-stopped dictionary. □

S21.3. **A strictly contractive quantum channel with infinite exact memory.** Let $K = \mathbb{C}^2 \oplus \mathbb{C}|\perp\rangle$, let P project onto the qubit and let $\Omega = |\perp\rangle\langle\perp|$. With the qubit channel Φ_α and preparation from Section S20, define

$$\Psi_{\lambda,\alpha}(X) = \lambda\Phi_\alpha(PXP) + [\mathrm{tr}((I-P)X) + (1-\lambda)\,\mathrm{tr}(PX)]\Omega. \tag{S21.7}$$

The qubit output is embedded in its original block and cross-block coherences are discarded. Prepare the qubit and audit the qubit effect $E = (I+\sigma_z)/2$, extended by zero at $\perp$.

Theorem S21.5 (Uniform physical mixing versus exact response memory)**.** *Equation* (S21.7) *is a completely positive trace-preserving channel with unique stationary state* Ω*. For all density matrices* ρ, σ *and all* $n \ge 0$*,*

$$\|\Psi_{\lambda,\alpha}(\rho) - \Psi_{\lambda,\alpha}(\sigma)\|_1 \le \lambda\|\rho-\sigma\|_1, \qquad \|\Psi^n_{\lambda,\alpha}(\rho) - \Omega\|_1 \le 2\lambda^n.$$

Its prepared audit response is exactly (S21.5)*, so its exact stochastic observer memory is* (S21.6)*. In particular, at* $\alpha = 1$ *and the algebraic rotation* $(3+4i)/5$*, every* $0 < \lambda < 1$ *gives infinite exact memory despite strict trace-norm contraction and exponential convergence to one stationary state. For algebraic* λ *and algebraic preparation data, all primitive superoperator entries are algebraic.*

The squared-prediction embedding has protected dimension six, fixed dictionary $\{-1, 1\}$*, three actual audit outcomes and least expected cost* $1/p$*. In the specified irrational, internally undamped case, every finite live-memory class has an attained strictly larger risk than the full-history value* 2*. A prefix with* T *exact-output labels and one live zero-output tail label has the same cost and excess risk at most*

$$\frac{2\varepsilon p(q\bar{t}\lambda)^T}{1-q\bar{t}\lambda}, \qquad q = 1-p, \quad \varepsilon = 1/4. \tag{S21.8}$$

Proof. Write $\Xi(X) = \Phi_\alpha(PXP) + \mathrm{tr}((I-P)X)\Omega$ and $\mathcal{E}(X) = \mathrm{tr}(X)\Omega$. Both are channels: compression, the declared qubit channel and scalar replacement are completely positive, and their output traces sum to the input trace. Then $\Psi_{\lambda,\alpha} = \lambda\Xi + (1-\lambda)\mathcal{E}$. On a traceless Hermitian difference the second term vanishes. Trace-norm contraction of a channel follows directly by applying positivity and trace preservation to the positive and negative parts of that difference. This proves the contraction inequality. Since Ω is fixed, iteration gives the displayed bound and uniqueness.

Starting in the qubit, its surviving block after n calls is $\lambda^n\Phi^n_\alpha(\rho)$; the complementary mass is at $\perp$. Taking the effect gives (S21.5). The preceding theorem proves the memory claim. At $(3+4i)/5$ the unit phase is not a root of unity, by the rational algebraic-integer argument already proved. The spectral obstruction is in the decaying output-visible block, not at the peripheral spectrum of the entire trace-preserving channel.

The embedding of Theorem S25.1 doubles the protected dimension and retains its real squared audit. Its unique optimal probability kernel must reproduce u. Compactness of each fixed-memory controller class and the uniform geometric word tail give attainment; equality at a finite size would contradict the exact-memory theorem. For the approximation, store $u_0, \dots, u_{T-1}$ and thereafter output probability zero. Since $u_n \le \lambda^n$, summing the exact risk identity (S25.4) yields (S21.8). Nothing in this construction observes leakage or changes first-herald stopping. □

This is not the earlier unique-sign counterexample under depolarization. Here the required response is probability valued and its absolute nonconstant eigenvalues all lie strictly inside the unit disk. The impossibility comes from their phases relative to the visible decay rate. The stationary

state is on a face, so the interior hypothesis of the instrument-wide realization theorem is genuinely unavailable. With $\gamma = 1$ and $\varphi = \pi$, moreover, $u_0 = 0$ and $0 \le u_n \le \lambda$ for all $n \ge 1$. The one-label zero response is uniformly within λ of this law while exact memory remains infinite for every positive λ. Exact realization is not a robust approximate-memory assertion.

S21.4. **Finitely many physical models detect the obstruction.** The continuum of herald-flag parameters used to force an arbitrary full response can be replaced, against a prescribed finite-memory bound, by a finite set whose cardinality depends only on that bound and the target's linear order.

Theorem S21.6 (Finite rational-resolvent tests and fixed-dimensional witnesses)**.** *Suppose a target channel response has generating function $U(z) = P(z)/D(z)$ with $\deg P \le r-1$, $\deg D \le r$, and $D(z) \ne 0$ on $[0, q\bar{t}]$. Fix $M \ge 1$ and any $M + r$ distinct parameters $t_j \in [0, \bar{t}]$. In the squared embedding, an observer with at most M live labels has risk exactly 2 on the finite family (t_j, σ), $\sigma \in \{-1, 1\}$, if and only if it realizes the entire target response at every failure count.*

Consequently for the algebraic leaking qubit with $\lambda = 1/2$, $\alpha = \gamma = 1$, $e^{i\vartheta} = (3+4i)/5$ and $\varphi = 0$, the following holds for every M: a six-dimensional prepared squared experiment with only $2(M+3)$ rationally specified models, one unit-cost command and the fixed two-value dictionary has boundary value 2, but every at-most-M-label observer has risk at least $2 + \delta_M$, where $\delta_M > 0$ is an attained, computable real algebraic number. One may take $p = 1/2$, $\bar{t} = 1/2$ and $t_j = j/[2(M+2)]$, $0 \le j \le M+2$. Neither the physical dimension nor the target response order increases with M.

Proof. Boundary feasibility still forces first-herald commitment in this finite model family, since its physical herald probability is always p. An M-label rule has $B(z) = e_1^{\mathsf{T}}(I - zQ)^{-1}d = P_Q(z)/D_Q(z)$, where $\deg P_Q \le M-1$, $\deg D_Q \le M$ and $D_Q(z) \ne 0$ for $0 \le z < 1$. Smaller matrices may be padded by unreachable rows.

The two signs and (S25.4) make risk 2 equivalent to $B(qt_j) = U(qt_j)$ at every chosen parameter. The polynomial $P_Q D - P D_Q$ has degree at most $M + r - 1$ and vanishes at $M + r$ distinct points. It is therefore zero. Equality of the power series about zero gives every response coefficient. The converse follows immediately from the same risk identity. This is rational interpolation with known degree, not extrapolation from arbitrary fitted samples.

For the stated target,

$$U(z) = \frac{1}{2(1-\lambda z)} + \frac{1-\lambda c z}{2(1 - 2\lambda c z + \lambda^2 z^2)}, \qquad c = 3/5,$$

so $r = 3$ suffices and all coefficients are rational. The exact leakage theorem excludes every finite realization. Hence the finite tests exclude equality for the specified M. The full-history target kernel still attains 2 in every selected model. Compact controller simplexes and the geometric tail imply the fixed-M minimum is attained, so its gap δ_M is strictly positive.

Finally introduce real variables Q, d and vectors v_j satisfying $(I - qt_j Q)v_j = d$. These matrices are invertible for every stochastic Q, and the achieved risk is $2 + 2\varepsilon p \max_j |(v_j)_1 - U(qt_j)|$. All defining constraints are polynomial over the rationals. Real-algebraic elimination computes the attained minimum and an algebraic minimizing controller; in particular the positive gap is algebraic and computable. This is an exact finite procedure, not a polynomial-time claim or a numerical certificate inferred from local optimization. The general procedure is proved here; a full elimination at every M has not been executed in the finite diagnostics. □

Thus finite linear rank four does not bound the classical memory forced by a finite family of actual squared-loss tests, even on a fixed protected dimension. The number of models in this conclusion grows with the tested memory bound. It is not a claim that one fixed finite model list excludes every finite observer.

S21.5. **Positive background regularizes the output face.** The leakage obstruction is sensitive to the face on which the terminal effect vanishes. Replace the effect by $E_\eta = \eta I/2 + (1-\eta)E$, $0 < \eta < 1$, on the same qutrit. The response becomes

$$u_n^{(\eta)} = \eta/2 + (1-\eta)u_n.$$

This changes a real terminal measurement, not the protected channel or the observer's information.

Theorem S21.7 (Exact finite realization after background regularization)**.** *For all parameters of Theorem S21.3 and every $0 < \eta < 1$, the response $u^{(\eta)}$ has a finite exact stochastic realization. If $T \geq 1$ satisfies*

$$(1-\eta)\lambda^T \leq \eta/2, \tag{S21.9}$$

then at most $T + 2 + m(\lambda\alpha, \vartheta)$ live labels suffice. The construction uses a prescribed initial label and attains the squared-task optimum at the same cost $1/p$.

For fixed $\lambda \in (0,1)$, $\alpha = 1$ and irrational phase, the least exact size tends to infinity as $\eta \downarrow 0$, while the displayed upper is $O_\lambda(\log(1/\eta))$. No matching lower rate in η is asserted. For algebraic nondegenerate inputs this finite upper and the inherited fixed-size response test give a terminating search for a minimal exact controller.

Proof. Put $a = (1-\eta)\lambda^T/2$ and $\psi = T\vartheta + \varphi$. The suffix starting at T is the average of the two probability responses

$$f_k = \eta/2 + 2a\lambda^k, \qquad g_k = \eta/2 + 2a\gamma\alpha^T(\lambda\alpha)^k\cos(k\vartheta + \psi).$$

The first has a two-label transient/absorbing realization with decoder values $\eta/2 + 2a$ and $\eta/2$. For the second, use the normalized eigenvector construction of Theorem S20.1 at eigenvalue $\lambda\alpha e^{i\vartheta}$, with decoder $\eta/2 + 2a\gamma\alpha^T \operatorname{Re}(e^{i\psi}v_j)$. Condition (S21.9) gives $2a \leq \eta/2$, so all these decoder values lie in $[0, \eta] \subset [0,1]$.

Store a deterministic chain for $u_0^{(\eta)}, \ldots, u_{T-1}^{(\eta)}$. At its final failure transition enter the two disjoint suffix machines with weights $1/2, 1/2$. No additional initialization label is needed: the last prefix row makes and stores the choice within the counted blocks. At every later failure count the output is $(f_k + g_k)/2 = u_{T+k}^{(\eta)}$. This proves the exact count and preserves the physical stopping rule.

If exact sizes stayed bounded along $\eta \downarrow 0$, pad their stochastic matrices and decoders to one size and take a convergent subsequence in the compact row simplexes. Every finite coefficient is continuous in those variables. The limit would realize u at $\eta = 0$, contradicting Theorem S21.3. The logarithmic upper follows by solving (S21.9) with fixed λ. For algebraic input, the target has a supplied linear representation of dimension at most four. The finite word-equality criterion of Proposition S28.7, together with this size upper, terminates a minimal exact-size search. Algebraic controller encodings and exact sampling resources remain distinct from label count. □

S22. The finite Hankel determinant used by the new hierarchy

For completeness we verify the nonsingularity used by Theorem S25.2 without appealing to a positive-realization theorem. Put $h = k - 1$ and $u_n = a^n P(n)$, where P has degree $h - 1$ and nonzero leading coefficient, and $0 < a < 1$. The h polynomials $P(n), P(n+1), \ldots, P(n+h-1)$ are linearly independent. Indeed their successive forward differences have strictly decreasing degrees $h-1, h-2, \ldots, 0$ and nonzero leading coefficients; the change from translates to differences is triangular and invertible. Thus the columns $u_i, u_{i+1}, \ldots, u_{i+h-1}$ span the same evaluation space as $a^i, ia^i, \ldots, i^{h-1}a^i$.

On rows $i = 0, \ldots, h$, the matrix with columns $1, a^i, ia^i, \ldots, i^{h-1}a^i$ is nonsingular. To see this directly, begin with the Vandermonde matrix on distinct nodes $1, x_1, \ldots, x_h$, whose columns are their powers on these rows. Replace the last h columns by successive divided differences and let all x_j tend to a. The limiting columns are $\binom{i}{j}a^{i-j}$ for $j = 0, \ldots, h-1$, with determinant $(a-1)^h$

times the nonzero within-group confluent factor (equal to one for these divided-difference columns). Multiplication by nonzero powers of a and an invertible triangular change from falling powers of i to ordinary powers gives the claimed matrix. Equivalently its determinant is, up to the fixed ordering sign, $(a-1)^h a^{h(h-1)/2}\prod_{j=0}^{h-1} j!$, which is nonzero. The distinct-node Vandermonde formula itself follows by divisibility by every pairwise difference and comparison of its leading monomial.

For the Erlang survival sequence in the main theorem, $P(n)=\sum_{j=0}^{h-1}\binom{n}{j}((1-a)/a)^j$, whose leading coefficient is $((1-a)/a)^{h-1}/(h-1)!\neq 0$. Appending the constant column to its h translates therefore gives the nonsingular matrix A_u used in the finite physical witness. This also proves the exact full two-output Hankel rank: the displayed k columns give the lower, while the k-state stochastic chain gives the upper. No assertion that finite rank alone implies positive stochastic realization is involved.

Independent current developments

The following theorem bodies are retained in full, with their individual hypotheses. The new core does not depend on all of them. Cross-references distinguish the new main article from these independent complementary consequences; the source preservation map additionally points to the unchanged predecessor packet.

S23. Finite realization of uniformly interior instruments

The boundary decision theorems identify the observer memory needed for an optimal task. We now consider a distinct, stronger form of physical-model reduction: one finite hidden classical model must preserve every controlled report word, every stopping rule and every terminal measurement of a declared instrument family. Its hidden labels are not made available to the observer. The distinction is essential: a finite hidden-state realization does not in general make the posterior, or an optimal observer, finite-state.

Invariant positive cones are classical realization tools [48, 47]. Our purpose here is not another abstract cone criterion. We derive a finite invariant set from a quantitative condition on every actual report branch, give its dimension dependence and an explicit sparse implementation, and prove exact stopped-law and cost transfer. The commands need not commute, need not reset the input, and need not share a stationary state. The condition is sufficient, not necessary for every finite hidden realization. A matching noise exponent below concerns the expressly defined state-positive class, not every possible classical response representation.

S23.1. **A finite hull containing all uniformly mixed states.** Let $\mathcal{D}_d$ be the density matrices on $\mathbb{C}^d$, $d\geq 2$, and put

$$\mathcal{D}_d(\eta)=\{\rho\in\mathcal{D}_d:\rho\succeq \eta I/d\}=(1-\eta)\mathcal{D}_d+\eta I/d,\qquad 0<\eta\leq 1.$$

For unit vectors modulo phase use the metric

$$\operatorname{dist}([u],[v])=\sqrt{1-|\langle u,v\rangle|^2}=\tfrac{1}{2}\|uu^*-vv^*\|_1=\|uu^*-vv^*\|_{\mathrm{op}}.$$

In particular it satisfies the triangle inequality. All convex hulls below are hulls of actual positive density matrices, not signed coordinates.

Lemma S23.1 (An explicit interior hull)**.** *There are pure states $P_1,\dots,P_K$ such that*

$$\mathcal{D}_d(\eta)\subseteq \operatorname{conv}\{P_1,\dots,P_K\}\subseteq\mathcal{D}_d,\qquad K\leq\left\lceil (4d/\eta)^{d-1}\right\rceil. \tag{S23.1}$$

Any additional k prescribed initial states may be included as k further generators. The generators are shared by all maps satisfying the same interior margin.

Proof. For the unitarily invariant probability measure on complex projective space, a ball of radius $r \in [0,1]$ has mass r^{2d-2}. To verify the formula, normalize d independent standard complex Gaussian coordinates. Their squared moduli before normalization are independent exponential variables. Changing variables to their sum and normalized proportions gives constant density on the probability simplex. Thus $|u_1|^2$ has density $(d-1)(1-x)^{d-2}$ on $[0,1]$, and the probability of $|u_1|^2 > 1-r^2$ is r^{2d-2}. This also proves the formula at the endpoints by continuity.

Set $\varepsilon = \sqrt{\eta/d}$. A maximal set of rays with pairwise distances greater than ε is an ε-net. The radius-$\varepsilon/2$ open balls about its points are disjoint. Their equal positive masses bound the cardinality by $(4/\varepsilon^2)^{d-1}$. Such a finite maximal set exists by successively adding uncovered points; the same mass bound forces this process to stop.

Let A be any Hermitian matrix, with greatest and least eigenvalues a, b. Choose a net vector v within ε of a greatest-eigenvalue vector. Expanding in that eigenbasis gives

$$\operatorname{tr}(Avv^*) \geq a - (a-b)\varepsilon^2.$$

On the other hand,

$$\max_{\rho \in \mathcal{D}_d(\eta)} \operatorname{tr}(A\rho) = a - \eta(a - \operatorname{tr} A/d) \leq a - \eta(a-b)/d.$$

The net hull therefore has support function at least that of $\mathcal{D}_d(\eta)$ in every Hermitian direction. Finite-dimensional separation proves containment. The other inclusion and the additional-generator assertion follow from positivity and convexity. □

S23.2. **Report-resolved positive interpolation.** There is a finite set of globally legal commands $\mathcal{A}$ and, for each command, a finite report set Y_a. Write $\mathcal{I}_{a,y}$ for its completely positive report map and assume $\sum_y \mathcal{I}_{a,y}$ is trace preserving. A finite list $\rho^{(1)}, \ldots, \rho^{(k)}$ of allowed prepared states is specified. The initial preparation index is fixed before execution; it is not an unknown state revealed to the observer. Assume the following unnormalized inequality for every command, report and positive input:

$$\mathcal{I}_{a,y}(X) \succeq \frac{\eta}{d} \operatorname{tr}(\mathcal{I}_{a,y}(X)) I, \qquad X \succeq 0. \tag{S23.2}$$

A zero branch is allowed. No probability is divided out at zero. The hypothesis is on each recorded branch, not merely on the sum of the maps. It is satisfied when a depolarizing channel with mixing weight at least η is physically applied after each report branch. Merely assuming positive definite outputs at each nonzero input is not used to infer a uniform ratio at a vanishing branch.

A *state-positive realization* consists of density matrices $\tau_1, \ldots, \tau_M$, including the prescribed preparations as designated initial labels, and nonnegative matrices $Q_{a,y}$ satisfying

$$\mathcal{I}_{a,y}(\tau_i) = \sum_j Q_{a,y}(i,j)\tau_j, \qquad \sum_{y,j} Q_{a,y}(i,j) = 1. \tag{S23.3}$$

The second equation includes all reports of a fixed command. The simulator in state i, on receiving command a, draws the pair (y,j) with these probabilities, reports only y, and retains j privately. For a terminal positive-operator-valued measurement $\{E_z\}_z$, it reports z with probability $\operatorname{tr}(E_z\tau_j)$. A fixed finite decoder dictionary is handled by the unchanged external observer.

Theorem S23.2 (Uniform finite controlled realization)**.** *Under* (S23.2)*, a state-positive realization exists with*

$$M \leq k + \left\lceil (4d/\eta)^{d-1} \right\rceil. \tag{S23.4}$$

Each row of each $Q_{a,y}$ may be chosen with at most d^2 nonzero entries. The count is independent of the horizon, number of commands, number of reports, and choice of terminal measurement; those quantities still affect the transition-table size.

For every adaptive randomized command policy using only past declared reports and its own randomness, this one realization preserves all finite report laws and all adaptively chosen terminal

measurement laws. It also preserves the complete stopped law, including the event of never stopping. Every nonnegative cost that is a function of the executed command/report path has the same expectation, finite or infinite. If calls instead have specified unobserved nonnegative mean observables G_a, the assignment $c_a(i) = \operatorname{tr}(G_a\tau_i)$ preserves their total expected cost at every stopping rule. This latter assertion is an expectation statement, not a joint-law assertion about an unspecified random cost observation.

Proof. Take the hull of Lemma S23.1 and append the prepared states. For any generator τ_i, let $s = \operatorname{tr}\mathcal{I}_{a,y}(\tau_i)$. If $s = 0$, positivity makes the branch matrix zero, and set that whole row of $Q_{a,y}$ to zero. If $s > 0$, its normalized output belongs to $\mathcal{D}_d(\eta)$ and hence to the pure-state core hull. Choose convex coefficients there and multiply them by s. This proves the first identity. Taking its trace and summing in y proves row normalization.

A convex combination in the trace-one Hermitian affine space needs at most d^2 generators. Indeed any larger supported list is affinely dependent; subtract a scalar multiple of an affine dependence until one positive coefficient becomes zero, preserving nonnegativity, total mass and represented matrix, and iterate. Applying this finite reduction separately to each branch gives the sparsity claim.

For a fixed command/report history h, let X_h be its unnormalized physical state and p_h the simulator's unnormalized hidden-state row. Induction using (S23.3) gives the matrix identity

$$X_h = \sum_i p_h(i)\tau_i. \tag{S23.5}$$

It holds initially because the initial label is the prescribed preparation. Multiplying both sides by a report branch gives the next identity. Traces give the report probability, and pairing with any terminal effect gives its joint audit probability.

For an adaptive observer the command probabilities and commitment probabilities conditional on a declared history are the same in the two experiments. Condition on its private tape, apply the same identity at each finite history, and integrate over that tape. Equivalently keep one unnormalized identity for every observer-memory state. The simulator's hidden state is not an extra input to this policy. Thus all finite stopped cylinders agree. The events of survival through n calls decrease to nontermination; their equal probabilities give equality there as well. Bounded terminal measures follow by summing disjoint finite stopping events. Nonnegative path costs follow from equality of the path law and monotone convergence.

For mean observables, the expected cost contribution at a begun history is $\operatorname{tr}(G_aX_h) = \sum_i p_h(i)c_a(i)$. Sum these identities first to a finite horizon and then monotonically to the stopping time. No total-variation bound on bounded terminal scores is used to infer an unbounded waiting-cost bound. □

This theorem realizes the prepared experiment, not an arbitrary entangled quantum input channel. It gives no affine encoding of every density matrix into one simplex and does not grant the hidden simulator label to the original observer. For a single prescribed initial state one anchor suffices. If $d = 1$, the original scalar instrument already has one state. If $\eta = 1$, every nonzero output is I/d, so one core state plus the preparation anchors suffices without a projective net.

Corollary S23.3 (Same-policy resource and risk invariance)**.** *The original instrument and its hidden classical realization have identical feasible expected-cost budgets and identical risks for each common report-based observer. Consequently their optimal risk curves agree at every fixed observer memory bound and every fixed decoder dictionary, including feasible boundary budgets. The assertion also holds modelwise for an unknown family satisfying one uniform margin, with model-dependent hidden transition tables and the same report interface; the parameter is not supplied to the observer. If the prepared state varies with the model, it is represented by one designated model-dependent anchor and a common pure-state core.*

An observer with ℓ live labels coupled to an M-state simulator has at most ℓM joint live states. These are joint system states, not a claim that M is the observer's optimal memory. Giving the hidden label to the observer changes the statistical experiment and invalidates the claimed same-policy comparison.

Proof. Theorem S23.2 equates the cost and risk of every admissible observer without changing its transition rows, dictionary or report information. Taking the same constrained infimum gives the assertion, even when the infimum is not attained in that memory class. For a model family use the same core hull in every model and solve its branch identities separately; each modelwise equality holds for the same observer. Taking the supremum over the unknown model therefore preserves the robust objective and constraints. The product count records the two distinct retained coordinates. □

S23.3. **A matching exponent for state-positive universal realization.** Let $v_d(\eta)$ be the least number of arbitrary density matrices whose convex hull contains $\mathcal{D}_d(\eta)$. Mixed vertices are permitted. This number measures uniform state-positive representation, not the smallest hidden model for one selected scalar response.

Theorem S23.4 (Sharp noise exponent at fixed physical dimension)**.** *For $d \geq 2$ and $0 < \eta \leq 1/2$, put $s = \eta(1 - 1/d)$. Then*

$$s^{-(d-1)} \leq v_d(\eta) \leq \left\lceil (4d/\eta)^{d-1} \right\rceil. \tag{S23.6}$$

Thus the exponent $d - 1$ as $\eta \downarrow 0$ is necessary and sufficient at each fixed d. Constants in this comparison depend on d.

For every $0 < \eta \leq 1/4$ there is also a finite-command, unit-call-cost prepared instrument satisfying (S23.2) *for which every state-positive exact realization preserving all terminal effects needs at least*

$$(4/(5\eta))^{d-1} \tag{S23.7}$$

hidden states. Its command count may grow as the margin decreases. No lower bound of this exponent for unrestricted, possibly non-state-positive, response realizations is asserted.

Proof. The upper is Lemma S23.1. For the lower, for each unit u consider the state $(1-\eta)uu^* + \eta I/d$. Pairing a convex representation in density vertices τ_j with uu^* shows that at least one vertex satisfies $\langle u, \tau_j u\rangle \geq 1 - s$.

For any fixed density matrix τ, this inequality holds on a set of projective measure at most s^{d-1}. If its largest eigenvalue a is below $1 - s$, the set is empty. Otherwise $a \geq 1 - s > 1/2$ and its top eigenvector v is unique. With $x = |\langle u, v\rangle|^2$, all other eigenvalues are at most $1 - a$, so

$$\langle u, \tau u\rangle \leq ax + (1 - a)(1 - x).$$

To reach $1 - s$ one needs $x \geq (a - s)/(2a - 1) \geq 1 - s$; the last inequality is equivalent to $(1 - a)(1 - 2s) \geq 0$. The projective ball formula from the lemma now bounds the measure by s^{d-1}. The caps associated with all vertices cover every ray, so their measures sum to at least one. This proves the first lower.

For the finite-command assertion take a finite projective $\eta/4$-net of rays u_j and a command for each replacement channel

$$\Phi_j(X) = \operatorname{tr}(X)((1 - \eta)u_j u_j^* + \eta I/d).$$

These are valid completely positive, trace-preserving maps with the required margin, one deterministic report and unit cost. Exactness of every terminal effect after that command forces its output matrix to be a convex combination of the simulator's density vertices. Thus at each net point some vertex has expectation at least $1 - s$. For an arbitrary ray u choose a net point within $\eta/4$. Since the operator norm distance of the two projectors is at most $\eta/4$, pairing with any density matrix changes the expectation by at most $\eta/4$. The caps at level $1 - s - \eta/4$ therefore cover all rays. Here

$s + \eta/4 \le 5\eta/4 < 1/2$, and the preceding cap bound gives at least $(s + \eta/4)^{-(d-1)} \ge (4/(5\eta))^{d-1}$ vertices. This is a finite physical family, not an uncountable list disguised as a finite input. □

For unrestricted classical hidden models a single scalar response may admit non-quantum intermediate response vectors. The state-positive lower does not exclude that possibility. Conversely, a hidden decoder defined as a positive normalized affine functional of every terminal effect is necessarily of the form $E \mapsto \operatorname{tr}(E\tau)$ for a density matrix τ: extend the affine functional linearly from the effect interval to the Hermitian space, use the trace pairing, and test rank-one positive effects. This explains the positivity requirement in the count rather than hiding it in an abstract factorization.

S23.4. **Sparse algebraic construction and several-step interiority.**

Proposition S23.5 (Finite algebraic construction with sparse branch rows)**.** *For a finite instrument family and finite preparation list with real-algebraic entries and a supplied algebraic margin $\eta > 0$, a state-positive realization can be constructed by a terminating real-algebraic procedure. One may use the size bound* (S23.4)*, and all generator matrices and transition probabilities can be chosen algebraic. At most d^2 successors per command/report row are needed. The original margin condition is itself a finite quantified real-algebraic condition on pure inputs; a positive rational margin, if one exists, can be found by enumeration and such tests.*

This is a finite synthesis assertion, not a polynomial-time bound. The retained-state count, the number and algebraic encoding lengths of table entries, and the resources of exact random sampling remain different quantities.

Proof. The existence proof uses at most $K = \lceil (4d/\eta)^{d-1} \rceil$ pure core projectors and the stated anchors. Treat their real Hermitian coordinates and all branch coefficients as variables. Impose $P_j \succeq 0$, $\operatorname{tr} P_j = 1$, $P_j^2 = P_j$, the fixed anchor identities, nonnegative normalized rows, and the finitely many matrix equations (S23.3). These are polynomial conditions over the real algebraic field; positivity can be expressed by principal minors. The preceding theorem proves feasibility. Real-closed-field elimination and sample-point construction give an algebraic solution [15]. Row sparsity can be obtained afterwards by elimination of an affine dependence over the same algebraic field, or by a finite disjunction of supports of size at most d^2.

For margin testing, it suffices by linearity to quantify over pure inputs uu^* with $\|u\| = 1$; the output inequality is again a principal-minor condition in real and imaginary coordinates. If some real positive margin works, a smaller positive rational margin works. Enumeration of positive rational candidates with an exact test therefore eventually succeeds on the strict-interiority class. No termination is asserted when no positive margin exists. For $J = \sum_a |Y_a|$, a sparse implementation stores at most JMd^2 nonzero branch probabilities, besides Md^2 density coordinates and any chosen audit table. These are storage counts of mathematical entries, not bit bounds for their encodings. □

Theorem S23.6 (Interiority after a finite block, with its clock counted)**.** *Suppose instead that for some integer $L \ge 1$, every length-L command/report branch product satisfies* (S23.2) *with the same margin. Individual branches need not do so. Put $J = \sum_a |Y_a|$. Then an exact state-positive realization exists with at most*

$$\left(k + \left\lceil (4d/\eta)^{d-1} \right\rceil\right) \sum_{j=0}^{L-1} J^j \tag{S23.8}$$

hidden states. It preserves intermediate reports and stopping within the block; it does not replace L calls by one unpriced call.

Proof. Start with the core projectors and preparation anchors. For each, list its normalized nonzero images under all command/report products of length less than L, retaining the product as part of the hidden label. There are at most the displayed number. At a depth below $L - 1$, an actual branch goes to its listed child with its actual conditional probability. At depth $L - 1$, the next

product output is either zero or belongs to $\mathcal{D}_d(\eta)$; decompose it into the core and restart at depth zero. Zero branches have zero rows. These transitions satisfy the same matrix interpolation identity at every single call. Theorem S23.2's history induction therefore applies without suppressing any intermediate observation, audit or charge. The depth is encoded in the enlarged state label, not in an external clock. □

S23.5. **Why forgetting the report invalidates the premise.** The average channel can be completely depolarizing even though every recorded branch sends pure states to pure states. The Pauli instrument $\mathcal{I}_P(X) = PXP^*/4$ on a qubit is an example. Thus an average-state calculation cannot verify (S23.2).

A stronger example has, in addition, a recorded irrational rotation. Mix half of that Pauli instrument with the single recorded branch $\mathcal{I}_u(X) = UXU^*/2$, where

$$U = 5^{-1/2}\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}.$$

Its average sends every density matrix to at least $I/4$, but the report word u^n retains the pure state $U^n|0\rangle\langle 0|U^{*n}$. These states are all distinct: $e^{i\vartheta} = (3+4i)/5$ is not a root of unity, since $2\cos\vartheta = 6/5$ would otherwise be a rational algebraic integer. A finite state-positive simulator preserving all terminal effects would represent every one of these pure states as a convex combination of its finite density vertices. An extreme point of $\mathcal{D}_2$ can only be represented by identical density matrices, so every distinct orbit state would have to occur as a vertex. This is impossible. Thus average interiority does not even guarantee finite state-positive realization, while all finite report words here have well-defined nonzero probabilities.

The example is a boundary qualification of the positive theorem. It is not used in place of that theorem, and no unrestricted classical hidden-state impossibility is inferred from this particular extreme-state proof alone.

S23.6. **An explicit unrestricted memory obstruction at weak damping.** The preceding matching exponent is for state-positive realization. A different argument gives a quantitative, albeit non-sharp, obstruction for every stochastic realization of the fixed-angle scalar qubit response. It sharpens the qualitative divergence in Theorem S28.6 without changing its exact three-label result.

Lemma S23.7 (A finite near-peripheral spectral witness)**.** *If an $M \times M$ row-stochastic matrix has a nonzero eigenvalue $z = \alpha e^{i\theta}$, $0 < \alpha \le 1$, then either $1 - \alpha > 1/(2M)$ or some $1 \le l \le M$ satisfies*

$$|1 - e^{il\theta}| \le (4\sqrt{2M^{M+1}} + M)\sqrt{1-\alpha}. \tag{S23.9}$$

The assertion does not require irreducibility, reversible rows or a lower bound on all positive transition probabilities.

Proof. Normalize an eigenvector x so that its largest coordinate modulus is one, attained at i_0, and multiply its phase so $x_{i_0} = 1$. Let X_n be the Markov chain starting there. Then $\mathbb{E}x_{X_n} = z^n$ and

$$\mathbb{E}|x_{X_n} - z^n|^2 = \mathbb{E}|x_{X_n}|^2 - |z|^{2n} \le 1 - \alpha^{2n}.$$

Choose once for each state a successor with transition probability at least $1/M$, and follow this directed path. Its first n steps have probability at least M^{-n}, so their endpoint i_n obeys

$$|x_{i_n} - z^n| \le \sqrt{M^n(1-\alpha^{2n})} \le \sqrt{2nM^n(1-\alpha)}.$$

Two of $i_0, \dots, i_M$ agree, say $i_j = i_k$ with $j < k$. Their two bounds give $\alpha^j|1 - z^{k-j}| \le 2\sqrt{2M^{M+1}(1-\alpha)}$. If $1 - \alpha \le 1/(2M)$, Bernoulli's inequality gives $\alpha^j \ge 1/2$. Finally $|1 - e^{i(k-j)\theta}| \le |1 - z^{k-j}| + 1 - \alpha^{k-j}$ and $1 - \alpha^{k-j} \le M(1-\alpha) \le M\sqrt{1-\alpha}$. These inequalities prove the claim. □

Corollary S23.8 (A rational exclusion margin for every memory size)**.** *Fix* $\zeta = (3+4i)/5$ *and* $0 < \alpha < 1$*. If*

$$\tfrac{1}{2}[1+\alpha^n \operatorname{Re}(\zeta^n)] = e_1^{\mathsf{T}} Q^n d \quad (n \geq 0), \qquad Q\mathbf{1} = \mathbf{1}, \quad Q \geq 0, \quad 0 \leq d \leq 1,$$

with M *classical states, then*

$$1 - \alpha \geq \frac{1}{49M(25M)^M}. \tag{S23.10}$$

Consequently the exact memory in the fixed-angle calibration and common-squared-audit embeddings diverges at least on the order of $\log(1/(1-\alpha))/\log\log(1/(1-\alpha))$ *as* $\alpha \uparrow 1$*. This lower is not claimed to match the retained polygon upper or to solve the general-*M *frontier.*

Proof. The response has three distinct nonzero exponential modes $1, \alpha\zeta, \alpha\bar{\zeta}$. Applying the characteristic recurrence of Q to it and using Vandermonde independence forces $\alpha\zeta$ to be an eigenvalue. For every positive integer l, the nonzero Gaussian integer $(3+4i)^l - 5^l$ has modulus at least one, so $|1-\zeta^l| \geq 5^{-l}$. The nonzero assertion follows from the root-of-unity argument above.

If $1 - \alpha \leq 1/(2M)$, Lemma S23.7 gives

$$5^{-M} \leq (4\sqrt{2M^{M+1}} + M)\sqrt{1-\alpha} \leq 7M^{(M+1)/2}\sqrt{1-\alpha}.$$

Squaring proves (S23.10). In the other case that bound is weaker than $1/(2M)$ and is automatic. Taking logarithms gives the exact necessary inequality

$$\log \frac{1}{1-\alpha} \leq \log 49 + \log M + M \log(25M),$$

which implies the stated asymptotic order by elementary inversion. □

Noise-dependent classical-memory obstructions for quantum processes precede this result. In particular Fanizza, Lumbreras and Winter prove a worst-angle lower bound for noisy qubit processes, conditional on positive realizations for the whole phase family [50]. Their statement is not a lower rate for the one fixed angle in this corollary, nor does our weaker fixed-angle estimate replace it. The new general contribution of this section is the branch-resolved constructive realization and its stopped-resource identity, together with the state-positive dimension exponent. The established cone and spectral mechanisms are credited rather than renamed.

S24. Boundary stopping and finite stochastic realization

The weighted potential identity and the cost-active support criterion of Theorem S28.1 already allow model- and state-dependent reports. We now replace that sufficient coverage condition by a necessary-and-sufficient stopping criterion. We first characterize rigidity for prepared instruments with model- and state-dependent report probabilities. On the resulting boundary face, ties lead to a stochastic realization problem, not in general a deterministic residual quotient. The optimal-face and anchored convex realization theorems are inherited in full as Theorems S28.2 and S28.3. Their pointed-polytope formulation below extends them to the larger rigidity class; neither the convex realization method nor the original robust-tie classification is claimed as new in this continuation.

S24.1. **A primitive class and an exact boundary stopping criterion.** Let Θ be compact metric and let the protected physical space have finite dimension d. There is one globally legal command, a finite failure alphabet Y, and one productive herald s. Model θ has a prepared density matrix ρ_θ and completely positive maps $\mathcal{I}_{\theta,y}$ and $\mathcal{I}_{\theta,s}$ whose sum is trace preserving. All these data are continuous in θ. A nonnegative call cost has conditional mean $\operatorname{tr}(G_\theta \rho)$, with continuous Hermitian G_θ satisfying

$$0 < c_* I \preccurlyeq G_\theta \preccurlyeq c^* I < \infty.$$

A failure forbids commitment. A herald permits commitment to a member of one fixed finite dictionary C, or rejection and continuation from the actual post-herald state. The full report is the

only physical observation available to the controller. No extra cost observation, reset, state inspection or free clock is supplied. After commitment an unobserved prospective audit has continuous loss observables $0 \preccurlyeq F_{\theta,r} \preccurlyeq LI$, $r \in C$. A separately executed fixed final-audit cost is added to all budgets.

Put $\mathcal{T}_\theta = \sum_y \mathcal{I}_{\theta,y}$ and impose the primitive transience test

$$I - (\mathcal{T}_\theta^*)^d I \succ 0 \quad (\theta \in \Theta). \tag{S24.1}$$

This condition does not require scalar herald effects. Define

$$\delta = \min_\theta \lambda_{\min}\big(I - (\mathcal{T}_\theta^*)^d I\big) > 0, \qquad \kappa = c^* d/\delta, \qquad H_\theta = \sum_{n\ge 0} (\mathcal{T}_\theta^*)^n G_\theta.$$

For $T \ge 0$, set $\beta_T = (1-\delta)^{\lfloor T/d \rfloor}$, interpreting the zeroth power as one even when $\delta = 1$. For a failure word $w = y_1 \cdots y_n$, let $\mathcal{I}_{\theta,w} = \mathcal{I}_{\theta,y_n} \cdots \mathcal{I}_{\theta,y_1}$, with the identity map at the empty word. Its first-herald mass and terminal score are

$$\lambda_\theta(w) = \operatorname{tr}\big(\mathcal{I}_{\theta,s}\mathcal{I}_{\theta,w}(\rho_\theta)\big), \tag{S24.2}$$

$$g_\theta(w,r) = \operatorname{tr}\big(F_{\theta,r}\mathcal{I}_{\theta,s}\mathcal{I}_{\theta,w}(\rho_\theta)\big). \tag{S24.3}$$

In particular $0 \le g_\theta(w,r) \le L\lambda_\theta(w)$. These are unnormalized quantities; null reports are never divided out.

Lemma S24.1 (Uniform first-herald control from the instrument)**.** *First-herald commitment is proper from every physical state in every model, has conditional remaining cost at most κ, and has survival mass after T failures at most β_T. Its expected cost from the declared preparation is $t_\theta = \operatorname{tr}(H_\theta \rho_\theta)$, continuous in θ. The least robust feasible cost is*

$$N_0 = \max_{\theta\in\Theta} t_\theta.$$

Moreover, (S24.1) *is equivalent, modelwise, to absence of a nonzero subspace invariant under every failure Kraus operator and annihilated by the herald effect.*

Proof. For one model write $B = I - \mathcal{T}^* I = \mathcal{I}_s^* I$ and $O_n = I - (\mathcal{T}^*)^n I$. Then $O_{n+1} = B + \mathcal{T}^* O_n$. For Kraus operators L_j of $\mathcal{T}$, positivity gives $\ker O_{n+1} = \ker B \cap \bigcap_j L_j^{-1}(\ker O_n)$. The decreasing kernels stabilize at their first equality and can strictly decrease at most d times. Their stabilized space is exactly the failure-invariant dark subspace in the statement. This proves the equivalence without choosing a normalized posterior.

Continuity and compactness give $\delta > 0$. The inequality $(\mathcal{T}_\theta^*)^d I \preccurlyeq (1-\delta)I$ iterates, uniformly over θ, to the displayed survival bound. Summing in blocks of d calls bounds expected cost by $c^* d/\delta$. The cost series converges uniformly in operator norm; hence H_θ and t_θ are continuous. Every legal committed trajectory contains all calls through its first herald. Nonnegative additional costs therefore make its expectation at least t_θ in that model. First-herald commitment attains all these modelwise costs simultaneously, proving the formula for N_0 and its attainability. □

Let $\mathcal{W} = \{w : \lambda_\theta(w) > 0 \text{ for some } \theta\}$. Define the normalized unused-budget margin

$$a(w) = \inf_{\theta:\lambda_\theta(w)>0} \frac{N_0 - t_\theta}{\lambda_\theta(w)} \quad (w \in \mathcal{W}). \tag{S24.4}$$

This is a criterion on the primitive first-herald experiment, not a condition on an unknown optimal decision policy.

Theorem S24.2 (Necessary and sufficient boundary stopping rigidity)**.** *Every robust policy of expected cost at most N_0 commits at the first herald in every model, up to null histories, if and only if $a(w) = 0$ for every $w \in \mathcal{W}$.*

More precisely, if r_w is its conditional probability of rejecting its first herald after w, then

$$c_* r_w \le a(w). \tag{S24.5}$$

If $a(w) > 0$ for some word, there is a finite-label boundary-feasible policy that rejects that first herald with positive probability and then commits at the next herald. Thus failure of the criterion has an actual legal, paid implementation.

Proof. Before the first herald the sole command is fixed. Conditional on its full word, the private controller randomness is independent of the physical state and of the model: it changes no executed instrument. Consequently the rejection probability r_w is the same in all models. On that event at least one further call is required, whose conditional expected cost is at least c_*. Decomposing the nonnegative cost at the first herald gives

$$\mathbb{E}_\theta C_\tau \ge t_\theta + c_* \sum_w \lambda_\theta(w) r_w.$$

Keeping one word and using feasibility proves (S24.5). If every $a(w)$ vanishes, rejection occurs with probability zero in each model; countability of words suffices.

Conversely choose a word with $a(w) > 0$ and set $e = \min\{1, a(w)/(2\kappa)\} > 0$. Accept every first herald except that, after precisely this word, reject with probability e and then accept the next herald. The extra conditional cost after rejection is at most κ, from the actual post-herald state, by Lemma S24.1. Thus total cost in each model is at most $t_\theta + e\kappa\lambda_\theta(w) \le N_0$. A finite word matcher, a default accepting label and one post-rejection label implement this rule. Its only persistent information is those counted labels. This proves necessity and the realization assertion. □

Corollary S24.3 (Active preparation coverage)**.** *The rigidity condition holds if every $w \in \mathcal{W}$ has positive mass in at least one cost-active model θ with $t_\theta = N_0$. In particular it holds if one cost-active model has positive mass at every possible word. It also holds when the first-herald costs are equal across models. A positive definite herald effect and failure maps that send every nonzero positive input to a nonzero output give positive mass to every word, and therefore imply rigidity under* (S24.1)*.*

Proof. An active positive-mass model makes the ratio in (S24.4) zero. For the last assertion, a finite failure composition has nonzero positive output, and pairing it with a positive definite herald effect gives positive trace. Lemma S24.1 supplies an active model. □

The infimum in (S24.4) matters. Rigidity may also follow from models approaching the active set with both unused budget and word mass tending to zero, even if no active model gives that word positive mass. Conversely a word supported only where there is enough relative budget slack can permit boundary rejection. The theorem classifies stopping rigidity in this single-command uniformly transient class; it does not claim that a residual decision rule classifies multi-command boundary control.

Example S24.4 (Active coverage is sufficient but not necessary)**.** In dimension one take $\theta \in [0, 1/2]$, $h(0) = 0$, $h(\theta) = e^{-1/\theta}$ for $\theta > 0$, and herald probability $p_\theta = 1/(4 - h(\theta))$. Give failures $1, 0$ probabilities $(1 - p_\theta)\theta$ and $(1 - p_\theta)(1 - \theta)$, with unit calls. Then $N_0 = 4$ and the only cost-active model is 0, which gives no mass to words containing 1. Nevertheless for a word of length n containing j occurrences of 1,

$$\frac{N_0 - t_\theta}{\lambda_\theta(w)} = \frac{e^{-1/\theta}}{p_\theta(1 - p_\theta)^n \theta^j (1 - \theta)^{n-j}} \longrightarrow 0.$$

Indeed the factors other than θ^j have positive limits, and $e^{-1/\theta}/\theta^j \le (j + 1)!\theta$ by the exponential series. Words with no 1 already have positive active-model mass. The exact criterion therefore gives boundary rigidity even though active support coverage fails. All primitive data are continuous and the herald probability is uniformly positive. No algebraic-data claim is made for this example.

Example S24.5 (A paid rejection remains feasible at the robust least budget)**.** Take two scalar models with unit costs. Model A has herald probability $1/4$ and only failure 0; model B has herald probability $1/2$ and only failure 1. The robust least budget is 4. For the word consisting of one 1, its mass is $1/4$ in B and zero in A, so $a(1) = 8$. Accept the first herald except that after precisely this word reject with probability $1/2$, then accept the next herald. Its cost is 4 in A and $2 + (1/4)(1/2)2 = 9/4$ in B. A four-label matcher implements it. Thus a robust minimum-cost boundary need not eliminate stopping freedom merely because first-herald operation is modelwise cheapest.

S24.2. **The entire robust optimal face, including ties.** Assume now that $a(w) = 0$ on $\mathcal{W}$. At N_0 every feasible policy is a kernel $b : Y^* \to \Delta(C)$ with risk

$$\mathcal{R}_\theta(b) = \sum_{w,r} b(w, r) g_\theta(w, r).$$

Its values on globally null words can be completed arbitrarily. Define $V = \min_b \sup_\theta \mathcal{R}_\theta(b)$. For a probability measure μ on Θ, write $g_\mu(w, r) = \int g_\theta(w, r)\, d\mu(\theta)$ and $A_\mu(w) = \operatorname{argmin}_{r \in C} g_\mu(w, r)$.

Theorem S24.6 (Exact robust optimal face)**.** *Primal and dual optima are attained, and*

$$V = \max_{\mu \in \mathcal{P}(\Theta)} \sum_w \min_{r \in C} g_\mu(w, r). \tag{S24.6}$$

For any attaining μ_*, the complete set of robust optimal kernels is

$$\mathcal{B}_* = \{b : \ b(w, \cdot) \in \Delta(A_{\mu_*}(w)) \ (w \in Y^*), \quad \mathcal{R}_\theta(b) \le V \ (\theta \in \Theta)\}. \tag{S24.7}$$

The global inequalities in (S24.7) *cannot in general be omitted. Wordwise Bayes ties do not authorize an arbitrary robust decoder.*

Proof. The product of finite decision simplexes is compact convex. Every risk tail beyond length T is at most $L\beta_T$, uniformly in the kernel and model. Finite sums are jointly continuous, so the full payoff is continuous and bilinear in the kernel and model measure. Compact minimax, or finite-prefix separation followed by compact limits with this uniform tail, gives (S24.6) and both attainments [40]. The fixed-measure minimization separates wordwise. For an optimal kernel,

$$0 \le \sum_{w,r} b(w, r)\,[g_{\mu_*}(w, r) - \min_u g_{\mu_*}(w, u)] = \int \mathcal{R}_\theta(b)\, d\mu_* - V \le 0.$$

Every summand is nonnegative, so all positive decoder weights are on minimizing actions. No strictly positive common word law is needed: zero-mass words simply impose no nontrivial local restriction. The all-model inequalities are necessary; together they imply optimality by the definition of V. This proves the characterization. The physical constructions in Theorem S25.1 below show why local ties alone are insufficient. □

S24.3. **What replaces a finite deterministic residual quotient.** Let $\mathcal{F} = (\Delta(C))^{Y^*}$. Elements are full decoder-response functions, not hidden physical states. Define the affine shift $S_y f(z) = f(yz)$. A finite-dimensional polytope $P \subset \mathcal{F}$ is called invariant here when $S_y P \subset P$ for every failure letter. It is the convex hull of finitely many response functions; its extreme points are consequently finite. For $b \in P$ put

$$h(P, b) = |\operatorname{ext} P| + \mathbf{1}_{\{b \notin \operatorname{ext} P\}}.$$

The extra term records a prescribed pure initial label rather than an uncharged random initial state.

Theorem S24.7 (Exact finite stochastic-memory criterion with robust ties)**.** *The least live-label count attaining V at budget N_0 is exactly*

$$\min\{h(P, b) : b \in \mathcal{B}_*, \ b \in P \subset \mathcal{F}, P \text{ a finite invariant polytope}\}, \tag{S24.8}$$

with value infinity when no such pair exists. This equivalence includes stochastic transitions, randomized terminal decisions, impossible words, and all robust ties. It requires one common realization, not a different machine in each model.

For a deterministic kernel forced uniquely on every word, the criterion reduces to the number of distinct right residuals $z \mapsto b(wz)$, recovering the classical minimal output-machine count. Without uniqueness, a finite number of deterministic residual classes is neither the right requirement on an optimal randomized kernel nor a substitute for (S24.8)*.*

If a realizing observer has M live labels and m distinct actual decoder values, event stopping uses at most $\max\{M, m\}$ total labels and state stopping at most $M + m$. These are the established packing counts, not extra live memory or a hidden clock.

Proof. An M-label first-herald observer has row-stochastic matrices Q_y and decoder rows $D_i \in \Delta(C)$, from one prescribed initial label. Let $f_i(z)$ be its decision distribution on a future herald after failure word z, when initialized in label i. Then

$$f_i(\varnothing) = D_i, \qquad S_y f_i = \sum_j Q_y(i,j) f_j.$$

Thus $P = \operatorname{conv}\{f_1, \dots, f_M\}$ is invariant and contains its initial response b as one listed point. It has at most M extreme points. If b is not extreme, it is itself a listed nonextreme point, so $h(P, b) \le M$ still holds. Its optimality puts b in $\mathcal{B}_*$. This proves necessity and the lower count.

Conversely list the vertices of an admissible polytope, adding b as an initial label only when it is not already a vertex. Every shift of each listed function belongs to P, and hence has fixed convex coefficients in those vertices. Choose these coefficients as its row of Q_y. Decode at a herald using that function's value at the empty word. Induction on word length proves that the machine's complete response from each label is the listed function. In particular its initial response is b. This produces exactly $h(P, b)$ labels, commits at the first herald and attains V. No physical state or posterior is read. Pruning unreachable labels cannot invalidate either implication or a minimal count.

When b is deterministic, each distinct residual is an extreme point of the entire product $\mathcal{F}$: every coordinate is a vertex of $\Delta(C)$. Invariance puts every residual in P, so each must be one of its vertices. Conversely a finite set of distinct residuals is closed under the shifts and realizes the usual deterministic output machine. Its convex hull has precisely these vertices. This proves the reduction. The terminal counts follow by independently packing continuing and final roles as in Lemma 17.4. □

Theorem S24.7 is an exact geometric existence criterion in response-function space. It is not a finite numerical test for arbitrary data, and it does not assume that a minimal stochastic realization lies in the affine span of the reached residuals. Such an assumption could discard valid hidden-state realizations. Positive realization by invariant cones or convex sets, and the distinction between linear rank and positive realization, have a long history [48, 49]. The new mathematical use here is the explicit least-budget condition, the robust tied optimal face, the pure-initial-label accounting, and the physical realization and obstruction results which follow.

Proposition S24.8 (Fixed-memory attainment and boundary approximation)**.** *For every prescribed M, the least boundary risk among stochastic observers with at most M live labels is attained. If* (S24.8) *exceeds M, this risk is strictly larger than V. Nevertheless, a finite prefix of an optimal kernel, followed by one arbitrary decoder label, costs exactly N_0 in the robust sense and has excess risk at most $L\beta_T$. A sufficient live count is $1 + \sum_{n=0}^{T-1} |Y|^n$. Modelwise cost equals t_θ, not necessarily N_0 in every model.*

Proof. Transition matrices and decoder rows at fixed M range over compact simplexes. Finite-prefix risks are continuous and their uniform tail is at most $L\beta_T$. Thus worst-model risk is continuous and

takes its minimum. Equality with V is equivalent to an optimal realization by Theorem S24.7. A prefix tree storing words shorter than T, followed by a common tail label, changes only terminal mass bounded by β_T in every model. It still accepts the first herald and hence has the same modelwise costs t_θ. This proves all assertions without taking an interior-budget completion theorem to zero slack. □

S25. Squared-prediction realizations of robust stochastic memory

A local Bayes tie can conceal global constraints on the entire randomized decision law. We construct such constraints using a fixed finite-dimensional instrument and a genuine squared terminal audit, not a time-indexed loss oracle. The construction gives exact finite-memory thresholds of every integer order, finite physical witnesses for strict smaller-memory gaps, and a finite-rank quantum law which has no finite stochastic realization. It also permits an exact algebraic fixed-memory optimization on a declared compact-model class.

S25.1. **A physical embedding of a target channel response.** Fix a channel Ψ on a d_0-dimensional protected system, a preparation ρ, and an effect $0 \preccurlyeq E \preccurlyeq I$. Its target response is

$$u_n = \operatorname{tr}(E\Psi^n(\rho)) \in [0,1], \qquad n \geq 0.$$

Choose $0 < p < 1$, $q = 1-p$, $0 < \bar{t} < 1$, and $\varepsilon = 1/4$. The unknown fixed model is $(t,\sigma) \in [0,\bar{t}] \times \{-1,1\}$. Add one protected classical two-state register, initially alive. Its channel D_t discards coherences and maps the alive population to alive with probability t, otherwise to dead, while dead remains dead. Thus the real matrix entries of D_t depend polynomially on t. A failure applies $D_t \otimes \Psi$ with probability q and a herald applies the identity with probability p. There is one unit-cost command, one failure letter, and the fixed dictionary $C = \{-1,1\}$.

Let A be the alive projection and set $T = 2E - I$. Define two commuting Hermitian audit moments on the combined protected space by

$$v_\sigma = I - \sigma\varepsilon A \otimes T, \qquad m_\sigma = -\sigma\varepsilon(A \otimes I)/2.$$

For $h = 3/2$, use audit outcomes $z \in \{-h, 0, h\}$ with effects

$$B_{\sigma,+} = \frac{v_\sigma + h m_\sigma}{2h^2}, \qquad B_{\sigma,-} = \frac{v_\sigma - h m_\sigma}{2h^2}, \qquad B_{\sigma,0} = I - v_\sigma/h^2. \tag{S25.1}$$

The observer commits before this audit is performed and pays loss $(z-r)^2$. Its conditional loss observable is

$$F_{\sigma,r} = 2I + \sigma\varepsilon A \otimes (rI - T). \tag{S25.2}$$

Theorem S25.1 (A robust tied squared-task embedding)**.** *The displayed data define a legal prepared experiment with protected dimension $2d_0$, bounded squared loss, and least cost $1/p$. At that budget its robust risk is exactly 2. Its unique optimal decision kernel is*

$$b(n,+1) = u_n, \qquad b(n,-1) = 1 - u_n. \tag{S25.3}$$

For every fixed t, the equally weighted pair $(t,+1),(t,-1)$ is a least-favorable measure under which both actions are tied at every word. Nevertheless the all-model constraints force (S25.3)*, including its fractional values.*

For an arbitrary first-herald kernel $b_n = b(n,+1)$, its exact worst-model excess is

$$\sup_{t,\sigma} \mathcal{R}_{t,\sigma}(b) - 2 = 2\varepsilon p \sup_{0 \leq t \leq \bar{t}} \left| \sum_{n \geq 0} (qt)^n (b_n - u_n) \right|. \tag{S25.4}$$

Thus its exact finite stochastic-memory problem is the positive realization problem for the physically generated response u_n. Algebraic input channel, state, effect and constants give algebraic-polynomial primitive data.

Proof. One has $3I/4 \preccurlyeq v_\sigma \preccurlyeq 5I/4$ and $\|hm_\sigma\| \le 3/16$. Consequently all three effects in (S25.1) are positive and sum to I. Their first two moments are exactly m_σ, v_σ. Since $r^2 = 1$, the squared-loss observable is $v_\sigma - 2rm_\sigma + I$, which is (S25.2). The realized loss is bounded by $(h+1)^2 = 25/4$; its expectation is not an arbitrary signed score.

After n failures the alive mass is t^n, independently of the target state $\Psi^n\rho$. The expected squared loss at decision r is therefore $2 + \sigma\varepsilon t^n[r - (2u_n - 1)]$. First-herald commitment is forced at $1/p$, either by the indicator identity of Lemma S29.1 or by Theorem S24.2. Averaging the decision and summing its terminal mass pq^n gives (S25.4). The kernel (S25.3) makes every model's risk exactly 2, while each symmetric pair of models averages to 2 for every kernel. This proves the value and the stated least-favorable measures.

If a robust optimum differs at some coefficient, let k be its first index with $b_k - u_k \neq 0$. The power series in (S25.4) vanishes on the interval by optimality. Divide it by $(qt)^k$ and let $t \downarrow 0$. The remaining tail tends to zero because its coefficients are bounded, forcing $b_k - u_k = 0$, a contradiction. This proves uniqueness without assuming an analytic inverse oracle. The target physical channel is not accessible to the observer except through the final audit. Its response must be reproduced by the observer's actual retained labels. □

This theorem explains why the local simplexes in (S24.7) must be coupled by global risk inequalities. In this family every local simplex is the whole $\Delta(C)$ under the stated least-favorable measure, but there is only one robust optimal kernel.

The classical Erlang-chain rank mechanism is inherited from Theorem S28.5. The additions here are its realization by the same physical squared loss in every model and the finite physical witness theorem, which no longer needs continuum model uncertainty for the exact memory lower.

S25.2. **Every finite exact memory order occurs.** For an integer $k \ge 2$ and a rational $0 < a < 1$, take a classical target chain with $k-1$ transient stages and one absorbing dead state. At each transient stage it stays with probability a and advances to the next stage with probability $1-a$; advancing from the last transient stage reaches dead. It starts at the first stage. Let E indicate the transient states. Its survival response is

$$u_n = \sum_{j=0}^{k-2} \binom{n}{j} (1-a)^j a^{n-j}, \tag{S25.5}$$

where a term is zero for $j > n$.

Theorem S25.2 (A squared-loss realization of the exact memory hierarchy)**.** *The squared-task family of Theorem S25.1 built from this target has robust boundary value* 2 *and least exact live-label count* k*. It has protected physical dimension* $2k$*. A* k*-label stochastic observer realizes the optimum at exactly* $1/p$ *cost. No observer with fewer live labels, even with arbitrary real stochastic transitions and randomized readouts, attains it. No deterministic full-history decision rule attains it either.*

The full vector-valued Hankel response has rank exactly k*, supplying the matching lower. Moreover, a finite collection of rational physical model parameters gives an explicit positive risk gap for every observer with fewer than* k *live labels, as specified in Proposition S25.3.*

Proof. Use the target chain itself as the observer's failure-transition matrix; at a herald, decode $+1$ in transient stages and -1 in dead. Its probability of $+1$ after n failures is (S25.5), so it attains the value with k labels from a prescribed first label.

For any M-label stochastic observer with initial row e_1, failure matrix Q and probability-of-$+1$ column d, its response is $b_n = e_1 Q^n d$. The matrix with rows indexed by i and columns by suffixes j and decoder outcomes has rank at most M, since each entry factors through $e_1 Q^i$. Include in particular the constant column 1 and the columns u_{i+j} for $0 \le j \le k-2$, on rows $0 \le i \le k-1$.

These k columns are independent. Indeed $u_n = a^n P(n)$ for a polynomial P of degree $k-2$ with nonzero leading coefficient. Its minimal shift annihilator is $(X-a)^{k-1}$: applying shift minus a lowers

the polynomial degree by exactly one, until zero. Equivalently, the finite confluent Vandermonde matrix with columns $1, a^i, ia^i, \dots, i^{k-2}a^i$, $0 \le i < k$, is nonsingular because $1 \neq a$. A direct determinant proof is given in the supplement. The columns u_{i+j} span the last $k-1$ columns because their translated degree-$k-2$ polynomials are independent. Hence the displayed Hankel block is nonsingular and every exact observer has at least k labels. The construction supplies the upper.

At $n = k-1$, the survival probability lies strictly between zero and one: both no advances and all advances have positive probability. The unique optimal kernel is therefore fractional. A deterministic decision rule, even one with unlimited history, cannot equal it. The remaining assertion follows from the next proposition. □

Proposition S25.3 (Finite physical witnesses for a smaller-memory gap)**.** *Let $u_n \in [0,1]$ be any target response for which the $k \times k$ matrix*

$$A_u(i,0) = 1, \qquad A_u(i,j+1) = u_{i+j}, \quad 0 \le i < k, \quad 0 \le j < k-1$$

has nonzero determinant, where $k \ge 2$. Put $K = 2k-3$ and $\Delta = |\det A_u|/[k!(k-1)] > 0$. Let $W_{jn} = (j/K)^n$ for $0 \le j, n \le K$, taking $0^0 = 1$, and $C_0 = \|W^{-1}\|_\infty$. Choose $0 < r \le \min\{q\bar{t}, 1/2, \Delta/(4C_0)\}$, put $Z_{jn} = (rj/K)^n$, and $C_r = \|Z^{-1}\|_\infty$. Then every observer with fewer than k live labels has worst-model risk at least

$$2 + \varepsilon p \Delta / C_r. \tag{S25.6}$$

It suffices to take the finite models $t_j = rj/(Kq)$ and both signs σ. For rational target data and rational $p, \bar{t}$, all these constants and witnesses can be chosen rational. This is an explicit positive certificate, not an asserted sharp asymptotic memory rate.

Proof. The corresponding matrix A_b for an $M < k$ label observer has rank at most M and determinant zero. All entries lie in $[0,1]$. The Leibniz formula and telescoping a product of $k-1$ nonconstant entries give $|\det A_u - \det A_b| \le k!(k-1)\max_{0 \le n \le K} |u_n - b_n|$. Thus one coefficient difference $e_n = b_n - u_n$ has magnitude at least Δ.

Let $h(z) = \sum_{n \ge 0} e_n z^n$. At $z_j = rj/K$, its tail after degree K is at most $r^{K+1}/(1-r)$. Inverting the finite Vandermonde system gives

$$\max_{n \le K} |e_n| \le C_r \left(\max_j |h(z_j)| + \frac{r^{K+1}}{1-r} \right).$$

Since $Z = W \operatorname{diag}(1, r, \dots, r^K)$, one has $C_r \le C_0 r^{-K}$ for $r \le 1$. Our choice implies $C_r r^{K+1}/(1-r) \le 2C_0 r \le \Delta/2$. Consequently $\max_j |h(z_j)| \ge \Delta/(2C_r)$. Substitute this in the exact risk identity (S25.4), with both signs present at each node. The proof keeps the full infinite tail, not merely a fitted finite sample. □

Corollary S25.4 (Finite-model rational exact-memory hierarchy)**.** *For every integer $k \ge 2$, there is a finite family of algebraic, in fact rational-superoperator, prepared classical instruments with the same fixed two-value dictionary and a genuine squared terminal audit, whose least-budget optimum is 2 and whose exact least live-label count is k. A finite list of physical models gives an explicit strictly positive risk gap for every smaller stochastic observer. The first compulsory call and all subsequent calls remain charged. This finite-family statement does not assert uniqueness of the full-history optimal kernel.*

Proof. Take rational $p, \bar{t}, a$ in the preceding hierarchy, and retain just the finite parameter list of Proposition S25.3, with both signs at every node. Symmetry lower-bounds every policy's worst risk by 2. The k-label target chain attains 2 at every retained model. The finite-witness lower excludes every smaller memory. All maps and the finite audit effects have rational entries in the classical diagonal representation. First-herald commitment still has the common cost $1/p$, and the same predictable identity forces it at the boundary. The finite subfamily need not identify every coefficient of an arbitrary infinite decision kernel, which explains the final qualification. □

S25.3. **The inherited rank obstruction under a common squared audit.** Theorem S28.6 already establishes the rank-three obstruction for the signed calibration task. Here its same stochastic-realization obstruction is transported, with full proof retained, to one common squared-loss audit. The obstruction itself is not a newly discovered fact.

Theorem S25.5 (A rank-three protected quantum response with no finite realization)**.** *Take the target qubit with*

$$\rho = E = |0\rangle\langle 0|, \qquad U = \frac{1}{\sqrt{5}}\begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \qquad \Psi(A) = UAU^*.$$

Its response is $u_n = (1 + \cos(n\vartheta))/2$*, where* $e^{i\vartheta} = (3 + 4i)/5$*. The full two-outcome Hankel response has rank three, but no finite stochastic observer realizes it exactly.*

The embedding theorem therefore gives a compact uncertain-model squared-prediction family on protected dimension four whose unique full-history boundary optimum is randomized, whose least-favorable wordwise actions are all tied, and whose exact live-memory count is infinite. At each finite memory bound its optimal risk is strictly above 2*. A prefix of length* T*, followed by one tail label, has excess risk at most*

$$\frac{2\varepsilon p(q\bar{t})^T}{1 - q\bar{t}} \tag{S25.7}$$

and costs exactly $1/p$*.*

Proof. The response formula follows by the direct qubit rotation. Its three distinct exponential modes are $1, \zeta^n, \bar{\zeta}^n$, where $\zeta = (3 + 4i)/5$. Their nonzero coefficients and the Vandermonde determinant give Hankel rank three. The number ζ is not a root of unity: otherwise $\zeta + \bar{\zeta} = 6/5$ would be a rational algebraic integer and hence an integer.

If a finite stochastic matrix Q realized $u_n = e_1 Q^n d$, its generating function would be $e_1(I - zQ)^{-1}d$. But

$$\sum_{n\geq 0} u_n z^n = \frac{1}{2(1-z)} + \frac{1}{4(1-\zeta z)} + \frac{1}{4(1-\bar{\zeta}z)}.$$

The pole at $z = \zeta^{-1}$ forces ζ to be an eigenvalue of Q. Every unit-modulus eigenvalue of a finite row-stochastic matrix is a root of unity. To verify the needed fact, choose a complex eigenvector and a coordinate of maximal nonzero modulus. Equality in the stochastic triangle inequality forces all positive successors to have the same modulus and phase multiplied by that eigenvalue. Following positive edges reaches a directed cycle; the phase must return after its finite length. This gives a root of unity, a contradiction.

The embedding gives uniqueness of the optimal randomized kernel, and Proposition S24.8 gives strict attained gaps at fixed finite memory. Matching u_n for $n < T$ and assigning any tail probability changes each coefficient by at most one. Summing the tail in (S25.4) gives (S25.7). There is no change to first-herald stopping or its paid cost. □

Irrational quantum rotations and the separation of linear from positive stochastic realizations are established mechanisms, not new priority claims [48, 49]. The assertion here concerns their exact robust squared-prediction embedding, all-tied least-favorable actions, fixed-dictionary boundary optimality and real-valued stochastic-controller exclusion. It does not solve the sharp intermediate-memory risk curve for the different noisy-qubit sign task in Theorem S29.10.

S25.4. **Exact optimization at any prescribed finite memory.** The invariant-polytope criterion is not asserted effectively decidable in general. For a specified memory bound, however, the physical instrument gives a finite algebraic optimization without an unproved exact posterior algorithm.

Theorem S25.6 (Algebraic fixed-memory boundary optimization)**.** *Suppose* Θ *is a nonempty compact semialgebraic set over the real algebraic numbers, and the real Hermitian-coordinate entries*

of all prepared states, instruments, costs and loss observables are algebraic-polynomial functions of its parameters. Assume (S24.1) *and boundary rigidity. For each prescribed live bound M, the exact robust boundary optimum V_M is an attained real algebraic number and is computable by real-algebraic elimination. An attaining stochastic controller can be chosen with real algebraic entries.*

A completely primitive, decidable sufficient subclass for rigidity is $\mathcal{I}^*_{\theta,s} I \succ 0$ and $\mathcal{I}^*_{\theta,y} I \succ 0$ for every θ, y; compactness then also gives uniform first-step transience. These conditions allow model- and state-dependent report probabilities. Both squared-task embeddings above belong to this subclass.

For the same algebraic data, V is a computable real via finite-prefix games with the explicit $L\beta_T$ tail. Whenever $V_M > V$, finite computations eventually certify that strict inequality. No general terminating test for the equality $V_M = V$, no test that some finite M attains V, and no polynomial complexity for quantifier elimination is inferred.

Proof. Write the controller variables as row-stochastic Q_y and decoder probabilities $D_i(r)$, with initial label 1. For each model its exact score potentials are the unique Hermitian solution of

$$K_i = \mathcal{I}^*_{\theta,s}\left(\sum_r D_i(r) F_{\theta,r}\right) + \sum_y \mathcal{I}^*_{\theta,y}\left(\sum_j Q_y(i,j) K_j\right), \qquad 1 \le i \le M. \tag{S25.8}$$

Indeed the associated positive continuation map on block observables sends the identity tuple, at its nth power, to the tuple with common entry $(\mathcal{T}^*_\theta)^n I$. Its norm on Hermitian tuples is therefore at most β_n. Its Neumann series converges, giving uniqueness and $0 \preccurlyeq K_i \preccurlyeq LI$. The risk is exactly $\operatorname{tr}(\rho_\theta K_1)$. This proves the equation's meaning and boundedness from the raw instrument, not by assuming a certificate exists.

For a proposed upper v, impose stochastic row constraints and the statement

$$\forall \theta \in \Theta\ \exists (K_1, \dots, K_M): \quad \text{(S25.8)}, \quad 0 \preccurlyeq K_i \preccurlyeq LI, \quad \operatorname{tr}(\rho_\theta K_1) \le v.$$

In real coordinates these are polynomial equations and inequalities; positive semidefiniteness is expressed by principal minors. Projection and quantifier elimination over the real algebraic field produce the exact feasible value set [15]. Compact controller space and the uniform tail prove its lower endpoint is attained. A finite semialgebraic endpoint over that field is algebraic, and the nonempty optimal-controller fiber contains an algebraic point. This gives the value and a controller by a finite elimination procedure. The Md^2 modelwise potential coordinates are analytic variables in this computation, not runtime observer memory.

Positive definiteness of the primitive effects is a finite quantified algebraic condition. It implies positive mass at every word, and compactness bounds the herald effect below by a common positive multiple of I. Corollary S24.3 therefore gives rigidity. The general theorem's rigidity premise is not claimed algorithmically checked merely by writing the infinite-word criterion.

Finally, truncate the decision kernel to words shorter than T. The finite-prefix robust minimum is an algebraic quantified problem of the same kind, now with one decision row per word. Its value is below V and within $L\beta_T$ of it because any prefix can be completed by a first-herald tail decoder. A rational positive lower certificate for δ can be found by increasing-precision algebraic tests, so these are effective converging intervals. Once the exact algebraic value V_M is strictly above V, some finite upper interval endpoint for V is strictly below V_M. This certifies nonattainment at that fixed bound. In the equality case that search need not terminate. □

S26. A constructive finite-budget frontier from primitive instruments

The geometric transfer theorem identifies a geometric curve when its converse and construction certificates are available. We now give a different, intrinsic route for finite time-homogeneous experiments: derive a proper fallback from the raw instruments and compute converging bounds on the optimal task risk itself. Neither a reset state, a balancing group, a supplied reference member nor a posterior stored by the controller is assumed. The finite and all-initial-state qualifications

are essential. The qualitative invariant-subspace obstruction has a direct antecedent in quantum reachability [28, Theorem 3.6]; it is proved below to obtain the quantitative fallback required by the new budget-preserving approximation.

S26.1. **Primitive interface and universal termination.** The unknown model belongs to a nonempty finite set Θ. Model θ has physical space $\mathcal{H}_\theta$ of dimension d_θ. The nonempty finite command set $\mathcal{A}$ is globally legal. A command has completely positive report maps $\mathcal{I}_{\theta,a,y}$ with normalized sum. Reports range over finite sets, with a declared productive subset Y_a^+. After a productive report a controller may commit a decision from a fixed dictionary C or reject; after any rejected or nonproductive report the post-report state is the next state. No additional return or reset is available. An interface requiring extra physical return calls must first represent and charge those calls explicitly; the theorem is not applied by hiding them in an unpriced transition. Commitment occurs exactly once, before a future audit is observed. The audit loss for decision $r \in C$ is represented on the post-report state by

$$0 \preccurlyeq F_{\theta,a,y,r} \preccurlyeq LI, \qquad 0 < L < \infty.$$

Nonnegative call costs have observables $c_* I \preccurlyeq G_{\theta,a} \preccurlyeq c^* I$, with $0 < c_* \le c^* < \infty$. These are the precommit core costs. A final audit with one fixed separately charged cost is added to every total budget after the theorem; an action-dependent unmodelled terminal charge is not included by assertion. Policies share one set of command and decision rules across models and unknown initial density matrices. Let $R_C(N)$ be their least worst-model, worst-initial-state expected terminal loss at uniform expected cost at most N. Their precommit memory is unrestricted for this value; a finite-label implementation is derived, not assumed.

Fix any command distribution $\pi_a > 0$. The fallback chooses that row independently at every call and commits at the first productive report, using one fixed $r_0 \in C$. Its killed physical map is

$$\mathcal{T}_\theta = \sum_a \pi_a \sum_{y \notin Y_a^+} \mathcal{I}_{\theta,a,y}, \quad B_\theta = I - \mathcal{T}_\theta^* I, \quad O_{\theta,n} = I - (\mathcal{T}_\theta^*)^n I.$$

A common dark subspace in one model is a nonzero subspace annihilated by every productive effect and invariant under every Kraus operator of every nonproductive report map. This condition is independent of the Kraus representation: it says that every positive state on the subspace remains there on failure and has zero productive probability for every command.

Theorem S26.1 (Primitive fallback and its sharp depth test)**.** *The following are equivalent: there is one common policy proper at finite uniform expected cost from every model and initial density matrix; no model has a common dark subspace; and O_{θ,d_θ} is positive definite for every θ. In that case the displayed one-row fallback is proper. With*

$$\delta_\theta = \lambda_{\min}(O_{\theta,d_\theta}) > 0, \qquad \kappa = \max_\theta c^* d_\theta / \delta_\theta,$$

it costs at most κ from every current physical state in every model. One may replace this bound by the sharper certified value

$$\kappa = \max_\theta \left\| (I - \mathcal{T}_\theta^*)^{-1} \sum_a \pi_a G_{\theta,a} \right\|.$$

Only one live label and one output value are needed; these share a label in event stopping and require two labels in state stopping. Depth d cannot be replaced uniformly by $d-1$. There is no dimension-only upper bound on expected cost.

Proof. Work in one model. Put $K_n = \ker O_n$, so $K_0 = \mathcal{H}$ and $O_{n+1} = B + \mathcal{T}^* O_n$. For a Kraus representation $\mathcal{T}(A) = \sum_v L_v A L_v^*$, positivity gives

$$K_{n+1} = \ker B \cap \bigcap_v L_v^{-1} K_n.$$

The spaces decrease. Equality at two consecutive levels makes that space invariant under every L_v and annihilated by B, hence makes all later levels equal. There are at most d strict dimension drops, so K_d is stable. Since every π_a is positive, $\ker B$ is the intersection of the productive-effect kernels, and the L_v include nonzero multiples of every nonproductive Kraus operator. Thus a nonzero K_d is a common dark subspace. Conversely every such subspace lies in all K_n. Starting in it, no history policy can encounter a productive report: induction holds commandwise and reportwise. Such a policy cannot commit legally.

If $K_d = 0$, positivity yields $(\mathcal{T}^*)^d I \preccurlyeq (1-\delta)I$. Iteration bounds survival after kd fallback calls by $(1-\delta)^k$ for every normalized input. Summing blocks gives expected calls at most d/δ and expected cost at most c^*d/δ. The finite model set makes the maximum bound finite. The same estimate implies convergence of the positive cost series, whose sum is the stated resolvent; its largest eigenvalue is the exact uniform cost for the fixed row. The fallback does not read the model or physical state. Its single row and fixed decoder give the label counts.

For sharp depth, on basis $e_1, \dots, e_d$ take one nonproductive Kraus operator $L = \sum_{j=2}^{d} |e_{j-1}\rangle\langle e_j|$ and a productive effect $|e_1\rangle\langle e_1|$. Then O_n is the projection onto the first $\min(n, d)$ basis vectors. The input e_d requires exactly d unit-cost calls. For the lack of a cost bound, already in dimension one a productive probability $\epsilon > 0$ with unit costs has expected time $1/\epsilon$. This proves both limitations. □

The qualitative obstruction is inherited mathematical machinery, not a new discovery of invariant subspaces. Its purpose here is to supply an actual common fallback and its paid constant. Absence of an invariant dark subspace on a restricted preparation is a different problem; the all-initial-state hypothesis cannot be silently removed. Finiteness of Θ likewise cannot be replaced by modelwise termination in an arbitrary family without a uniform bound.

S26.2. **Finite problems and restoration of the exact budget.** For a horizon $T \geq 1$, use one common realization plan on the action–report tree, as in Lemma S31.1. At every productive report include either commitment to $r \in C$ or continuation. Let $C_{\theta,T}$, $S_{\theta,T}$ and $V_{\theta,T}$ be, respectively, its expected cost through the first T calls, its open effect after the depth-T decision, and its accumulated terminal-loss operator. They are affine functions of the common sequence weights, computed by composing the primitive maps. In particular $0 \preccurlyeq S_{\theta,T} \preccurlyeq I$ and $0 \preccurlyeq V_{\theta,T} \preccurlyeq L(I - S_{\theta,T})$.

Define the finite semidefinite value

$$q_{C,T}(N) = \min\left\{q : \; C_{\theta,T} + c_* S_{\theta,T} \preccurlyeq NI, \quad V_{\theta,T} \preccurlyeq qI \quad (\theta \in \Theta)\right\}. \tag{S26.1}$$

The minimum is over the common plan and $0 \leq q \leq L$. The reserve $c_* S$ is not a cutoff declaration of success: every open branch owes another paid call. For $N \geq \kappa$, prefixes of the actual fallback show feasibility. Compactness gives attainment. Set

$$D_T(N) = \min\{1, (N/c_* - 1)/T\}, \qquad e_T = (\kappa - c_*) D_T(N).$$

Theorem S26.2 (Same-budget constructive risk frontier)**.** *Under the preceding primitive conditions with no common dark subspace, the finite values are nondecreasing in T and satisfy $q_{C,T}(N) \leq R_C(N)$. For every $N > \kappa$,*

$$0 \leq R_C(N) - q_{C,T}(N) \leq L D_T(N) + \frac{L e_T}{N - \kappa + e_T}. \tag{S26.2}$$

The upper is realized by a common proper finite-label observer with expected cost at most the original N in every model and from every initial state. Thus $q_{C,T}(N) \uparrow R_C(N)$ with an explicit $O(T^{-1})$ bound at each fixed $N > \kappa$. No reset or scalar posterior dynamics are required.

Writing $a = |\mathcal{A}|$ and y for the largest report count, a sufficient live-label bound is

$$2 + \sum_{t=0}^{T-1} (ay)^t.$$

The event-stopped total is at most the maximum of this bound and $|C|$; the state-stopped total is at most their sum. These are existence bounds, not optimal memory, running-time or program-bit bounds.

At $N = \kappa$ the completed finite plan still gives cost at most $N + e_T$ and loss at most $q_{C,T}(N) + LD_T(N)$. The same-budget claim at that boundary is not inferred from the displayed fraction. If $\kappa = c_*$, no budget excess is present and the same-budget bound $LD_T(N)$ holds for every $N \geq \kappa$.

Proof. A genuine cost-N policy restricted to depth T satisfies the reserved cost inequality, because after every open path it must make at least one further call. Its truncated terminal loss is no greater than its full nonnegative loss. This proves the lower. A feasible plan at $T + 1$ restricts to one at T: the new calls cost at least $c_* S_T$, new terminations reduce open mass, and their loss contributions are nonnegative. The reserve and risk inequalities therefore survive restriction, proving monotonicity.

For any feasible prefix, normalization and the conditional cost floor give

$$C_T \succeq c_* \sum_{t=0}^{T-1} S_t \succeq c_*[I + (T-1)S_T], \qquad S_0 = I.$$

Here each difference of successive open effects is positive. With the reserve this gives $S_T \preccurlyeq D_T(N)I$ in every model. Complete an optimizing prefix by executing the actual fallback at every open leaf. The fallback bound holds at every post-report state, including states never reached by a reference execution. Its cost and loss satisfy

$$C^{\mathrm{comp}} \preccurlyeq C_T + \kappa S_T \preccurlyeq (N + e_T)I, \qquad V^{\mathrm{comp}} \preccurlyeq V_T + LS_T \preccurlyeq (q_{C,T} + LD_T)I.$$

It is proper because the finite prefix is followed by a proper fallback. There is no assertion that the original policy's expected remaining cost is uniformly small.

For $N > \kappa$, choose the completed policy with probability $p = (N - \kappa)/(N + e_T - \kappa)$ and the fallback with probability $1 - p$, storing the branch in the ensuing live role. Its cost is at most $p(N+e_T)+(1-p)\kappa = N$. Its risk is at most $p(q_{C,T}+LD_T)+(1-p)L \leq q_{C,T}+LD_T+Le_T/(N-\kappa+e_T)$. If $e_T = 0$ choose the completed policy directly. This proves the upper and convergence. The boundary statements follow from the unmixed completion.

A prefix stores its current action–report node, at most $\sum_{t<T}(ay)^t$ live roles; its current report is within-call input. One fallback role and one initial mixture role suffice. At that initial call, sample the branch and its command jointly; after the report either commit or enter the selected branch's counted next role. If both branches chose the same command, their conditional branch weights are proportional to their command probabilities, independent of the physical state, since the executed report instrument is identical. Equivalently use the joint within-call coin directly. No branch bit survives outside the retained label. Commitment uses the same fixed dictionary in both branches. Lemma 17.4 gives the two total counts. □

Corollary S26.3 (Compact decisions at a fixed output width)**.** *Suppose the decision carrier D is compact metric and the raw audit losses obey the common decision modulus ω of* (S33.1)*. Fix width m. For a finite η-net $D_\eta \subset D$, let $q_{m,T,\eta}(N)$ be the least value in* (S26.1) *among dictionaries in D_η of size at most m. Then, for $N > \kappa$,*

$$[q_{m,T,\eta}(N) - \omega(\eta)]_+ \leq R_m(N) \leq q_{m,T,\eta}(N) + LD_T(N) + \frac{Le_T}{N - \kappa + e_T}.$$

The constructed observer uses at most m terminal values. As $T \to \infty$ and $\eta \downarrow 0$ this gives a two-parameter constructive characterization of the unrestricted-history finite-output risk, not just a verifier for a supplied balancing program. With algebraic primitive data and polynomial score dependence on algebraic net points, every finite test is a finite real-algebraic problem; exact real samplers and program arithmetic remain separate implementation resources.

Proof. Round each point of any competing dictionary to the same chosen nearest net point at every history. Neither stopping nor cost changes, and at most m values remain. Its expected loss changes by at most $\omega(\eta)$, because terminal events have total mass one. Restrict this rounded program to a finite prefix to obtain the lower. For the upper, choose an attaining net dictionary and apply Theorem S26.2; the fallback uses a point of this same dictionary. There are finitely many such dictionaries, so the optimization is finite. With the stated algebraic inputs all coefficients of the finite plan are algebraic, and semidefinite inequalities can be expressed by principal minors. This gives decidability via the already credited real-algebraic method, not an efficient uniform algorithm for arbitrary real input data. □

S26.3. **Removing the crude fallback threshold on every interior budget.** The cost of an arbitrary positive command row may greatly exceed the least feasible cost. It is not an intrinsic resource threshold. The following preliminary optimization removes that restriction without assuming the optimal policy is already known.

Let κ_0 be the uniform cost bound of the one-row fallback and let H_θ be its exact cost operator, so $0 \preccurlyeq H_\theta \preccurlyeq \kappa_0 I$. Ignore terminal loss for this preliminary step and commit one fixed dummy decision. Write $N_{\min}$ for the infimum of uniform expected cost among proper common policies. For a finite prefix define

$$W_{\theta,T}(H_\theta) = \sum_{h:\,|h|=T} \gamma_{\mathrm{open}}(h)\, \mathcal{I}^*_{\theta,h}(H_\theta).$$

Here γ_{open} includes the decision to continue after the depth-T report, and $\mathcal{I}_{\theta,h}$ is its raw path map. This is the exact cost of executing the fallback at the open leaves. Both W and the earlier C_T, S_T are affine in the one common realization plan. Set

$$a_T = \min\{a : C_{\theta,T} + c_* S_{\theta,T} \preccurlyeq aI \quad (\theta \in \Theta)\},$$
$$u_T = \min\{u : C_{\theta,T} + W_{\theta,T}(H_\theta) \preccurlyeq uI \quad (\theta \in \Theta)\}.$$

Theorem S26.4 (Constructive feasibility threshold and all interior budgets)**.** *Both finite problems attain their values, and*

$$c_* \le a_T \uparrow N_{\min}, \qquad \kappa_0 \ge u_T \downarrow N_{\min},$$

with the quantitative enclosure

$$0 \le u_T - a_T \le (\kappa_0 - c_*) \min\{1, (\kappa_0/c_* - 1)/T\}. \tag{S26.3}$$

The least common proper cost $N_{\min}$ is attained, though no finite-memory attainment is concluded at that boundary.

For every $N > N_{\min}$ a finite search through these upper problems eventually exhibits $u_{T_0} < N$. Its optimizing prefix completed by the one-row fallback is an actual finite-label fallback of cost $\kappa = u_{T_0} < N$, valid from every initial state and model. Using it in Theorem S26.2 gives the same-budget risk enclosure for every strictly interior feasible budget*, not only budgets above the original one-row cost. If this improved fallback has B live labels, the sufficient live bound becomes*

$$1 + B + \sum_{t=0}^{T-1} (ay)^t.$$

All decisions may still use one prescribed dictionary; the cost search is independent of its values. Algebraic primitive data give finite real-algebraic upper tests, but no uniform efficient bit-complexity claim is made.

Proof. The prefix of a genuine proper policy satisfies the one-call reserve, so $a_T \le N_{\min}$. Restriction to shorter prefixes proves a_T is nondecreasing. Every prefix completed by the fallback is proper and has exactly the cost $C_T + W_T(H)$; hence $u_T \ge N_{\min}$. Prefixes of the fallback itself give $u_T \le \kappa_0$. Extending an optimizing upper prefix by one fallback call reproduces its exact completed cost, by

the identity $H_\theta = g_\theta + \mathcal{T}_\theta^* H_\theta$ and first-productive commitment. Therefore $u_{T+1} \le u_T$. Compactness of the finite plan space and continuity of its cost operators prove both attainments.

For a lower optimizer, the reserve proof gives $S_{\theta,T} \preccurlyeq \min\{1, (a_T/c_* - 1)/T\} I$. Positivity and $H_\theta \preccurlyeq \kappa_0 I$ give $W_{\theta,T}(H_\theta) \preccurlyeq \kappa_0 S_{\theta,T}$. Complete that very prefix. Its cost is at most

$$a_T + (\kappa_0 - c_*) \min\{1, (a_T/c_* - 1)/T\}.$$

Since $a_T \le N_{\min} \le \kappa_0$, this proves (S26.3). The squeeze proves both limits. For boundary attainment, the compact product of all finite nodewise decision simplexes has the finite-intersection property for the closed prefix cost constraints at budget $N_{\min}$: feasibility at the largest of finitely many horizons follows from $a_T \le N_{\min}$. A common infinite policy therefore obeys all reserved constraints. Their survival bound tends to zero, and monotone convergence gives properness and cost at most $N_{\min}$. This is the same compactness mechanism as the inherited synthesis theorem, not a finite-memory assertion.

If $N > N_{\min}$, convergence of u_T implies that the strict inequality $u_{T_0} < N$ holds at some finite level. An upper optimizer is itself the finite prefix and a physically executable fallback, not a model-dependent analytic certificate without a common policy. Its start-state uniformity permits it at every later open leaf of a time-homogeneous experiment. Substitute its cost and its counted live labels in the completion and initial-mixture proof; every inequality remains unchanged, except that its B live roles replace the one fallback row. A fixed dummy decision may be replaced by any point of the eventual task dictionary without changing precommit calls or cost. Exact algebraic tests decide a strictly feasible upper threshold when the coefficients and proposed budget have the stated algebraic representation. General real data still require an exact oracle or further numerical certification. □

For example, two orthogonal nonzero failure projections have one-row fair fallback cost $\kappa_0 = 2$, but fair-first alternation has uniform cost $3/2$ and stops by call two. The cost upper problem at depth two therefore supplies a fallback below every budget $N > 3/2$, including $3/2 < N < 2$. Thus the preliminary optimization removes a genuine restriction of the crude fallback, rather than only renaming its constant.

S26.4. **Robustness and scope of the construction.** If two fallback killed maps on the same space satisfy $\|\mathcal{T}^* - \widetilde{\mathcal{T}}^*\|_{\infty\to\infty} \le \epsilon$ on Hermitian matrices, telescoping powers gives $\|O_d - \widetilde{O}_d\| \le d\epsilon$: each positive subunital adjoint is a contraction in this norm. Hence $\delta - d\epsilon > 0$ certifies a perturbed proper fallback and cost at most $c^* d/(\delta - d\epsilon)$, when the perturbed calls obey the same cost upper bound. This is a quantitative stability certificate for the fallback, not a claim that a terminal-law distance controls unbounded cost. The sharp ladder concerns depth; the scalar rare-success example shows why a spectral margin must enter any numerical certificate.

This section treats finite known candidate models, not statistical learning of an unrestricted unknown kernel. It constructs a common policy without a lower witness, but does not prove that its risk is always the geometric profile $\mathcal{Q}$. The geometric transfer and this primitive synthesis are complementary routes: one gives interpretable sharp resource laws under its certificates, the other supplies a convergent constrained optimum and finite implementation on its finite interface. Quantum reachability, sequence form, potential arguments and semidefinite separation are established ingredients; the present assertion is their explicit paid, same-budget, fixed-output synthesis with a derived fallback and all-model/all-initial-state quantifiers.

S27. Boundary attainment and finite optimality witnesses

An upper construction and a lower inequality do not by themselves characterize which strategies are optimal. We now close the constrained risk frontier at its feasible boundary, derive its exact variational dual at a fixed dictionary, and identify finite witnesses for each compact-model prefix problem. Cost and task witnesses are allowed to be different physical preparations in different

models. Their common-program Bellman certificate is necessary and sufficient at the finite level, and, together with the preceding paid completion, certifies arbitrarily accurate strategies at every interior budget.

The convex-analytic ingredients are classical: finite-support semi-infinite duality is developed in [33, 34], and compactness of strategic and occupation measures has an extensive control-theoretic history [11, 35]. We prove the needed statements in this experiment's coordinates. We do not claim a new general separation or Carathéodory theorem. The distinction between a fixed dictionary and a hard bound on the number of freely chosen decoder values is essential: the latter can destroy convexity even in physical dimension one.

S27.1. **The exact frontier, including its left endpoint.** For this subsection only, the model family need not be compact, time homogeneous, or finite-dimensional. It suffices that all models share a countable, finitely branching action–report tree with nonempty finite legal action sets at open nodes. Costs have finite conditional means at each finite history, are nonnegative, and have the same positive conditional mean floor c_*. Terminal losses lie in $[0, L]$. Initial preparations may be absorbed into the model index. One common observer must commit exactly once and satisfy its expected-cost bound in every model. The decision set D is compact metric and the terminal losses have a common decision modulus. These conditions are satisfied by the primitive interfaces above; they specify the wider scope used here without assuming a free posterior or a common physical reset.

Use the common sequence weights of Lemma S31.1. A fixed ordered decoder tuple C may have repeated entries. Let $\mathcal{P}_N(C)$ denote the common infinite plans of uniform expected cost at most N. Let $q_{C,T}(N)$ be the infimum of worst truncated terminal loss over common depth-T plans with the one-call reserve $C_{\theta,T} + c_* S_{\theta,T} \le N$ in every model and preparation. For an operator interface this inequality means the corresponding Loewner inequality. Define $q_{m,T}$ by optimizing the decoder tuple in D^m, and likewise define R_m. A finite infeasible problem has value $+\infty$.

Theorem S27.1 (Closed constrained frontier and boundary convergence)**.** *Every nonempty $\mathcal{P}_N(C)$ is compact in common-plan coordinates. Terminal-risk functions converge uniformly to their depth-T truncations on this set, with error at most*

$$L \min\{1, (N/c_* - 1)/T\} \qquad (T \ge 1).$$

Thus the optimal risk is attained at every feasible finite budget, including the least feasible budget when one exists. The same assertion holds with a compact decoder tuple of prescribed width m.

A budget is feasible if and only if all its finite reserved problems are feasible. At every feasible N,

$$q_{C,T}(N) \uparrow R_C(N), \qquad q_{m,T}(N) \uparrow R_m(N). \tag{S27.1}$$

Both risk functions are nonincreasing and right-continuous on their feasible budget ray. For a fixed dictionary, R_C is convex and hence continuous at every interior feasible budget. No convexity of R_m is asserted. Boundary convergence gives neither a computable rate nor a finite-memory optimum at the least budget.

Proof. The infinite common-plan space is a closed subset of the countable product of the intervals $[0, 1]$, specified by the common flow equalities. It is compact and metrizable. Every prefix cost, survival mass and truncated terminal loss is continuous in its finitely many sequence coordinates for each model. Intersecting all prefix reserve constraints over every model and every depth gives a closed set.

On that intersection, the cost floor and monotonicity of survival give $C_T + c_* S_T \ge c_*[1 + T S_T]$. Hence the displayed uniform survival bound holds, and survival tends to zero. Nonnegative prefix costs increase to the actual cost; their bounds imply a proper common plan of cost at most N. Conversely a proper cost-N plan satisfies every reserve because each open path owes a further paid

call. This identifies the closed intersection with $\mathcal{P}_N(C)$, without assuming continuity of total cost or a uniform expected remaining-cost tail.

The nonnegative omitted terminal loss is at most L times survival. A finite prefix loss is uniformly continuous in its finite plan coordinates, uniformly over models: its coefficients are bounded by L, and there are finitely many terminal paths. For a decoder tuple, the common decision modulus supplies the additional uniform continuity. Thus the full worst-model risk is a uniform limit of continuous functions on the compact cost-bounded set. It attains its minimum. The product with D^m is compact as well; output indices, not unknown model-dependent values, are the terminal choices.

A later feasible prefix restricts to an earlier one while preserving its reserve and decreasing its accumulated nonnegative loss. Thus the finite values are nondecreasing and bounded above by the actual value. Conversely choose optimizers at depths tending to infinity, extend them by arbitrary legal rows, and take a coordinatewise convergent subsequence, also in D^m when needed. At each fixed earlier depth the limiting reserve and terminal-loss constraints hold in every model. The limit is proper by the preceding argument, and its terminal risk is at most the increasing limit of the finite values. This proves (S27.1). The same compact finite-intersection argument without the loss constraints proves the feasibility equivalence. Choosing a sequence of feasible budgets decreasing to their infimum also proves attainment of the least finite feasible cost.

For right-continuity, use optimizers at budgets $N_j \downarrow N$, all bounded by $N + 1$, and take a convergent subsequence in that compact cost-bounded set. Its limit is feasible at N, and uniform terminal-tail control identifies the limiting risk. Monotonicity supplies the reverse inequality. Finally, two common plans with the same dictionary may be mixed in sequence coordinates. Their costs and losses mix linearly before taking worst models, so the worst cost and risk mix sublinearly. This proves convexity of R_C. Such mixing is a common behavioral policy by Lemma S31.1; a finite implementation must count its live branch roles as in the supplement. Mixing two optimized width-m dictionaries can use $2m$ values, so this argument does not prove convexity of R_m. □

For a fixed dictionary, write $\mathfrak{C}(P)$ and $\mathfrak{V}(P)$ for a plan's worst expected cost and worst terminal risk, and let $\mathcal{P}_{\mathrm{fin}}$ contain the proper common plans of finite uniform cost. The following is a scalar resource dual, not an interchange of a physical model and an observer.

Corollary S27.2 (Exact resource-price dual)**.** *If some finite budget is feasible, then for every* $N \geq N_{\min}$*,*

$$R_C(N) = \sup_{\lambda \geq 0} \inf_{P \in \mathcal{P}_{\mathrm{fin}}} \{\mathfrak{V}(P) + \lambda[\mathfrak{C}(P) - N]\}. \tag{S27.2}$$

At $N > N_{\min}$ *the supremum is attained. If a common plan of cost at most* $\kappa < N$ *is known, an attaining multiplier satisfies* $0 \leq \lambda \leq L/(N - \kappa)$*. Every primal optimum then minimizes the penalized expression and obeys* $\lambda[\mathfrak{C}(P) - N] = 0$*. At* $N = N_{\min}$ *the supremum remains exact but need not be attained by a finite multiplier. These statements concern one fixed dictionary, not its nonconvex width-constrained union.*

Proof. Weak duality follows by testing any feasible plan. On the interior the finite convex nonincreasing function R_C has a supporting line of slope $-\lambda \leq 0$ at N. For every finite-cost plan of cost n, its risk is at least $R_C(n)$, which is at least $R_C(N) - \lambda(n - N)$. A primal optimum gives the reverse bound and complementary slackness. Applying the supporting inequality at κ gives $\lambda(N - \kappa) \leq R_C(\kappa) - R_C(N) \leq L$. Supporting lines exist because the left and right secant slopes of a finite convex function are ordered and locally bounded. At the left endpoint choose $N_j \downarrow N_{\min}$ from the interior. The attaining interior multiplier gives a dual value at $N_{\min}$ at least $R_C(N_j)$. Right-continuity and weak duality prove endpoint equality without a bounded-multiplier assumption. □

S27.2. **A complete decision comparison of performed stopped laws.** Fix the actual decoder dictionary C, rather than only its cardinality. Let Ω be the measurable disjoint union of terminal action–report words, committed values in C, and a prospective audit outcome in a common standard Borel space. The audit kernel after each such word and commitment is fixed in each model. Write P_θ^P for the complete performed stopped law of a legal plan P. It includes the stopping call and the raw visible word, not just the final audit marginal. For a specified target family of probability laws T_θ on this same space define

$$\delta_{N,C}(\mathsf{T}) = \inf_{P\in\mathcal{P}_N(C)} \sup_\theta \mathrm{TV}(\mathsf{P}_\theta^P, \mathsf{T}_\theta), \qquad \mathrm{TV}(\mu,\nu) = \sup_A |\mu(A)-\nu(A)|.$$

The target is data, not an extra executable oracle. Its feasibility in the source interface is the question being tested.

Theorem S27.3 (Paid complete-score randomization criterion)**.** *Under the common finite-branching and positive-cost hypotheses of Theorem S27.1, at any feasible finite N, the set of families $\{\mathsf{P}_\theta^P\}_\theta$ is compact and convex in the uniform total-variation metric. Hence the infimum defining $\delta_{N,C}$ is attained, and*

$$\delta_{N,C}(\mathsf{T}) = \sup_{\pi,g}\left[\min_{P\in\mathcal{P}_N(C)} \sum_{\theta\in\operatorname{supp}\pi} \pi_\theta \mathbb{E}_{\mathsf{P}_\theta^P} g_\theta - \sum_{\theta\in\operatorname{supp}\pi} \pi_\theta \mathbb{E}_{\mathsf{T}_\theta} g_\theta\right]. \tag{S27.3}$$

Here π ranges over finitely supported model probabilities and each $g_\theta : \Omega \to [0,1]$ is a finite-valued measurable score. The inner policies remain budget-feasible in every *model, not only the support of π. In particular exact common realization of the target at the prescribed budget and dictionary is equivalent to its dominating the optimal source score in every such finite decision test.*

On the compact primitive interface, let one common finite-label fallback have conditional cost at most $\kappa < N$. For every $T \geq 1$ and every $P \in \mathcal{P}_N(C)$ there is a common finite-label $P_T \in \mathcal{P}_N(C)$ satisfying

$$\sup_\theta \mathrm{TV}(\mathsf{P}_\theta^{P_T}, \mathsf{P}_\theta^P) \leq D_T + \frac{e_T}{N-\kappa+e_T}, \qquad D_T = \min\{1, (N/c_* - 1)/T\}, \quad e_T = (\kappa - c_*)D_T. \tag{S27.4}$$

The prior finite-prefix and fallback live-label bounds apply. Therefore finite-label laws are dense at the same budget at every strictly interior feasible budget, and finite-score lower witnesses and actual finite-label realizations approximate the exact deficiency. This is a comparison of the declared performed laws, not of every quantum channel with arbitrary ancillary experiments.

Proof. At depth at most T there are finitely many terminal action–report words and output indices. Their contributions to a stopped law are fixed physical audit measures multiplied by common sequence weights. For two plans, the total-variation difference of these finite contributions is at most the sum of the absolute differences of those weights, uniformly over models, since each physical measure has mass at most one. The two omitted positive tails have mass at most D_T each on the budget set. First choose T large and then make the finitely many coordinates close. This proves continuity of the law-family map into uniform total variation. Compactness of $\mathcal{P}_N(C)$ gives compactness of its image; linearity in sequence weights gives convexity. Distance to any fixed target family is a continuous function in this metric, so its minimum is attained. This argument does not need continuity of total expected cost or uniform integrability of its tail.

For equal-mass probability measures, $\mathrm{TV}(\mu,\nu) = \sup_A[\mu(A)-\nu(A)]$, since complements reverse the sign. Thus each test functional $F_{\theta,A}(P) = \mathsf{P}_\theta^P(A) - \mathsf{T}_\theta(A)$ is continuous and affine on the compact convex plan set. The right side of (S27.3) is at most the deficiency by the elementary minimax inequality. If $r < \delta_{N,C}$, the closed sets $\{P : F_{\theta,A}(P) \leq r\}$ have empty total intersection. Compactness gives a finite subcollection with empty intersection. Its attainable test vectors form a compact convex set disjoint from the lower orthant $(-\infty, r]^q$. Strict separation gives nonnegative weights summing to one, with the weighted test value greater than r for every feasible plan. Nonnegativity

follows because a negative coefficient would make the supremum on that lower orthant infinite. Grouping tests with the same model yields a finitely supported π and a convex combination of finitely many indicators as g_θ. The latter is finite-valued. This proves the reverse inequality by letting r increase to the deficiency. Zero deficiency implies exact realization by attainment, not only approximation.

For (S27.4), keep P unchanged through the depth-T commitment decision, and execute the real fallback on its open paths. Coupling the original and completed experiments with the same calls and current coins until then makes their complete stopped laws identical unless the original path is still open. The total-variation discrepancy is therefore at most D_T, including all raw history and audit coordinates. The completed cost is at most $N + e_T$, because its reserved prefix already paid c_* for each open path. Mix this completion with the common fallback with probability $p = (N - \kappa)/(N - \kappa + e_T)$. The expected cost is at most N in every model. Mixing changes a probability law by at most $1 - p$ in total variation. The triangle inequality proves the displayed bound. Its finite implementation counts the actual finite tree, the fallback and the initial branch role; the same dictionary is used throughout. The compact-family cost theorem provides such a fallback at every $N > N_{\min}$. Applying this approximation to an attaining deficiency minimizer proves the final assertion. □

The decision-separation principle belongs to the Blackwell–Le Cam tradition [1, 37]. The additional work here is the compactness of the *performed* law family under a uniform paid stopping constraint, together with finite-label approximation without increasing that constraint. It requires all bounded measurable scores of that performed outcome; localized affine-square tests alone do not suffice. The conclusion does not grant a free stateful simulator, an uncharged audit on rejected calls, or control of unbounded expected duration by total variation. The latter is controlled by the separate budget throughout. If the dictionary itself varies, laws recording different numerical readouts need not vary continuously in total variation; the general-loss width theorem and the fixed-dictionary comparison are distinct statements.

S27.3. **Sparse spectral certificates on compact uncertainty.** Return to the continuous compact finite-dimensional interface of Section S32. Fix a dictionary and horizon. Let $\mathcal{X}_T$ be the common finite sequence-plan polytope and let k_T be its affine dimension, after eliminating flow equalities. The same statement applies to a common tree with compulsory terminal nodes, or to a prescribed face of its plan polytope; k_T then denotes the affine dimension of that polytope. For $z = (\theta, u)$ with $\theta \in \Theta$ and $\|u\| = 1$, put

$$c_z(x) = \langle u, (C_{\theta,T}(x) + c_* S_{\theta,T}(x))u\rangle, \qquad v_z(x) = \langle u, V_{\theta,T}(x)u\rangle.$$

These are continuous families of affine functions of x; $0 \le v_z(x) \le L$. Testing all such z is equivalent to the operator constraints. Write $q_T(N)$ for the resulting compact-model finite optimum.

Theorem S27.4 (Exact finite witness and complementary slackness)**.** *Suppose some $x_0 \in \mathcal{X}_T$ satisfies $c_z(x_0) \le \kappa < N$ for every z. Then there are risk atoms z_j, cost atoms w_i, and nonnegative weights β_j, α_i such that*

$$\sum_j \beta_j = 1, \qquad \lambda := \sum_i \alpha_i \le \frac{L}{N - \kappa}, \qquad \#\{j : \beta_j > 0\} + \#\{i : \alpha_i > 0\} \le k_T + 1, \tag{S27.5}$$

$$q_T(N) = \min_{x \in \mathcal{X}_T} \left\{ \sum_j \beta_j v_{z_j}(x) + \sum_i \alpha_i [c_{w_i}(x) - N] \right\}. \tag{S27.6}$$

For any fixed optimal plan x_* these atoms can be chosen active there:

$$\beta_j > 0 \Rightarrow v_{z_j}(x_*) = q_T(N), \qquad \alpha_i > 0 \Rightarrow c_{w_i}(x_*) = N.$$

Their preparations are pure; active vectors are respectively eigenvectors of the loss operator at q_T and the reserved-cost operator at N. Conversely, a feasible plan, such active atoms and minimization of the displayed affine expression certify optimality. The total atom bound is sharp for each affine dimension k_T. It concerns dual witnesses, not the number of controller labels or a fixed horizon-independent adversarial prior.

Proof. Set $v(x) = \max_z v_z(x)$ and $c(x) = \max_z c_z(x)$. Compactness gives attained maxima. The finite value as a function of its budget is convex and nonincreasing. A supporting line at N, which is interior because $\kappa < N$, yields a multiplier $\lambda \geq 0$ for which x_* minimizes $v + \lambda(c - N)$ and $\lambda(c(x_*) - N) = 0$, exactly as in Corollary S27.2.

For completeness, the finite polytope argument underlying its stationarity is proved here. The directional derivative of a compact maximum of affine functions is the maximum of their active gradients: take a maximizing index along a sequence of positive steps, pass to a convergent subsequence, and compare with every active index. Let A be the compact convex sum of the active risk-gradient hull and λ times the active cost-gradient hull; omit the second term if $\lambda = 0$. If A did not intersect the negative normal cone of $\mathcal{X}_T$ at x_*, separation of a compact convex set and a closed convex cone would give a tangent direction h with $\max_{a \in A} a \cdot h < 0$. For a polytope, every vector in this tangent cone gives a feasible sufficiently short step: impose its finitely many active linear inequalities, then choose the step short enough for the inactive ones. The directional derivative would be negative, a contradiction. Thus there are active finite convex combinations whose sum g satisfies $g \cdot (x - x_*) \geq 0$ for all $x \in \mathcal{X}_T$. Convex combinations in a finite-dimensional compact hull can be chosen finite by eliminating affine dependencies.

In the k_T-dimensional affine coordinates, append one coordinate to the active gradients. Represent $(g, 1)$ as a conic sum of $(\nabla v_z, 1)$ for risk atoms and $(\nabla c_w, 0)$ for cost atoms. If more than $k_T + 1$ generators have positive coefficients, a linear dependence permits subtracting a suitable multiple of that dependence until one coefficient vanishes and none becomes negative. Repeat. This preserves g and risk mass one, and all remaining atoms stay active. Their affine combination is minimized at x_*. Its value there is q_T, proving (S27.6). Evaluating that minimum at x_0 gives $q_T \leq L - \lambda(N - \kappa)$, hence the bound in (S27.5), even if compression changed the original multiplier.

For sufficiency, at any feasible x the expression in braces is at most $v(x)$, while at the specified active minimizing plan it equals its risk. This sandwiches the optimum. If a positive semidefinite matrix $NI - C$ has zero quadratic form at a unit vector, it annihilates that vector; the spectral claims follow, and similarly for risk. Finally take a compulsory one-call scalar experiment with $k + 1$ actions and $k + 1$ models, unit cost, one decoder and loss 1 exactly when action and model indices coincide. With commitment required after that call, its plan polytope is the k-simplex and its risk is $\max_i x_i$. At any $N > 1$, cost atoms cannot be active. The value $1/(k + 1)$ requires the risk weights all equal $1/(k + 1)$, since the dual minimum is their smallest weight. Every one of the $k + 1$ atoms is necessary. This proves sharpness within physical experiments, not just an abstract cone. □

Finite-support duality and the conic reduction are classical [34, Theorem 4.4]. The operational content is their use with the actual common plan, the open-path reserve, distinct spectral cost and score preparations, and a completion at the same resource budget. An arbitrary coarse grid is not a replacement for this certificate: active models may lie in the interior of a parameter set, as Example S32.10 shows.

Proposition S27.5 (Common Bellman evaluation and equality of used rows)**.** *For any finite atom lists with risk mass one, the right side of* (S27.6) *is evaluated exactly by a backward recursion on the single action–report tree. At each productive report it minimizes between a decision in the fixed dictionary and continuation; nonproductive reports must continue. The open depth-T cost is the actual one-call reserve. There is a deterministic finite plan minimizing this dual expression, but it need not satisfy the original robust budget and risk constraints. A randomized plan minimizes it if and only if every choice of positive sequence weight has zero backward slack at its reached node.*

Proof. Propagate, for each atom, its unnormalized physical state along each raw action–report word, without multiplying by controller probabilities. At a pre-call word h, the immediate cost for action a is $\sum_i \alpha_i \operatorname{tr}(G_{w_i,a}\rho_{w_i,h})$. At its productive report y, stopping with r contributes $\sum_j \beta_j \operatorname{tr}(F_{z_j,a,y,r}\rho_{z_j,hay})$. Continuation contributes the next node's recursively computed value. At an open leaf of depth T, assign $c_* \sum_i \alpha_i \operatorname{tr}\rho_{w_i,h}$. Add over reports and minimize over the common action. Subtract λN once at the root. This evaluates the desired affine functional by backward induction; the full formula is recorded in the supplement.

Each minimization admits a deterministic choice because its alternatives are finite. Keeping the chosen current command or report never reveals the model: the propagated matrices are offline analytic coefficients for a common tree. At every row, its randomized value minus its minimum is a nonnegative slack. Backward substitution expresses the root excess as the sum of those slacks times the corresponding nonnegative sequence weights. It vanishes precisely when every positively used alternative has zero slack. A dual-minimizing plan alone may violate the individual model constraints; Theorem S27.4 requires their separate primal verification. □

Corollary S27.6 (Finite certificates for the full stopped optimum)**.** *Fix $N > N_{\min}$ and a dictionary C on the compact primitive interface. For every $\eta > 0$ there is a finite horizon, at most $k_T + 1$ spectral witness atoms, and a common proper finite-label observer at cost at most N such that the Bellman-evaluated dual lower and that observer's worst risk differ by at most η. The same actual dictionary is used at every history and by the fallback. With compact optimized output width, attainment still holds, but this dual is conditional on the dictionary; global certification requires the separate dictionary optimization or a certified decision net. No minimax swap over dictionaries is authorized.*

Proof. The compact-family cost hierarchy supplies a finite fallback of uniform cost $\kappa < N$. Its prefix is strictly feasible for every reserved finite problem. Apply Theorem S27.4. The compact-model problem imposes all model constraints, so completion and initial mixing introduce no model-cover error. With $D_T = \min\{1, (N/c_* - 1)/T\}$ and $e_T = (\kappa - c_*)D_T$, its realized risk is at most $q_T + LD_T + Le_T/(N - \kappa + e_T)$. This difference tends to zero, while every dual value lower-bounds every feasible full-history strategy. Choose T to make the difference at most η. The prior completion proof counts the finite prefix, fallback and initial-mixture roles. For variable dictionaries use the preceding decision-net or algebraic optimization, not convexification of their union. □

Theorem S27.7 (Simultaneous expected resources)**.** *The compact-model certificate and completion extend to finitely many additive nonnegative resource costs. Write $C^{(\ell)}_{\theta,T}$ for the prefix cost of resource $\ell = 0, \dots, r$, assume conditional lower and upper bounds $0 \le c_\ell I \preccurlyeq G^{(\ell)}_{\theta,a} \preccurlyeq c^*_\ell I$, and require $c_0 > 0$. Fix a budget vector $\mathbf{N}$ and one common finite fallback with uniform resource bounds $\kappa_\ell < N_\ell$ simultaneously. Each finite constraint is*

$$C^{(\ell)}_{\theta,T} + c_\ell S_{\theta,T} \preccurlyeq N_\ell I.$$

Every finite optimal plan admits an exact certificate with risk mass one and cost atoms indexed by a resource, a model and a pure preparation. If λ_ℓ is the total multiplier mass for resource ℓ, then

$$\sum_{\ell=0}^{r} \lambda_\ell (N_\ell - \kappa_\ell) \le L, \qquad \text{total positive cost and risk atoms} \le k_T + 1.$$

Thus the support bound depends on the affine plan dimension, not directly on the number of resource constraints. The common Bellman recursion adds all weighted immediate resource costs; its reserve sums c_ℓ times the weighted open mass of the cost atoms for resource ℓ.

Let $q_T(\mathbf{N})$ denote the finite lower and put

$$D_T = \min\{1, (N_0/c_0 - 1)/T\}, \quad e_{\ell,T} = (\kappa_\ell - c_\ell)D_T, \quad p_T = \min_{0 \le \ell \le r} \frac{N_\ell - \kappa_\ell}{N_\ell - \kappa_\ell + e_{\ell,T}}.$$

There is a common proper finite-label observer satisfying every original budget and having worst loss at most

$$q_T(\mathbf{N}) + LD_T + L(1 - p_T).$$

The difference tends to zero. At any feasible finite vector, primal attainment and monotone finite lower convergence also hold, without asserting a finite boundary implementation. The costs may represent separately charged calls, time, energy or communication, when they obey these additive hypotheses; a hard program-size, scratch-space or dictionary-cardinality constraint is not such a cost by declaration.

Proof. For the finite problem set $c_\ell(x) = \max_{\theta,u}\langle u, (C^{(\ell)}_{\theta,T} + c_\ell S_{\theta,T})u\rangle$ and $v(x) = \max_{\theta,u}\langle u, V_{\theta,T}u\rangle$. The upward closed set of achievable vectors $(c_0(x) - N_0, \dots, c_r(x) - N_r, v(x))$ with coordinatewise slack is convex and closed, since x lies in a compact polytope. A supporting hyperplane at $(0, \dots, 0, q_T)$ has nonnegative coefficients. Its risk coefficient is positive: if it were zero, the jointly strictly feasible x_0 would give a strictly smaller pairing. Normalize that coefficient to one. This yields multipliers $\lambda_\ell \geq 0$, minimization of $v + \sum_\ell \lambda_\ell(c_\ell - N_\ell)$ and complementary slackness.

Repeat the active-gradient and normal-cone argument with one cost-gradient hull for each positive multiplier. All cost generators still have appended coordinate zero, and risk generators have appended coordinate one. The conic reduction in dimension $k_T + 1$ therefore proves the same total support bound. Evaluating the resulting affine minimum at the jointly strict fallback prefix gives the displayed weighted margin bound. At each node the immediate scalar cost is simply the sum over resource-indexed atoms, which proves the recursion claim.

Resource 0 bounds survival by D_T. Completing an optimizer by the one common fallback costs at most $N_\ell + e_{\ell,T}$ in resource ℓ and adds at most LD_T risk. Choose this completed plan with probability p_T and the fallback otherwise. For every ℓ its cost is at most $p_T(N_\ell + e_{\ell,T}) + (1 - p_T)\kappa_\ell \leq N_\ell$. The risk bound follows exactly as for one resource. Strict positive margins make $p_T \to 1$. Common-tree and fallback labels are counted as before. On a general feasible vector, intersect all prefix reserve constraints; the single coercive resource 0 forces properness and uniform bounded-loss tails. The compactness and finite-intersection proof of Theorem S27.1 then applies with all the additional closed constraints. Separate fallbacks safe for separate resources would not justify this proof: joint feasibility of one common fallback is essential. □

S27.4. **Distinct witnesses and the price of hard output width.**

Example S27.8 (A shared adversarial prior can give a strict dual gap)**.** There are two commands A, B, one productive report, one decoder, and two scalar models. Costs are charged but are not extra observations; the visible action–report history has the same law in both models. Model 1 has costs $(1, 2)$ and losses $(0, 0)$; model 2 has costs $(1, 1)$ and losses $(1, 0)$. Commitment may occur only after a call. At budget $N = 3/2$, writing x for the probability that the final command is B, every plan costs at least $1 + x$ in model 1 and has risk $1 - x$ in model 2. One-call mixing at $x = 1/2$ attains the exact value $1/2$. The dual certificate has one unit cost atom at model 1 and one unit risk atom at model 2.

If one incorrectly forces both atom lists to be proportional to the same probability $(t, 1 - t)$, the one-call dual becomes

$$\sup_{0\leq t\leq 1,\lambda\geq 0} \min_{0\leq x\leq 1} \{(1 - t)(1 - x) + \lambda(tx - 1/2)\} = \tfrac{3}{2} - \sqrt{2} < \tfrac{1}{2}.$$

For $t \leq 1/2$ its best value is zero; for $t > 1/2$ the best multiplier is $(1 - t)/t$, giving $3/2 - t - 1/(2t)$, maximized at $t = 1/\sqrt{2}$. All quantities come from real cost and loss tables. A single least-favorable prior for one scalarized task is not a substitute for independently imposed robust cost and risk constraints.

Proposition S27.9 (An exact hard-width convexification gap)**.** *There is a one-model, one-dimensional physical experiment with two commands, compact decision set $D = \{0, 1\}$ and bounded nonnegative loss, such that for $1 \le N \le 2$,*

$$R_1(N) = \min\{1/2, 2 - N\}, \qquad R_2(N) = (2 - N)/2.$$

Thus R_1 is attained and right-continuous but not convex. Its scalar Lagrange relaxation over all width-one programs equals R_2; at $N = 3/2$ it has a gap $1/4$. Randomizing two different one-value dictionaries pays for a two-value dictionary, not for an uncharged initial seed.

Proof. Both commands always give one productive report, with costs $1, 2$. Decision 0 has loss $1/2$ after either command. Decision 1 has loss 1 after the cheap command and zero after the costly one. These are permissible prospective audit losses: the future audit can be the deterministic command symbol with the displayed loss table. This is a bounded general-loss example, not a claim about a squared-loss realization.

For the sole decoder 0 every proper program has loss $1/2$. For the sole decoder 1, if its final command is costly with probability x, expected total cost is at least $1 + x$ and risk is $1 - x$. One-call mixing realizes all these pairs, so optimization gives R_1. With both values, the smallest loss is $1/2$ after a cheap last command and zero after a costly one. The same cost lower and one-call realization give R_2.

Over width-one programs the penalized infimum is $\min\{1/2 + \lambda(1 - N), \lambda(2 - N)\}$: the cheap decoder-1 endpoint is dominated. Maximizing over $\lambda \ge 0$ gives $(2 - N)/2$, with equality of the two lines at $\lambda = 1/2$. At $N = 3/2$ the primal width-one value is $1/2$ and the relaxation is $1/4$. In the randomized implementation attaining the latter, a cheap call uses decoder 0 and a costly call uses decoder 1, so two distinct final values are genuinely required. □

The preceding results supply an exact variational and finite-witness completion of the primitive risk route. They do not establish that every causal family admits the geometric reference certificates, a common finite unknown-model predictive quotient, or optimal computational resources. Boundary primal attainment is compatible with the failure of uniform expected cost-tail convergence and with lack of a finite boundary implementation. Those independent questions are not resolved by a success certificate for a finite optimization problem.

S28. Weighted boundary decisions and positive stochastic realization

The deterministic residual quotient of Section S29 does not classify a robust problem with tied Bayes actions. In that case the object to realize is a probability-valued decision function, and its residuals need not take finitely many values. We first derive that object without assuming model-independent report probabilities. We then characterize its finite stochastic implementation by an invariant convex family, and give exact memory classifications for a random-horizon calibration task. The convex realization machinery is classical in origin: linear representations and positive realizations are different notions [45, 46, 47]. The new assertions concern the physical minimum-cost face, its robust optimal decision face, and their counted stochastic implementation.

S28.1. **State-dependent reports and a minimum-cost potential.** Fix a compact metric model space Θ, a finite-dimensional physical space K, a nonempty finite failure alphabet Y, and a fixed nonempty finite decoder dictionary C. All primitive matrices below depend continuously on θ. The prepared density is ρ_θ. There is one globally legal command, with completely positive branches $\mathcal{S}_\theta$ for a productive herald and $\mathcal{K}_{\theta,y}$ for failure y, whose sum is trace preserving. A rejected herald retains the actual post-$\mathcal{S}_\theta$ state; a failure retains the actual post-$\mathcal{K}_{\theta,y}$ state. No replacement preparation, inspection of the protected state, or additional unpriced operation is available. Calls have nonnegative cost with mean observable

$$c_* I \preccurlyeq G_\theta \preccurlyeq c^* I, \qquad 0 < c_* \le c^* < \infty.$$

Commitment is allowed only on a herald and precedes the prospective audit. Its loss at decoder r has observable $0 \preccurlyeq F_{\theta,r} \preccurlyeq LI$ on the post-herald state. An independently executed final audit cost must be added separately. The declared reports are the whole observer input; charged costs do not supply an additional observation channel. These are single-command hypotheses, not a finite-memory criterion for a general controlled POMDP.

Put $\mathcal{T}_\theta = \sum_y \mathcal{K}_{\theta,y}$ and $d = \dim K$. Assume

$$I - (\mathcal{T}_\theta^*)^d I \succ 0 \qquad (\theta \in \Theta). \tag{S28.1}$$

This is the primitive depth test already proved in Theorem S26.1, now applied to the one command. Compactness gives a common $\delta > 0$ with the left side at least δI. Thus first-herald survival after n calls is at most

$$b_n = (1-\delta)^{\lfloor n/d \rfloor}.$$

The uniformly convergent resolvent

$$H_\theta = (I - \mathcal{T}_\theta^*)^{-1} G_\theta, \qquad c_* I \preccurlyeq H_\theta \preccurlyeq K_0 I, \quad K_0 = c^* d/\delta$$

is continuous in θ. Write $h_\theta = \operatorname{tr}(H_\theta \rho_\theta)$, $N_0 = \max_\theta h_\theta$ and $\Theta_0 = \{\theta : h_\theta = N_0\}$. For $w = y_1 \cdots y_n$, let $\mathcal{K}_{\theta,w} = \mathcal{K}_{\theta,y_n} \cdots \mathcal{K}_{\theta,y_1}$ and

$$\lambda_\theta(w) = \operatorname{tr} \mathcal{S}_\theta \mathcal{K}_{\theta,w}(\rho_\theta).$$

The empty word is included. Words may have zero probability. We use the following support condition:

$$\lambda_\theta(w) > 0 \text{ for some } \theta \quad \Longrightarrow \quad \lambda_{\theta'}(w) > 0 \text{ for some } \theta' \in \Theta_0. \tag{S28.2}$$

This is a sufficient coverage condition for robust boundary rigidity, not a claimed necessary condition for every such rigidity. In particular it holds if one cost-maximizing model has $\mathcal{S}^* I \succ 0$ and $\mathcal{K}_y^* I \succ 0$ for every y. These inequalities ensure that every finite word has positive mass in that model; neither its posterior nor the model label is supplied to the observer.

Theorem S28.1 (Weighted herald rigidity)**.** *For every proper finite-cost observer in a fixed model,*

$$\mathbb{E}_\theta C_\tau = h_\theta + \mathbb{E}_\theta \sum_{i \le \tau:\, \text{herald } i \text{ rejected}} \operatorname{tr}(H_\theta \rho_i^+), \tag{S28.3}$$

where ρ_i^+ is its normalized post-herald physical state. Consequently the least robust budget is N_0. Under (S28.2)*, every budget-N_0 observer commits at the first herald in every model. Model- and state-dependent failure and herald probabilities are permitted.*

Proof. At a reached pre-call state ρ, the expected cost plus the potential after continuing failures is $\operatorname{tr}[(G_\theta + \mathcal{T}_\theta^* H_\theta)\rho] = \operatorname{tr}(H_\theta \rho)$. Continuing after a herald adds exactly its post-herald potential, weighted by the actual rejection decision. Conditional summation to a deterministic horizon gives the displayed identity with cost and rejected-herald sums truncated, minus the remaining live potential. That remainder is bounded by $K_0 \Pr_\theta(\tau > n)$ and tends to zero. Finite cost and the positive cost floor imply proper survival decay; monotone convergence applies to the two nonnegative sums. This proves (S28.3). First-herald commitment realizes h_θ simultaneously, so its largest value is the least robust budget.

At every $\theta' \in \Theta_0$, a boundary observer has zero residual. Since $H_{\theta'} \succeq c_* I$, it rejects no herald with positive probability. Before the first herald, the distribution of its private memory conditional on the failure word depends only on that word and its own coins: there is one command, and the physical report maps do not depend on those coins. In particular its rejection probability at the first herald after w is model independent. If w is possible anywhere, (S28.2) makes it possible at a cost-maximizing model, where this probability is zero. Thus the probability is zero in every model. The sufficient full-support test follows by successively taking traces of positive, nonzero branch outputs. □

The coverage qualification cannot be dropped. With unit calls, let one model have geometric herald rate $1/3$ and only failure letter a. Let another have a prepared two-state system whose first call fails with letter b, enters a state with deterministic heralds, and thereafter remains there. Both pass (S28.1) (use a two-dimensional identity extension in the first model). Their first-herald costs are three and two. A policy accepting all a-histories but rejecting the first herald after b still has robust cost three. The word b is absent from the expensive model. This failure of coverage licenses an actual extra stopping decision; it is not a zero-probability technicality.

S28.2. **The optimal face with arbitrary robust ties.** Assume (S28.2). Define the unnormalized loss coefficients

$$f_\theta(w,r) = \operatorname{tr}(F_{\theta,r}\mathcal{S}_\theta\mathcal{K}_{\theta,w}(\rho_\theta)), \quad 0 \le f_\theta(w,r) \le L\lambda_\theta(w).$$

A boundary rule is $b = (b_w)_{w\in Y^*} \in \mathcal{D} := \prod_{w\in Y^*}\Delta(C)$, independent of θ, with risk $R_\theta(b) = \sum_{w,r} b_w(r) f_\theta(w,r)$. For a probability measure μ on Θ put $f_\mu(w,r) = \int f_\theta(w,r)\,d\mu(\theta)$ and $A_\mu(w) = \arg\min_{r\in C} f_\mu(w,r)$.

Theorem S28.2 (Weighted minimax face and tied optimality)**.** *The full-history boundary value and a least-favorable measure are attained, and*

$$V = \min_{b\in\mathcal{D}} \max_{\theta} R_\theta(b) = \max_{\mu\in\mathcal{P}(\Theta)} \sum_{w} \min_{r\in C} f_\mu(w,r). \tag{S28.4}$$

For any attaining μ_* the complete optimal face is

$$\mathcal{B}_* = \{b\in\mathcal{D} : R_\theta(b) \le V \ (\theta\in\Theta), \quad \operatorname{supp} b_w \subseteq A_{\mu_*}(w) \ (w\in Y^*)\}. \tag{S28.5}$$

Words with zero μ_*-averaged mass impose no extra decoder restriction. In general the support restrictions alone are insufficient: the inequalities for the individual models remain essential.

Proof. The terms beyond length $n-1$ have sum at most Lb_n, uniformly in b and θ. Finite partial sums are jointly continuous, and this bound proves joint continuity on the compact convex rule space and on the compact model space. Passing from models to their probability measures makes the payoff bilinear and continuous. Compact minimax gives (S28.4), with both extrema attained; equivalently apply finite-dimensional separating minimax to finite word truncations and then use the uniform tail and compactness. Minimization for fixed μ is wordwise, since each coordinate ranges over a simplex.

For an optimal b, its μ_*-average is both at most V and at least the Bayes value V. Subtraction yields the nonnegative identity

$$0 = \sum_{w,r} b_w(r)[f_{\mu_*}(w,r) - \min_u f_{\mu_*}(w,u)].$$

Every summand vanishes. This gives the support restriction even without positive model-independent word masses; if a whole word has zero averaged mass, every coefficient there is zero and all actions are allowed. Conversely the model inequalities make b robustly optimal, since V is the infimum. This proves the full face, not sufficiency of Bayes ties alone. For example two constant loss vectors $(0,1)$ and $(1,0)$ have fair robust optimum $1/2$, while both pure actions are tied under the equal model mixture and each has worst risk one. □

S28.3. **What replaces a deterministic residual congruence.** View each $g\in\mathcal{D}$ as a probability-valued response function on future words. Its left shift is $(L_y g)(z) = g(yz)$. An *anchored finite convex realization* of b consists of $g_1 = b, g_2, \dots, g_M \in \mathcal{D}$ with

$$L_y g_s \in \operatorname{conv}\{g_1,\dots,g_M\} \quad (s = 1,\dots,M,\ y\in Y). \tag{S28.6}$$

The distinguished initial response g_1 is counted even if it is not an extreme point of this convex hull. This anchor reflects a prescribed initial live label; it must not be replaced by a free random initial state. The auxiliary g_s need not themselves be robust optimal from a new preparation.

Theorem S28.3 (Exact stochastic boundary memory with ties)**.** *Under the weighted rigidity hypotheses, an exact robust boundary optimum with at most M live labels exists if and only if some $b \in \mathcal{B}_*$ admits an anchored convex realization with at most M generators. The least such generator count is the least exact live-label count. This criterion permits all robust ties, state-dependent word masses and stochastic transitions.*

For a specified response b, the block Hankel matrix

$$\mathsf{H}_b(w,(z,r)) = b_{wz}(r)$$

has rank at most the generator count. Finite Hankel rank alone is not sufficient. Event-stopped implementations use at most $\max(M,|C|)$ total labels, and state-stopped implementations at most $M+|C|$; unused decoder values can be removed from these upper counts.

Proof. From (S28.6) choose stochastic coefficients $Q_y(s,t)$ representing each shifted generator. Set the decoder row $D_s = g_s(\varnothing)$. Induction on word length gives

$$g_s(y_1 \cdots y_n) = e_s^{\mathsf{T}} Q_{y_1} \cdots Q_{y_n} D.$$

The machine starts in label one, updates on failures with these matrices, and on the first herald draws its decoder from D_s. It uses no physical-state input and realizes b exactly, so (S28.5) proves optimality.

Conversely, for each reached label of any stochastic machine define $g_s(z)$ to be its terminal decoder distribution after processing z and then a herald. Its rows are probabilities. The Markov property gives $L_y g_s = \sum_t Q_y(s,t) g_t$, and the initial function is b. Unreachable labels may be removed; any unused rows can be completed arbitrarily. Thus the same number of reached labels gives the required generators, proving both directions and minimality. Factor the Hankel matrix as the row of reached state probabilities after w times the column of future response probabilities from each state after z to obtain the rank lower bound. The established event/state role packing gives the total-label upper counts. □

This is a positive realization criterion on the optimal decision face, not a claim to originate residual automata or finite-dimensional linear realization. Rabin–Scott and the Moore/Myhill–Nerode tradition characterize deterministic finite residual functions [44]; observable-operator and multiplicity-automaton theories separate linear rank from stochastic realization [45, 46]. Here exact unique Bayes actions recover the deterministic criterion of Section S29. With ties, a convex family can have infinitely many different residual points, and selecting positive successors no longer preserves the required output probabilities. The next application exhibits both phenomena exactly.

For finite primitive models the criterion also has a direct operator verification form. Given stochastic rows Q_y, D, define the killed map on the direct sum of M physical blocks by

$$(\mathcal{T}_{\theta,Q} X)_t = \sum_{s,y} Q_y(s,t) \mathcal{K}_{\theta,y}(X_s), \qquad A_{\theta,s} = \sum_r D_s(r) \mathcal{S}_\theta^* F_{\theta,r}.$$

The physical survival bound also bounds this map, independently of the memory rows. Its unique score potential is

$$Z_\theta - \mathcal{T}_{\theta,Q}^* Z_\theta = A_\theta, \qquad 0 \preccurlyeq Z_\theta \preccurlyeq LI. \tag{S28.7}$$

Its initial risk is $\operatorname{tr}((Z_\theta)_1 \rho_\theta)$. Consequently, when an exact target value V is supplied, stochastic rows satisfying this risk at most V in every model are a finite necessary-and-sufficient certificate for exact M-label optimality. The equations follow by summing the absolutely convergent killed score series; normalization bounds its sum by LI, and transience proves uniqueness. With finitely many algebraic models and algebraic V, this is a finite real-algebraic existence test. It does not assume that the unknown optimum of an arbitrary problem is itself algebraic.

S28.4. **Robust Born-probability replication at an unknown random horizon.** Consider a protected system prepared in ρ, one trace-preserving channel Φ, and an effect $0 \preccurlyeq E \preccurlyeq I$. On each call a failure has probability t and applies Φ; a herald has probability $1-t$ and leaves the system unchanged. The fixed unknown parameter ranges over $[t_-, t_+] \subset (0,1)$, with $t_- < t_+$. Calls cost one. On commitment output $r \in C = \{0,1\}$ before observing the Bernoulli audit Z with effect E. There are two audit-loss models $\sigma \in \{-1,1\}$, neither affecting report probabilities, with loss

$$\ell_\sigma(r,Z) = \tfrac{1}{2}[1+\sigma(r-Z)] \in [0,1]. \tag{S28.8}$$

The unknown sign enforces two-sided calibration rather than choosing one favorable signed discrepancy. This is a replication task for the Born probability at a random horizon, not the squared-sign prediction task of the previous qubit theorem. The observer has no access to the protected system except the failure/herald reports.

Write $\beta_n = \operatorname{tr}(E\Phi^n\rho)$ and let b_n be the probability of output one after n failures. The endpoint t_+ has the largest cost and full word support, so Theorem S28.1 forces first-herald commitment at the least budget $N_0 = 1/(1-t_+)$. A boundary policy has

$$\sup_{t,\sigma} R_{t,\sigma}(b) = \tfrac{1}{2} + \tfrac{1}{2} \sup_{t\in[t_-,t_+]} \left| (1-t)\sum_{n\ge 0} t^n (b_n - \beta_n) \right|. \tag{S28.9}$$

Theorem S28.4 (Unique randomized optimum and exact realization memory)**.** *The full-history boundary value is* $1/2$*, and its unique response kernel is* $b_n = \beta_n$ *for all* $n \ge 0$*. An exact finite-memory optimum exists precisely when*

$$\beta_n = e_1^{\mathsf{T}} Q^n d \quad (n \ge 0) \tag{S28.10}$$

for a finite row-stochastic matrix Q *and a vector* $d \in [0,1]^M$*; its least size is the exact live memory. Under a least-favorable measure putting equal weight on the two signs at any fixed* t*, every wordwise action is Bayes-tied. Thus uniqueness of the robust randomized kernel does not imply uniqueness of any wordwise Bayes action.*

Put $g(t) = (1-t)\sum_{n\ge 0} t^n \beta_n$*. The exact one-live-label value is*

$$V_1 = \tfrac{1}{2} + \tfrac{1}{4}(\max_t g(t) - \min_t g(t)). \tag{S28.11}$$

For every fixed M *the best risk is attained. Whenever* (S28.10) *is impossible at that size, its risk is strictly greater than* $1/2$*. A* T*-label unary prefix chain gives a boundary-feasible risk at most* $1/2 + t_+^T/2$*.*

Proof. A direct expectation of (S28.8) gives (S28.9). The choice $b_n = \beta_n$ proves value $1/2$, which is also a lower bound from the two opposite signs. At equality the power series $\sum_n (b_n - \beta_n)t^n$ vanishes on a nonempty interval inside its unit disk of convergence. The analytic identity theorem makes all its coefficients zero. For completeness, a nonzero convergent power series is locally its first nonzero power times a nonvanishing analytic factor at any interior zero; hence its zeros cannot accumulate in its disk unless it is identically zero. This proves uniqueness.

Theorem S28.3 in the unary alphabet gives exactly (S28.10). Averaging the two loss signs makes every conditional loss $1/2$, proving the least-favorable tie assertion. A one-label rule is a constant output probability u. Its discrepancy is $u - g(t)$; the midpoint of the minimum and maximum of g minimizes the uniform absolute error, with radius half their difference. This proves (S28.11).

For fixed M, the compact set of stochastic rows and decoder values has continuous risk: each finite sum is continuous and the tail is uniformly at most t_+^T. Attainment and unique optimality prove the strict gap. Store labels $0, \dots, T-1$, advance until the final label and repeat it thereafter, with outputs $\beta_0, \dots, \beta_{T-1}$. It is exact for $n < T$ and differs by at most one later. Formula (S28.9) bounds its risk excess by $t_+^T/2$. Its stopping is unchanged, so this is the same least budget, not an enlarged-budget approximation. □

Theorem S28.5 (Every finite exact memory dimension occurs)**.** *For every $k \geq 2$ and algebraic $a \in (0,1)$ there is a k-state protected classical channel in this task whose least exact stochastic live memory is exactly k. Take an initial chain of $k-1$ transient states: each stays with probability a and advances with probability $1-a$; state k is absorbing. Audit whether the state is still transient. Then*

$$\beta_n = \sum_{j=0}^{k-2} \binom{n}{j} (1-a)^j a^{n-j}, \qquad g(t) = 1 - \left(\frac{(1-a)t}{1-at}\right)^{k-1}. \tag{S28.12}$$

The exact one-label excess is $[g(t_-) - g(t_+)]/4$. For $k = 2$, infinitely many different probabilistic residuals are implemented by just two stochastic labels.

Proof. The number of advances before time n is binomial until absorption, proving the formula for β_n with $\binom{n}{j} = 0$ for $j > n$. An independent observer copy of the transition probabilities, using only private coins and k labels, gives the required decoder probability; it neither reads nor copies the actual protected state. This proves the upper memory bound.

As a function of n, $\beta_n = a^n P(n)$ for a polynomial P of degree exactly $k-2$. The shifted polynomials $P(n), \ldots, P(n+k-2)$ span all polynomials of degree at most $k-2$: successive finite differences lower the degree by exactly one, with nonzero leading coefficient. The scalar Hankel row/column rank is therefore $k-1$. Including both output probabilities $\beta_n, 1-\beta_n$ adds the constant function, which is independent because every exponential-polynomial term tends to zero. Thus the probability-valued Hankel rank is k, giving the matching lower by Theorem S28.3. Summing the binomial series, using $\sum_{n \geq j} \binom{n}{j} x^{n-j} = (1-x)^{-j-1}$, gives g. It decreases with t, giving the stated one-label formula. For $k = 2$ the residuals have first coordinate a^n, all distinct, although the displayed chain has two labels. □

S28.5. **A sharp positive-realization transition for the protected qubit.** Use the algebraic rotation from the preceding section,

$$U = 5^{-1/2} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \quad c = \cos\vartheta = 3/5, \; s = \sin\vartheta = 4/5,$$

with $\rho = |0\rangle\langle 0|$, $E = |0\rangle\langle 0|$ and $\Phi_\alpha(A) = \alpha U A U^* + (1-\alpha)\,\mathrm{tr}(A) I/2$, $0 \leq \alpha \leq 1$. For each fixed known retention factor α, the required response is

$$\beta_n = \tfrac{1}{2}[1 + \alpha^n \cos(n\vartheta)], \tag{S28.13}$$

with $\beta_0 = 1$ also at $\alpha = 0$.

Theorem S28.6 (Exact qubit realization phases)**.** *For the robust calibration task, the following hold.*

(a) *At $\alpha = 1$ no finite classical memory attains the boundary optimum, although the probability-valued Hankel rank is three. At $\alpha = 0$ the least exact live memory is two.*

(b) *At every $0 < \alpha < 1$ an exact stochastic optimum exists with at most*

$$M_\alpha = \max\{3, \lceil \pi / \arccos\alpha \rceil\}$$

live labels. The least number tends to infinity as $\alpha \uparrow 1$.

(c) *Put $\alpha_3 = 5/(3+4\sqrt{3})$. The least exact live memory is three precisely when $0 < \alpha \leq \alpha_3$. It is at least four for $\alpha_3 < \alpha < 1$. The three-label assertion is global over all stochastic transition matrices and randomized decoder rows, not only circulant candidates.*

Proof. For $\alpha > 0$, the three distinct modes 1, $\alpha e^{i\vartheta}$ and $\alpha e^{-i\vartheta}$ have nonzero coefficients in (S28.13). Vandermonde independence gives Hankel rank three. Cayley–Hamilton applied to any representation (S28.10) shows that its transition matrix must have the two latter modes as eigenvalues: substituting its characteristic recurrence and using independence of the three exponential sequences makes the characteristic polynomial vanish at each mode.

Every unit-modulus eigenvalue z of a finite stochastic matrix is a root of unity. Indeed choose an eigenvector coordinate of maximal modulus. Equality in the triangle inequality for its stochastic row forces every positive-probability successor to have the same maximal modulus and its value to equal z times the current value. Follow positive successors to a repeated state. The directed cycle gives $z^j = 1$ for some positive integer j. But $e^{i\vartheta}$ is not a root of unity: otherwise $2c = 6/5$ would be a rational algebraic integer and hence an integer. This proves (a) at $\alpha = 1$. At $\alpha = 0$ a starting label with output one followed by an absorbing fair-output label realizes the sequence, whereas one label cannot change its initial output.

For (b), take the regular M_α-gon with vertices v_j on the unit circle, including $v_0 = (1, 0)$. Its inradius is at least α, so αR_ϑ maps every vertex into the polygon. Choose barycentric coefficients $Q(j, l)$ of each image in these vertices and let $d_j = (1 + (v_j)_1)/2$. Then Q is stochastic, all $d_j \in [0, 1]$, and the expected vertex after n transitions from vertex zero is $(\alpha R_\vartheta)^n v_0$. This proves (S28.13) exactly. If least exact memories stayed bounded along $\alpha \uparrow 1$, compactness of their padded transition/decoder rows would give, by taking a subsequence and each finite coefficient limit, a finite realization at $\alpha = 1$. This contradicts (a).

For (c), define a circulant stochastic matrix with first row

$$u_0 = \frac{1 + 2\alpha c}{3}, \qquad u_+ = \frac{1 - \alpha c + \sqrt{3}\alpha s}{3}, \qquad u_- = \frac{1 - \alpha c - \sqrt{3}\alpha s}{3}.$$

Its other rows are the cyclic translates. All entries are nonnegative exactly up to the asserted threshold. Its complex nonconstant eigenvalues are $\alpha(c \pm is)$, and decoder $(1, 1/4, 1/4)$ from the first state realizes (S28.13). The rank lower makes three minimal.

Conversely any three-state realization has eigenvalues $1, \alpha(c + is), \alpha(c - is)$. For $B = I - Q$ put $a_i = B_{ii} = \sum_{j \neq i} Q_{ij} \geq 0$. Its two nonzero eigenvalues are $u \pm iv$, where $u = 1 - \alpha c$ and $v = \alpha s$. Its trace and second characteristic coefficient give

$$\sum_i a_i = 2u, \qquad u^2 + v^2 = \sum_{i<j} (a_i a_j - Q_{ij} Q_{ji}) \leq \sum_{i<j} a_i a_j \leq \tfrac{1}{3}(2u)^2.$$

Therefore $\sqrt{3}\alpha s \leq 1 - \alpha c$, exactly $\alpha \leq \alpha_3$. No assumption of irreducibility, symmetry or deterministic decoding entered this lower bound. □

The damped channel has no strict limiting preference between the two signed calibration losses, and its exact optimal decoder law remains nonconstant at arbitrarily late times. Positive stochastic realization nevertheless succeeds for every $\alpha < 1$. This does not contradict the earlier impossibility for unique squared-sign prediction: the fixed losses, the robust optimal face and the output law differ. The unit-circle obstruction and invariant positive cones belong to classical positive-realization theory [47]; the task-specific exact count, the global three-label threshold and the same-boundary physical implementation are proved here. No sharp asymptotic memory rate is inferred from the polygon upper.

Proposition S28.7 (Finite algebraic certificates and minimal synthesis)**.** *Suppose a probability-valued response has a supplied real-algebraic linear representation $b_w = a^{\mathsf{T}} T_w B$ of dimension r on a finite alphabet. Existence of an exact stochastic realization with M live labels is equivalent to the finite conditions*

$$Q_y \geq 0,\ Q_y \mathbf{1} = \mathbf{1}, \quad D \geq 0,\ D\mathbf{1} = \mathbf{1}, \qquad e_1^{\mathsf{T}} Q_w D = a^{\mathsf{T}} T_w B \quad (|w| \leq M + r - 1).$$

These are real-algebraic feasibility conditions, hence decidable at each prescribed M, and a feasible instance has algebraic controller entries. For the qubit family with algebraic $0 < \alpha < 1$, the finite upper from Theorem S28.6 makes successive size tests a terminating synthesis of a minimal exact stochastic observer. No efficient bit bound or general decision procedure for unbounded positive realization is asserted.

Proof. Use the block diagonal representation with transition $\operatorname{diag}(Q_y, T_y)$, initial row $(e_1^{\mathsf{T}}, -a^{\mathsf{T}})$ and final columns $(D, B)^{\mathsf{T}}$. Its reachable row spaces grow from dimension at most one in ambient dimension $M + r$. Equality of successive spaces implies invariance under every letter, so the space generated by words of length at most $M + r - 1$ contains all word rows. Annihilating the final columns on these finitely many words therefore annihilates them on every word. The reverse implication is immediate. The conditions are polynomial in the controller entries, with real-algebraic coefficients, so real-closed-field elimination decides feasibility and yields an algebraic sample when nonempty [15]. The polygon construction supplies a finite upper; one can obtain a certified integer bound by testing $\alpha < \cos(\pi/M)$ with increasing integers, using algebraic comparisons. Thus finitely many prescribed-size tests find the first feasible size. This is a complete terminating construction for this family, not a claim about the complexity of general probabilistic automata. □

S29. Finite-memory optimality at the least feasible budget

Boundary attainment in Theorem S27.1 concerns the compact space of full-history plans. It does not assert that the optimum belongs to any finite-label class. We resolve that distinction on a class of prepared heralded channel experiments. The result is a necessary-and-sufficient memory criterion, an explicit finite-dimensional quantum obstruction, and quantitative approximation at the same boundary budget. It is not an additional assumption of the general compactness theorem.

S29.1. **Prepared heralded channels and a boundary rigidity identity.** Let $0 < p < 1$, put $q = 1 - p$, and fix a nonempty finite failure alphabet Y with probabilities $\pi_y > 0$ summing to one. There is one globally legal unit-cost command. Model θ has a prepared density matrix ρ_θ on a fixed finite-dimensional space and channels $\Phi_{\theta,y}$. Its report instrument is

$$\mathcal{I}_{\theta,s}(A) = pA, \qquad \mathcal{I}_{\theta,y}(A) = q\pi_y \Phi_{\theta,y}(A) \quad (y \in Y). \tag{S29.1}$$

The herald s is productive; a failure y forbids commitment. Rejection of s leaves the physical state unchanged. The observer cannot inspect the protected physical state. After commitment to r in one fixed nonempty finite dictionary C, the future audit has conditional loss observable $0 \preccurlyeq F_{\theta,r} \preccurlyeq LI$. Any separately charged fixed final-audit cost is added to all budgets below. No unpriced channel, phase input or clock is supplied to the observer.

For $w = y_1 \cdots y_n \in Y^*$, including the empty word, set

$$\rho_\theta(w) = \Phi_{\theta,y_n} \cdots \Phi_{\theta,y_1}(\rho_\theta), \quad \lambda(w) = pq^{|w|} \prod_{i=1}^{|w|} \pi_{y_i}, \quad \ell_\theta(w, r) = \operatorname{tr}(F_{\theta,r}\rho_\theta(w)).$$

Every word has positive mass, $\sum_w \lambda(w) = 1$, and $\sum_{|w| \geq T} \lambda(w) = q^T$. Probabilities of visible failures are independent of both the hidden model and the private controller coins. The channels themselves may change the prepared state noncommutatively.

Lemma S29.1 (The least budget forces first-herald commitment)**.** *The least feasible precommit expected-call budget is $N_{\min} = 1/p$. For every proper finite-cost policy, let J count productive reports which it rejects. Then, in every model,*

$$p\mathbb{E}_\theta \tau = 1 + \mathbb{E}_\theta J. \tag{S29.2}$$

Consequently a policy is feasible at $N_{\min}$ if and only if it commits at the first herald, up to null histories. At that budget the only choice is its decision distribution $b_w \in \Delta(C)$ after each failure word w, and its risk is

$$\mathcal{R}_\theta(b) = \sum_w \lambda(w) \sum_{r \in C} b_w(r)\ell_\theta(w, r). \tag{S29.3}$$

Proof. The event that call i is made is predictable, and the conditional herald probability at that call is p. Summing the nonnegative indicators gives $\mathbb{E}_\theta \sum_{i\le\tau} \mathbf{1}_{\{s_i=s\}} = p\mathbb{E}_\theta\tau$. Exactly one of these heralds is accepted; all others are rejected. This proves the identity without an optional-stopping integrability assumption. It implies the lower $1/p$, attained by first-herald commitment. Equality forces the nonnegative integer J to vanish almost surely. Conversely that rule has a geometric call count of mean $1/p$. Conditional on a failure word, the physical state is the displayed channel product and is independent of the private tape. The independent, unobserved audit is performed only after the decision. This gives the risk formula. □

S29.2. **A robust decision quotient and its exact memory content.** Assume the model space Θ is compact metric and the prepared states, channels and audit observables are continuous in θ. A singleton is permitted; an arbitrary prior is not part of the data. Write $V = \min_b \sup_\theta \mathcal{R}_\theta(b)$ at $N_{\min}$. For a probability measure μ on Θ put

$$\ell_\mu(w,r) = \int \ell_\theta(w,r)\, d\mu(\theta), \quad a_\mu(w) = \min_{r\in C} \ell_\mu(w,r), \quad V(\mu) = \sum_w \lambda(w) a_\mu(w).$$

Theorem S29.2 (Boundary minimax and the regularity criterion)**.** *The primal optimum and a least-favorable measure are attained, and*

$$V = \max_{\mu\in\mathcal{P}(\Theta)} V(\mu). \tag{S29.4}$$

For an attaining μ_*, every robust optimum has zero expected decision slack:

$$\int \mathcal{R}_\theta(b)\, d\mu_*(\theta) - V = \sum_w \lambda(w) \sum_r b_w(r)[\ell_{\mu_*}(w,r) - a_{\mu_*}(w)] \ge 0. \tag{S29.5}$$

Suppose the minimizer $a(w) \in C$ of $\ell_{\mu_*}(w,\cdot)$ is unique for every word. Then a is the unique robust optimal decision rule, and a finite stochastic autonomous observer attains V if and only if the relation

$$w \equiv_a v \quad\Longleftrightarrow\quad a(wz) = a(vz) \text{ for every } z \in Y^* \tag{S29.6}$$

has finite index. The least number of live labels attaining V is exactly that index. Random transitions and randomized readouts cannot reduce this exact count.

At each prescribed finite live-label bound M, the best boundary risk is attained. If the index is infinite, that best risk is strictly larger than V for every finite M, although finite-label policies converge to V at the same budget. A prefix implementation uses at most $1 + \sum_{j=0}^{T-1} |Y|^j$ live labels and has excess risk at most Lq^T. Its output dictionary remains C. For any implementation with M live labels and at most m actual decoder values, event stopping uses at most $\max(M,m)$ total labels and state stopping uses at most $M+m$, by Lemma 17.4.

Proof. The product $\mathcal{B} = \prod_{w\in Y^*} \Delta(C)$ is compact and convex. The tail of every risk series is at most Lq^T, uniformly in b and θ. Each truncated expression is jointly continuous and affine in the appropriate variable. The full payoff on $\mathcal{B}\times\mathcal{P}(\Theta)$ is therefore continuous and bilinear, with both factors compact convex. The elementary compact minimax theorem applies [40]. For completeness, it can also be obtained here from the finite-prefix games: truncate at T, separate the convex hull of the finite-dimensional model loss vectors from a strict lower orthant, and take compact subsequential limits of priors and plans. The uniform error Lq^T passes both inequalities to the limit. Minimizing for a fixed measure chooses a minimizing decision separately at every word and gives $V(\mu)$. This proves equality and both attainments. Nonnegative summation proves the slack identity. At a robust optimum its left side is at most zero and hence is zero. Since every $\lambda(w) > 0$, uniqueness forces b_w to be the point mass at $a(w)$. Existence of a robust optimum then proves that this Bayes rule is itself robustly optimal; a general Bayes minimizer need not have this property when ties occur.

At the boundary an M-label stochastic observer is represented by one stochastic transition matrix Q_y for each failure letter and one decoder distribution $d_s \in \Delta(C)$ at each live label. Its initial live label is fixed. Unreachable rows can be completed arbitrarily. Zero slack says that every label reached with positive probability after w decodes $a(w)$ with probability one. Choose one positive-probability successor at each pair (s, y) and the uniquely prescribed output on each reached label. This gives a deterministic observer with no more labels and the same output at every word: its path always had positive probability in the stochastic observer. Thus exact random realization implies deterministic realization.

If a deterministic machine sends w, v to the same label, every common continuation sends them through identical transitions and gives the same output. Hence they are equivalent in (S29.6). Any exact machine has at least its number of equivalence classes. Conversely this relation is a right congruence: $w \equiv_a v$ implies $wy \equiv_a vy$. Its classes, with transition $[w] \mapsto [wy]$ and decoder $a(w)$, form an exact deterministic observer whenever the index is finite. This is the standard residual-function construction for finite automata, applied to the optimal decision rule rather than to the full predictive state.

For fixed M, all Q_y and d_s lie in a finite product of compact simplexes. Each finite risk sum is continuous in their entries. The same uniform geometric tail makes full risks, and their supremum over Θ, continuous. A minimum is attained. It can equal V only if an exact machine exists, proving the strict gap for infinite index. Finally follow an optimal full-history rule on words of length less than T, storing that finite tree, and use one fallback decoder label thereafter. The policy still stops at the first herald and costs exactly $1/p$. Only the terminal mass q^T can change loss, proving the approximation and state bound. This argument uses no interior budget slack. □

The quotient in (S29.6) is specific to this boundary task and this fixed dictionary. It is not a common predictive quotient preserving every experiment or future audit. The minimax argument is classical; the new content is its combination with exact minimum-cost commitment, zero-slack determinization, and a necessary-and-sufficient autonomous memory count. Finite-index residual characterizations and unary periodicity are established automata theory [43]; their use does not establish a new priority claim for those tools.

Proposition S29.3 (Ties for one preparation, and why robust ties differ)**.** *For a single prepared model, without uniqueness, an exact finite stochastic observer exists if and only if there is a finite-index right congruence $\sim$ on Y^* such that*

$$\bigcap_{w \in A} \arg\min_{r \in C} \ell(w, r) \neq \varnothing \quad \textit{for every equivalence class } A.$$

The least number of live labels is the least number of classes among such congruences. For several unknown models the same zero-slack condition is necessary for an optimal stochastic observer under a least-favorable measure, but deterministic selection from tied Bayes actions need not preserve robust optimality.

Proof. For one preparation, every wordwise minimizing rule is globally optimal. Zero slack makes the decoder support of any reached stochastic label a subset of every optimal-action set at its reaching words. Choose one such output and a positive successor for each failure letter. Determinization preserves optimality. The partition of words by its reached deterministic label is a right congruence with the displayed nonempty intersections. Conversely choose an output in each intersection and run the congruence-class observer. Minimizing the number of classes proves the count.

For the distinction, take two models with identical channels and reports and losses $(0, 1)$ and $(1, 0)$ on a two-point dictionary, independent of the word. The robust value is $1/2$, attained with one live label and a fair terminal decision. Under the equally weighted least-favorable measure both actions are tied at loss $1/2$. Either constant deterministic selection has worst loss one. Thus Bayes zero slack alone cannot license robust determinization in the presence of ties. □

Proposition S29.4 (A finite distinguishing-depth lower certificate)**.** *In the unique-action case of Theorem S29.2, suppose the index is infinite and $m = |C| \geq 2$. Fix M. There is a finite H_M such that every deterministic M-label observer disagrees with a at a word of length at most H_M. Put*

$$\gamma_H = \min_{|w|\leq H,\ r\neq a(w)} [\ell_{\mu_*}(w, r) - \ell_{\mu_*}(w, a(w))] > 0, \qquad \pi_{\min} = \min_y \pi_y.$$

Every stochastic M-label boundary observer obeys

$$\sup_\theta \mathcal{R}_\theta - V \geq \frac{p\gamma_{H_M}}{m} \left(\frac{q\pi_{\min}}{M}\right)^{H_M} > 0. \tag{S29.7}$$

This is a finite distinguishing certificate, not a sharp asymptotic formula for the optimal intermediate-memory risk.

Proof. There are finitely many deterministic transition and output tables on M named labels. Each table fails at some finite word because the residual index is infinite. The maximum of their first distinguishing depths is finite. From any stochastic table choose a largest-probability successor for each (s, y), of probability at least $1/M$, and a largest-probability output at each label, of probability at least $1/m$. The resulting deterministic table fails at a word w of length $n \leq H_M$. In the stochastic table the selected internal path and output occur with probability at least M^{-n}/m conditional on this word. Their slack is at least γ_{H_M}. The word's terminal mass is at least $p(q\pi_{\min})^n$. Keep just that term in (S29.5); since $q\pi_{\min}/M < 1$, lowering it to the exponent H_M gives the bound. Worst-model risk is no smaller than the least-favorable average. □

S29.3. **An algebraic qubit with no finite-memory boundary optimum.** A quantum orbit supplies a finite-dimensional physical realization, not an arbitrary nonregular string placed in an external clock. Take one failure letter and a single prepared qubit,

$$\rho_0 = |0\rangle\langle 0|, \qquad U = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & -1 \\ 1 & 2 \end{pmatrix}, \qquad \Phi(A) = UAU^*, \qquad C = \{-1, 1\}. \tag{S29.8}$$

On commitment the future audit measures $Z = \mathrm{diag}(1, -1)$, with outcomes $z \in \{-1, 1\}$, and scores $(z - r)^2$. Thus $F_r = 2I - 2rZ$ and $L = 4$. Write ϑ for the Bloch rotation angle, so $\cos\vartheta = 3/5$, $\sin\vartheta = 4/5$, and put $c_n = \cos(n\vartheta)$. In particular $c_0 = 1$, $c_1 = 3/5$, and $c_{n+1} = (6/5)c_n - c_{n-1}$.

Theorem S29.5 (Boundary nonattainment by every finite classical memory)**.** *For every $0 < p < 1$, the least budget is $1/p$ and the full-history risk there is*

$$V_p = 2 - 2p \sum_{n=0}^{\infty} q^n |c_n|. \tag{S29.9}$$

Its unique optimal rule is $a_n = \operatorname{sgn} c_n$. It has infinite residual index. Hence no finite stochastic autonomous observer attains V_p at this budget. This remains true with arbitrary real transition probabilities and randomized terminal decisions.

Let $V_{p,M}$ be the optimal risk with M live labels and the same two-point dictionary. Each such minimum exists, and

$$0 < V_{p,M} - V_p \leq 4q^M. \tag{S29.10}$$

The exact one-live-label optimum is

$$V_{p,1} = 2 - \frac{2p(1 - 3q/5)}{1 - 6q/5 + q^2}. \tag{S29.11}$$

At $p = 1/2$ it is $12/13$, while the full-history value is the strictly smaller convergent series in (S29.9)*. All specified primitive entries are algebraic. The bound* (S29.7) *is an explicit positive lower with $m = 2$, $\pi_{\min} = 1$, and $\gamma_H = 4 \min_{0\leq n\leq H} |c_n|$.*

Proof. After n failures the prepared state is $U^n\rho_0U^{*n}$, whose audit mean is c_n. The expected squared loss of decision r is $2-2rc_n$. Lemma S29.1 forces first-herald commitment, so independent pointwise minimization gives (S29.9).

The number $e^{i\vartheta}$ is not a root of unity. Otherwise $2\cos\vartheta = 6/5$ would be a rational algebraic integer, hence an integer, a contradiction. Thus ϑ/π is irrational; no c_n is zero, and each residue subsequence $(n_0 + jr)\vartheta$ is dense modulo 2π for every positive integer r. The latter elementary rotation fact follows, for example, by the pigeonhole approximation of a nonzero arbitrarily small rotation and its successive multiples. Each residue subsequence therefore visits both open semicircles on which cosine has opposite signs. The optimal sign sequence cannot be ultimately periodic. A finite deterministic unary observer always follows a finite initial path and then a cycle, so it produces an ultimately periodic decision sequence. Theorem S29.2 now excludes every finite stochastic observer and proves the strict gap and its attainment at fixed M.

Use M live labels to store $n = 0, \dots, M-1$, incrementing on failure until the last label, which thereafter repeats. Decode the correct sign at these M labels and commit at the first herald. The rule is exact for $n < M$ and its excess loss is bounded by four on the tail of mass q^M. Its cost is still exactly $1/p$. This proves the upper in (S29.10).

For one live label the expected output is one constant $b \in [-1, 1]$. Its risk is $2 - 2bA$, where

$$A = p\sum_{n\ge0} q^n c_n = p\,\mathrm{Re}\,\frac{1}{1-qe^{i\vartheta}} = \frac{p(1-3q/5)}{1-6q/5+q^2} > 0.$$

Thus $b = 1$ is optimal and gives (S29.11). Substituting $p = q = 1/2$ gives $12/13$. The failure and herald maps are genuine completely positive trace-nonincreasing maps with trace-preserving sum. They use one declared command and a protected terminal audit; the proof never reads that audit early or turns the qubit into observer-owned memory. □

Corollary S29.6 (A compact uncertain-preparation family)**.** *Replace the known initial qubit by $\rho_\theta = (I + \theta Z)/2$, with unknown fixed $\theta \in [\eta, 1]$ and $0 < \eta \le 1$. Keep the same instrument, dictionary and audit. The robust boundary value is*

$$V_{p,[\eta,1]} = 2 - 2\eta p\sum_{n\ge0} q^n|c_n|.$$

A least-favorable preparation is ρ_η, and the unique robust optimal rule is again $\operatorname{sgn} c_n$*. It is unattainable by every finite stochastic autonomous memory, while the same-budget prefix approximation remains valid. Thus the obstruction is not tied to a chosen Bayesian prior or to one pure initial preparation. It concerns this declared preparation family, not all qubit density matrices.*

Proof. At preparation θ the conditional mean is θc_n. The sign rule minimizes loss word by word for every positive θ, and its total risk decreases with θ. Its worst value is therefore the displayed one at η. Every common policy is lower-bounded by its risk in that same preparation, whose unique minimizer is the sign rule. This proves the robust value, the least-favorable measure δ_η, and uniqueness without an assumed prior. Apply Theorem S29.2. If $\eta < 1$, the family includes both mixed and pure states; no inference is made at $\eta = 0$, where the least-favorable audit is a fair bit and uniqueness fails. □

Corollary S29.7 (Exact classification on unitary orbits)**.** *For any finite-dimensional unitary failure channel, fixed prepared state, fixed finite dictionary and audit observables, suppose the minimizing decision along the orbit is unique at every nonnegative integer time. A finite-label boundary optimum exists if and only if the optimal decision sequence is periodic from time zero. Its least live-label count is the least positive period.*

Proof. The residual theorem first gives equivalence with ultimate periodicity. If the sequence has eventual period r, there are integers $k_j \to \infty$ with $U^{rk_j} \to I$ in operator norm, by compactness of

the finite-dimensional unitary torus. For each fixed n, the states at times $n + rk_j$ and $n + r + rk_j$ converge to those at n and $n + r$. Uniqueness among finitely many decisions gives a strict gap at each of these two limiting states, so their optimal decisions eventually agree with the limiting decisions. Eventual r-periodicity makes these decisions equal. Therefore $a_n = a_{n+r}$ for every $n \geq 0$. A cycle of length the least period realizes the rule, and every finite unary machine has an eventual cycle at least that long. □

S29.4. **Dissipation and the strict decision-gap condition.** Finite memory is not confined to periodic unitary orbits. A stable decision at the limiting state gives a different positive realization mechanism.

Proposition S29.8 (Uniform contraction with a strict limiting decision gap)**.** *Consider a single prepared heralded model with finitely many failure letters. Suppose there is a density matrix ρ_∞ and constants $C_0 < \infty$, $0 < \alpha < 1$ such that*

$$\|\rho(w) - \rho_\infty\|_1 \leq C_0\alpha^{|w|} \quad (w \in Y^*).$$

Suppose one dictionary value r_* has a strict limiting gap

$$g = \min_{r \neq r_*} \operatorname{tr}[(F_r - F_{r_*})\rho_\infty] > 0.$$

For any nonnegative integer T with $2LC_0\alpha^T < g$, there is an exact optimal boundary observer with at most $1 + \sum_{n=0}^{T-1} |Y|^n$ live labels, where the empty sum is zero. All words of length at least T use the same decision r_*. Ties at earlier words are harmless in this single-model statement. In particular, dissipative channels with a certified contraction and a strict limiting task gap can have an exact finite-memory optimum at the least feasible cost.

Proof. For every $r \neq r_*$ and every word of length n, trace duality gives

$$\ell(w, r) - \ell(w, r_*) \geq g - \|F_r - F_{r_*}\|_\infty \|\rho(w) - \rho_\infty\|_1 \geq g - 2LC_0\alpha^n.$$

This is strictly positive at $n \geq T$. Store all shorter words in a finite tree and merge their depth-T continuations into one label decoding r_*. Choose a least-loss value separately at each shorter word. Every decision minimizes its actual conditional loss, and the rule stops at the first herald, so it attains the single-model optimum at cost $1/p$. The displayed number counts the entire prefix tree and the common tail label. No limiting state is supplied to the observer during execution; it is an offline certificate of its finite program. □

The limiting gap is a real restriction, not a consequence of mixing. The following family keeps a strictly positive contraction rate but retains the infinite exact-memory obstruction.

Corollary S29.9 (Noisy rotating qubits: contraction is not sufficient)**.** *In Theorem S29.5, replace the failure channel by*

$$\Phi_\alpha(A) = \alpha UAU^* + (1 - \alpha)\operatorname{tr}(A)I/2, \qquad 0 \leq \alpha \leq 1.$$

For $0 < \alpha \leq 1$, the exact boundary value and one-live-label value are

$$V_{p,\alpha} = 2 - 2p\sum_{n \geq 0} (q\alpha)^n |c_n|,$$

$$V_{p,\alpha,1} = 2 - \frac{2p(1 - 3q\alpha/5)}{1 - 6q\alpha/5 + q^2\alpha^2}.$$

No finite stochastic observer attains $V_{p,\alpha}$, while

$$0 < V_{p,\alpha,M} - V_{p,\alpha} \leq \frac{4p(q\alpha)^M}{1 - q\alpha}.$$

For $0 < \alpha < 1$ the failure channel is uniformly trace-norm contractive to $I/2$. At $\alpha = 0$, however, one live label is exactly optimal, with value $2 - 2p$. Thus a positive mixing margin alone cannot replace the strict limiting decision gap in Proposition S29.8.

Proof. Depolarization commutes with unitary conjugation, so $\rho_n = I/2 + \alpha^n U^n(\rho_0 - I/2)U^{*n}$ and its audit mean is $\alpha^n c_n$. The unique minimizing decision is therefore the same nonperiodic sign sequence whenever $\alpha > 0$. Apply Theorem S29.2. A wrong decision at time n has excess loss at most $4\alpha^n$; summing the tail from M gives the displayed bound. For a constant decision, replace q by $q\alpha$ in the elementary cosine series, retaining the factor p, to obtain the one-label value. Trace differences of states have zero trace and contract by exactly the factor α under this channel. At $\alpha = 0$ every positive-length failure word has state $I/2$ and both decisions are tied, while at the empty word the decision $+1$ is optimal. The constant $+1$ rule attains $2-2p$. At $I/2$ the two limiting losses agree, explaining why the preceding positive-gap theorem does not apply. □

S29.5. **The exact two-label frontier for a noisy quantum orbit.** The qubit application also admits a complete nontrivial finite-memory optimization. The next formula allows every rotation angle and damping strength, rather than fixing the algebraic example's angle.

Theorem S29.10 (Exact two-live-label quantum prediction law)**.** *Consider the same prepared qubit, herald and squared Pauli audit, with a Bloch rotation angle ϑ, $c = \cos\vartheta \in [-1, 1]$, and depolarizing strength $0 \le \alpha \le 1$. Put $x = q\alpha$ and*

$$f_c(t) = \frac{1 - ct}{1 - 2ct + t^2}, \qquad A = pf_c(x).$$

The optimal one-live-label risk is $2 - 2A$. The exact optimum among all stochastic observers with at most two live labels is $2 - 2S_2$, where the following cases determine S_2 and an attaining transition matrix. On a herald the first label decodes $+1$ and the second -1, except in the constant-decision cases.

If $x = 0$, then $S_2 = p$ and one label suffices. For $x > 0$ and $0 \le c \le 1$, let $s = \sqrt{1 - c^2}$ and $t_* = c/(1 + s)$. Then

$$S_2 = \begin{cases} A, & x \le t_*, \\ p(1 + s)/s - A, & x > t_*. \end{cases}$$

The second case never occurs at $c = 1$. In the first case use a constant $+1$ decision. In the second case an optimal failure matrix is

$$Q = \begin{pmatrix} t_*/x & 1 - t_*/x \\ 0 & 1 \end{pmatrix}.$$

For $x > 0$ and $-1 \le c < 0$, write $k = -c$, $s = \sqrt{1 - k^2}$, and

$$t = \min\left\{x, \max\left\{0, \frac{x + k - s\sqrt{1 + 2kx + x^2}}{1 + kx}\right\}\right\}.$$

Then

$$S_2 = A + \frac{2[pf_c(-t) - A]}{1 + t/x}, \qquad Q = \begin{pmatrix} 0 & 1 \\ t/x & 1 - t/x \end{pmatrix}$$

is attaining. All formulas include ties and degenerate transitions. The first compulsory call is included in the cost, which remains exactly $1/p$ in every case.

In particular, for the irrational algebraic orbit with $c = 3/5$, $p = 1/2$ and $\alpha = 1$, the exact two-live-label risk is $43/52$, attained by $Q = \left(\begin{smallmatrix} 2/3 & 1/3 \\ 0 & 1 \end{smallmatrix}\right)$. For this same angle, two live labels improve on one if and only if $q\alpha > 1/3$. This is a phase boundary for the declared random-horizon prediction task, not the unresolved intermediate-memory deterministic-release law.

Proof. At the least budget, Lemma S29.1 reduces every feasible observer to first-herald commitment. Its two-state failure matrix has the form $Q = \left(\begin{smallmatrix} 1-a & a \\ b & 1-b \end{smallmatrix}\right)$, $0 \le a, b \le 1$, from initial label zero. Write $u_i \in [-1, 1]$ for the expected decoded sign at label i. Its risk is

$$2 - 2v \cdot u, \qquad v = p \sum_{n \ge 0} x^n \cos(n\vartheta) e_0 Q^n.$$

The series is absolutely convergent because $x < 1$. If $a + b = 0$, then $v = (A, 0)$. Otherwise put $\lambda = 1 - a - b$ and $z = a/(a + b)$. The elementary two-state decomposition gives

$$v_0 = (1 - z)A + zB, \quad v_1 = z(A - B), \qquad B = p f_c(x\lambda) > 0.$$

Also $A > 0$, so $v_0 > 0$. Optimizing the two decoder signs therefore gives reward A if $B \le A$, and $A + 2z(B - A)$ otherwise. For a fixed λ, the largest feasible z is one when $\lambda \ge 0$ and $1/(1 - \lambda)$ when $\lambda < 0$.

For $c \ge 0$, differentiation gives $f_c'(t) = (c - 2t + ct^2)/(1 - 2ct + t^2)^2$. Its maximum on $[-x, x]$ occurs at $\min\{x, t_*\}$. When $x \le t_*$, $B \le A$ for every matrix. Otherwise the maximizing $\lambda = t_*/x$ is nonnegative and permits $z = 1$. Thus the upper $2 \max B - A$ is attainable, and $f_c(t_*) = (1 + s)/(2s)$ gives the formula.

For $c = -k < 0$, nonnegative λ is optimized at zero. For negative λ set $t = -x\lambda \in [0, x]$ and choose $z = 1/(1 + t/x)$. The improvement divided by two is

$$\frac{px}{1 + 2kx + x^2} \frac{x + k - (1 + kx)t}{1 - 2kt + t^2}.$$

Its derivative has the sign of $(1 + kx)t^2 - 2(x + k)t + 2k(x + k) - (1 + kx)$. The larger root exceeds x because $(x + k)/(1 + kx) > x$ for $k > 0$. Consequently the maximum on $[0, x]$ is at the smaller root clipped to this interval; that root is the displayed t. The endpoint zero includes the best nonnegative-λ choice. This proves global optimality over all two-state matrices and all randomized decoders. Substitution gives $A = 7/13$, $\max B = 9/16$ in the specified algebraic case, hence $2 - 2(2 \cdot 9/16 - 7/13) = 43/52$. The strict improvement condition follows from the positive-angle cases. No numerical optimization is used in this proof. □

S29.6. **A finite-precision synthesis bound for unary rational channels.** There is also a positive computational conclusion at the boundary, rather than only a nonattainment theorem. This statement concerns one known prepared channel and a fixed finite dictionary; the full robust synthesis problem is not asserted to have the same complexity.

Theorem S29.11 (Polynomial prefix synthesis at a bounded herald rate)**.** *Suppose Y has one letter, the prepared density matrix and the channel superoperator in real Hermitian coordinates have rational entries, and the m loss observables have rational entries and lie in $[0, LI]$. Let p and a requested $0 < \epsilon < L$ be rational. From these data one can construct an autonomous observer of exact cost $1/p$ and risk at most $V + \epsilon$, using*

$$T = \max\left\{1, \left\lceil \frac{\log(L/\epsilon)}{-\log(1 - p)} \right\rceil\right\}$$

live labels. The integer T may instead be chosen by exact successive rational tests of $L(1 - p)^T \le \epsilon$, without evaluating a logarithm.

After a common-denominator preprocessing, suppose the data entries have denominator dividing D and numerators and D have at most B bits; include the rational herald rate, loss bound and tolerance in the input size. The synthesis uses $O(T(d^4 + md^2))$ rational coordinate operations on integers with $O(TB + \log d)$ bits, in addition to the scalar stopping test. A direct implementation has $O(d^2(TB + \log d))$ bits of working storage, apart from its output program, and a transition/readout table of $O(T(\log(T + 1) + \log(m + 1)))$ bits. Thus, for a herald rate bounded away from zero, this is a polynomial bit-time construction in the rational input size and $\log(1/\epsilon)$, using standard integer

arithmetic. It is not a polynomial bound uniform as $p \downarrow 0$, a matching lower bound, or a complexity claim for continuum-model robust optimization.

Proof. Compute $\rho_n = \Phi^n(\rho_0)$ for $0 \le n < T$. At each step evaluate the m rational numbers $\operatorname{tr}(F_r\rho_n)$ and choose a least one, resolving ties by the fixed dictionary order. Use a chain of T live labels; on failure advance until the last label, which repeats. At a herald commit the decision stored at that label. The rule is optimal on all terminal words of length less than T. Its excess loss is at most L on the remaining terminal mass q^T, while its call count is still geometric with mean $1/p$. This proves the guarantees.

A channel application is a matrix–vector product in d^2 real Hermitian coordinates, requiring $O(d^4)$ scalar operations. Evaluating all decisions costs $O(md^2)$ more. Use real and imaginary matrix-entry coordinates, with off-diagonal entries paired, so all physical state coordinates have absolute value at most one. After n applications their denominator divides D^{n+1}. Keeping this common denominator explicitly gives integer numerators of $O((n+1)B)$ bits; a loss inner product adds only $O(B + \log d)$ bits. The same size bound follows directly during multiplication and summation, and comparison of decisions uses integer comparisons at the shared denominator. Store two state vectors and the current best decision; the stated working bound includes the growing state numerators. The input itself is read-only and not included in working storage. The output table stores a successor and a dictionary index per live label, with no stored floating phase or external clock. Standard integer addition and multiplication have polynomial bit cost, giving the final claim. If the exact scalar test is used, the denominator and numerator of q^T also have $O(TB)$ bits. The displayed bound on T is logarithmic in precision only when p has the stated positive lower bound. □

The qubit in Theorem S29.5 belongs to this rational-superoperator class: although U has algebraic entries, its conjugation map has rational coefficients with denominator five. Its exact optimum requires unbounded classical memory, but its boundary-optimal risk has efficiently computable finite-memory approximants. The stored number of live-label bits is $\lceil \log_2 T \rceil$; this does not replace the separate program and arithmetic storage accounts above.

The algebraic rotating-qubit idea and nonperiodic quantum acceptance events have established antecedents in quantum automata [43]. We do not claim priority for that mechanism. Here it is compiled into a paid, single-command prediction experiment whose least feasible cost is attained, whose full-history bounded-risk optimum is attained, and whose exact risk is nevertheless unattainable with every fixed finite classical memory. The same boundary admits exponentially accurate finite-label approximations. The transition-table account excludes the fixed numerical dictionary and primitive description, whose encoding lengths must also be added when measuring total program size. These facts distinguish optimum attainment, fixed-memory gaps, and density, rather than inferring one from another. They do not settle the sharp intermediate-memory phase-type timing curve of Theorem S30.1, the general boundary-density question, or a universal unknown-model predictive quotient.

S30. A timing task with an exact attainment threshold

The complete stopped law can distinguish controllers whose terminal outputs are identical. A basic instance is a delayed-release protocol: a stateless physical device emits the same productive report at every unit-cost call, and the observer is asked to commit at a prescribed call. There is one legal command, a constant decoder and a constant audit. An autonomous observer has no free clock, call counter or persistent random tape. Its retained label is its only intercall state. Thus finite live-state implementations are discrete phase-type stopping mechanisms. Phase-type timing and its optimal variability have classical antecedents; the continuous-time coefficient-of-variation theorem of Aldous–Shepp [39] is a different objective from the complete-law, discrete-call approximation considered here.

Fix an integer $K \geq 2$ and let the target stopped law be commitment exactly at call K. For a proper observer with stopping time $\tau \geq 1$, its total-variation discrepancy from this target is $1 - \Pr(\tau = K)$. Write $\Delta_{N,M}(K)$ for the smallest discrepancy at expected-call budget $N \geq 1$ with at most M autonomous live labels. Event stopping may reuse one live label for the constant decoder. State stopping requires an additional disjoint terminal label. Let $\Delta_{N,\infty}(K)$ denote the full-history optimum. This notation counts *live* labels, not total labels, in both conventions.

Theorem S30.1 (Delayed release: budget curve, exact state threshold and pumping gap)**.** *For every $K \geq 2$ and $N \geq 1$,*

$$\Delta_{N,\infty}(K) = \frac{(K-N)_+}{K-1}. \tag{S30.1}$$

For one live label, put $n = \min\{N, K\}$. The exact optimum is

$$\Delta_{N,1}(K) = 1 - \frac{1}{n}\left(1 - \frac{1}{n}\right)^{K-1}. \tag{S30.2}$$

At every $N > 1$, the least live-label count attaining (S30.1) *is exactly K. Consequently its least total count is K for event stopping and $K+1$ for state stopping. At $N = 1$, immediate commitment with one live label already attains the discrepancy one.*

More quantitatively, for $M < K$, every feasible autonomous observer with $s = \Pr(\tau = K)$ obeys

$$s + \frac{s^2}{M^{2K-2}} \leq 1, \qquad (K-1)s + \frac{Ks^2}{M^{2K-2}} \leq N - 1. \tag{S30.3}$$

These explicit bounds force a strict gap from the unrestricted optimum at every $N > 1$. For $N \geq K$ they give the budget-independent bound $\Delta_{N,M}(K) \geq 1/(4M^{2K-2})$. The one-label curve and K-label attainment threshold are exact; no sharpness of the intermediate-M quantitative bound is asserted.

Proof. Put $s = \Pr(\tau = K)$. Since $\tau \geq 1$, $\mathbb{E}\tau \geq 1 + (K-1)s$. Thus $s \leq \min\{1, (N-1)/(K-1)\}$. This is attained by stopping at call one with probability $1 - s$ and at call K with probability s. It is an actual causal construction: at the first call either commit or enter a finite deterministic chain, and then advance one counted label per call until commitment. It uses K live labels when $s > 0$, with no external clock or surviving seed. For $N \geq K$ use the deterministic chain. This proves (S30.1) and its upper state count. Equivalently, the bounded score $\mathbf{1}_{\{\tau \neq K\}}$ witnesses this complete-law discrepancy in Theorem S27.3; the constant terminal-output marginal would not witness it.

With one live label the post-report halt probability is one constant $p > 0$, so τ is geometric on the positive integers. The budget says $p \geq 1/N$, and its target mass is $p(1-p)^{K-1}$. Its derivative on $(0, 1)$ has the sign of $1 - Kp$. The maximum over $[1/N, 1]$ is therefore at $p = 1/\min(N, K)$. The endpoints give the same formula, proving (S30.2).

For the lower state count and the quantitative bound, let Q be the substochastic transition matrix on the live labels and let $h = (I - Q)\mathbf{1}$ be its halt vector. The initial live label is prescribed. A path that halts at call K contains K live labels before its calls and at most M^{K-1} possible label sequences. If $s > 0$, one such path has probability $q \geq s/M^{K-1}$. When $M < K$, its label sequence repeats a label. The intervening live cycle has length $\ell \geq 1$ and transition-product probability r. Every other factor in the original path is at most one, so $r \geq q$. Traverse that same cycle once more and follow the original remainder to halt. This is a legal path of length $K + \ell$, of probability $qr \geq q^2$. It is disjoint from all length-K paths. Normalization yields $1 \geq s + q^2$, proving the first inequality. The identity $\mathbb{E}\tau = 1 + \mathbb{E}(\tau - 1)$ yields

$$\mathbb{E}\tau \geq 1 + (K-1)s + (K + \ell - 1)q^2 \geq 1 + (K-1)s + Kq^2,$$

which proves the second. Zero-probability paths need no conditioning, and the case $s = 0$ is immediate.

For $1 < N < K$, the positive quadratic term makes the largest permitted s strictly smaller than $(N-1)/(K-1)$; for $N \geq K$ the first inequality makes it strictly smaller than one. Thus fewer

than K live labels cannot attain the unrestricted curve, even as an infimum. If $s \geq 1/2$, the first inequality gives $1 - s \geq 1/(4M^{2K-2})$; if $s < 1/2$, that bound is weaker than $1 - s > 1/2$. This gives the last assertion. The globally typed state-stopped interface has a separate terminal label, whereas the final event permits reuse in event stopping, yielding the two total counts. □

This temporal lower bound is not a command-domain incompatibility count: every live context has the very same legal command. It comes from repeated autonomous state and the extra observable stopping histories forced by a positive cycle. A clock supplied as a physical input, a nonconstant report encoding elapsed time, or a persistent external seed changes the interface and invalidates that lower. The result is an exact delayed-release consequence with matching budget and attainment counts, not a solution of the entire intermediate-state phase-type optimization or a claim to settle a previously recognized open timing problem.

S31. An intrinsic hierarchy for paid causal synthesis

A comparison inequality does not create the acquisition program used by its upper bound. We give an intrinsic criterion over all classical feedback programs on a finite-branching causal interface. It applies to unknown-model classical experiments and finite-dimensional quantum instruments, with one program shared by every model and initial state. Exact unrestricted-history synthesis and exact fixed-memory synthesis remain different conclusions.

S31.1. **Primitive data and the controller class.** A visible history records executed commands and reports. At each open decision node h, the legal next-command set $\mathcal{A}(h)$ is finite and nonempty. After a productive report there may also be a finite set $\mathcal{O}(h)$ of legal terminal routes. There is no terminal route at the root or after a nonproductive report. Each command has finitely many reports, so the tree is locally finite and countable. Legal domains are common to the admitted models and are part of the interface. A physical phase determined by visible history is not an additional free input to a finite-memory observer.

The unknown model θ belongs to an arbitrary nonempty set Θ, not necessarily finite, compact or equipped with a prior. At each history the hidden quantum state lies in a specified finite-dimensional matrix algebra. Command a has completely positive maps $\mathcal{I}_{\theta,h,a,y}$ with

$$\sum_y \mathcal{I}^*_{\theta,h,a,y}(I) = I.$$

Dimensions may depend on the model and finite history. The initial identity is I_θ. Nonnegative classical transition matrices are the diagonal special case. There is no positive-likelihood or eigenvalue floor.

Costs are nonnegative. Their conditional expected-cost observables satisfy the uniform floor

$$G_{\theta,h,a} \succeq \underline{c} I, \qquad \underline{c} > 0. \tag{S31.1}$$

A pathwise floor implies this assumption, but only the displayed conditional floor is needed here. The cost C_τ is the core cost through commitment. A separately executed preparation or audit is charged in addition or represented as a compulsory call; it is not silently included without its physical operation.

Fix finitely many bounded terminal scores $j = 1, \ldots, J$. Route o at h has a Hermitian score observable $F_{\theta,h,o,j}$ on the current state, with

$$\alpha_j I \preccurlyeq F_{\theta,h,o,j} \preccurlyeq \beta_j I, \qquad -\infty < \alpha_j \leq \beta_j < \infty. \tag{S31.2}$$

The route is chosen before observing its future audit. For example, F is the adjoint of the declared audit instrument applied to a bounded score of its outcome and the chosen route. These are primitive unnormalized observables, not assumed posteriors. Prescribed Hermitian targets on the initial state are denoted by $A_{\theta,j}$; constant targets are scalar multiples of I_θ.

A controller uses full visible history and independent private randomness, but neither the unknown model nor an unknown initial density matrix. It must commit exactly once almost surely and satisfy

$$\sup_{\theta,\rho} \mathbb{E}_{\theta,\rho} C_\tau \le N, \qquad \mathbb{E}_{\theta,\rho}\Phi_j = \operatorname{tr}(\rho A_{\theta,j}) \quad (\theta \in \Theta,\ \rho \succeq 0,\ \operatorname{tr}\rho = 1). \tag{S31.3}$$

This is classical feedback over the specified quantum experiment, not an uncharged coherent controller memory. Uniformity over initial states is distinct from uniformity over rooted continuations of a history-dependent experiment; both are required before a program may be repeated after rejection. Restricted preparation faces and deadlocks are treated explicitly in the current supplement.

S31.2. **One realization plan, rather than modelwise policies.** At a horizon $T \ge 1$, include decisions following the first T reports. Continuing at depth T leaves an open branch; its next physical call has not occurred. The weight $\gamma(h)$ is the product of controller choice probabilities along h, excluding physical report probabilities. Write $\gamma(h,d)$ after a command or terminal decision. The common sequence-form polytope $\mathcal{P}_T$ is given by

$$\gamma(\varnothing) = 1, \quad \gamma(h,d) \ge 0, \quad \sum_d \gamma(h,d) = \gamma(h), \quad \gamma(h,a,y) = \gamma(h,a) \tag{S31.4}$$

for every present report child. All variables lie in $[0,1]$, and the same variables are used for every model. Sequence form itself is classical [10]; its role is to preserve the common-policy quantifier.

Let $\mathcal{I}_{\theta,h}$ be the composition of the primitive maps along h, without controller weights. Define affine functions on this polytope by

$$\mathsf{B}_{\theta,T} = \sum_{\substack{|h|\le T \\ o\in\mathcal{O}(h)}} \gamma(h,o)\mathcal{I}^*_{\theta,h}(I), \qquad \mathsf{S}_{\theta,T} = I_\theta - \mathsf{B}_{\theta,T}, \tag{S31.5}$$

$$\mathsf{V}_{\theta,j,T} = \sum_{\substack{|h|\le T \\ o\in\mathcal{O}(h)}} \gamma(h,o)\mathcal{I}^*_{\theta,h}(F_{\theta,h,o,j}), \tag{S31.6}$$

$$\mathsf{C}_{\theta,T} = \sum_{\substack{|h|< T \\ a\in\mathcal{A}(h)}} \gamma(h,a)\mathcal{I}^*_{\theta,h}(G_{\theta,h,a}). \tag{S31.7}$$

Every coefficient is a raw finite-path matrix. No report probability is divided out.

Lemma S31.1 (One plan for all models)**.** *The polytope is nonempty and compact. Its elements are exactly the finite-prefix laws of common randomized feedback controllers, with expectations given by traces against the displayed operators, simultaneously for every model and initial state. Arbitrary private random memory does not enlarge these laws relative to visible-history policies using fresh randomness. Moreover* $\mathsf{B}_{\theta,T}, \mathsf{S}_{\theta,T} \succeq 0$ *and their sum is* I_θ*.*

Proof. A behavioral policy gives the product weights. Conversely, at a node of positive weight choose d with probability $\gamma(h,d)/\gamma(h)$; at a zero-weight node choose any legal row. Induction recovers every specified weight. The finite linear constraints form a closed bounded set, and any choice of legal rows proves nonemptiness.

To include arbitrary private memory, fix a visible action–report path and integrate over all private random tapes producing those controller choices when supplied that report path. This probability is independent of the physical model and initial state. The raw path instrument is the same for every compatible tape, because the executed commands and reports have been fixed. Thus it factors from the tape integral, and the integrated choice weights obey the same equations. The ratio construction reproduces the laws in all models at once. It uses full visible-history storage; it does not make a tape into free finite memory.

Multiply the raw path effects by these weights. Summing report effects uses normalization of each instrument, and summing controller choices uses the plan equation. Induction shows that

terminated and open effects sum to the incoming effect, starting from I_θ. Every such effect is positive. The cost and score formulas follow by summing their actual finite-prefix contributions. □

S31.3. **The necessary cost reserve and terminal-score intervals.** For a finite $F \subset \Theta$, let $\mathcal{H}_T(F, N)$ be the feasibility problem in $\gamma \in \mathcal{P}_T$ defined by

$$\mathsf{C}_{\theta,T} + \underline{c}\mathsf{S}_{\theta,T} \preccurlyeq N I_\theta, \tag{S31.8}$$

$$\alpha_j \mathsf{S}_{\theta,T} \preccurlyeq A_{\theta,j} - \mathsf{V}_{\theta,j,T} \preccurlyeq \beta_j \mathsf{S}_{\theta,T} \quad (\theta \in F,\ 1 \le j \le J). \tag{S31.9}$$

These are linear inequalities in the classical case and linear matrix inequalities in the quantum case. They are not implemented by pretending that open paths terminate at the cutoff. Those paths owe at least one further paid call, and their remaining bounded scores lie in the displayed interval.

Lemma S31.2 (Restriction and the sharp budget tail)**.** *A feasible plan at level $T+1$ restricts to a feasible plan at level T. Every feasible level with $N \ge \underline{c}$ satisfies*

$$0 \preccurlyeq \mathsf{S}_{\theta,T} \preccurlyeq \delta_T I_\theta, \qquad \delta_T = \min\{1, (N/\underline{c} - 1)/T\}. \tag{S31.10}$$

If $N < \underline{c}$, there is no feasible proper program.

Proof. Write $D_T = \mathsf{B}_{T+1} - \mathsf{B}_T \succeq 0$ for the new terminal effect. Normalization gives $\mathsf{S}_T = D_T + \mathsf{S}_{T+1}$. New score increments lie between $\alpha_j D_T$ and $\beta_j D_T$. Adding them to the next-level remainder intervals proves the current-level intervals. Every path still open after depth T executes another call before the next decision. Therefore $\mathsf{C}_{T+1} - \mathsf{C}_T \succeq \underline{c}\mathsf{S}_T$. Together with the next-level cost constraint this implies the current one.

Let S_t denote the open effect after t calls, so $S_0 = I$ and $S_t \succeq S_T$ for $t < T$. The conditional cost floor gives $\mathsf{C}_T \succeq \underline{c} \sum_{t=0}^{T-1} S_t \succeq \underline{c}[I + (T-1)S_T]$. Adding the reserve gives $\underline{c}[I + TS_T] \preccurlyeq NI$. Positivity and $S_T \preccurlyeq I$ prove the claimed bound. Every proper program must execute its first paid call, proving the infeasible boundary below the floor. □

Theorem S31.3 (Finite-obstruction characterization of paid causal synthesis)**.** *Under the stated finite-branching, positive-cost and bounded-score hypotheses, the following are equivalent:*

(i) *One common proper visible-history program satisfies* (S31.3) *for every model and every initial state.*

(ii) *$\mathcal{H}_T(F, N)$ is feasible for every finite model set $F \subset \Theta$ and every integer $T \ge 1$.*

Consequently nonexistence is witnessed by a finite horizon and a finite set of operating models. Feasibility at arbitrarily many tested levels alone is not a finite certificate for the infinite condition. No fixed bound on controller memory is asserted for the limiting policy.

Proof. For necessity take the prefix of the actual program. Every open path must make at least one additional call before commitment, so its conditional future expected cost is at least $\underline{c}$. Nonnegative conditional summation proves the reserved cost inequality. Scores on the disjoint future terminal events lie between α_j and β_j times their total open mass. Almost-sure commitment makes that mass exactly S_T, proving the score intervals. These same prefix weights work in every model.

For sufficiency consider the compact product of all nodewise probability simplexes on the countable history tree. A finite-prefix plan can be extended to an element of this product by arbitrary legal rows at later nodes. Its realization weights are finite products of coordinates, hence continuous. Each finite collection of the cost and score constraints is closed. For such a collection take the union of its finitely many model indices and its greatest horizon. Assumption (ii) and the restriction lemma give one feasible plan for every constraint in the collection. Extend it to all nodes. The closed constraints therefore have the finite-intersection property in a compact space, and one common infinite policy satisfies them all.

For this policy (S31.10) gives uniformly vanishing survival probabilities, including the exact boundary $N = \underline{c}$. Thus it commits almost surely in every model and initial state. Its partial costs are nonnegative and at most N; monotone convergence yields the actual total cost bound. The

norm of the score remainder is at most $\max(|\alpha_j|, |\beta_j|)\delta_T$. It tends to zero, so the bounded terminal scores have exactly the prescribed operator expectations. The program is common because the product coordinates were common throughout. Finally, negating the equivalence proves existence of a finite obstruction when no such program exists. □

Corollary S31.4 (Attainment of the least finite cost)**.** *If any common proper program satisfies the prescribed targets at finite uniform cost, then the infimum of such cost budgets is attained.*

Proof. Let feasible budgets decrease to their finite infimum N_*. For each fixed finite model set and horizon, the corresponding nonempty compact feasible sets decrease with the budget. Their intersection is nonempty and is feasible at N_*. Every finite problem at N_* is therefore feasible; the theorem gives a proper program at that budget. The first-call floor ensures $N_* \geq \underline{c}$. □

S31.4. **A strict dual certificate at an obstructing level.** List all left-minus-right matrix violations of (S31.8)–(S31.9) as $K_l(\gamma)$, $1 \leq l \leq L_0$. Thus feasibility is $K_l \preccurlyeq 0$ for every l. Define

$$v = \min_{\gamma \in \mathcal{P}_T} \max\{0, \max_l \lambda_{\max}(K_l(\gamma))\}.$$

Compactness makes this minimum attained and $v > 0$ exactly when the finite problem is infeasible.

Theorem S31.5 (Strict finite dual obstruction)**.** *The exact violation value is*

$$v = \max_{\substack{W_l \succeq 0 \\ \sum_l \operatorname{tr} W_l \leq 1}} \min_{\gamma \in \mathcal{P}_T} \sum_l \operatorname{tr}[W_l K_l(\gamma)]. \tag{S31.11}$$

The maximum is attained. An infeasible level has a positive-matrix separating family with strictly positive value. The inner minimization is a finite linear program in the common realization weights. No Slater hypothesis or inverse covariance is required.

Proof. For a fixed collection of violations, maximizing the displayed pairing over positive matrices with total trace at most one gives their largest positive eigenvalue, or zero by taking all matrices zero. Taking a minimizer defining v gives the weak upper bound for every family W.

For the converse when $v > 0$, fix $0 \leq \eta < v$. In the finite direct sum of Hermitian matrix spaces, the point $(\eta I_l)_l$ does not belong to the closed convex set

$$\{(K_l(\gamma))_l : \gamma \in \mathcal{P}_T\} + \prod_l \{D_l \succeq 0\}.$$

This set is closed because its first summand is compact. Finite-dimensional strict separation gives a nonzero linear functional whose infimum on that set exceeds its value at the excluded point. The functional must be nonnegative on the positive cone, or its infimum along a cone ray would be minus infinity. Hence it has the form of pairing with positive W_l. Their total trace is positive; normalize it to one. Separation yields an inner minimum greater than η. Let $\eta \uparrow v$ and take a convergent subsequence in the compact normalized positive-matrix set. The inner minimum is continuous in W, being the minimum of a continuous bilinear function over compact $\mathcal{P}_T$. Its limiting value is at least v. If $v = 0$, a feasible plan gives the upper bound zero and $W = 0$ attains zero. This proves equality and attainment in all cases. □

The finite dual is a genuine obstruction, not an assertion that the first obstructing horizon is uniformly bounded or effectively discoverable for arbitrary real primitive data. Existing quantum decision-process and reachability theories already contain substantial control, duality and computability results [12, 13, 14]. Generic quantum MDPs, sequence form and convex separation are not claimed as new ingredients.

S31.5. **Paid finite-prefix completion.** Suppose every open leaf reachable by a selected finite prefix admits one common legal fallback that commits with conditional expected cost at most $\kappa_0 \geq \underline{c}$, uses at most B_0 live roles, and has terminal scores in the same $[\alpha_j, \beta_j]$ intervals. Its command and physical phase compatibility must be certified at these leaves.

Corollary S31.6 (A paid finite-prefix completion)**.** *A feasible level-T plan completed by this fallback has, for every model and initial state,*

$$\mathbb{E}C \leq N + (\kappa_0 - \underline{c})\delta_T, \tag{S31.12}$$

$$-(\beta_j - \alpha_j)\delta_T I \preccurlyeq A_{\theta,j} - \mathbb{E}[\Phi_j] \preccurlyeq (\beta_j - \alpha_j)\delta_T I. \tag{S31.13}$$

Here the score expectation is its operator on the initial state. If the finite prefix uses L_T live node roles and the completed terminal decoder has t actual values, a sufficient state-stopped label count is $L_T + B_0 + t$ and an event-stopped count is $\max(L_T + B_0, t)$. Program constants, arithmetic and the possible exponential growth of L_T are separate costs.

Proof. The completed cost is at most $\mathsf{C}_T + \kappa_0 \mathsf{S}_T$. Subtract the reserved cost inequality and use the survival bound. Both the prescribed remaining score and the fallback's remaining score lie between $\alpha_j \mathsf{S}_T$ and $\beta_j \mathsf{S}_T$. Their difference lies between the stated opposite multiples of S_T, proving the error bound. A finite prefix is implemented by storing its current node; the fallback uses its declared live roles. The exact role-packing lemma gives the two sufficient counts. These are existence counts, not a claim that a real-valued finite plan has an exact finite-bit sampler. □

Proposition S31.7 (Sharpness of the uniform tail rate)**.** *For unit cost per call, every $b \geq 1$ and integer $T \geq 1$ with $b - 1 \leq T$ admit a proper finite-branching program of mean cost b and survival probability $(b-1)/T$ after T calls.*

Proof. With probability $p = (b-1)/T$ commit on call $T+1$, and otherwise on call one. Then $\mathbb{E}\tau = 1 + Tp = b$ and $\Pr(\tau > T) = p$. Its prefix has cost $1 + (T-1)p$ and reserve p. Thus the reserve and the boundary $b = 1$ are exact. If $b - 1 > T$, the universal tail bound one is attained by a deterministic sufficiently long path. This concerns the survival bound, not a uniform expected cost-tail estimate. □

S31.6. **Synthesis of a finite marked squared-task reference.** For a finite virtual mark set X, fix masses ν_x, target means $f(x) \in \mathbb{R}^k$ and nonnegative residuals $w(x)$ of a genuinely prepared reference law. Assume bounded audit coordinates. Use as terminal scores

$$\mathbf{1}_{\{X=x\}}, \quad \mathbf{1}_{\{X=x\}} Z_i \ (1 \leq i \leq k), \quad \mathbf{1}_{\{X=x\}} \|Z\|^2,$$

with respective scalar-identity targets ν_x, $\nu_x f_i(x)$ and $\nu_x[w(x) + \|f(x)\|^2]$. They are computed from the declared terminal audit instrument and route.

Corollary S31.8 (Intrinsic synthesis of finite-mark squared-task balance)**.** *There is one common proper finite-branching acquisition program with these unnormalized marked moments and cost at most N for all models and initial states if and only if every finite problem of the hierarchy for this finite score list is feasible. Such a program satisfies*

$$\mathbb{E}[q(X)\|Z - r(X)\|^2] = \sum_x \nu_x q(x)[w(x) + \|f(x) - r(x)\|^2]$$

for every selected readout. If, in addition, the same return rule is legal and the same certificate holds on every actual rooted continuation after rejection, it gives exact task balance for repeated acquisition. The hierarchy alone does not supply a finite live-memory bound.

Proof. The first equivalence is the hierarchy theorem applied to the explicit bounded score list. Expand the squared loss in each mark and substitute the three unnormalized moment identities. This proves the task equation without division by ν_x, so zero-mass marks are harmless. Conversely

the displayed marked moment requirement is exactly that finite target list. Applying this program repeatedly requires its hypotheses on the actual later roots; no equality of initial-state matrices can replace missing history-dependent continuation data. A finite prefix with a legal fallback, or the separate finite-memory theorem, supplies an explicitly charged implementation when its additional hypotheses hold. □

S32. Compact uncertainty: certified risk at the unchanged budget

The preceding primitive construction uses finitely many operating models. A finite grid of an unknown continuum is not itself a robust model family: a policy that meets its budget on the grid can overspend between grid points. We now control that discrepancy from the raw instruments, including the unbounded waiting cost after completion. The resulting finite programs enclose the optimal risk for a compact model family and produce a common finite-label observer at the *original* budget. No model index is revealed to the observer. The proof replaces a worst-horizon accumulation by the actual pre-call occupation of the tested policy.

Robustness and approximation of partially observed control have substantial prior theory [29, 30, 14]. The particular assertion here concerns an undiscounted stopping task, a uniform expected-cost constraint, finite output width, and a physically executable completion which restores that same constraint. It is not priority for model approximation or perturbation bounds in general.

S32.1. **A compact family of primitive instruments.** Use the finite command/report interface of Section S26, but let the unknown, *fixed throughout a run*, model θ range over a nonempty compact metric space Θ. All models act on the same d-dimensional physical space. Productive report sets and legal commands are common. Each normalized instrument is $\{\mathcal{I}_{\theta,a,y}\}_y$, each nonnegative call cost has observable $G_{\theta,a}$, and the prospective terminal loss at $r \in D$ has observable $F_{\theta,a,y,r}$. Here D is compact metric, and

$$c_* I \preccurlyeq G_{\theta,a} \preccurlyeq c^* I, \qquad 0 \preccurlyeq F_{\theta,a,y,r} \preccurlyeq L I, \qquad 0 < c_* \le c^* < \infty,$$

with $L > 0$. An observed cost or phase used by a policy must be included in the declared report; it is not another free input. Additional preparation, return and terminal-audit charges use the same explicit convention as before.

For two models define symmetric primitive discrepancies

$$\epsilon_I(\theta,\phi) = \frac{1}{2} \max_a \sup_{\rho \succeq 0,\, \operatorname{tr}\rho = 1} \sum_y \|\mathcal{I}_{\theta,a,y}(\rho) - \mathcal{I}_{\phi,a,y}(\rho)\|_1, \tag{S32.1}$$

$$\epsilon_g(\theta,\phi) = \max_a \|G_{\theta,a} - G_{\phi,a}\|, \qquad \epsilon_F(\theta,\phi) = \sup_{a,y \in Y_a^+, r \in D} \|F_{\theta,a,y,r} - F_{\phi,a,y,r}\|.$$

Assume these discrepancies tend uniformly to zero with the model distance. Assume also a common modulus $\omega(\eta) \downarrow 0$ for terminal losses as decision distance tends to zero. Joint continuity on the compact model and decision carriers is one sufficient primitive condition. The report-resolved trace distance in (S32.1) is for classical feedback on the specified physical system; a coherent external reference or quantum controller is not included by calling it a diamond distance.

A finite set $F \subset \Theta$ is an $(\epsilon_I, \epsilon_g, \epsilon_F)$-cover when each $\theta \in \Theta$ has a $\phi \in F$ with all three discrepancies bounded by these displayed numbers. The bounds, not just the grid points, are the certificate. Compactness and the assumed continuity provide arbitrarily fine finite covers. An *effective* cover requires supplied representations and certified moduli; compactness alone does not provide an algorithm on unspecified real data.

Lemma S32.1 (A uniform common fallback from compact primitive data)**.** *There is a proper common policy of finite uniform expected cost for every model and every initial density matrix if and*

only if no model has a common dark subspace. In this case any one fixed full-support command row, stopped at its first productive report, is a uniform fallback. With the notation of Theorem S26.1,

$$\delta = \min_{\theta\in\Theta} \lambda_{\min}[I - (\mathcal{T}_\theta^*)^d I] > 0, \qquad \kappa_0 = c^* d/\delta$$

is a valid cost bound. If a finite cover has instrument error at most ϵ_I, then

$$\delta \geq \min_{\phi\in F} \lambda_{\min}[I - (\mathcal{T}_\phi^*)^d I] - d\epsilon_I. \tag{S32.2}$$

A positive right side certifies the common fallback on the entire family, not only on the cover.

Proof. A dark subspace in even one model prevents any all-initial-state policy from committing there, by the commandwise argument of Theorem S26.1. If none exists, that theorem makes every displayed depth-d observable positive definite. Its entries depend continuously on θ. Compactness and continuity of the least eigenvalue therefore give a strictly positive minimum. The same block-survival proof bounds every expected cost by $c^* d/\delta$, using the one fixed row and no model-dependent choice.

For the quantitative inequality, if $0 \preccurlyeq A_y \preccurlyeq HI$, instrument normalization permits subtraction of $HI/2$ in every report component. Trace duality gives, for every density matrix,

$$\left| \sum_y \operatorname{tr}\left[A_y(\mathcal{I}_{\theta,a,y} - \mathcal{I}_{\phi,a,y})(\rho)\right] \right| \leq H\epsilon_I(\theta,\phi). \tag{S32.3}$$

Indeed the sum of the traces of the differences is zero, and the remaining terms are at most $H/2$ times the sum of their trace norms. To compare killed maps on $0 \preccurlyeq A \preccurlyeq I$, take $A_y = A$ on nonproductive reports and zero on productive ones, and average the common command row. Telescope the d powers. Positive subunital maps preserve the interval $[0, I]$ and contract the Hermitian operator norm, so each of the d differences has norm at most ϵ_I. The least-eigenvalue inequality gives (S32.2). This argument uses bounded effects, not a normalized posterior or a positive likelihood floor. □

S32.2. **Perturbation controlled by actual occupation.** For a common depth-T realization plan, write $C_{\theta,T}$, $S_{\theta,T}$ and $V_{\theta,T}$ as before. After the decision following report T, the open branches have not yet executed another call. For a fixed initial density ρ, put

$$n_{\phi,T}(\rho) = \mathbb{E}_{\phi,\rho} \min(\tau, T).$$

This is the actual number of begun calls in the finite prefix under ϕ. The next lemma requires a budget only in this source model, not in a hybrid execution.

Lemma S32.2 (Occupation-sensitive prefix comparison)**.** *For every common depth-T plan, every initial density matrix and every $\kappa \geq 0$,*

$$|\operatorname{tr}\rho(S_{\theta,T} - S_{\phi,T})| \leq \epsilon_I n_{\phi,T}(\rho), \tag{S32.4}$$

$$\left|\operatorname{tr}\rho\left[(V_{\theta,T} + LS_{\theta,T}) - (V_{\phi,T} + LS_{\phi,T})\right]\right| \leq L\epsilon_I n_{\phi,T}(\rho) + \epsilon_F, \tag{S32.5}$$

$$\left|\operatorname{tr}\rho\left[(C_{\theta,T} + \kappa S_{\theta,T}) - (C_{\phi,T} + \kappa S_{\phi,T})\right]\right| \leq n_{\phi,T}(\rho)[\epsilon_g + \epsilon_I(c^*(T-1) + \kappa)]. \tag{S32.6}$$

Here the errors may be replaced by any certified bounds for this pair of models. In particular, a source reserve $C_{\phi,T} + c_* S_{\phi,T} \preccurlyeq NI$ implies $n_{\phi,T}(\rho) \leq N/c_*$ for all ρ. Thus the loss error need not grow as $T\epsilon_I$ on a fixed-cost prefix, whereas the cost comparison still has an explicit remaining-horizon factor.

Proof. Use backward cost or score observables on the finite history tree. At a pre-call node h with $t < T$ calls already made, average its common command row. For the loss padded by L on open depth-T leaves, the target-model post-report continuation observable is a convex combination of the terminal score and the subsequent padded score; it lies in $[0, LI]$. For the survival indicator the corresponding observable lies in $[0, I]$. For cost padded by κ on open leaves it lies in $[0, (c^*(T-t-1) + \kappa)I]$. These are deterministic bounds for the finite prefix, valid before any comparison of its stopping laws.

Subtract the two backward recursions by writing

$$\mathcal{I}_\theta^* A_\theta - \mathcal{I}_\phi^* A_\phi = (\mathcal{I}_\theta^* - \mathcal{I}_\phi^*) A_\theta + \mathcal{I}_\phi^* (A_\theta - A_\phi).$$

At each node, (S32.3) bounds the first term by the corresponding continuation bound times ϵ_I. Iterate the second term with the *source* path instruments and source controller weights. Pairing at the initial state expresses the accumulated discrepancy as a sum over actual source pre-call occupation. The total of these masses is $n_{\phi,T}(\rho)$, including all failures and all private random choices.

For cost, the immediate observable difference adds at most ϵ_g per source call. Replacing $T-t-1$ by $T-1$ gives (S32.6). For terminal loss, the differences of the actual terminal observables add at most ϵ_F times source commitment mass through the cutoff, which is at most one. They are not charged at every rejected report. The padded terminal value at an open cutoff is the same L in both models. These facts prove (S32.5); survival has no terminal-observable discrepancy. The positive source cost floor gives the last assertion.

One can equivalently obtain the recursion by first fixing all private choices at each visible history and then integrating them. The common behavioral realization is justified by Lemma S31.1. The proof does not reveal ϕ to the controller, condition on a future audit, run a competitor on a fictitious reference past, or divide by a possibly zero report mass. □

S32.3. **A finite certificate at precisely the required budget.** Fix a nonempty finite dictionary $C \subset D$ and a common fallback which can start from *every* current physical state in every θ, commits with a value in C, has uniform cost at most κ, and uses B live labels. Initially Lemma S32.1 supplies one with $B = 1$; the cost hierarchy below supplies cheaper ones. Assume $N > \kappa$.

Let $q_{F,C,T}(N)$ be the finite semidefinite value (S26.1) with constraints only at the finitely many cover models $\phi \in F$, but with the same common realization plan. Define

$$D_T = \min\{1, (N/c_* - 1)/T\},$$

$$e_{F,T} = (\kappa - c_*) D_T + \frac{N}{c_*}[\epsilon_g + \epsilon_I (c^*(T-1) + \kappa)], \tag{S32.7}$$

$$z_{F,T} = L D_T + L \frac{N}{c_*} \epsilon_I + \epsilon_F, \qquad \Gamma_{F,T} = z_{F,T} + \frac{L e_{F,T}}{N - \kappa + e_{F,T}}. \tag{S32.8}$$

A proper fallback must make at least one call, so $\kappa \geq c_*$. These errors are nonnegative.

Theorem S32.3 (Same-budget risk enclosure for compact uncertainty)**.** *Under the primitive hypotheses just stated, the finite problem is feasible and attains its value, and*

$$q_{F,C,T}(N) \leq R_{\Theta,C}(N) \leq \min\{L, q_{F,C,T}(N) + \Gamma_{F,T}\}. \tag{S32.9}$$

The upper is realized by one proper finite-label controller of expected cost at most N for every $\theta \in \Theta$ *and every initial density matrix. It uses one dictionary, not a modelwise family of dictionaries. With a commands and at most y reports per command, sufficient live labels are*

$$1 + B + \sum_{t=0}^{T-1} (ay)^t. \tag{S32.10}$$

Event-stopped total labels are the maximum of this bound and $|C|$; state-stopped total labels are their sum.

For nested covers and increasing horizons the finite lower values are nondecreasing. If $T \to \infty$, $T\epsilon_I \to 0$, and $\epsilon_g, \epsilon_F \to 0$, their limit equals $R_{\Theta,C}(N)$. At a fixed slack $N - \kappa > 0$, taking $T \asymp \epsilon^{-1/2}$ when all primitive errors are at most ϵ gives an explicit $O(\sqrt{\epsilon})$ enclosure. This is an upper modulus, not a claim that its exponent is optimal.

Proof. The actual fallback has cost below N in all models, so its prefixes prove feasibility on F. A finite plan lies in a compact sequence-form polytope, and the constraints are closed with $0 \le q \le L$. The minimum is attained. Restriction of any feasible full policy on Θ to the finite models and horizon proves the lower. Adding cover models or extending the horizon cannot weaken that lower, by the restriction lemma.

Take a finite optimizer. At every $\phi \in F$, the reserve gives $S_{\phi,T} \preccurlyeq D_T I$ and source call occupation at most N/c_*. For any θ choose a covering ϕ. Complete the same prefix on the actual experiment with the declared fallback at every open leaf. Positivity bounds the completed cost by $C_{\theta,T} + \kappa S_{\theta,T}$ and the completed loss by $V_{\theta,T} + LS_{\theta,T}$. At the covering model these are respectively at most $N + (\kappa - c_*)D_T$ and $q_{F,C,T}(N) + LD_T$. Lemma S32.2 now gives completed cost at most $N + e_{F,T}$ and loss at most $q_{F,C,T}(N) + z_{F,T}$ in *every* model and initial state. The fallback is applied on the actual post-prefix state, not the state at the covering model. Its all-state guarantee is what makes this completion legal and proper.

Before the first physical call, choose the completed prefix with probability

$$p = \frac{N - \kappa}{N - \kappa + e_{F,T}}$$

and the fallback alone otherwise. The choice is independent of the unknown model and input. Its robust expected cost is at most $p(N + e_{F,T}) + (1-p)\kappa = N$. Its loss is at most

$$p(q_{F,C,T} + z_{F,T}) + (1-p)L \le q_{F,C,T} + z_{F,T} + (1-p)L.$$

This is the claimed gap. The actual loss is always at most L, giving the truncation of the upper. Both branches commit almost surely, so does their mixture. The sole initial choice is compiled into one initial row whose within-call branch choice routes to prefix or fallback labels; it is not an uncounted persistent seed. The finite tree has at most the displayed sum of live node roles, and the fallback uses B. The role-packing lemma proves both total-label counts.

The explicit errors tend to zero under the stated joint refinement. Compact continuity supplies arbitrarily fine model covers, so such refinements exist. For the rate, substitute $D_T = O(T^{-1})$ and bound the mixture denominator below by $N - \kappa$. The gap is at most a constant times $T^{-1} + \epsilon T + \epsilon$, yielding the asserted choice. None of these estimates assumes independence of the physical states between calls or a mean-cost tail converging uniformly over feasible policies. □

Corollary S32.4 (Preserving an optimized finite output width)**.** *Let $R_{\Theta,m}(N)$ optimize over all dictionaries of at most m points in D. For a finite η-net D_η, minimize $q_{F,C,T}(N)$ over tuples $C \in D_\eta^m$ with repetition, and call the result $q_{F,m,T,\eta}(N)$. The fallback may use any one value of the chosen dictionary without changing its costs. Then*

$$[q_{F,m,T,\eta}(N) - \omega(\eta)]_+ \le R_{\Theta,m}(N) \le \min\{L, q_{F,m,T,\eta}(N) + \Gamma_{F,T}\}.$$

The common observer has at most m outputs. In particular model and decision approximation do not create a hidden continuum-sized decoder or a stored model estimate.

Proof. Round each point of any feasible competitor's single dictionary to one net point, identically at every history. Its costs and stopping law do not change. Bounded terminal-observable error changes expected loss by at most $\omega(\eta)$ because the terminal events have total mass one. Restricting gives the lower. A finite net has finitely many tuples, so choose a minimizer and apply Theorem S32.3 for the upper. Repetitions can only reduce the actual width. □

S32.4. **The least feasible cost and every interior budget.** The primitive fallback cost κ_0 can be far from optimal. It must not become an artificial admission threshold. For each finite cover and horizon, let

$$a_{F,T} = \min\{a : C_{\phi,T} + c_* S_{\phi,T} \preccurlyeq aI \ (\phi \in F)\},$$

ignoring task loss and committing a fixed dummy value. This is an attained finite semidefinite problem with $c_* \le a_{F,T} \le \kappa_0$. Put $D_T^0 = \min\{1, (\kappa_0/c_* - 1)/T\}$ and

$$E_{F,T} = (\kappa_0 - c_*)D_T^0 + \frac{\kappa_0}{c_*}[\epsilon_g + \epsilon_I(c^*(T-1) + \kappa_0)], \qquad U_{F,T} = a_{F,T} + E_{F,T}. \tag{S32.11}$$

Here a cover needs only instrument and cost errors. Let $N_{\min}(\Theta)$ be the least uniform expected cost of a common proper policy.

Theorem S32.5 (Constructive compact-family feasibility threshold)**.** *One has*

$$a_{F,T} \le N_{\min}(\Theta) \le U_{F,T}.$$

The upper is the certified cost bound of an actual common finite-label policy. Along nested covers and horizons with $T\epsilon_I \to 0$ and $\epsilon_g \to 0$, the lower bounds increase to $N_{\min}(\Theta)$ and the running minima of the upper bounds decrease to the same number. The least cost is attained by a full-history common policy.

For every $N > N_{\min}(\Theta)$, a sufficiently fine finite cover and sufficiently large horizon exhibit $U_{F,T} < N$. Their completed policy is a finite-label fallback with certified $\kappa = U_{F,T} < N$. Consequently Theorem S32.3 and Corollary S32.4 apply at every strictly interior feasible budget*, with no supplied balancing program or reference witness. For effective data and certified covers this is a terminating search for a strictly feasible fallback, not a dimension-only efficient synthesis bound. Finite-memory attainment at $N = N_{\min}(\Theta)$ is not asserted.*

Proof. Every actual proper policy supplies a feasible reserved prefix on the cover, proving the lower. Complete a lower optimizer with the original uniform fallback. At a covering model, its open mass is at most $D_T(a_{F,T}) \le D_T^0$, its expected prefix calls at most $a_{F,T}/c_* \le \kappa_0/c_*$, and its padded cost at most $a_{F,T} + (\kappa_0 - c_*)D_T^0$. Apply (S32.6) with padding κ_0 to every covered model. This proves the upper and its operational realization. It bounds the cost on arbitrary current states, since the prefix problem and perturbation bound were uniform in their initial density matrices.

The error in (S32.11) tends to zero along the stated refinement. Restriction and addition of models prove lower monotonicity; the running minimum construction gives upper monotonicity without incorrectly claiming that the raw error bounds themselves decrease. The squeeze proves both limits.

For attainment, choose proper common policies with uniform costs decreasing to $N_{\min}$. The countable product of finite nodewise probability simplexes is compact. Every finite-prefix reserved cost constraint at each fixed model and state is closed in these coordinates. A finite collection of such constraints at budget $N_{\min}$ is feasible: take a convergent subnet of the selected policies' coordinates and use continuity of the finite-prefix operators and convergence of their budgets. Compactness then gives one policy satisfying all these constraints simultaneously. Its survival is at most $D_T(N_{\min})$ after T calls, and the positive floor and monotone convergence prove properness and total cost at most $N_{\min}$ in every model. No uniform expected-cost tail estimate was used.

At a strictly larger requested budget, the upper sequence eventually falls below it. The completed policy attaining that certificate supplies exactly the fallback required for the risk theorem. Relabelling its dummy output to a chosen task decision changes no call, so it works for any eventual dictionary. Its finite live-state count is retained in (S32.10). Effective cover and primitive arithmetic hypotheses are additional inputs for performing these tests; the proof does not infer them from topological compactness. □

Corollary S32.6 (Effective synthesis from compact algebraic primitives)**.** *Suppose $\Theta \subset \mathbb{R}^p$ and $D \subset \mathbb{R}^k$ are nonempty compact semialgebraic sets defined over the real algebraic numbers. Suppose the real and imaginary entries of the primitive instrument maps and costs are polynomial in θ, and the terminal-loss entries are polynomial in (θ, r). Assume the common finite interface, positive cost floor and absence of dark subspaces as above. Then $N_{\min}(\Theta)$ is a computable real. For every*

algebraic $N > N_{\min}(\Theta)$, integer $m \geq 1$ and positive algebraic accuracy ξ, a terminating finite procedure produces a common finite-label observer of cost at most N whose risk differs from $R_{\Theta,m}(N)$ by at most ξ, with algebraic controller probabilities and decoder values.

The procedure also produces rigorous lower and upper value bounds. It need not be computationally efficient; the statement concerns a supplied finite polynomial description, not learning the experiment from data. Exact bounded-bit sampling, minimal program size and finite-memory optimality at the boundary are separate questions.

Proof. All assertions about positivity of the finite-dimensional Hermitian primitive matrices, normalization, and $O_{\theta,d} \succeq \delta I$ are first-order polynomial assertions over real algebraic coefficients. Represent complex entries by their real and imaginary parts, and use principal minors for positive semidefiniteness. Quantifier elimination over real closed fields [15] therefore verifies these assertions and exhibits an algebraic $\delta > 0$ valid for all θ when it exists. Compactness and the no-dark lemma guarantee existence. The positive lower cost bound and finite upper cost/loss bounds may also be chosen algebraic: compactness gives such bounds, and the same polynomial tests verify them. This gives an algebraic bound κ_0 for the common one-row fallback.

At a fixed horizon, use common sequence weights and a tuple of m decoder values as variables. Composing a finite number of polynomial primitive maps gives polynomial coefficients in θ. The constraints $C_{\theta,T} + c_* S_{\theta,T} \preccurlyeq NI$ and $V_{\theta,T} \preccurlyeq qI$ for all $\theta \in \Theta$ are thus a finite first-order real-algebraic formula. Compactness of the finite plan and decoder space gives an attained minimum. Its value and some minimizing variables are algebraic, by quantifier elimination and the existence of algebraic points in nonempty semialgebraic sets defined over the real algebraic numbers. This is a robust finite-horizon optimization, not necessarily an ordinary finite-list semidefinite program or a convex optimization in the decoder variables.

The same method computes the uniform reserved cost minimum $a_{\Theta,T}$. Complete its optimizer with the known fallback. With no model-grid error, its cost is at most

$$a_{\Theta,T} + (\kappa_0 - c_*) \min\{1, (\kappa_0/c_* - 1)/T\}.$$

This explicit interval converges to $N_{\min}$ and can be made narrower than any rational tolerance, proving computability. If $N > N_{\min}$, enumeration of these finite uniform problems eventually certifies a completed fallback cost $\kappa < N$; it has finite labels and algebraic probabilities. The uniform risk problem just described and the completion-mixture proof then have gap at most

$$LD_T(N) + \frac{L(\kappa - c_*)D_T(N)}{N - \kappa + (\kappa - c_*)D_T(N)}.$$

Choose a finite T making this less than ξ, compute an algebraic optimizer, and mix its completion with the certified fallback. The mixing probability is algebraic and restores the original budget. Its one common decoder tuple has at most m values. This proves every claimed termination and approximation assertion without supposing a finite grid represents a continuum exactly. Real-algebraic elimination may be extremely expensive, and equality at the limiting feasibility boundary is not decided by an unbounded strict-threshold search. □

S32.5. **Two-sided robustness and boundaries of the theorem.**

Corollary S32.7 (Robustness of the optimal constrained risk)**.** *Let Θ_1, Θ_2 be two compact families on the same interface. Suppose each member of either family is within $(\epsilon_I, \epsilon_g, \epsilon_F)$ of some member of the other. Suppose one common finite-label fallback costs at most $\kappa < N$ on their union, with the same cost and loss bounds. Then, for fixed C or optimized width m,*

$$|R_{\Theta_1}(N) - R_{\Theta_2}(N)| \leq \Gamma_{F,T} \quad (T \geq 1),$$

where the right side is the explicit expression (S32.7)–(S32.8) *evaluated at the pairwise error bounds; no finite cover is needed in this corollary. Hence it is at most the infimum of these bounds over*

T. In particular it tends to zero with the primitive errors at a fixed positive budget slack. The same-budget $O(\sqrt{\epsilon})$ upper modulus follows when all three errors are at most ϵ.

Proof. Take any feasible policy on Θ_1 within ξ of its optimal risk and truncate it at depth T. Its reserve and nonnegative loss bounds hold at every member of Θ_1. For each target in Θ_2, use its close source member in the occupation comparison. Complete and mix with the common fallback exactly as in Theorem S32.3. This gives a single policy feasible at N on Θ_2 with risk at most $R_{\Theta_1}(N) + \xi + \Gamma_{F,T}$. For optimized width, its one source dictionary is used on every target and by the fallback. Let $\xi \downarrow 0$, then reverse the roles. This compares the optimal risks through feasible strategies and does not exchange a minimization and a maximization. □

Example S32.8 (Risk need not be continuous at the feasibility boundary)**.** In physical dimension one, both commands A, B always report a productive result. There is one retained decoder value. Let its prospective losses be zero after A and one after B. At model $\theta = 0$, both call costs are one; at $\theta = \epsilon > 0$, they are respectively $1 + \epsilon$ and one. There is no instrument or loss perturbation, only cost error $\epsilon_g = \epsilon$, and there is no dark state. At budget $N = 1$, any positive probability of choosing A or continuing after the first report costs strictly more than one in the perturbed model. Thus $R_{\{0\}}(1) = 0$ while $R_{\{\epsilon\}}(1) = 1$. Both instances remain feasible. Therefore the strict interior/slack condition cannot simply be deleted from a general continuity theorem. The example can be realized by command-dependent deterministic future audits and squared loss at the same decoder.

Example S32.9 (Why the remaining-horizon cost term is real)**.** In dimension one, command A costs one and is immediately productive at the source; at the target it fails with probability $\epsilon \in (0, 1/2]$ and is productive otherwise. Command W costs one and is never productive. Use a prefix that first calls A, commits on success, and after failure makes $T - 1$ calls to W before leaving the cutoff open. Source prefix occupation is exactly one. Instrument discrepancy is ϵ, while costs and task observables have not changed. The target's cost padded by κ on the open leaf is exactly

$$1 + \epsilon(T - 1 + \kappa),$$

compared with source value one. Repeated A is a genuine fallback of cost at most two. This attains the instrument term of (S32.6) when $c^* = 1$. It proves the need for a remaining-horizon factor for arbitrary tested prefixes; it does *not* prove sharpness of the optimized-risk square-root modulus.

Example S32.10 (Even affine primitive families cannot be certified at vertices)**.** Let $\theta \in [0, 1]$, physical dimension one, and one unit-cost command whose productive report is a Bernoulli(θ) bit. At commitment a fresh independent Bernoulli(θ) bit is the future audit. Fix $C = \{0, 1\}$ with squared loss. Both the report instruments and the post-report loss observables are affine in θ. Nevertheless, for every budget $N \geq 1$,

$$R_{\{0,1\},C}(N) = 0, \qquad R_{[0,1],C}(N) = \tfrac{1}{2}.$$

For the two endpoint models, committing the first report as prediction has zero loss. For the full interval this rule has risk $2\theta(1 - \theta) \leq 1/2$ at cost one. Under $\theta = 1/2$, the prospective audit is fair conditional on any finite stopping history and any selected decision in C; every proper policy therefore has loss exactly $1/2$. This gives the matching lower against arbitrary history and randomization. It is a continuum-uncertainty obstruction even with a uniform one-call fallback and no hidden dynamical state. A convex parameter set and affine primitive entries do not justify testing only its vertices: finite-path coefficients and adaptive terminal scores need not be affine in the fixed unknown parameter.

Example S32.11 (Pointwise termination and model switching are different)**.** The noncompact scalar family with productive probability $p \in (0, 1]$ terminates in every model but has no finite uniform cost. Its compact closure adds the dark model $p = 0$. Compactness and a positive certified margin are therefore substantive.

Nor is the model allowed to change adversarially during a run. With one command on a qubit, let model 1 have failure Kraus operator $|0\rangle\langle 1|$ and model 2 have $|1\rangle\langle 0|$, with the complementary productive effects. Each fixed model terminates within two calls from every initial state. Alternating models 1 and 2 from state $|1\rangle$ causes failure forever. The compact finite union satisfies the present fixed-model theorem but not a switching-adversary theorem. Such uncertainty requires a different joint survival certificate.

This compact-family result is an intrinsic constrained-risk approximation on its stated finite physical interface, not an intrinsic characterization of when the geometric profile $\mathcal{Q}$ is exact. It supplies no uncharged model-identification phase and requires no normalized-filter contraction. Its cost threshold, risk certificate, finite output and finite observer implementation come from the same primitive data. The separate exact geometric, spectral, reset-communication and predictive-realization theorems retain their own hypotheses and research questions.

S33. The exact task defect at a prescribed output width

The raw matrix order of Section S39 is exact for localized affine-square tests. Ordinary squared prediction tests form a smaller family. Requiring the stronger order can therefore discard valid task comparisons, while taking a supremum over arbitrarily many readouts can overcharge a finite decoder. We identify the least additive defect for the actual task and its output width.

S33.1. **Finite dictionaries and an attained optimal defect.** Let Y, Z be standard Borel spaces, D a nonempty compact metric decision space, and $\ell : Z \times D \to [0, L]$ Borel. Assume a uniform decision modulus

$$|\ell(z, r) - \ell(z, s)| \le \omega(d(r, s)), \qquad \lim_{\delta \downarrow 0} \omega(\delta) = 0, \tag{S33.1}$$

with ω nondecreasing. For finite positive measures P, Q on $Y \times Z$, put $\lambda = P_Y + Q_Y$. The unnormalized conditional loss densities $g_P(y, r), g_Q(y, r)$ relative to λ have jointly Borel versions continuous in r. Indeed disintegrate $P+Q$ over λ, multiply its conditional law by the Radon–Nikodym densities of P, Q relative to $P + Q$, and integrate ℓ. Off one completed null set,

$$|g_P(y, r) - g_P(y, s)| \le p(y)\omega(d(r, s)), \quad |g_Q(y, r) - g_Q(y, s)| \le q(y)\omega(d(r, s)),$$

where $p = dP_Y/d\lambda$, $q = dQ_Y/d\lambda$ and $p + q = 1$. Zero total mass causes no difficulty. Put $F(y, r) = g_P(y, r) - g_Q(y, r)$ and

$$\Delta_C^{\ell}(P \mid Q) = \int \max\{0, \max_{r \in C} F(y, r)\}\, d\lambda$$

for a nonempty finite dictionary $C \subset D$.

Theorem S33.1 (Width-resolved optimal task defect)**.** *For every integer $m \ge 1$, the following equal quantities are attained:*

$$\Delta_m^{\ell}(P \mid Q) = \max_{(r_1, \dots, r_m) \in D^m} \int \max\{0, F(y, r_1), \dots, F(y, r_m)\}\, d\lambda, \tag{S33.2}$$

$$= \sup_{\substack{a: Y \to [0,1]\ mBorel \\ r: Y \to D\ mBorel,\ |r(Y)| \le m}} \int a(y)\ell(z, r(y))\, (P - Q)(dy, dz). \tag{S33.3}$$

Thus this is the smallest additive error valid for every selected width-m task. An optimum uses a deterministic report event and readout from one fixed dictionary; independent current randomization cannot improve it. The value is independent of the dominating measure and satisfies

$$\Delta_1 \le \Delta_m \le m\Delta_1, \quad \Delta_{m+n} \le \Delta_m + \Delta_n, \quad \frac{\Delta_n}{n} \le \frac{\Delta_m}{m} \quad (1 \le m \le n). \tag{S33.4}$$

Moreover

$$\Delta_\infty^\ell(P \mid Q) := \int [\sup_{r\in D} F(y,r)]_+ \, d\lambda = \lim_{m\to\infty} \Delta_m^\ell(P \mid Q). \tag{S33.5}$$

If a δ-net of D has n points, then

$$0 \le \Delta_\infty - \Delta_n \le \omega(\delta)[P(Y\times Z) + Q(Y\times Z)]. \tag{S33.6}$$

In particular $\Delta_1 = 0$ if and only if $\Delta_\infty = 0$. The width hierarchy distinguishes positive defect magnitudes, not different zero-defect orders.

Proof. Fix an ordered dictionary. At each report choose the first index maximizing its positive loss-density difference, and accept exactly when that maximum is positive. These decisions are Borel and give the displayed envelope. They pointwise dominate every other selection and every randomized readout in that dictionary. Every finite-image rule uses some such dictionary, proving equality of the suprema.

Moving every tuple coordinate by at most δ changes the envelope by at most $(p+q)\omega(\delta)$ almost everywhere. Integration makes the objective continuous on compact D^m, with repetitions allowed. It attains its maximum, and the selected deterministic test attains the second supremum. Averaging over an independent current seed cannot increase that value. A seed choosing dictionaries whose union exceeds m does not describe a width-m decoder.

The test expression gives independence of the dominating measure. Inclusion gives monotonicity. A nonnegative maximum is bounded by the sum of the single-center positive parts, proving $\Delta_m \le m\Delta_1$. Partition a dictionary into two groups to obtain subadditivity. For $m \le n$, choose an optimal n-tuple and average over its m-element index subsets. A selected maximizing positive index is retained with probability m/n. Thus the average subdictionary envelope is at least m/n times the original one. Integrate and take one subdictionary at least as good as its average. This proves the last inequality, including repeated entries.

Continuity makes a countable dense decision set determine the pointwise supremum. The positive envelopes of its finite initial segments increase to that supremum; monotone convergence proves the limit and measurability. A finite net gives the quantitative estimate by the same modulus calculation. Finally, $\Delta_1 = 0$ makes $F(y,r) \le 0$ almost everywhere for each point in the countable dense set. Intersect these full-measure sets and use continuity to obtain the inequality for all decisions on one common set. Hence $\Delta_\infty = 0$; the converse is immediate. □

For fixed P_0, P_1, the map $t \mapsto \Delta_m^\ell(tP_0 \mid P_1)$ is convex, nondecreasing and $LP_0(Y\times Z)$-Lipschitz on $[0,\infty)$. It is a supremum of affine functions of t with slopes between zero and that bound. The positive-variation and measurable-selection ingredients are classical. The decision-cardinality restriction should not be confused with Blackwell k-decision orders [8]: the present localized fixed-loss order at zero defect already follows from width one.

S33.2. **The exact squared-task order.** Take $D = H \subset \mathbb{R}^k$ compact convex and $\ell(z,r) = \|z-r\|^2$, with bounded audits. Relative to a common report measure, write the signed moment densities of $P_1 - tP_0$ as

$$a = \frac{d(P_1 - tP_0)_Y}{d\lambda}, \quad b = \frac{d\int z(P_1 - tP_0)(dy,dz)}{d\lambda}, \quad c = \frac{d\int \|z\|^2(P_1 - tP_0)(dy,dz)}{d\lambda}.$$

The corresponding quadratic is $\psi_y(r) = a(y)\|r\|^2 - 2b(y)\cdot r + c(y)$.

Theorem S33.2 (Necessary and sufficient squared-task certificate)**.** *The least unrestricted-width defect is*

$$\Delta_\infty^{\mathrm{sq}}(tP_0 \mid P_1) = \int [-\min_{r\in H} \psi_y(r)]_+ \, d\lambda. \tag{S33.7}$$

Localized squared-task order with zero defect holds for all finite dictionaries if and only if $\min_H \psi_y \geq 0$ *on one common full-measure set. No inverse covariance, equal matrix support or positive eigenvalue floor is required. For* $a > 0$ *the minimum is*

$$c - \|b\|^2/a + a \operatorname{dist}(b/a, H)^2; \tag{S33.8}$$

for $a = 0$ *it is* $c - 2\sup_H b \cdot r$. *On the radius-*R *closed Euclidean ball,*

$$\min_H \psi = \begin{cases} c - \|b\|^2/a, & a > 0,\ \|b\| \leq aR, \\ c + aR^2 - 2R\|b\|, & \textit{otherwise.} \end{cases} \tag{S33.9}$$

The second line includes the nonpositive-mass and zero-radius boundary cases whenever the first does not apply.

Proof. Expanding the square makes the directed loss-density difference for (tP_0, P_1) equal to $-\psi$. The preceding theorem proves the integral and common-null-set statements. Completing the square proves the first formula; $a = 0$ gives a linear optimization. On a ball with $a > 0$, either the unconstrained minimizer b/a lies inside or its radial projection minimizes. If $a \leq 0$, the least value at radius s is $c + as^2 - 2\|b\|s$, nonincreasing for $0 \leq s \leq R$. Its minimum is at R. These arguments include ties, $b = 0$ and $R = 0$. □

Report mass is a separate resource test: its least selected-event defect in $P_Y \leq uQ_Y + e$ is $\int [p - uq]_+ d\lambda$. It is not implied by one arbitrary task inequality. Equality of the entire task family has a sharper conclusion.

Proposition S33.3 (Task equality and its exact moment content)**.** *Suppose the affine hull of* H *has dimension at least one. Localized squared-task equality for* P, Q *at every constant readout in* H *is equivalent to equality of their report masses, their first-moment measures projected onto the direction space of that affine hull, and their full centered squared-norm measures* $\int \|z - h_0\|^2 P(dy, dz)$ *for an origin* h_0 *in the affine hull. Full second-moment matrix measures need not agree. For a singleton carrier the mass must instead be tested separately.*

Proof. Choose a countable dense set of readouts and intersect their full-measure equality sets. The common modulus extends equality to all of H. Translate by h_0 and restrict the quadratic difference to the direction space V of the affine hull. A compact convex set with this affine hull has nonempty relative interior. A polynomial vanishing there vanishes on V: restrict successively to line segments in a relative open ball, where a one-variable polynomial with an interval of zeros is identically zero, and identify its coefficients. The quadratic coefficient is the scalar mass difference times the identity on V, so that difference is zero because $\dim V \geq 1$. The linear coefficient is the projected centered first-moment difference, and the constant coefficient is the full centered squared-norm difference. They vanish. Conversely these three equalities make every expanded squared-task difference zero. When H is a singleton, only one constant loss measure is tested, so a zero-loss audit does not identify mass. □

Example S33.4 (Identical tasks with incompatible matrix supports)**.** There is one report and the decision carrier is the unit ball in $\mathbb{R}^2$. Let P_0 audit the two signs of e_1 fairly and P_1 the two signs of e_2 fairly. Every ordinary squared decision has loss $1 + \|r\|^2$ under both laws. Their affine-square moment matrices are respectively $\operatorname{diag}(1, 1, 0)$ and $\operatorname{diag}(1, 0, 1)$ in the coordinates $(1, z_1, z_2)$. No two-sided positive relative matrix order can compare their different kernels. Thus exact squared-task comparison is strictly broader than the sufficient positive-matrix criterion, without contradicting that criterion's exact statement for its larger test family.

Theorem S33.5 (Sharp width amplification in every finite dimension)**.** *For every integer* $k \geq 1$ *there are probability report–audit laws with identical report masses and squared decisions in the unit*

ball of $\mathbb{R}^k$ *such that*

$$\Delta_m^{\mathrm{sq}}(P_0 \mid P_1) = \frac{\min(m,k)}{5k} \quad (m \geq 1). \tag{S33.10}$$

Both the linear width growth and the discrepancy between fixed-width and unrestricted defects are sharp in this family.

Proof. Let the report i be uniform on $\{1,\dots,k\}$. Under P_0 the audit is zero. Under P_1, conditional on i, it is e_i with probability $4/5$ and $-e_i$ with probability $1/5$. The conditional loss difference is $(6/5)r_i - 1$. It is positive only when $r_i > 5/6$. A unit-ball vector cannot meet this condition for two different coordinates, since $2(5/6)^2 > 1$. Its positive contribution in its one possible coordinate is at most $1/5$. A dictionary of m vectors therefore contributes at most $\min(m,k)/(5k)$. Choosing that many distinct basis vectors and accepting their corresponding reports attains the bound. The construction has bounded audits, exact report-mass equality, and no asymptotic dimension assumption. □

S33.3. **The same dictionary across actual histories.** Let u range over a standard Borel context space with finite positive occupation measure μ. Let P_u, Q_u be finite kernels with integrable total masses; assume the same bounded decision-continuous loss. Form the finite measures $\mathbf{P} = \mu(du)P_u(dy,dz)$ and $\mathbf{Q} = \mu(du)Q_u(dy,dz)$, where the full report is (u,y).

Proposition S33.6 (Exact fixed-dictionary occupation defect)**.** *One has*

$$\Delta_m^{\ell}(\mathbf{P} \mid \mathbf{Q}) = \max_{C \in D^m} \int \Delta_C^{\ell}(P_u \mid Q_u)\,\mu(du) \leq \int \Delta_m^{\ell}(P_u \mid Q_u)\,\mu(du). \tag{S33.11}$$

The maximum uses a single dictionary for every context. The inequality can be strict by an arbitrarily large factor as the number of contexts grows. Neither side asserts that a prescribed occupation measure is causally realizable.

Proof. For a fixed dictionary, take the positive envelope of its unnormalized loss densities on the product context/report space. Disintegration and integration give exactly the integral of the rowwise fixed-dictionary envelopes. This can also be checked without selecting contextwise densities: a finite partition of reports and rational acceptance/readout choices approximate every bounded selected task in the finite measure $P_u + Q_u$. A countable algebra generating the standard Borel report space, together with a countable dense decision set, suffices for these approximations. Hence the rowwise suprema are measurable. The uniform decision modulus and integrable total masses make the integrated fixed-dictionary envelope continuous on compact D^m, so it attains its maximum. Applying the width-defect theorem to the product measures yields the equality. Moving the maximum inside the integral gives the inequality.

For strictness use the preceding sharp example with context $u = i$ and no additional report. Every row has one-center defect $1/5$, whereas a single common width-m dictionary over uniform contexts has defect $\min(m,k)/(5k)$. Thus even individually exact rowwise errors may overcharge a common finite decoder. In a stopped experiment μ must be the actual pre-call occupation of the competitor; this algebraic identity does not optimize or fabricate that occupation. □

S34. Width-sensitive transfer for a general acquired loss

The exact defects just constructed can be inserted into the paid causal proof without enlarging the observer's decoder. We state the resulting theorem for bounded nonnegative decision-continuous losses. Squared prediction is its principal example, not an implicit restriction on its quantifiers.

Let D be a nonempty compact metric decision space and let $\ell : Z \times D \to [0,L]$ satisfy (S33.1). A genuinely prepared comparison law $K_0(dx,dz)$ has report law ν and conditional task functions

$g_r(x) = \mathbb{E}_0[\ell(Z,r) \mid x]$. Disintegration supplies jointly Borel versions with the same decision modulus. Define

$$\mathcal{Q}^{\ell}_m(b) = \min_{\substack{(r_1,\dots,r_m)\in D^m \\ 0\le h\le b,\ \int h\,d\nu=1}} \int h(x) \min_j g_{r_j}(x)\,\nu(dx), \qquad b \ge 1. \tag{S34.1}$$

Repetition in the tuple permits fewer than m outputs. Values at infeasible width or cap are $+\infty$.

Fix one actual lower model, the same actual-history occupation and effort convention as in Section 2, and the same lower report-mass comparison with constants u_-, a_-. The full-report comparison rows must satisfy

$$\mathbb{E}^0[\ell(Z,r) \mid Y = y] = g_r(\pi y) \qquad (r \in D) \tag{S34.2}$$

on one common full-measure set. It is enough to verify this on a countable dense decision set; the uniform modulus extends it to all decisions. This excludes an unaccounted informative component of the report.

At a reached lower context $u = (h,a)$ let

$$e_{-,m}(u) = \Delta^{\ell}_m(l_- P^0_u \mid P_u), \qquad 0 < l_- \le 1. \tag{S34.3}$$

These optimal defects may be replaced by measurable certified upper bounds. Suppose every robustly feasible budget-N competitor has actual occupation μ for which

$$\Delta^{\ell}_m(l_-\mu(du)P^0_u \mid \mu(du)P_u) \le E_{N,m}. \tag{S34.4}$$

The common terminal dictionary is fixed across these contexts. Proposition S33.6 shows that it suffices to bound $\int e_{-,m}\,d\mu$; the displayed global certificate can be sharper. A uniform row bound $e_{-,m} \le \bar{e}_m$ gives $E_{N,m} \le (N/\kappa_-)\bar{e}_m$. The killed-potential lemma also applies to nonuniform certified row errors. The occupation is that of each actual competing program. It is not a freely chosen measure or the occupation of a fictitious comparison execution.

For the upper, retain one common acquisition program $\mathcal{G}$, its conditional whole-attempt cost κ_+, live count B, actual rooted-continuation validity, and export multiplicity D_0. Its report acceptance obeys the same lower mass certificate with l_+, a_+. For every virtual dictionary of width at most m, assume

$$\mathbb{E}[q(X)\ell(Z, T_J r(X))] \le u_+ \int q(x) g_{r(x)}(x)\nu(dx) + e_{+,m}, \qquad u_+ \ge 1, \tag{S34.5}$$

conditionally on every surviving history in every model. This is one program and one virtual dictionary, not modelwise optimized decisions. The exact defect theorem computes the least possible $e_{+,m}$ whenever the compared effective loss family has the stated common decision modulus. For example, uniformly Lipschitz exports on a compact carrier preserve such a modulus. The two measures can be placed on a disjoint union of their audit carriers with the corresponding loss definitions; equality of hidden audit coordinates is not needed. Otherwise (S34.5) itself is the required task certificate.

Theorem S34.1 (Paid transfer at the actual decoder width)**.** *Assume the prepared-kernel, commitment, cost, mass, effort, occupation and resource conditions just stated. Use exactly the lower and upper widths $m_-(M), m_+(M)$ of* (16.1)*, with the event upper requiring $M \ge B$. Put $b_\pm = N/\kappa_\pm$, $t = (1 - A_N)_+/u_-$ and suppose $b_- \ge 1$ and $m_-(M) \ge 1$. If $t > 0$, then*

$$R^{\Theta,\ell}_i(N,M) \ge \left[l_- t \mathcal{Q}^{\ell}_{m_-(M)}(b_-/t) - E_{N,m_-(M)} \right]_+ . \tag{S34.6}$$

For $t = 0$ the lower is zero, without evaluating a profile at infinity. Empty legal classes retain value $+\infty$.

If $b_+ \ge 1$, $m_+(M) \ge 1$ and

$$\beta = \frac{l_+ b_+}{1 + b_+ a_+} \ge 1,$$

there is one common Borel finite-role construction with actual expected cost at most N and

$$R_i^{\Theta,\ell}(N,M) \le \frac{u_+(1+b_+a_+)}{l_+}\,\mathcal{Q}^{\ell}_{m_+(M)}(\beta) + b_+ e_{+,m_+(M)}. \tag{S34.7}$$

It commits almost surely in every model. If $\beta < 1$, the first-attempt upper is $u_+\,\mathcal{Q}^{\ell}_{m_+(M)}(1)+e_{+,m_+(M)}$, at cost at most κ_+. Exact matched certificates with identity exports give the checkpoint and event/state-stopped equalities of Corollary 16.3, with $\mathcal{Q}^{\ell}$ replacing $\mathcal{Q}$.

Optimal row defects are necessary and sufficient for their declared selected-task inequalities. Sharpness of those defects does not, by itself, assert sharpness of the entire stopped risk bound when other mass, cost or implementation certificates are not matched.

Proof. First the profile minimum exists. The density set is weak-star compact in $L^\infty(\nu)$, and D^m is compact. Moving its decision tuple by at most δ changes its pointwise minimum loss by at most $\omega(\delta)$. Split an objective difference into this uniform change and weak-star pairing with the bounded fixed limiting loss. The objective is therefore continuous on the product, so the minimum is attained. This argument uses product nets if necessary, not unjustified sequential compactness. A first nearest conditional-loss index is Borel.

Fix any competing observer. Its terminal values form one dictionary C with at most $m_-(M)$ values, by the exact stopping-role lemma. Adjoin independent current coins to the full report, and write the current acceptance and readout as functions of the actual context and report. In the lower comparison, summing their contributions against actual pre-call occupation gives exactly the two product measures in (S34.4). The defining defect inequality yields

$$\mathbb{E}_*\ell(Z,\widehat{r}) \ge l_- \sum_{c\in C}\int g_c\,d\xi_c - E_{N,m_-(M)},$$

where the accepted comparison measures ξ_c are constructed as in (17.4). Condition (S34.2) is what identifies their conditional tasks after selection. The actual model commits exactly once; the same report-mass proof gives $\xi = \sum_c \xi_c \le b_-\nu$ and $\xi(X) \ge t$. All these measures are finite by the actual effort inequality.

For $s = \xi(X) \ge t > 0$, scale to $\xi' = (t/s)\xi$. Nonnegative loss and pointwise minimization over C give

$$\sum_c \int g_c\,d\xi_c \ge \int \min_{c\in C} g_c\,d\xi' \ge t\,\mathcal{Q}^{\ell}_{m_-(M)}(b_-/t).$$

The normalized density $t^{-1}d\xi'/d\nu$ has unit mass and the asserted cap. Taking the worst-model risk and then the infimum proves the lower without exchanging infimum and supremum. Nonnegativity and the empty-class convention give its boundary cases.

For the upper take an attained optimal tuple and density at width $m_+(M)$ and cap β. In each attempt accept with probability $h(X)/\beta$, choose the first minimizing conditional-loss dictionary value, and apply its declared export. The conditional acceptance probability is at least $l_+/\beta - a_+ = 1/b_+$. Induction bounds survival through n attempts by $(1-1/b_+)^n$; no physical renewal or independent-attempt law is assumed. Thus commitment is almost sure and the expected number of begun attempts is at most b_+. Summing the conditional whole-attempt cost gives actual expected cost at most N.

The conditional unnormalized terminal loss of an attempt is at most $u_+\mathcal{Q}^{\ell}_{m_+(M)}(\beta)/\beta + e_{+,m_+(M)}$. Summing on the disjoint actual terminal events and the same survival bound proves the displayed upper. Its actual decoder has at most $D_0 m_+(M)$ values and its live roles are those of $\mathcal{G}$. The role-packing lemma gives the claimed event or state counts, with no extra retained mark or free seed. When $\beta < 1$, use $q = 1$ in the first attempt instead. Finally, in the exact matched case the lower and upper profiles agree at the stated respective widths. All assertions follow. □

Corollary S34.2 (Acquired Bregman geometry)**.** *Let H be compact convex in $\mathbb{R}^k$, and let Φ be continuously differentiable and convex on a neighborhood of H. For the Bregman loss*

$$D_\Phi(z,r) = \Phi(z) - \Phi(r) - \nabla\Phi(r)\cdot(z-r),$$

the general theorem applies, with

$$g_r(x) = w_\Phi(x) + D_\Phi(f(x),r), \quad f(x) = \mathbb{E}_0[Z \mid x], \quad w_\Phi(x) = \mathbb{E}_0[\Phi(Z) \mid x] - \Phi(f(x)) \geq 0. \tag{S34.8}$$

If additionally Φ is twice continuously differentiable and $aI \preccurlyeq \nabla^2\Phi \preccurlyeq AI$ on H, with $0 < a \leq A$, then

$$\frac{a}{2}\mathcal{Q}_m^{\mathrm{sq}}(b) \leq \mathcal{Q}_m^{D_\Phi}(b) \leq \frac{A}{2}\mathcal{Q}_m^{\mathrm{sq}}(b).$$

Without the curvature lower bound the squared-loss exponents are not asserted for every Bregman loss.

Proof. Expand the conditional expected loss and insert f. The affine term cancels, leaving the displayed identity; Jensen's inequality gives $w_\Phi \geq 0$. This conditional-expectation property has a classical Bregman-prediction antecedent [9]. Compactness and continuity give a bounded nonnegative loss and a common decision modulus, so the general theorem applies. Under the Hessian bounds, integration of the second derivative along the segment between z and r gives $(a/2)\|z-r\|^2 \leq D_\Phi(z,r) \leq (A/2)\|z-r\|^2$. Integrate for every feasible dictionary and selected density and then take the infima. The profile comparison follows with the same acquisition and width constraints. □

S35. The exact zero-risk resource spectrum

The mass of one favorable atom gives a sufficient resource threshold, not in general a necessary one. Other zero-loss atoms may be available to the same decoder. We determine the complete finite-width threshold, for the general acquired loss first and then for squared prediction. This statement is about a genuinely acquired probability law; converting it to a physical budget still requires the exact transfer hypotheses.

Use the compact decision space and nonnegative conditional losses $g_r(x)$ of Section S34. For a dictionary C, put $d_C(x) = \min_{r\in C} g_r(x)$ and define

$$z_m = \max_{C\subset D,\, 1\leq|C|\leq m} \nu\{d_C = 0\}.$$

Theorem S35.1 (Exact finite-width zero-loss criterion)**.** *The maximum defining z_m is attained. For every finite $m \geq 1$ and finite $b \geq 1$,*

$$\mathcal{Q}_m^{\ell}(b) = 0 \quad \Longleftrightarrow \quad bz_m \geq 1. \tag{S35.1}$$

When the inequality holds, an optimal procedure accepts only reports in a zero-loss set of an attaining dictionary. When it fails, the attained profile is strictly positive. Neither implication requires positive residual variance, a dimension assumption, a density, or an absence of atoms.

Proof. Represent a dictionary by a tuple in compact D^m, allowing repetitions. Conditional losses are continuous in each decision on one common full-measure set. For a convergent sequence of tuples, the nonnegative minima converge pointwise. The indicator of the closed condition $d_C(x) = 0$ is upper semicontinuous in the tuple. Reverse Fatou for bounded functions therefore makes its integral upper semicontinuous; compactness gives a maximizing tuple.

If $v = \nu\{d_C = 0\} \geq 1/b$, the density $h = v^{-1}\mathbf{1}_{\{d_C=0\}}$ has unit mass and cap at most b, with objective zero. Conversely, take an attained optimal tuple and density for a zero profile, whose existence was proved in Theorem S34.1. Nonnegativity implies $h(x)d_C(x) = 0$ almost everywhere. Thus its unit mass is supported on $\{d_C = 0\}$ and is at most $b\nu\{d_C = 0\} \leq bz_m$. If $bz_m < 1$, no optimizer has zero value; nonnegativity and attainment then make the value strictly positive. □

For squared prediction, form the finite measure on H

$$\mu_0(A) = \nu\{x : w(x) = 0,\ f(x) \in A\}.$$

List its positive atomic masses in nonincreasing order as $a_1 \geq a_2 \geq \cdots$, padding with zeros if there are finitely many, and set $a_j = 0$ for every j if there are none. The total mass is at most one. Repeated values in this list belong to distinct atom locations.

Theorem S35.2 (Atomic zero-risk spectrum and exact resource thresholds)**.** *For every $m \geq 1$,*

$$z_m = \sum_{j=1}^{m} a_j, \qquad \mathcal{Q}_m(b) = 0 \quad \Longleftrightarrow \quad b\sum_{j=1}^{m} a_j \geq 1. \tag{S35.2}$$

The nonatomic part of μ_0 contributes nothing at a fixed finite width, even when $w = 0$ everywhere. All finite-width zero-risk acquisition thresholds are therefore determined by the ordered atomic masses of the zero-residual predictive law, not by its matrix rank or its ambient dimension.

Under the exact matched certificates with identity exports, the checkpoint threshold at width m is $N = \kappa/z_m$ when $z_m > 0$, and no finite budget yields zero risk when $z_m = 0$. For the event-stopped equality, replace m by the total width $M \geq B$. For the state-stopped equality with a sharp mandatory live count, use total labels $B + m$. These conclusions are necessary and sufficient within those declared exact-transfer classes, not for arbitrary experimental interfaces.

Proof. For squared loss, $d_C = w + \operatorname{dist}(f, C)^2$ is zero exactly when $w = 0$ and $f \in C$. Hence $\nu\{d_C = 0\} = \mu_0(C)$. A finite set receives mass only from atoms. There are at most countably many positive atoms: for each positive integer n, at most n atoms can have mass exceeding $1/n$. A nonempty collection of remaining positive masses has a largest member. Indeed if its supremum is $a > 0$, only finitely many members can exceed $a/2$, and one of this finite set attains the supremum. Successive removal therefore defines the displayed ordering. Choosing the locations of its first m positive masses maximizes $\mu_0(C)$ among all dictionaries of size at most m. Missing locations may be filled by arbitrary repeated points in H. This proves the first identity; Theorem S35.1 proves the equivalence. Finally substitute $b = N/\kappa$ into the respective exact transfer equalities, keeping their different memory counts. □

Example S35.3 (Why one atom does not determine the necessary threshold)**.** Let the acquired marked law put mass $1/4$ at $(0, 0)$ and mass $3/4$ at $(1, 0)$, in a carrier containing $[0, 1]$. Treating only the first atom would suggest budget cap four. In fact $z_1 = 3/4$ and $z_2 = 1$. Thus $\mathcal{Q}_1(b) = 0$ exactly for $b \geq 4/3$, while $\mathcal{Q}_2(b) = 0$ already at $b = 1$. The remainder has zero residual variance, which is precisely why the single-atom necessary threshold does not apply. Proposition S40.5 now explicitly states the additional hypothesis needed for its one-atom necessary threshold; the remainder of that proposition is unchanged.

S35.1. **An exact variational form, including ties.** The following fixed-dictionary formula records the classical selection mechanism explicitly. It is a population trimming or bounded-density variational principle, not a new invention of a trimming functional [2, 3]. Its role here is to keep boundary mass and the physical acquisition cap in the same proof.

Proposition S35.4 (Selected-mass dual and optimal selection)**.** *For $0 \leq d \leq L$ measurable on a probability space and $b \geq 1$,*

$$\min_{0 \leq h \leq b,\ \int h\, d\nu = 1} \int h d\, d\nu = \max_{0 \leq t \leq L} \left(t - b \int (t - d)_+ \, d\nu \right). \tag{S35.3}$$

There is a threshold t satisfying $\nu(d < t) \leq 1/b \leq \nu(d \leq t)$. An optimizer has density b on $\{d < t\}$, zero on $\{d > t\}$ and a constant in $[0, b]$ on the tie set, chosen to give unit mass. More generally these saturation rules characterize every optimum for an attaining threshold, up to null sets. Applying

this formula with $d = d_C$ and then minimizing over $C \in D^m$ gives the exact acquired profile; no interchange of the dictionary minimum and threshold maximum is asserted.

Proof. For every feasible h and every t, pointwise minimization of $h(d-t)$ over $[0, b]$ gives $h(d-t) \geq -b(t-d)_+$. Integrating and using unit mass proves weak duality. Take a lower $1/b$ quantile of d in $[0, L]$; elementary continuity of probability measures from below and above gives the two quantile inequalities, including $b = 1$ and atoms at the endpoints. The prescribed density exists because the missing mass lies between zero and $b\nu(d = t)$. It attains equality in the pointwise inequality, proving equality and both attainments. The nonnegative difference in that inequality is $h(d-t)$ on $d > t$ and $(b-h)(t-d)$ on $d < t$. Its integral vanishes exactly under the stated saturation rules. On $d = t$ it is zero for every allowed density. This also proves the claim about all optimizers and ties. □

S36. Exact finite-memory synthesis on reachable quantum supports

The preceding hierarchy characterizes full-history strategies. We now fix the observer alphabet and give a finite certificate that respects its persistent memory. The physical models in this section are time-homogeneous and finite-dimensional, with finite command and report sets and globally legal command rows. Restricted preparation faces and physical command domains have an explicit extension in the supplement. A finite model family is required for the finite algebraic synthesis conclusion. The analytic certificate itself can be imposed modelwise on any declared family, always with the same controller variables.

S36.1. **The actual continuing instrument.** Fix ℓ live classical labels, a prescribed initial label s_0, and t terminal decoder values r_u in a compact carrier D. In a model θ, the hidden physical algebra is $M_{d_\theta}(\mathbb{C})$. A command a has completely positive report maps $\mathcal{I}_{\theta,a,y}$ with normalized report sum, and expected-cost observable $G_{\theta,a} \succeq \underline{c} I$, $\underline{c} > 0$. A terminal audit following a productive report has bounded score observable $F_{\theta,a,y,j}(r_u)$ on the post-report state. Every terminal choice is made before that audit.

The observer uses common command probabilities $p_{s,a}$. After command a and report y, it uses common probabilities $q^0_{s,a,y,v}$ to continue at live label v, or $q^1_{s,a,y,u}$ to commit to decoder value r_u. These nonnegative probabilities sum to one. Nonproductive reports have all $q^1 = 0$. Only the current command, report and fresh randomness are available within that call; there is no external clock, port, persistent seed or hidden controller state.

In the event-stopped interface, live and terminal roles may share labels, and $\max(\ell, t)$ total labels suffice. In the state-stopped interface their types are disjoint and $\ell + t$ labels are required. Any fixed overlap or halt partition can instead be declared explicitly. The physical quantum state is part of the experiment, not a free coherent controller memory.

Use the block-diagonal live algebra $\mathcal{A}_\theta = \bigoplus_{s=1}^{\ell} M_{d_\theta}(\mathbb{C})$. The actual continuing map and one-call observables are

$$(\mathcal{T}_\theta X)_v = \sum_{s,a,y} p_{s,a} q^0_{s,a,y,v} \mathcal{I}_{\theta,a,y}(X_s), \tag{S36.1}$$

$$(g_\theta)_s = \sum_a p_{s,a} G_{\theta,a}, \tag{S36.2}$$

$$(f_{\theta,j})_s = \sum_{a,y,u} p_{s,a} q^1_{s,a,y,u} \mathcal{I}^*_{\theta,a,y}(F_{\theta,a,y,j}(r_u)). \tag{S36.3}$$

The continuing map is completely positive and trace-nonincreasing. Its complementary one-call terminal effect is $b = I - \mathcal{T}^*(I)$. Normalization and the score bounds give

$$g \succeq \underline{c} I, \qquad \alpha_j b \preccurlyeq f_j \preccurlyeq \beta_j b. \tag{S36.4}$$

We suppress θ in modelwise arguments. Let $\iota(\rho)$ insert an initial physical matrix in block s_0. The cost through n calls is the pairing with $\sum_{k<n}(\mathcal{T}^*)^k g$, and the terminal score through n calls is the pairing with $\sum_{k<n}(\mathcal{T}^*)^k f_j$. These formulas count the actual continuing paths, including failures.

Lemma S36.1 (Finite determination of the reachable support)**.** *Put $D_\theta = \ell d_\theta$ and*

$$R_\theta = \sum_{n=0}^{D_\theta - 1} \mathcal{T}_\theta^n \iota(I), \qquad P_\theta = \operatorname{supp} R_\theta. \tag{S36.5}$$

This is the least block projection whose initial block is I and whose support is invariant under the continuing instrument:

$$\mathcal{T}(P) = P\mathcal{T}(P)P. \tag{S36.6}$$

Every actual live state is supported on P. Invariance is a support statement, not the equality $\mathcal{T}(P) = P$.

Proof. Let $R_n = \sum_{k=0}^{n} \mathcal{T}^k \iota(I)$. Its supports increase. If the supports of R_n and R_{n+1} agree, then the support of $\mathcal{T}(R_n) = \sum_{k=1}^{n+1} \mathcal{T}^k \iota(I)$ lies there. Since R_n is strictly positive on its support, it dominates a positive multiple of that support projection. Positivity shows that the projection's image also lies there. The support is therefore invariant and all later terms remain inside it. Otherwise its rank strictly increases. Starting with nonzero rank in dimension D_θ proves stabilization by $D_\theta - 1$.

Any invariant block projection containing the initial block contains every summand by positivity and hence their support. This proves leastness. Every raw positive reachable path operator is bounded by an appropriate positive sum in some iterate from $\iota(I)$, so its support lies in the least invariant projection. Equivalently apply the Kraus representation along the path. Thus all actual live states are contained in P. □

S36.2. **An exact cost and task certificate.** For a block projection P satisfying invariance, let $\mathcal{T}_P$ be the restriction to its supported algebra, with adjoint $\mathcal{T}_P^* X = P\mathcal{T}^*(X)P$. All inverses below act on this finite-dimensional supported algebra, not on unreachable blocks of the full system.

Theorem S36.2 (Reachable-support potential certificate)**.** *For one fixed common observer, the following are equivalent:*

(a) *In every model and from every initial density matrix it commits almost surely, has expected cost at most N, and has the prescribed terminal-score operators $A_{\theta,j}$.*

(b) *In every model there are a block projection P, a positive $H = PHP$, and Hermitian $K_j = PK_jP$, with $P_{s_0} = I$,* (S36.6)*, and*

$$P(H - \mathcal{T}^*H - g)P \succeq 0, \qquad H_{s_0} \preccurlyeq NI, \tag{S36.7}$$

$$P(K_j - \mathcal{T}^*K_j - f_j)P = 0, \qquad (K_j)_{s_0} = A_{\theta,j}, \tag{S36.8}$$

$$\alpha_j P \preccurlyeq K_j \preccurlyeq \beta_j P. \tag{S36.9}$$

One may choose the explicitly computed least reachable projection in (S36.5)*. On it, proper finite-cost operation is equivalent to*

$$\operatorname{spr}(\mathcal{T}_P) < 1. \tag{S36.10}$$

Under this condition the unique exact cost and score solutions are

$$H^{\min} = (I - \mathcal{T}_P^*)^{-1}(PgP), \qquad K_j = (I - \mathcal{T}_P^*)^{-1}(Pf_jP). \tag{S36.11}$$

Every positive cost superpotential dominates $H^{\min}$ on this support. Testing its initial block and those of the K_j is therefore necessary and sufficient, not merely a Lyapunov sufficient condition. Model-dependent certificates are analytic objects and are not stored by the common observer.

Proof. Assume (b) and work on the invariant support. Iterating the cost inequality gives

$$\sum_{k=0}^{n-1}(\mathcal{T}_P^*)^k(PgP)+(\mathcal{T}_P^*)^nH \preccurlyeq H.$$

Trace against an initial density matrix, discard the nonnegative remainder and use the initial budget bound. The left partial sum is the actual cost through n calls. Monotone convergence gives total expected cost at most N. The floor $PgP \succeq \underline{c}P$ bounds $\underline{c}\sum_{k<n}\Pr(\tau > k)$ by the same constant. Hence commitment is almost sure. The same argument gives finite cost from every state on P, with its bound supplied by H.

Iteration of the score equation yields

$$K_j = \sum_{k=0}^{n-1}(\mathcal{T}_P^*)^k(Pf_jP)+(\mathcal{T}_P^*)^nK_j.$$

The remaining trace is bounded in magnitude by $\max(|\alpha_j|,|\beta_j|)\Pr(\tau > n)$ and tends to zero. The finite sum records the scores on disjoint terminal events. Bounded scores and proper operation identify its limit with the actual terminal expectation. The initial block equation proves (a).

For necessity fix a model, let $X_0 = \iota(I)$, take the least reachable projection and write $D = \ell d$. Assumption (a) gives

$$\sum_{k\geq 0}\langle g, \mathcal{T}^kX_0\rangle \leq dN.$$

The corresponding series starting at $\mathcal{T}^nX_0$ is a nonnegative tail of this series. Sum these tails for $0 \leq n < D$ to obtain

$$\sum_{k\geq 0}\langle g, \mathcal{T}^kR\rangle \leq DdN. \tag{S36.12}$$

Since R is strictly positive on its support, every positive supported matrix is bounded by a finite multiple of R. Positivity proves finite expected cost from every such matrix. This includes states which need not occur as individual pure conditional states on any one path.

The positive partial cost potentials increase, and their pairing with R is bounded by (S36.12). Strict positivity of R on its finite-dimensional support bounds their norms. They converge to a positive $H^{\min}$ solving $H^{\min} - \mathcal{T}_P^*H^{\min} = PgP$. Its initial block is precisely the cost operator and is at most NI.

The cost floor also makes $\sum_k(\mathcal{T}_P^*)^kP$ converge in norm. In particular $(\mathcal{T}_P^*)^nP \to 0$. Every supported Hermitian matrix lies between $-\|X\|P$ and $\|X\|P$; positivity makes its image under these powers tend to zero. Decompose any complex matrix into Hermitian parts. In finite dimension, convergence of all these powers forces every eigenvalue to have modulus less than one. Otherwise an eigenvector of modulus at least one would not tend to zero. This proves (S36.10) and its norm-convergent Neumann inverse.

The score series converges absolutely in norm by (S36.4) and the summable survival series. Normalization gives $PbP = P - \mathcal{T}_P^*P$, whose series sums to P. Consequently the score series sums to a K_j between α_jP and β_jP, solves its equation and has the prescribed initial block. This proves (b) and both resolvent formulas.

Conversely, spectral radius less than one gives finite cost through the same Neumann series and proper operation through normalization. Budget and targets are then exactly the initial-block conditions. Finally any positive cost superpotential dominates every partial cost potential by the first telescoping inequality, and thus dominates their limit. More precisely, its nonnegative drift slack $D_H = P(H - \mathcal{T}^*H - g)P$ satisfies

$$H - H^{\min} = (I - \mathcal{T}_P^*)^{-1}D_H \succeq 0.$$

This proves both minimality and the exact excess-certificate formula. □

S36.3. **An attained finite-memory algebraic frontier.** For synthesis the common probabilities, decoder values and modelwise certificate matrices become variables. The state-stopped interface is a finite union over permitted halt partitions. The event-stopped interface permits declared live/output reuse. Both keep a fixed initial live label.

Corollary S36.3 (Finite algebraic synthesis at an exact memory budget)**.** *Assume finitely many physical models with real algebraic primitive coefficients, a compact semialgebraic decoder carrier, and audit-score observables polynomial in decoder coordinates. Fix a total label budget M and algebraic cost budget N. Existence of a common observer with the stated exact targets and cost at most N is equivalent to feasibility of a finite system of polynomial equalities and positive-semidefinite inequalities. It is decidable by real-algebraic quantifier elimination. A feasible instance has a common controller and certificates with real algebraic entries.*

For bounded continuous nonnegative terminal loss, the worst-model, worst-initial-state loss at budget N attains its minimum whenever the feasible fixed-memory class is nonempty. Under the algebraic hypotheses its optimum is real algebraic. These are exact existence statements, not polynomial-time bounds, rational-optimum claims or finite-bit exact implementation of all algebraic samplers.

Proof. Use polynomial normalization and positivity for the probability rows. Encode Hermitian matrices by their real diagonal and real and imaginary off-diagonal coordinates. Positive semidefiniteness is equivalent to nonnegativity of every principal minor. Encode $P = P^* = P^2$, $P_{s_0} = I$, invariance, support of H, K_j, and the cost and score conditions of the theorem. All expressions in (S36.1)–(S36.3) and the certificate are polynomial in these variables. There are finitely many choices of live and terminal role counts and halt partitions under M. Every model retains its own certificate matrices but the same controller variables. For a loss upper bound R, replace its exact initial score target by $(K_{\mathrm{loss}})_{s_0} \preccurlyeq RI$. The theorem identifies this exactly with the required worst-state loss bound. Classical real-algebraic elimination proves decidability and algebraic witnesses [15]; no new elimination algorithm is claimed.

Attainment needs a separate argument because the auxiliary potentials may be unbounded away from their initial block. The probability rows and decoder tuples form a compact space. Every finite-prefix cost and score is continuous in these parameters. A limit of observers with cost at most N has every finite-prefix cost at most N. Monotone convergence and the positive floor give the same total budget and proper operation for the limit. The uniform bound $\Pr(\tau > T) \le N/[\underline{c}(T+1)]$ makes bounded terminal-score tails uniformly vanish on this budget class. Therefore full terminal scores are continuous there, as are the finitely many largest-eigenvalue worst-model losses. Exact score constraints are closed. The loss minimum on a nonempty compact feasible set is attained.

The achievable loss-bound set is semialgebraic over the real algebraic numbers by the first part. An attained finite endpoint of such a one-variable set is algebraic. Its nonempty fiber also has real algebraic coordinates. This proves the last statement without imposing a false compactness bound on the certificate matrices or claiming rational optimal probabilities. □

Finite-state POMDP synthesis, potential equations, spectral transience, quantum decision processes and real-algebraic methods have established antecedents [16, 17, 12, 13, 14]. The present claim is the precise common-model, all-initial-state, paid certificate with exact event/state semantics, its necessary reachable-support construction and its connection to acquired task balance. Those classical ingredients separately are not presented as new.

Example S36.4 (An unreachable recurrent block is not an obstruction)**.** There are two live labels and one physical state. From the prescribed start 1, a productive cost-one call commits immediately. Label 2 is never reached and has a cost-one self-loop. The full continuing matrix is $\mathrm{diag}(0, 1)$, with spectral radius one. Nevertheless the actual observer is proper and has cost one. Its reachable projection is $\mathrm{diag}(1, 0)$; on that support the continuing matrix is zero and $H^{\min} = P$. Thus an

unrestricted inverse of $I - \mathcal{T}$ or a global strict-transience requirement would incorrectly reject it. Scalar blocks embed this example in any quantum dimension.

Corollary S36.5 (Same-task balance with a certified finite live count)**.** *Apply the finite-memory certificate to the localized mass, first-moment and squared-norm scores in Corollary S31.8. A feasible certificate gives one actual finite-label common balancing program at its certified cost. Conversely every such observer on the declared time-homogeneous interface yields the certificate. If the return rule and all actual rooted continuations are also certified, the program's actual live roles and exported decoder values may be used in the upper widths of the transfer theorem. Failure at a prescribed M is an obstruction for that observer class, not merely failure of a searched single-call mixture.*

Proof. The score list is finite and its quadratic expansion is exactly the unnormalized squared-task identity. The potential theorem is necessary and sufficient for the finite observer producing it. The common p, q, r variables are the observer itself, so no uncounted history tape or model index is used. Its live labels and costs are those of its certified transitions. The rooted-continuation condition supplies the required uniformity when attempts repeat. The supplement gives the post-report compilation: prospective audits are not observed on rejected branches, and a logical subroutine return is not mistaken for an irrevocable outer halt. The transfer theorem then uses these same actual resource counts. □

S37. A finite test for reusable causal task balance

The synthesis theorems produce an acquisition routine, but its first invocation is not the whole stopped experiment. A rejected report changes the physical state. Checking only the average returned state loses the report history on which rejection depends. Requiring task equality on every density matrix on the reachable Hilbert support, on the other hand, can be unnecessarily strong. The correct object for exact linear task identities is the report-resolved real linear span of the unnormalized returned states.

Fix a time-homogeneous routine with a finite visible return alphabet X. At a routine start, the physical state is a density matrix on a finite-dimensional carrier K. The routine is legal and proper on that carrier, as certified for example by Theorem S36.2; all its physical operations and live roles are counted. On return x, a specified legal rejection operation produces the unnormalized next-start state

$$\mathcal{R}_x(\rho).$$

The maps $\mathcal{R}_x$ are completely positive and $\sum_x \mathcal{R}_x$ is trace preserving. The rejection operation is trace preserving conditional on that return, so it does not change the return probability. It is performed only when the caller rejects and is charged separately when it has a physical cost. Its phase must be in the counted controller labels. There is no implicit physical reset.

For finitely many bounded prospective audit scores, let $F_{x,j}$ be the Hermitian start-state observable of the unnormalized score on return x, before the caller's commitment decision. The future audit is not observed on rejection. Put $F_{x,0} = \mathcal{R}_x^*(I)$ for report mass. Fix constants $m_{x,j}$, including $m_{x,0} = \nu_x$, with $\sum_x \nu_x = 1$. The initial preparation family $\mathcal{S}$ is a nonempty set of density matrices on K; it need not contain every such matrix. All data may be indexed by an unknown model θ, but the actual routine and caller rules are the same in every model. The following statements are imposed modelwise without allowing the controller to read θ.

Let $d = \dim K$, let $\mathrm{Herm}(K)$ be the real vector space of Hermitian matrices, and define

$$W_0 = \mathrm{span}_{\mathbb{R}}\, \mathcal{S}, \qquad r_0 = \dim W_0, \tag{S37.1}$$

$$W_{n+1} = W_n + \sum_{x \in X} \mathcal{R}_x(W_n), \qquad n_0 = d^2 - r_0. \tag{S37.2}$$

The additions are sums of real linear subspaces, not sums of channels or support projections.

Theorem S37.1 (Finite report-resolved continuation criterion)**.** *One has $W_{n_0} = W_{n_0+1} =: W_\infty$. This is exactly the span of all positive, unnormalized routine-start states*

$$\mathcal{R}_{x_n} \cdots \mathcal{R}_{x_1}(\rho), \qquad \rho \in \mathcal{S}, \quad n \geq 0.$$

The following conditions are equivalent.

(i) *After every finite positive-probability history of rejected returns, from every preparation in $\mathcal{S}$, the next routine has conditional marked audit moments $m_{x,j}$, including its report masses ν_x.*

(ii) *Every residual $F_{x,j} - m_{x,j} I$ annihilates W_∞ under the trace pairing.*

(iii) *Those residuals annihilate W_{n_0}.*

These conditions imply the same identities for every caller using the past returned marks and private randomness to choose acceptance, provided rejection uses exactly the declared return maps. They are also necessary when all finite sequences of rejections are legal caller choices. No unseen future audit, fresh model-dependent program or stored physical posterior is used by the caller.

Proof. An equality $W_n = W_{n+1}$ makes W_n invariant under every $\mathcal{R}_x$, so all subsequent subspaces agree. Before equality each dimension increases by at least one, starting at r_0 and never exceeding d^2. Stabilization therefore occurs by n_0. Induction expands W_n as the span of word images of length at most n of W_0; taking the invariant limit proves the first assertion.

For a word $w = (x_1, \ldots, x_n)$ and $\rho \in \mathcal{S}$, write $A_w = \mathcal{R}_{x_n} \cdots \mathcal{R}_{x_1}(\rho)$. Complete positivity gives $A_w \succeq 0$. If its trace is positive, the conditional next-start state is $A_w / \operatorname{tr} A_w$. Its marked expectation is $\operatorname{tr}(F_{x,j} A_w)/\operatorname{tr} A_w$, so equality to $m_{x,j}$ is precisely $\operatorname{tr}[(F_{x,j} - m_{x,j} I) A_w] = 0$. If its trace is zero, positivity gives $A_w = 0$, and the same unnormalized equality is automatic. Thus (i) is equivalent to annihilation of all word images, hence to (ii). Stabilization gives (iii).

A caller's randomized rejection decisions contribute scalar nonnegative weights to each fixed visible word, because its choices use only that word and private randomness, not the hidden matrix. Conditioning also on its private tape changes these weights but not the physical word map. Integrating over compatible tapes preserves every displayed linear identity. Therefore the identities hold at every actual surviving caller history. Conversely deterministic rejection of the first n returns makes each positive-probability word of length n a possible start history, proving necessity. These arguments use finite histories; conditional-hazard summation, rather than a claim of renewal, supplies the infinite stopping conclusion below. □

Corollary S37.2 (A finite-dimensional certificate for repeated finite-mark balance)**.** *Take the mass, first-moment and squared-norm score targets of Corollary S31.8. Suppose their residuals annihilate the report-resolved span in every model, and one invocation including its executed rejection return costs conditionally at most κ on all actual start states. For any finite dictionary and any feasible $0 \leq h \leq b$ with $\sum_x h(x)\nu_x = 1$, accept return x with probability $h(x)/b$. The resulting routine commits almost surely, costs at most κb, and has exact terminal task value*

$$\sum_x h(x)\nu_x \left[w(x) + \|f(x) - r(x)\|^2\right].$$

The same finite observer implements every attempt, with no added intercall history register. A rejection return performed within the productive call adds no live role; an extra physical return subroutine must have its calls and live roles counted. This yields the upper construction needed by the transfer theorem from a finite report-resolved continuation test, rather than leaving its history-uniform task identity as an unverified premise.

Proof. The three marked moment identities and expansion of the square give the same unnormalized task identity conditional on every actual start history by the theorem. Thus the conditional acceptance probability is $1/b$ and the unnormalized loss is b^{-1} times the displayed value. Induction gives survival $(1 - 1/b)^n$ through n invocations; summing begun invocations gives expected number b. Summing their conditional costs and disjoint terminal losses proves both assertions. For a final

accepted invocation the rejection operation is not performed, and nonnegative cost only improves its budget. The decision is made at the final productive report before the prospective audit. The existing routine's live labels are reused on rejection, while its finite terminal decoder is read only on commitment. This proves the memory statement with the explicit return-operation qualification. □

The cost assumption in this corollary is not silently converted into a linear-span equality. A positive cost superpotential on the reachable support is a sufficient, checkable bound. If the exact whole-invocation cost observable H is available and constant cost κ is required at every reachable start, the condition is exactly $(H - \kappa I) \perp W_\infty$ by the same proof. A one-sided inequality on only the reachable positive states need not be equivalent to a Loewner inequality on their whole support; no such equivalence is asserted.

Example S37.3 (Averaging is too weak; support equality is too strong)**.** On a qubit let $\rho_0 = I/2$ and

$$\mathcal{R}_0(A) = \tfrac{1}{2}\operatorname{tr}(A)|0\rangle\langle 0|, \qquad \mathcal{R}_1(A) = \tfrac{1}{2}\operatorname{tr}(A)|1\rangle\langle 1|.$$

The average return sends ρ_0 to itself, so iterating only that average generates the one-dimensional span of I. But the report-resolved span is the entire diagonal Hermitian subspace. The observable σ_z has zero expectation at the initial and averaged returned state, and expectations $+1, -1$ at the two actual conditional starts. An initial or pooled calibration of that task therefore fails the required repeated identity.

Conversely σ_x annihilates the complete report-resolved diagonal span, though it is a nonzero operator on its full two-dimensional Hilbert support. A test insisting on the operator identity $\sigma_x = 0$ on that support would reject an exact reachable-task identity. Both examples can be realized by a fair visible coin, a prospective Pauli audit at commitment, and a charged conditional replacement preparation on rejection. They distinguish a span of density matrices from a Hilbert-space support and do not make replacement preparations free.

Proposition S37.4 (Finite exact arithmetic verification)**.** *For a fixed finite-dimensional routine whose return maps, task observables and a finite spanning preparation list have algebraic real and imaginary parts, the continuation criterion is decidable by finitely many exact linear-algebra operations. At most $d^2 - r_0$ span enlargements are needed; each enlarges a space of dimension at most d^2. For a finite family of physical models the same test is performed in every model on the one fixed controller. This is a verification result for the supplied routine, not a polynomial-time synthesis theorem for its controller parameters or a complexity bound for evaluating arbitrary algebraic numbers.*

Proof. Represent Hermitian matrices by d^2 real coordinates and every return map by a real matrix on those coordinates. Starting with a basis of the supplied preparation span, append all its images under all return maps and retain a basis by exact elimination. The dimension argument in Theorem S37.1 bounds the number of enlargements. At stabilization check the finitely many trace pairings with every residual. All operations and zero tests are exact in the finite algebraic field generated by the input entries. This implements the equivalence of the theorem. □

Finite reachable linear spans and word-equivalence tests have classical probabilistic-automata antecedents [22]; reachable Hilbert supports and quantum termination also have established theory [23, 24]. Those algorithms and the dimension argument are not claimed as new. The assertion used here is the exact report-resolved test for repeated marked tasks, its distinction from pooled calibration and support-wide identities, and its paid post-report compilation into the same causal resource theorem.

S37.1. **Continuous reports and a sharp finite-depth obstruction.** The finite-return restriction can be removed from the exact continuation test without asserting general effective synthesis. Let X now be standard Borel. A rejected return is described by a completely positive instrument

$D \mapsto \mathcal{R}(D)$, countably additive in the finite-dimensional space of linear maps, with $\mathcal{R}(X)$ trace preserving. Let $F_j(D)$ be finite Hermitian operator-valued prospective task measures and $m_j(D)$ their prescribed finite scalar measures; include report mass as $F_0(D) = \mathcal{R}(D)^* I$ and $m_0 = \nu$. Each bounded score gives such a finite measure. Define

$$W_{n+1} = W_n + \operatorname{span}_{\mathbb{R}}\{\mathcal{R}(D)A : D \in \mathcal{B}(X),\ A \in W_n\}, \qquad E_j(D) = F_j(D) - m_j(D)I.$$

The operational identity below is an almost-sure identity at actual reached histories, not a condition imposed at individually null continuous reports.

Theorem S37.5 (Finite-depth criterion for standard Borel return instruments)**.** *With $W_0 = \operatorname{span}_{\mathbb{R}} \mathcal{S}$ of dimension r_0 and physical dimension d, the span stabilizes by depth $d^2 - r_0$. Exact marked task identities after every finite report-dependent rejection history hold if and only if*

$$\operatorname{tr}[E_j(D)A] = 0 \quad (A \in W_{d^2-r_0},\ D \in \mathcal{B}(X),\ \text{every } j). \tag{S37.3}$$

If an identity fails, there is a witness using an allowed initial preparation, at most $d^2 - r_0$ rejected returns, a rectangular cylinder of report events of positive probability, and one subsequent task event.

There exists a finite Borel partition Π of the return space with at most $2^{d^2-r_0}$ cells such that the maps $\{\mathcal{R}(C) : C \in \Pi\}$ generate exactly the same reachable span as all Borel events. This is an existence statement about exact instrument measures. An arbitrary coarse partition or numerical binning is not such a certificate, and testing task-measure equality still requires its exact measure data.

Proof. Dimension growth and invariance give stabilization exactly as before. Induction identifies W_n with the span of the matrices $\mathcal{R}(D_n)\cdots\mathcal{R}(D_1)\rho$ for words of length at most n and $\rho \in \mathcal{S}$. These are positive unnormalized states for rectangular cylinders. If a residual does not annihilate their span, it does not annihilate one such generator. That generator is positive and nonzero, so its trace is positive. This proves the bounded-depth witness assertion and necessity of the displayed condition.

For sufficiency, fix an initial preparation and a finite depth. Iterating the instrument gives a finite positive operator-valued measure on X^n. Pairing it with a fixed residual $E_j(D)$ gives a finite signed scalar measure. Its value on every rectangular cylinder is zero by the invariant-span condition. Rectangles form a generating pi-system, so uniqueness of finite signed measures makes it zero on every measurable history event. A bounded nonnegative report-dependent rejection weight can be integrated against this zero measure; approximation by simple functions proves that the conditional task discrepancy is zero at the actual surviving histories. To obtain one common almost-sure set for all next task events, first use a countable generating algebra of the standard Borel report space, intersect its full-measure sets for the finitely many scores, and then use uniqueness of finite measures at each remaining history. Private caller randomness is integrated in the same way. This gives the asserted operational identities, including measurable rejection depending on the full report history rather than only on rectangles.

For the finite partition, start with $V = W_0$. If V is not invariant under all event maps, choose an event D and an $A \in V$ such that $\mathcal{R}(D)A \notin V$. Add this event to a list and replace V by the invariant span generated by all listed event maps and W_0. Its dimension increases by at least one. After at most $d^2 - r_0$ such choices, the resulting space is invariant under every event map and hence equals W_∞. Take the common Boolean refinement of the listed events. Every selected map is the sum of the maps of its partition cells. The invariant span under the partition maps therefore contains W_∞, while each cell is a Borel event and so cannot generate a vector outside W_∞. Equality follows. At most $2^{d^2-r_0}$ cells are needed. Nothing in this argument supplies an effective representation of those selected Borel events for arbitrary input measures. □

Theorem S37.6 (Sharp and physically realizable continuation depth)**.** *For every physical dimension $d \geq 2$, there is a completely positive trace-preserving return channel, one initial full-rank preparation, one visible return mark and a bounded two-outcome prospective audit such that the prescribed*

squared-task balance holds after n rejections for every $0 \le n < d^2-1$, but fails after d^2-1 rejections. All real and imaginary parts of primitive matrix entries can be chosen real algebraic. Thus the witness depth $d^2 - r_0$ is sharp already for $r_0 = 1$, even with one return mark and uniformly finite paid invocation cost.

Proof. Put $q = d^2 - 1$. Choose a Hilbert–Schmidt orthonormal real basis $F_1, \dots, F_q$ of traceless Hermitian matrices with algebraic entries. A generalized matrix-unit basis gives one, and $\|F_i\|_{\mathrm{op}} \le 1$. Choose distinct numbers

$$\lambda_i = \frac{i}{2dq(q+1)}, \qquad \varepsilon = \frac{1}{2dq}, \qquad \rho_0 = I/d + \varepsilon \sum_{i=1}^{q} F_i.$$

The norm bound makes $\rho_0 \succeq I/(2d)$ and its trace is one. Define

$$\Phi(A) = \operatorname{tr}(A) I/d + \sum_{i=1}^{q} \lambda_i \operatorname{tr}(F_i A) F_i.$$

Its Choi matrix in the unnormalized convention is

$$J_\Phi = I \otimes I/d + \sum_i \lambda_i F_i^{\mathsf{T}} \otimes F_i.$$

Since $\sum_i \lambda_i = 1/(4d)$, the perturbation has norm at most $1/(4d)$. Thus $J_\Phi \succeq 3I/(4d)$, which proves complete positivity. Taking traces proves trace preservation, and $\Phi(I) = I$, $\Phi(F_i) = \lambda_i F_i$. Consequently

$$\rho_n = \Phi^n(\rho_0) = I/d + \varepsilon \sum_i \lambda_i^n F_i.$$

The $q+1 = d^2$ matrices $\rho_0, \dots, \rho_q$ are linearly independent by the Vandermonde determinant on the distinct nodes $1, \lambda_1, \dots, \lambda_q$. There is therefore a Hermitian L annihilating $\rho_0, \dots, \rho_{q-1}$ but not ρ_q. This already proves the claimed depth. We give exact coefficients to make the physical realization explicit.

Set $x_0 = 1$, $x_i = \lambda_i$ and $w_i = \prod_{j \ne i} (x_i - x_j)^{-1}$. The coefficient of degree q in Lagrange interpolation of x^n at these $q+1$ nodes gives

$$\sum_{i=0}^{q} w_i x_i^n = 0 \ (0 \le n < q), \qquad \sum_{i=0}^{q} w_i x_i^q = 1.$$

Let $L_0 = w_0 I + \varepsilon^{-1} \sum_{i=1}^{q} w_i F_i$ and $K = |w_0| + \varepsilon^{-1} \sum_{i=1}^{q} |w_i| > 0$. Then $L = L_0/K$ is Hermitian with operator norm at most one, and $\operatorname{tr}(L\rho_n) = 0$ for $n < q$, while $\operatorname{tr}(L\rho_q) = 1/K > 0$. The effects $(I+L)/2, (I-L)/2$ give a valid two-outcome audit $Z \in \{-1, 1\}$ with mean $\operatorname{tr}(L\rho)$ and second moment one. All specified matrices and constants are algebraic; the square roots needed for a Kraus implementation of their positive Choi matrix may also be chosen algebraic by finite-dimensional spectral calculus.

One routine emits a deterministic productive mark without changing the state. On acceptance, commit a readout and perform this prospective audit. On rejection, execute the physical channel Φ and return to the start. Charge the productive operation and rejected return separately; each invocation has cost at most two unit calls and a fixed finite implementation. After n rejected invocations the actual start state is ρ_n. For the genuinely prepared initial reference, $f = 0, w = 1$, and the squared loss at readout $r \in [-1, 1]$ is $1 + r^2$ at every $n < q$. At $n = q$ it is $1 + r^2 - 2r/K$, which differs for any nonzero r. Thus all marked mass, mean and squared-norm identities hold up to the asserted depth and the task identity then fails. No report of zero probability, infinitely expensive routine, free physical reset or numerical approximation is used. □

The Hermitian reachable-space method itself, including its distinction from Hilbert-space reachability, is expressly present in Fu et al. [24]; it is not introduced here. The extensions proved above

are the standard Borel report-event formulation for reusable marked tasks, its finite-coarsening statement, and the matching physically realizable $d^2 - 1$ witness depth connected to paid rejection. Neither the finite witness bound nor its sharp example constitutes a general decision procedure for arbitrary continuous instrument measures.

S38. A spectral memory law for two quantum detectors

We apply the finite-memory framework without a balancing group action or reset to a prescribed reference state. The measurements disturb the state: this is first detection, not estimation of an untouched pre-command parameter. Quantum first-detection laws, dark states and probing-time optimization have substantial antecedents [19, 20]. Our question is the exact paid-call versus classical-label frontier when the observer chooses between two specified detectors.

Let P, Q be orthogonal projections on a finite-dimensional Hilbert space. Detector A has failure map $\rho \mapsto P\rho P$ and success map $\rho \mapsto (I-P)\rho(I-P)$; detector B uses Q in the same way. Every call costs one. There is no intervening evolution, additional measurement, reset or coherent controller memory. A failure does not permit commitment. The first success requires immediate productive commitment, with one constant terminal decoder value. The initial density matrix is arbitrary and unknown. An autonomous controller has a prescribed initial live label, fixed command rows and post-failure transitions, with no free clock or persistent external seed. Write V_j for the least worst-initial-state expected call count using at most j live labels, and V_* for the infimum over all visible-history policies.

Theorem S38.1 (Exact two-detector resource law)**.** *Suppose $P, Q \neq 0$ and $c = \|PQ\| < 1$. In every finite dimension,*

$$V_* = 1 + \frac{1}{2(1-c)}. \tag{S38.1}$$

It is attained by choosing the first detector fairly and then switching after every failure. The one-live-label value is

$$V_1 = 1 + \frac{1}{1-c} = 2V_* - 1, \tag{S38.2}$$

and a fair independent detector choice at every call attains it. With a fixed initial label, three live labels are necessary and sufficient to attain V_*. The two-live-label optimum is attained and satisfies

$$V_* < V_2 \leq 1 + \frac{1}{1-c^2}; \tag{S38.3}$$

its exact value is determined in Theorem S38.3 below.

If an explicitly randomized initial distribution over counted labels is instead part of the interface, two live labels are necessary and sufficient to attain V_*. Under event stopping the stated live counts are also total-label counts, since the constant terminal value may reuse a live label. State stopping requires one additional disjoint terminal label.

If either failure projection is zero, one call to that detector gives value one. If $c = 1$, their ranges have a common nonzero dark subspace and no policy is proper for every initial state. The robust cost is then infinite.

Proof. We give the required spectral reduction explicitly. Diagonalize PQP on $\operatorname{ran} P$. For an eigenvector u with eigenvalue $q \in (0,1)$, put $v = Qu/\sqrt{q}$. Then $\|v\| = 1$ and $\langle u, v\rangle = \sqrt{q}$. Their span is invariant under both projections, which restrict to the rank-one projections onto u, v. Distinct orthonormal eigenvectors give orthogonal such two-dimensional blocks, including after resolving multiplicities. The zero eigenspace gives $\operatorname{ran} P \cap \ker Q$; the remaining part of $\operatorname{ran} Q$ is orthogonal to P. The orthogonal complement of the two ranges lies in both kernels. Eigenvalue one is exactly a common failure range. This is the classical two-subspace geometry [18], proved here in the form used below. The largest nontrivial squared overlap is $q = c^2$ when $c > 0$.

After a failure of A, a second immediate A fails with probability one and leaves the state unchanged. The analogous statement holds for B. Such a repetition gives no information. In a full-history policy it can be deleted, with its private coins and deterministic failure simulated internally, leaving all subsequent physical states and detector choices coupled and weakly reducing calls. Infinite repetitions give infinite cost and cannot improve the optimum. Thus, after its first choice, a pointwise optimal policy switches after every failure.

On a block of squared overlap q, the post-failure pure state switches between u and v, and every further failure has probability q. Its expected remaining cost is $1/(1-q)$. On the whole P failure range this continuation cost is the positive operator $H_P = (I_P - PQP)^{-1}$, and similarly $H_Q = (I_Q - QPQ)^{-1}$. These inverses exist because $c < 1$. A first choice of A with probability r therefore has cost operator bounded below by

$$I + rPH_PP + (1-r)QH_QQ, \tag{S38.4}$$

with equality for subsequent alternation. On a principal block its largest eigenvalue is

$$1 + \frac{1 + \sqrt{1 - 4r(1-r)(1-q)}}{2(1-q)}. \tag{S38.5}$$

It increases with q and is minimized uniquely at $r = 1/2$. Unpaired parts have the corresponding zero-overlap bound, and the common kernel of P, Q succeeds on the first call. The largest block thus gives (S38.1). If $c = 0$, use a unit vector from each of the two nonzero orthogonal failure ranges; the same formula with $q = 0$ applies.

With one live label, the same probability r of choosing A is used after every failure. A persistent shared seed would be extra memory and is not admitted. On a block, let x, y be the remaining costs after A, B fail. For $0 < r < 1$,

$$x = 1 + rx + (1-r)qy, \qquad y = 1 + rqx + (1-r)y.$$

Solving and returning to the initial state gives $I + aP + bQ$ on that block, where

$$a = \frac{r/(1-r) + q}{1 - q^2}, \qquad b = \frac{(1-r)/r + q}{1 - q^2}.$$

The larger eigenvalue is $1 + [a + b + \sqrt{q(a+b)^2 + (1-q)(a-b)^2}]/2$. Since $a + b \geq 2/(1-q)$, it is at least $1 + (1 + \sqrt{q})/(1-q) = 1 + 1/(1 - \sqrt{q})$, with equality at $r = 1/2$. The deterministic endpoints have an undetectable failure state. The largest principal block proves (S38.2); the fair row realizes that operator bound in every block.

For the fixed-start unrestricted optimum, its first command row must be fair. On a largest principal block, both possible first failures have positive probability from the top vector of the fair cost operator. Every positive-probability repetition of the last failed detector strictly increases the expected cost there. Consequently an optimum needs a pure B row after A fails and a pure A row after B fails. These are two distinct rows, neither equal to the initial mixed row. Three live labels are necessary; those three rows and their obvious switching transitions suffice. This is a rigidity statement on the extremal block, not a claim about behavior on irrelevant physical subspaces.

The two-label parameter space is compact. For a sequence of proper two-label observers with uniformly bounded worst-state costs, every finite-prefix cost is continuous in the parameters and remains bounded at the limit. Monotone convergence gives a finite full cost there; unit costs imply proper operation. A minimizing sequence therefore has a proper minimizing limit. Such a limit cannot attain V_* by the preceding row argument, proving the strict lower in (S38.3). Deterministic alternation from a fixed first detector uses two labels and has worst cost $1 + (1 - c^2)^{-1}$, proving the upper.

If the interface permits a randomized initial distribution, start fairly in the two pure switching labels. This stores the first choice in a counted label and attains V_*. One live label cannot attain it by (S38.2). The event/state counts follow from their exact halt semantics. Finally, a zero failure

projection always succeeds. In finite dimension $\|PQ\| = 1$ implies a nonzero vector in both failure ranges by the equality case for orthogonal projections. It remains there under every permitted command and is never detected. This proves the degenerate cases. □

Corollary S38.2 (A primitive least-favorable preparation)**.** *Choose unit failure vectors u, v in a largest principal-angle block, with $\langle u, v\rangle = c$, and put*

$$\psi_* = (u+v)/\sqrt{2(1+c)}, \qquad \rho_* = |\psi_*\rangle\langle\psi_*|.$$

For $c = 0$ choose any unit vectors in the two nonzero orthogonal failure ranges. This one pure state gives expected cost at least V_* against every visible-history policy. Fair-first alternation has cost at most V_* against every initial state. Thus it is an explicit minimax saddle without a policy-dependent preparation or a minimax exchange. The same preparation gives the one-live-label lower V_1 against every one-label row.

Proof. The first failure probability from ρ_* is $(1+c)/2$ for either detector. Its post-failure state is respectively u or v, whose least remaining cost is $1/(1-c^2)$ by the switching argument. Every first command mixture therefore costs at least $1+(1+c)/[2(1-c^2)] = V_*$. The uniform upper was proved above. For a one-label row, the cost at ρ_* is $1+(a+b)(1+c)/2$. The bound $a+b \geq 2/(1-c^2)$ gives V_1, attained by the fair row. This preparation is a witness for the stated detection task, not an automatically recurrent reference row after arbitrary rejected experiments. □

S38.1. **The exact two-live-label frontier.** Put $q = c^2$. For $0 < r \leq 1$ define

$$a_q(r) = r + q, \qquad b_q(r) = (1-r)(r^{-1}+q), \tag{S38.6}$$

$$F_q(r) = 1 + \frac{a_q(r) + b_q(r) + \sqrt{(a_q(r)-b_q(r))^2 + 4q a_q(r) b_q(r)}}{2(1-q^2)}. \tag{S38.7}$$

Theorem S38.3 (Exact two-label optimum and a golden-ratio transition)**.** *In the nondegenerate class above,*

$$V_2(c) = \min_{0<r\leq 1} F_{c^2}(r). \tag{S38.8}$$

A minimizing observer has a fixed initial label S choosing A with probability r and B otherwise, and a label T choosing B purely. After an A failure it uses T; after a B failure it uses S. Success commits immediately. It realizes the optimum for every finite-dimensional pair with the same largest overlap c.

Let $q_* = (\sqrt{5}-1)/2$. The minimizer is unique and is characterized by

q	r_{opt}	V_2
0	q_*	$(1+\sqrt{5})/2$
$0 < q < q_*$	*unique* $r \in (0,1)$ *with* $F_q'(r) = 0$	$F_q(r_{\mathrm{opt}})$
$q_* \leq q < 1$	1	$1 + (1-q)^{-1}$.

(S38.9)

In the interior case, put $a = a_q(r)$, $b = b_q(r)$, $D = \sqrt{(a-b)^2 + 4qab}$ and $b' = -r^{-2} - q$. The unique equation is explicitly

$$(1+b')D + (a-b)(1-b') + 2q(b+ab') = 0. \tag{S38.10}$$

Squaring does not license extraneous roots. Together with the one-label and unrestricted laws, this determines the exact fixed-start cost at every integer live-memory budget: V_1 at one, V_2 at two, and V_* at three or more. State-stopped total counts add one terminal label; event-stopped counts do not.

Proof. First prove a converse against every two-label controller, including randomized post-failure transitions. Restrict the initial state to a largest principal block with failure states u, v and squared overlap q. For $q = 0$ use any unit vectors in the two nonzero orthogonal ranges. Relabel detector names if necessary so that the initial label has the larger probability r of choosing A. The other label has probability t, with $0 \leq t \leq r \leq 1$. Proper worst-state operation requires $r > 0$ and $t < 1$.

Even allowing the controller to choose whichever of its two labels it prefers after a failure, the least remaining costs at u, v are

$$x = \frac{(1-t)^{-1} + q/r}{1-q^2}, \qquad y = \frac{r^{-1} + q/(1-t)}{1-q^2}. \tag{S38.11}$$

Here is a direct verification for arbitrary subsequent choices, not an assumed stationary optimum. One has $x - qy = (1-t)^{-1}$ and $y - qx = r^{-1}$. For any command probability $z \in [t, r]$,

$$1 + zx + (1-z)qy - x = \frac{z-t}{1-t} \geq 0, \qquad 1 + zqx + (1-z)y - y = 1 - \frac{z}{r} \geq 0.$$

These are lower potential inequalities for every allowed label and randomized label choice. Iteration to a finite horizon bounds actual expected calls below by the initial potential minus a remaining term bounded by $\max(x, y)$ times survival. The remainder vanishes for a proper policy. An improper policy has infinite call cost. Thus x, y are universal lower bounds. They are attained by using the t-row after an A failure and the r-row after a B failure, as their equations verify.

From the prescribed initial label, the first call therefore has cost operator at least $I + rxP + (1-r)yQ$ on this block. For fixed r, both x, y are nondecreasing in t, so this operator is minimized at $t = 0$. Its coefficients then are

$$rx = \frac{r+q}{1-q^2}, \qquad (1-r)y = \frac{(1-r)(r^{-1}+q)}{1-q^2}.$$

Its largest eigenvalue is $F_q(r)$. This proves the lower without assuming deterministic transitions, a prior, or an uncounted initial seed.

For the matching upper use the displayed S, T observer. Its exact block costs are (S38.11) at $t = 0$. In a smaller principal block with squared overlap $q' \leq q$, both nonnegative coefficients $a_{q'}(r)/(1-q'^2)$ and $b_{q'}(r)/(1-q'^2)$ increase with q'. The larger eigenvalue of $aP + bQ$ is nondecreasing in either nonnegative coefficient by positive matrix order, and in overlap by its square-root formula. Thus the worst block is the largest. Unpaired ranges have no greater than the corresponding zero-overlap bound, and the common kernel succeeds immediately. The same observer therefore has cost at most $F_q(r)$ for every initial state and attains it on the largest block. Its finite post-failure costs prove proper operation. Equality in (S38.8) follows.

For $q = 0$, $F_0(r) = 1 + \max\{r, r^{-1} - 1\}$. The increasing and decreasing terms meet uniquely at $r^2 + r - 1 = 0$, giving the first line of (S38.9).

For $0 < q < 1$, use one fixed principal block. Its numerator matrix is $a_q(r)P + b_q(r)Q$, with a_q affine and $b_q''(r) = 2/r^3 > 0$. Its largest eigenvalue is strictly convex. Indeed the convex-combination matrix at two different parameters exceeds the matrix at the combined parameter by a positive multiple of Q. A top eigenvector of the latter has positive Q-expectation since $q > 0$ and $a_q > 0$. Evaluation at that vector proves strict convexity. The top eigenvalue is simple, so the function is differentiable, and it diverges as $r \downarrow 0$.

At $r = 1$ the numerator is $(1+q)P$. Its top vector is u, with Q-expectation q. Differentiation at this simple eigenvector gives

$$(1-q^2)F_q'(1) = 1 - q - q^2.$$

For $q \geq q_*$ this is nonpositive, so strict convexity makes $r = 1$ the unique minimum. For $q < q_*$ it is positive; divergence at zero and strict convexity give a unique interior stationary point. Differentiating the square root gives (S38.10). This proves every case. The already proved live-role counts give the full integer frontier. □

The golden-ratio threshold concerns the squared principal-angle cosine and the optimal controller row, not a universal physical constant. At small overlap one continuing row is pure and the initial row is mixed. At and above the threshold the optimal two-label policy is fixed-start deterministic alternation. A third live label restores a fair initial choice without contaminating either continuing row and strictly improves the worst-state cost.

S38.2. **A sharp memory-constrained minimax gap.** The exact finite-memory frontier also identifies a limitation of replacing a uniform converse by an optimized single-prior calculation. Let $\mathcal{C}_2$ be the proper fixed-start observers with at most two live labels in this detection interface.

Corollary S38.4 (Minimax gap at two live labels)**.** *For $P, Q \neq 0$ and $c < 1$,*

$$\inf_{\pi\in\mathcal{C}_2}\sup_{\rho}\mathbb{E}^{\pi}_{\rho}\tau = V_2(c) > V_*(c) = \sup_{\rho}\inf_{\pi\in\mathcal{C}_2}\mathbb{E}^{\pi}_{\rho}\tau. \tag{S38.12}$$

In particular no single initial preparation supplies the sharp two-label robust lower V_2. At zero overlap the gap is $(\sqrt{5}-2)/2$. Permitting a randomized initial distribution over the two counted pure labels removes this gap and gives robust value V_*. No analogous gap occurs at one live label or at three or more for this experiment.

Proof. The left equality and strict inequality are the memory theorems. For any fixed initial state, choose the better of the two deterministic first detectors and then alternate. Both choices are proper two-label observers. Their better cost is no larger than their average, which is the cost of fair-first alternation and is at most V_*. Hence the maximin is at most V_*. The single state ρ_* in Corollary S38.2 gives cost at least V_* against every policy, proving the reverse inequality. When $c = 0$, $V_2 = (1+\sqrt{5})/2$ and $V_* = 3/2$, giving the displayed gap. A randomized initial label realizes fair-first alternation directly. For one label, the same least-favorable state and fair independent row form a saddle at V_1; for three or more the full-history saddle applies. □

The policy in the inner maximin may depend on the fixed preparation; the robust policy on the left may not. The fixed-start two-label class is not closed under the convexification that would mix the two pure initial rows while remembering which continuing row to use. A free random initial distribution changes precisely that resource. Thus a minimax exchange would erase a real memory constraint. This is an exact result for the stated detector model, not a failure of the general sufficient transfer theorem or a claim that every finite-memory problem has such a gap.

S39. Raw task order and changing-rank stability

The central theorem is formulated in measures because selection acts before normalization. This section gives a verification principle from primitive kernels and a singular extension of that principle. It does not assume a covariance or information matrix in the experiment.

A measure determined by executable affine-square tests. For a report–audit law $P(dy, dz)$ on $Y \times H$ define the finite positive matrix measure

$$J_P(D) = \int_{D\times H} v(z)v(z)^{\mathsf{T}}P(dy, dz), \qquad v(z) = (1, z_1, \ldots, z_k)^{\mathsf{T}}. \tag{S39.1}$$

For finite positive matrix measures write $A \preccurlyeq B$ when $B(D) - A(D)$ is positive semidefinite for every Borel D. This definition is independent of the choice of dominating scalar measure.

Proposition S39.1 (An exact order criterion for localized affine-square tests)**.** *For two joint laws P, P^0 on the same full report space and $0 < l \le 1 \le u$, the following assertions are equivalent:*

$$lJ_{P^0} \preccurlyeq J_P \preccurlyeq uJ_{P^0}; \tag{S39.2}$$

for every Borel report event D and every $a \in \mathbb{R}^{k+1}$,

$$l\int_D |a^{\mathsf{T}}v(z)|^2 dP^0 \le \int_D |a^{\mathsf{T}}v(z)|^2 dP \le u\int_D |a^{\mathsf{T}}v(z)|^2 dP^0. \tag{S39.3}$$

It suffices to test rational vectors a. These equivalent conditions imply, for all Borel $q \in [0,1]$ and finite-image Borel $r : Y \to H$, both the report-mass and the squared-task inequalities with zero additive defect and these same factors. A violation of (S39.2) *is witnessed by a single Borel report event and a rational affine-square test.*

Proof. The integral in (S39.3) is $a^{\mathsf{T}} J_P(D) a$. The definition of positive semidefiniteness gives the equivalence. Continuity of the quadratic form gives sufficiency of rational vectors and a rational witness for a strict violation. Taking $a = (1, 0, \dots, 0)$ gives mass order. For a fixed readout r, sum the k quadratic tests with vectors $(-r_j, e_j)$, obtaining the order of $\|z - r\|^2$. Partition Y into the finitely many cells where r is constant. For a simple nonnegative q, apply order on intersections with these cells and add. Bounded monotone approximation extends to arbitrary $q \in [0, 1]$. All measures are finite and all tested losses bounded. □

Thus the criterion is necessary and sufficient for the declared task-rich affine-square family. It is a sufficient condition for the narrower squared-prediction task of the central theorem. It is not claimed necessary for that narrower task, or for the existence of an arbitrary causal balancing strategy. The matrix is built from raw joint audit laws, not from an assumed parameterization. The upper construction may use its virtual report space after averaging its current command; the lower criterion must use the full report seen by a competitor, conditional on that competitor's chosen command.

Theorem S39.2 (Order on measurable singular ranges)**.** *Let J_0, J_1 be finite positive matrix measures of size $k + 1$. With $\lambda = \operatorname{tr} J_0 + \operatorname{tr} J_1$, take measurable positive semidefinite densities A_0, A_1 with respect to λ. Then*

$$l J_0 \preccurlyeq J_1 \preccurlyeq u J_0 \tag{S39.4}$$

is equivalent, λ-almost everywhere, to equality of kernels $\ker A_0 = \ker A_1$ and to

$$l P_0 \preccurlyeq A_0^{\dagger/2} A_1 A_0^{\dagger/2} \preccurlyeq u P_0, \tag{S39.5}$$

where P_0 is the orthogonal projection onto $\operatorname{ran} A_0$ and $\dagger$ denotes the Moore–Penrose inverse. On the zero range both sides are zero. All these maps are measurable, even when rank changes. No positive lower bound on a nonzero eigenvalue of A_0 is required. In particular every zero affine-square task on an acquired report event remains exactly zero under the order.

Proof. A finite signed matrix measure is positive if and only if its density is positive semidefinite almost everywhere. To see the nontrivial direction, test the density against all rational vectors; a negative set of positive measure for one such quadratic form contradicts positivity of its integral on that set. The countable intersection of the resulting full-measure sets handles all rational vectors, hence all real vectors by continuity. Thus (S39.4) is equivalent to $l A_0 \preccurlyeq A_1 \preccurlyeq u A_0$ almost everywhere.

For two such matrices, a vector in $\ker A_0$ has zero quadratic form under A_1, and positivity implies that it is in $\ker A_1$. The lower inequality gives the converse. On their common range A_0 is positive definite; congruence by its inverse square root yields (S39.5). Conversely equality of kernels makes A_1 supported on that range. Congruence of (S39.5) by $A_0^{1/2}$ gives the original two matrix inequalities, extended by zero on the common kernel.

Measurability follows from finite-dimensional spectral functional calculus: P_0 is the pointwise limit of $A_0(A_0 + n^{-1} I)^{-1}$, and the scalar function equal to $x^{-1/2}$ on $x > 0$ and zero at zero is Borel. Applying it spectrally gives a Borel matrix function on each rank stratum, hence on their finite measurable union. We use these matrices pointwise; their unbounded inverse norms are never taken as a uniform error constant. Finally a nonnegative task of zero integral under J_0 has integral at most u times zero under J_1, proving preservation of zero tasks. □

There is an important distinction between this assertion and a rank formula for risk. Rank is allowed to change; the rate still comes from the acquired masses and coverings of Section S40. For example a nonzero matrix density with arbitrarily small positive eigenvalues can obey the order with fixed l, u, while an arbitrarily small absolute perturbation can create a new positive direction in its kernel and fail the order. This is precisely the difference relevant to a zero-loss stratum.

Proposition S39.3 (Positive-kernel verification and relative command calibration)**.** *Suppose raw positive input states satisfy $ls_0 \le s \le us_0$ in an ordered vector space and a fixed positive normalized report–audit instrument sends an input to a joint probability law. Its output laws and task matrices satisfy the same order. More generally let $\mathcal{I}_j$ be positive instruments and let a nominal probability vector (λ_j^0) satisfy a desired joint-law identity after averaging. If*

$$(1-\eta)\lambda_j^0 \le \lambda_j \le (1+\eta)\lambda_j^0 \quad \text{for every } j, \qquad 0 \le \eta < 1,$$

then the actual averaged joint law lies between $1-\eta$ and $1+\eta$ times the nominal law, for every input. Successive relative input and instrument bounds multiply their lower factors and upper factors. They do not add an acquisition-amplified defect.

Proof. Positivity sends the two positive differences $s-ls_0$ and us_0-s to positive measures. Integration of $v(z)v(z)^{\mathsf{T}}$ against a positive measure preserves positive matrix order. The command assertion follows by multiplying each nonnegative instrument measure by its nonnegative weight difference and summing. Applying the two inequalities successively multiplies the factors. These arguments do not divide by any report probability. □

Actual-mass-weighted predictive stability. The following is the trace/base-norm contraction principle for positive normalized instruments; in the quantum case compare [31, Corollary 3.40]. We include its proof to retain the unnormalized and zero-mass conventions used here.

Proposition S39.4 (Continuity through likelihood zeros and pure states)**.** *A classical stochastic instrument on signed measures, or a finite-dimensional completely positive normalized quantum instrument, maps normalized inputs s,t to unnormalized post-report states A_y, B_y. Use any common finite dominating measure for these two outputs and let $p_y = u(A_y)$, $q_y = u(B_y)$ be their masses. Then*

$$\int \|A_y - B_y\|_1 dy \le \|s-t\|_1, \tag{S39.6}$$

$$\|p-q\|_{\mathrm{TV}} \le \tfrac{1}{2}\|s-t\|_1, \tag{S39.7}$$

$$\int \min(p_y, q_y) \left\| \frac{A_y}{p_y} - \frac{B_y}{q_y} \right\|_1 dy \le 2\|s-t\|_1. \tag{S39.8}$$

Terms of zero mass are zero. The same unnormalized and report-law inequalities hold for a complete stopped instrument under one common legal policy which terminates almost surely under both inputs.

Proof. Write $s-t = D_+ - D_-$ with positive parts. Positivity, the triangle inequality and preservation of integrated mass give $\int \|\mathcal{I}_y(D_+) - \mathcal{I}_y(D_-)\|_1 dy \le u(D_+) + u(D_-) = \|s-t\|_1$. Taking masses gives (S39.7). If $p \le q$, then

$$p\|A/p - B/q\|_1 \le \|A-B\|_1 + (q-p).$$

Interchanging the roles for $q \le p$, integrating, and using $|p-q| \le \|A-B\|_1$ proves (S39.8). A finite causal continuation with a fixed policy is again a positive normalized instrument. Truncate a stopped continuation at a deterministic horizon, retaining a distinct surviving outcome. Apply the finite inequality, then let the horizon increase. The surviving masses vanish under both inputs by almost-sure termination, and the sum over terminal times converges monotonically. No expected-time assertion is deduced from these norm inequalities. □

The upper finite-state construction never stores a normalized posterior. It uses the acquired report and a finite decoder. Consequently stability of the raw instrument, rather than a globally contracting posterior recursion, suffices for the singular realizations below. More general closed-loop optimal-control claims would require their own strategy comparison and stability; they are not consequences of comparing the same fixed policy.

S40. Acquired geometry at regular, stratified and atomic scales

All masses in this section refer to the actually prepared pushforward $\sigma = (f, \sqrt{w})_{\#}\nu$. They are not volumes on a parameter space. Conditional expectation of any feasible density given $(f, \sqrt{w})$ preserves its mass, density cap and objective for every dictionary; conversely a density on this pushforward lifts by composition. Thus the profile can equivalently be optimized on this actual marked law. The transfer theorem compares operating task measures with this physical acquisition law. In the relative-order realizations it also bounds actual upper-program report measures above and below by fixed multiples of ν. Thus the comparison remains tied to acquired mass, even though perturbed conditional means need not coincide with f.

Proposition S40.1 (A non-power geometric comparison)**.** *For $r \geq 0$ let*

$$\Omega_m(r) = \max_{C \subset H,\ 1 \leq |C| \leq m} \nu\{x : w(x) + \operatorname{dist}(f(x), C)^2 \leq r^2\}.$$

For all $b, m \geq 1$,

$$\sup_{r \geq 0} r^2[1 - b\Omega_m(r)]_+ \leq \mathcal{Q}_m(b) \leq \inf\{r^2 : \Omega_m(r) \geq 1/b\}. \tag{S40.1}$$

In particular a verified estimate $b\Omega_m(r_-) \leq 1/2$ and a verified covering by m distortion sublevel sets of total acquired mass at least $1/b$ at radius r_+ imply $r_-^2/2 \leq \mathcal{Q}_m(b) \leq r_+^2$. The assertion requires no integer local dimension or isotropic geometry.

Proof. For a fixed dictionary and a density cap b, at most $b\Omega_m(r)$ of the selected mass lies in its radius-r sublevel set. The remaining mass costs at least r^2, proving the lower. To see that the maximum defining Ω_m is attained, use an ordered tuple in H^m. For every report, the indicator of the closed sublevel condition is upper semicontinuous in that tuple. Reverse Fatou for these bounded indicators makes its integral upper semicontinuous. Compactness gives a maximizing tuple. If the corresponding sublevel set has mass $v \geq 1/b$, the density $\mathbf{1}_D/v$ is feasible and has loss at most r^2. Taking the infimum over such radii gives the upper. This proof keeps the density cap and actual mass; a cover of the formal reachable set alone provides neither. □

The proposition can be used with anisotropic boxes, different local scales, or non-power small-ball laws, provided both the upper mass estimate and the required acquired cover are verified. It is a finite-scale statement, not a claim that an unspecified anisotropic dimension determines risk. We give explicit closed curves for a useful class and its finite stratifications.

Assume σ is supported on $K \times [0, 1]$, where $K \subset H$ is compact, and for some $d, \alpha > 0$, $0 < r_0 \leq 1$ and $c, C > 0$,

$$cr^d s^\alpha \leq \sigma(B(x, r) \times [0, s]) \leq Cr^d s^\alpha \quad (x \in K,\ 0 < r \leq r_0,\ 0 < s \leq 1). \tag{S40.2}$$

These are global estimates on the stated component, including small reliability mass. Singular Ahlfors-regular location measures of noninteger d are permitted.

Proposition S40.2 (The marked acquired profile)**.** *Under* (S40.2)*, uniformly for $b \geq 1$ and integer $m \geq 1$,*

$$\mathcal{Q}_m(b) \asymp F(b, m) := \max\{b^{-2/\alpha}, (bm)^{-2/(d+\alpha)}\}. \tag{S40.3}$$

The constants depend only on the displayed data and the bounded task carrier. This geometric calculation is inherited from the stopped-experiment and predictable-geometry work retained in [21]*; its proof is included for independence of the central theorem.*

Proof. Cover K by finitely many balls of radius $r_0/2$. Their upper bounds yield $\sigma(K \times [0, s]) \leq C_0 s^\alpha$. Under density cap b, choose $s = c_0 b^{-1/\alpha}$ sufficiently small so that at most half the selected mass has reliability at most s. This gives a risk lower $s^2/2$.

For a dictionary with m centers, its distortion sublevel at radius r lies in the union of m location balls of radius r with reliability at most r. A center outside K whose ball intersects K can, for this

mass bound, be replaced by one intersection point and the radius doubled. For $2r \le r_0$, the union has acquired mass at most $C_1 m r^{d+\alpha}$. Set $r = c_1(bm)^{-1/(d+\alpha)}$ with one sufficiently small constant c_1 so that the restriction and the half-mass bound both hold for all $b, m \ge 1$. Proposition S40.1 gives the other lower in (S40.3).

For the upper, the location marginal has two-sided d-dimensional ball bounds by taking $s = 1$. A maximal $4r$-separated set covers K by $4r$-balls. The marginal upper bound then forces at least $c_2 r^{-d}$ points for small r. Its radius-r balls are disjoint. Choose constants $c_3 > 0$ small and C_2 large such that, for all sufficiently large b, the integer

$$j = \min\{m, \lfloor c_3 b^{d/\alpha} \rfloor\} \ge 1, \qquad r = C_2(bj)^{-1/(d+\alpha)}$$

satisfies $r \le \min(r_0/8, 1)$, $j \le c_2 r^{-d}$, and $cjr^{d+\alpha} \ge 1/b$. The last condition fixes a sufficiently large C_2; after fixing it, a sufficiently small c_3 enforces the packing condition because $jr^d = C_2^d j^{\alpha/(d+\alpha)} b^{-d/(d+\alpha)}$. Increasing the lower threshold for b then enforces the radius condition and $j \ge 1$. Use j separated centers. The disjoint rectangles $B(x_i, r) \times [0, r]$ have union mass at least $1/b$ and distortion to those centers at most $2r^2$. Selecting a unit mass under cap b from that union proves the upper, and substitution of j gives (S40.3). For b in the remaining bounded interval, always accept and use one center. Its bounded loss is controlled by a fixed multiple of $b^{-2/\alpha}$, uniformly over arbitrarily large m. This proves all claimed uniformities. □

Corollary S40.3 (All-scale resource curves under fixed relative calibration)**.** *Suppose* (S40.2) *and the relative hypotheses of Corollary 16.2 hold. On a set of feasible budgets suppose the positive effort ratio κ_+/κ_- and positive width ratio m_-/m_+ are bounded above and below by fixed constants. Then*

$$R_i^{\Theta}(N, M) \asymp F(N/\kappa_-, m_-(M)) \tag{S40.4}$$

throughout that set, including arbitrarily fine resolution. The constants may depend on $l_\pm, u_\pm$, but these factors need not approach one as N, M grow.

More generally the same conclusion holds with absolute defects if $A_N \le A_0 < 1$, $b_+ a_+ \le A_1 < \infty$, $E_N \le c_0 F(b_-, m_-)$ and $b_+ e_+ \le C_0 F(b_-, m_-)$, for a sufficiently small $c_0 > 0$ depending on the fixed geometric and order constants.

Proof. For pure relative order and $l_+ b_+ \ge 1$, both effective budgets and widths in (16.5) differ from b_-, m_- by bounded positive factors. Apply Proposition S40.2. If $1 \le b_+ < 1/l_+$, then b_- lies in a bounded interval. The first-attempt upper is bounded, while the lower and $F(b_-, m_-)$ are bounded below by a fixed positive constant independently of m_-. This treats the boundary regime. With absolute defects, t_N is bounded below by $(1 - A_0)/u_-$ and the corrected upper budget differs by fixed factors. The geometric lower therefore exceeds cF before subtracting E_N. Take $c_0 \le c/2$. The upper defect has the stated order, yielding the result. A fixed positive absolute defect need not satisfy these scale conditions; relative order and absolute proximity are distinct hypotheses. □

Theorem S40.4 (A mixed-dimensional acquired law)**.** *Suppose the acquired marked law decomposes as a finite probability mixture $\sigma = \sum_{j=1}^J \lambda_j \sigma_j$, with $\lambda_j > 0$, $\sum_j \lambda_j = 1$. The components are distributions of the actual task coordinates, not the pushforwards of unspecified hidden conditional means. Each σ_j satisfies* (S40.2) *with its own d_j, α_j and fixed geometric constants. Define the feasible acquired-mass simplex*

$$\mathcal{S}_b = \{(s_j) : s_j \ge 0,\ \sum_j s_j = 1,\ s_j \le b\lambda_j\}.$$

For $m \ge 2J$ and all $b \ge 1$,

$$\mathcal{Q}_m(b) \asymp \min_{s \in \mathcal{S}_b} \sum_{j : s_j > 0} s_j \max\left\{ \left(\frac{b\lambda_j}{s_j}\right)^{-2/\alpha_j}, \left(\frac{b\lambda_j m}{s_j}\right)^{-2/(d_j+\alpha_j)} \right\}. \tag{S40.5}$$

A zero s_j contributes zero. The constants depend on the finite component data and J, not on b, m. Applying Theorem 16.1 to this profile gives the corresponding two-sided causal resource law; no stratum is renormalized without retaining its selected mass s_j.

Proof. For any feasible density and dictionary, set $s_j = \lambda_j \int h \, d\sigma_j$. The masses belong to $\mathcal{S}_b$. When $s_j > 0$, the normalized density $\lambda_j h / s_j$ on the jth component is bounded by $b\lambda_j / s_j$. Its dictionary still has at most m elements. Thus the full objective is at least

$$\sum_{j : s_j > 0} s_j \mathcal{Q}_m^j(b\lambda_j / s_j).$$

This proves the lower after Proposition S40.2.

Conversely fix feasible s and allocate $m_j = \lfloor m/J \rfloor$ centers to every component with positive mass. Their total is at most m and $m_j \geq m/(2J)$. Choose an optimal component density h_j at cap $b\lambda_j / s_j$ and dictionary C_j. The measure $\zeta = \sum_{s_j > 0} s_j h_j \sigma_j$ has mass one and is bounded by $b\sigma$. Use its Radon–Nikodym density with respect to σ and the union of the C_j. Nearest-center distortion to that union is no greater than distortion to the component's own dictionary. This gives the upper $\sum_j s_j \mathcal{Q}_{m_j}^j(b\lambda_j / s_j)$, within a fixed factor of the expression in (S40.5). This is a measurable existence construction, not automatically a finitely computable classifier. If a component label is not observed, this construction still uses only the density of ζ on the actual marked coordinates; no latent component choice is made by the observer. The displayed objective is continuous at $s_j = 0$ by its positive powers of s_j, and the feasible simplex is compact and nonempty. Its minimum is attained. Taking this minimum completes the proof. □

For overlapping abstract components, the Radon–Nikodym selection used in this proof is an ideal Borel construction. A physical finite classifier for it requires the separate effective partition and integration data; it does not follow merely from the mixture decomposition.

The mixture theorem permits different local dimensions and reliability exponents, including noninteger singular dimensions. It is more than a statement that strata may be supplied through an interface: it gives their coupled acquisition–memory optimization and proves both sides with the original masses. Proposition S40.1 remains available when an anisotropic component has no single power-law description.

Proposition S40.5 (An acquired zero-loss atom)**.** *Suppose $\sigma = \lambda \delta_{(x_*, 0)} + (1 - \lambda)\sigma_0$, $0 < \lambda < 1$, where σ_0 has no mass at that atom. Then $\mathcal{Q}_m(b) = 0$ for every $m \geq 1$ if $b\lambda \geq 1$. If $s = 1 - b\lambda > 0$, then*

$$s\mathcal{Q}_m^0\left(\frac{b(1-\lambda)}{s}\right) \leq \mathcal{Q}_m(b) \leq s\mathcal{Q}_{m-1}^0\left(\frac{b(1-\lambda)}{s}\right) \quad (m \geq 2). \tag{S40.6}$$

For $m = 1$ only the displayed lower applies; a valid upper fixes its center at x_* and optimizes the remaining selected mass with that center. If $w > 0$ almost surely on the remainder, then every finite-width profile is positive below threshold. Under pure relative transfer and the additional condition $w > 0$ almost surely on the remainder, zero risk is impossible when $u_b_\lambda < 1$. Without that additional condition, the displayed atom still supplies the sufficient zero-risk bound when $l_+ b_+ \lambda \geq 1$ and the upper width is feasible, but other zero-loss atoms may lower the true threshold. If $w > 0$ almost surely on the remainder, the exact matched threshold is $N = \kappa/\lambda$. The complete finite-width threshold without this restriction is Theorem S35.2.

Proof. Above threshold accept only the atom and decode x_*. Below threshold at least mass s must come from the remainder. Scale any selected remainder measure down to mass s; it remains dominated by $b(1 - \lambda)\sigma_0$ and nonnegative loss cannot increase. Normalize it to obtain the lower in (S40.6). For the upper retain all available atom mass $b\lambda$, allocate one center to it, and optimize the mass-s remainder with $m - 1$ centers. With one center there is no zero-center optimization, which explains the separate statement. If the remainder has $w > 0$ almost surely, any positive selected

probability has positive residual loss; attainment from Lemma 17.1 makes the profile minimum strictly positive. Finally substitute the atom criterion into the two effective caps of (16.5). Relative calibration preserves zero tasks but may change the required acquisition threshold by the stated factors; it does not preserve the numerical threshold without exact matching. □

Nonuniform defects and metastable preparation.

Lemma S40.6 (Killed-potential occupation bound)**.** *Let a nonnegative predictable defect d_i be charged at raw call i, and suppose, for every legal command before commitment, nonnegative integrable potentials obey*

$$d_i + \mathbb{E}_*[V_i \mathbf{1}_{\{i<\tau\}} \mid \mathcal{F}_{i-1}, a_i] \le V_{i-1} + \gamma \mathbb{E}_*[c_i \mid \mathcal{F}_{i-1}, a_i].$$

Here V_0 is integrable, $\gamma \ge 0$, and after commitment the potential is killed. Then every budget-N competitor satisfies $\mathbb{E}_* \sum_{i\le\tau} d_i \le \mathbb{E}_* V_0 + \gamma N$. Potentials may depend on the hidden physical state; this is an analytic certificate, not uncharged observer memory.

Proof. Multiply by the predictable indicator of making call i, integrate and sum to a deterministic horizon n. All interior potentials cancel. The remaining final potential is nonnegative and may be dropped. Bound the expected partial cost by N, then use monotone convergence for the nonnegative defect sum. Randomized commands are handled by integrating the commandwise inequality. No optional stopping with unverified uniform integrability is needed. □

Corollary S40.7 (Relative order with rare defective modes)**.** *Suppose relative order holds on good raw histories, while visible conditional additive defects are bounded by the conditional expectations of $a_b \mathbf{1}_{\{D_{i-1}=1\}}$ and $e_b \mathbf{1}_{\{D_{i-1}=1\}}$ for a hidden mode $D \in \{0,1\}$. At every legal call, given the physical past, assume*

$$\mathbb{P}_*(D_i = 1 \mid D_{i-1} = 0, \cdot) \le u, \qquad \mathbb{P}_*(D_i = 0 \mid D_{i-1} = 1, \cdot) \ge v > 0.$$

Let $\pi_0 = \mathbb{P}_*(D_0 = 1)$ be the actual prepared mass and $c_i \ge c_* > 0$. The central theorem applies with

$$A_N = \frac{a_b}{v}\pi_0 + \frac{a_b u}{v c_*} N, \qquad E_N = \frac{e_b}{v}\pi_0 + \frac{e_b u}{v c_*} N.$$

These quantities may replace rowwise worst-case defects even if bad-mode errors are not small and the residence time $1/v$ is large.

Proof. Use $V(D) = (a_b/v)\mathbf{1}_{\{D=1\}}$. In a good mode its expected increase is at most $a_b u/v$; in a bad mode its expected decrease is at least a_b and pays for the defect. Killing at commitment only improves the inequality. Since each raw call costs at least c_*, Lemma S40.6 gives the first bound. Counting only productive calls makes it smaller. Replace a_b by e_b for the task bound and use conditional expectation to pass from physical modes to the visible-history defects. This retains actual preparation and selection-biased residence, rather than substituting an invariant-mode distribution. □

S40.1. **An explicit six-regime allocation diagram.** Here we solve the variational comparator in Theorem S40.4, rather than claim that its implicit comparison constants are one. Take two components with $(d_1, \alpha_1, \lambda_1) = (1, 1, 1/2)$ and $(d_2, \alpha_2, \lambda_2) = (2, 2, 1/2)$, and total dictionary width $m = 4$. If s is the selected mass from the first component, the comparator is

$$J_b(s) = \max\left\{\frac{4s^3}{b^2}, \frac{s^2}{2b}\right\} + \max\left\{\frac{2(1-s)^2}{b}, \frac{(1-s)^{3/2}}{\sqrt{2b}}\right\},$$

with $\max(0, 1-b/2) \le s \le \min(1, b/2)$ and $b \ge 1$. Its minimum describes the risk up to the inherited component-dependent constants. It is not an exact formula for every acquired law having these exponents.

These component hypotheses can arise from an actual bounded preparation: take the first location uniform on a unit segment in $\mathbb{R}^2$ and the second uniform on a unit square, with reliability variables having distribution functions t and t^2, respectively. With a fresh fair sign E, audit $Z = Y + TEe_1$ and report (Y, T). Then $f = Y$ and $w = T^2$ in a common compact convex carrier. The mixture has the stated acquired dimensions and exponents. The diagram still concerns the theorem's comparator rather than replacing its comparison constants by exact ones.

Proposition S40.8 (Exact selected-mass phases for the comparator)**.** *Put*

$$b_1 = 6 - 2\sqrt{6}, \quad b_2 = (16\sqrt{6} - 24)/5, \quad b_3 = 8(\sqrt{2} - 1), \quad b_4 = 32/5, \quad b_5 = 36/5.$$

The minimizer is unique and continuous in b. It is

$$s_*(b) = \begin{cases} 1 - b/2, & 1 \le b \le b_1, \\ (\sqrt{b^2 + 12b} - b)/6, & b_1 \le b \le b_2, \\ 1 - b/8, & b_2 \le b \le b_3, \\ \text{the unique } s \in (0,1) \text{ with } 128s^4 = b^3(1-s), & b_3 \le b \le b_4, \\ b/8, & b_4 \le b \le b_5, \\ (\sqrt{81b^2 + 288b} - 9b)/16, & b \ge b_5. \end{cases} \tag{S40.7}$$

In particular there are two genuine intervals where the minimizer lies at a nonsmooth crossover, not just isolated changes between differentiable stationary formulas. The initially decreasing allocation to the first component eventually increases to one.

Proof. Each summand is strictly convex on its nonnegative mass interval, and hence J_b is strictly convex. Both components change from their location-limited term to their reliability-limited term when their own selected mass crosses $b/8$. On a differentiable interval the minimizing equation is equality of their marginal costs.

When both are reliability-limited, this equation is $12s^2/b^2 = 4(1-s)/b$, or $3s^2 = b(1-s)$. Its solution first enters the feasible interval at $s = 1 - b/2$, giving b_1, and reaches the second component's crossover $1 - s = b/8$ at b_2. At that crossover the second marginal-cost interval is $[3/8, 1/2]$. The first marginal cost $12(1 - b/8)^2/b^2$ decreases through that interval as b runs from b_2 to b_3. Thus $s = 1 - b/8$ is the minimizer throughout that interval.

With the first component reliability-limited and the second location-limited, marginal equality is $12s^2/b^2 = 3\sqrt{1-s}/(2\sqrt{2b})$, equivalent to the displayed quartic. Its left side is strictly increasing in s and its right side strictly decreasing. The root reaches $s = b/8$ at b_4. At this first-component crossover its own marginal-cost interval is $[1/8, 3/16]$. The second marginal cost $3\sqrt{1 - b/8}/(2\sqrt{2b})$ decreases from $3/16$ to $1/8$ between b_4 and b_5, giving the second nonsmooth interval. Thereafter both terms are location-limited; equality becomes $8s^2 = 9b(1-s)$, with the final displayed root. Substitution verifies the active conditions and the original mass caps in each interval. Strict convexity proves uniqueness; matching endpoint values proves continuity. The last equation implies $1 - s_*(b) \sim 8/(9b)$. □

S41. Classical and quantum realizations from raw kernels

The two realizations below are verified after the general theorem. Recurrent witness preparation, when present, belongs to the charged productive primitive; it is not a reset made freely available to other models. Neither supplies a premise used to prove it. Their conditional physical evolution between calls may be noncontracting, and their acquired laws may be singular. Both tasks score the post-command audit. A target which must remain unchanged is a different experiment; the nonintervening directional result is retained in the companion.

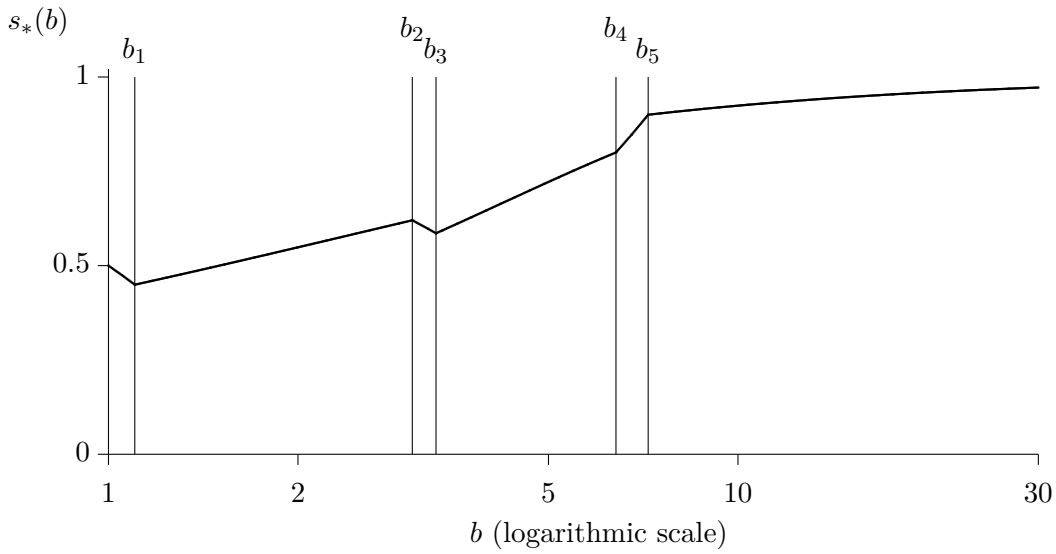


FIGURE 1. The exact minimizer of the displayed comparator, not a fitted risk curve. Vertical lines mark the five algebraic transition budgets. The two crossover intervals are part of the optimum; smoothing a maximum before differentiation would miss them. The plotted values are checked against independent bounded scalar minimization.

A common paid physical interface. The gate port G costs one per call and has success probability at least $p_0 > 0$, conditional on every physical and visible past. A success enables exactly one productive call at port A, also of cost one. Commitment is forbidden at G and allowed after a productive report. Rejection at A returns the protocol to G; it need not reset the hidden state. The two ports have disjoint physical command domains and no port tag is a free intercall input to an autonomous observer. Suppose the actual lower witness has gate success probability at most $p_1 \le 1$. Then

$$\kappa_- = 1 + 1/p_1, \qquad \kappa_+ = 1 + 1/p_0, \qquad B_0 = B = 2. \tag{S41.1}$$

Let G_τ and A_τ count the gate and productive calls through commitment. Every productive call consumes a gate success. A terminating path has no unconsumed success. Predictable counting gives $\mathbb{E}_* A_\tau \le p_1 \mathbb{E}_* G_\tau$, so $\mathbb{E}_* C_\tau \ge \kappa_- \mathbb{E}_* A_\tau$. The probability of j failures after any gate start is at most $(1-p_0)^j$, giving conditional expected gate count at most $1/p_0$ and whole-attempt cost at most κ_+. Both mandatory ports are visited before commitment, and Lemma 17.4 gives two live roles. Consequently the construction requires at least two event labels or three state labels when it has one output. If the witness has constant success $p_0 = p_1$, then $\mathbb{E}_* C_\tau = \kappa \mathbb{E}_* A_\tau$ with $\kappa = 1 + 1/p_0$. In that matched case budget feasibility is exactly $N \ge \kappa$ with the respective state thresholds. Even when $p_0 = 1$, the cost is two, not one.

Theorem S41.1 (A relatively calibrated classical realization)**.** *Let a hidden symbol E take values in a finite set of size q. A finite group $\mathcal{G}$ acts transitively on that set. The allowed productive commands apply $g \in \mathcal{G}$ and then, within the same call, a fixed positive normalized joint report–audit kernel $J_{gE}(dy, dz)$ with bounded audit in H. There is no unknown intervening operation between that command and its audit instrument. Evolution between port interactions is otherwise arbitrary and causal.*

The genuinely prepared comparison has uniform input and joint law $K_0 = q^{-1} \sum_e J_e$. Its full report is its mark. Suppose the operating family contains a single witness whose conditional input

weights, after every reached pre-command history, satisfy

$$(1-\eta_r)/q \le \mathbb{P}_*(E=e \mid h) \le (1+\eta_r)/q, \qquad 0 \le \eta_r < 1.$$

Suppose one parameter-independent implementation chooses its command independently of the hidden current symbol conditional on its visible past, with conditional weights

$$(1-\eta_u)/|\mathcal{G}| \le \lambda_g \le (1+\eta_u)/|\mathcal{G}|, \qquad 0 \le \eta_u < 1,$$

uniformly over all models and surviving histories. Use the gate interface above. Then Theorem 16.1 holds with identity exports, zero additive defects, and

$$l_- = 1-\eta_r, \quad u_- = 1+\eta_r, \qquad l_+ = 1-\eta_u, \quad u_+ = 1+\eta_u. \tag{S41.2}$$

The comparison need not be an operating member. Fixed nonzero $\eta_r, \eta_u < 1$ preserve the all-scale resource curve of Corollary S40.3 whenever its acquired geometry holds.

Proof. A chosen permutation preserves the two componentwise bounds on the witness weights. Multiplying by each positive J_e and summing gives the full command-conditioned joint-law order $(1-\eta_r)K_0 \le P_{h,g} \le (1+\eta_r)K_0$. Take $P^0_{h,g} = K_0$ and the identity mark. Conditional moments given its full report are exactly (2.1), so all lower certificates follow from Proposition S39.1. This holds for every chosen command and every competing policy, not merely for a randomizing policy.

For any fixed incoming symbol, uniform averaging over a transitive group gives the uniform distribution of its image: each image has the same number of preimages in the group. Averaging also over any incoming distribution preserves that fact. The actual componentwise command bounds and positivity therefore put the full joint upper-program law between $1-\eta_u$ and $1+\eta_u$ times K_0. Independence of the command from the hidden current input is needed for this conditional averaging; it is stated as a physical implementation property, not inferred from marginal coin frequencies. The upper accepts on the resulting report, not on a discarded group choice. Exports are the identity because the audit is the post-command audit. Group selection is within-call work, and only the gate or productive routing label or final decoder value persists. The gate proof establishes the same-witness cost lower, common cost upper and both live roles. The general theorem now applies. □

For a concrete acquired geometry take $q = 2$, with identity and sign reversal. Independently draw (Y, T) from a fixed law on $K \times [0, 1]$, report these coordinates, and audit

$$Z = Y + TEe_1 \tag{S41.3}$$

in a sufficiently large compact convex carrier. The uniform preparation gives $f = Y, w = T^2$. Choose a d-regular location measure and $\mathbb{P}(T \le s) = s^\alpha$ to verify (S40.2) directly; choose a finite mixture of such laws to verify Theorem S40.4; add a physically prepared point mass at $(x_*, 0)$ to obtain Proposition S40.5. A biased fresh witness and imperfect command probabilities satisfy (S41.2) with fixed nonzero constants. On $T = 0$ the task is exactly deterministic even under these errors. On $T > 0$ the two-point audit moment matrix has rank two, while at $T = 0$ it has rank one. This is an actual changing-rank task measure, not a formal limiting parameter set.

The command group and its bit-level sampler are part of the program description and workspace account. A high-cardinality group does not provide free randomization or a free persistent seed. In the binary realization a single fresh sign coin suffices within a productive call. Conditional posterior continuity of the classical raw kernels is governed by Proposition S39.4, including likelihood zeros.

Theorem S41.2 (A relatively calibrated noncommuting quantum realization)**.** *Let the productive physical system have finite dimension $d \ge 2$. Its available commands include the d^2 Weyl conjugations, followed within the same productive call by a fixed positive normalized quantum report–audit instrument $\mathcal{J}$, with classical joint law $\mathcal{J}(\rho)(dy, dz)$ and $z \in H$. The physical task is again the post-command audit. Put $\omega = I/d$ and $K_0 = \mathcal{J}(\omega)$.*

Suppose one actual witness obeys, conditionally before every selected productive command,

$$(1-\eta_r)\omega \preccurlyeq \rho \preccurlyeq (1+\eta_r)\omega, \qquad 0 \le \eta_r < 1.$$

Suppose the common implementation uses current Weyl weights between $(1-\eta_u)/d^2$ and $(1+\eta_u)/d^2$, independently of the current hidden input conditional on the past, with $0 \le \eta_u < 1$. With the same gate certificates, all conclusions of Theorem S41.1 hold with the constants (S41.2)*. The operating witness need not be exactly maximally mixed. Incoming states for the upper may be pure, and post-report ranks may change on positive acquired mass. No common eigenbasis or normalized-filter contraction is assumed.*

Proof. Let $W_{p,q}$ be the shift by q followed by the phase $\exp(2\pi ipj/d)$ on the jth basis vector. On a matrix unit $|j\rangle\langle k|$, averaging the phase kills off-diagonal terms; averaging shifts makes the diagonal uniform. Hence

$$d^{-2}\sum_{p,q} W_{p,q}\rho W_{p,q}^* = I/d$$

for every density matrix. This elementary randomization identity is standard [6]; we use its complete positive-instrument consequence, not a new state-randomization claim.

Each chosen conjugation fixes ω and preserves the witness Loewner order. Positivity of the full report–audit instrument therefore gives lower joint-law order relative to K_0 with factors $1 \pm \eta_r$. Positivity means that the image of every positive operator is a positive classical measure, including both report and future audit. It consequently compares the full observable report and task, not just the report marginal.

For the upper, the componentwise weight bounds give

$$(1-\eta_u)\omega \preccurlyeq \sum_{p,q} \lambda_{p,q} W_{p,q}\rho W_{p,q}^* \preccurlyeq (1+\eta_u)\omega$$

for every incoming state, including pure states. Apply the same positive instrument. Proposition S39.1 supplies the task certificates, and identity export, counted live roles and the gate costs finish the verification. The lower witness has a relative input bound around ω; we do not describe that hypothesis as absence of an input-rank condition. The absence of a floor concerns the post-report states and task strata, where the following example has genuine rank changes. □

Here is a raw qubit instrument giving the same acquired geometry as (S41.3) but noncommuting predictive states. Draw (y,t) with any stated law ν_0 on $K \times [0,1]$, choose an equatorial unit vector $n = (\cos\phi, \sin\phi, 0)$ and $0 \le \gamma(y,t) \le 1$, and set

$$E_v = (I + v\gamma n\cdot\sigma)/2, \qquad L_v = \sqrt{E_v}, \qquad v = \pm 1.$$

The report is (y,t,v) and the unnormalized post-report state is $L_v\rho L_v$. At commitment measure σ_z, obtain $e = \pm 1$, and score $Z = y + tee_1$. Under $\omega = I/2$, each v has mass $1/2$ and the normalized posterior is E_v. Its z component is zero, so the future audit sign is fair. Thus the reference report law is ν_0 times a fair v, and $f = y, w = t^2$, derived directly from the instrument.

The eigenvalues of E_v are $(1 \pm \gamma)/2$. Taking $\gamma = 1$ on a positive-mass set and $\gamma < 1$ on another gives actual pure and full-rank posterior strata. Varying ϕ produces noncommuting effects. For a continuous compact realization one may take $\gamma(t) = 1$ for $t \le 1/2$ and $\gamma(t) = 3/2 - t$ for $t \ge 1/2$, and a continuous nonconstant angle. The two-sided task order remains uniform across these strata; Proposition S39.4 supplies actual-mass-weighted predictive-update and report continuity. A point mass at $t = 0$ also gives a changing-rank task matrix and an exact zero-loss acquired component.

The qubit is part of the physical experiment, not a quantum memory register owned by the finite classical observer. The pulse and instrument form one productive call in this convention; their duration, finite sampling precision and arithmetic are separately charged. An interface requiring a separate pulse call must add its cost and any intervening routing state. Both realizations retain a strictly positive paid gate floor p_0.

Predictive realization rather than an assumed parameter manifold. For the finite classical kernels and finite-dimensional quantum instruments whose entries depend continuously on compact report coordinates, the primitive continuity criterion in Proposition S43.2 supplies compact positive-evidence history patches. Their executable predictive quotient is therefore Borel by Theorem S43.1. Fixed continuous causal evolutions between calls are included and need not contract. For a completely arbitrary merely Borel hidden evolution, the risk theorem still uses the stated raw certificates, but we do not assert the compact-exhaustion quotient theorem without its regularity hypotheses. The unknown model's family of posteriors is never stored by the common upper program.

S42. Finite computation, calibration and causal composition

The inequalities proved so far concern Borel existence with a declared physical acquisition program. A finite source description and an executable precision certificate are additional mathematical data. We state exactly where they enter.

Proposition S42.1 (Numerical implementation at a restored physical budget)**.** *Suppose a chosen ideal density and dictionary at cap β have a finite implementation whose acceptance probability, as a function of the current virtual report, differs pointwise from h/β by at most $\eta \geq 0$. Suppose its actual exported decoder values differ from the ideal exported values by at most δ_{num}, with no larger output width. These finite operations use the same legal acquisition program and counted live roles. Put $b = N/\kappa_+$ and*

$$\beta = \frac{l_+ b}{1 + b(a_+ + \eta)} \geq 1.$$

If the chosen ideal pair has objective at most $\mathcal{Q}_m(\beta) + \xi$, the implemented program terminates almost surely at actual expected core cost at most N and risk at most

$$\left[\sqrt{\frac{u_+(1 + b(a_+ + \eta))}{l_+}(\mathcal{Q}_m(\beta) + \xi) + b(e_+ + L\eta)} + \delta_{\mathrm{num}}\right]^2. \tag{S42.1}$$

Here ξ is a certified design/integration suboptimality, not a local optimizer's unverified gap. A uniform bound T_{attempt} on conditional expected physical and arithmetic time of a whole attempt yields expected time at most bT_{attempt}. Program bits and scratch are separately those of the certified implementation.

Proof. Acceptance rounding reduces the conditional hazard by at most η. The chosen cap gives $l_+/\beta - a_+ - \eta = 1/b$. The same conditional-survival summation as in the main theorem yields at most b attempts and cost at most N. Before perturbing decoder values, the one-attempt unnormalized loss changes by at most $L\eta$ under the actual joint law, because loss is bounded by L. Its other contribution is at most $u_+(\mathcal{Q}_m(\beta) + \xi)/\beta + e_+$. Summation gives the expression under the square root. At the actual committed report, the implemented and ideal exported values are within δ_{num}; Minkowski's inequality in the actual stopped L^2 law adds this distance to root-mean-square loss. It does not multiply the decoder error by the waiting time. The time claim is a separate predictable sum of the stated whole-attempt bound. □

In the preceding display T_{attempt} denotes a declared time quantity, not a free clock input to the observer. An implementation based on m fixed centers in a k-dimensional bounded box, p-bit coordinate certificates and sequential squared-distance comparisons has sufficient read-only data $O(mkp)$ bits, within-call scratch $O(kp + \log m)$ bits and $O(mkp^2)$ bit operations per report, in addition to its primitive measurement and coordinate-evaluation costs. These are upper accounts, not optimal computational lower bounds. General Borel classifiers need not have such certificates. The complete dyadic implementation in the companion gives explicit rational thresholds, separate read-only/scratch counts and an independently proved profile lower.

Corollary S42.2 (Finite-description matching upper under effective primitives)**.** *In the pure relative regular class, suppose additionally that the actual upper-program report laws are bounded above by $\bar{u}\nu$ on every surviving history. Assume effective compactness of the dictionary carrier and certified finite approximations of exported readouts inside H. Assume the calibrated report law has an effective sequence of finite physical partitions which approximates bounded finite decision vectors in $L^1(\nu)$, with certified integration of their masses and bounded task losses. Primitive classifiers and their finite costs are supplied as part of this effective interface. Then a finite-description upper with the same live and terminal role counts has risk at most a fixed multiple of $F(b,m)$ at actual expected cost at most $\kappa_+ b$, for all feasible b, m. Every such construction has finite, separately charged program, scratch and execution accounts. No polynomial-time discovery or matching bit-complexity lower is asserted.*

Proof. For $b \geq 4/l_+$ choose the ideal cap $\beta = l_+ b/2$. Its conditional acceptance is at least $2/b$. By Proposition S40.2, its profile is at most a fixed multiple of $F(b,m)$. Represent its finite decision as functions $q_c(x)$, one for each dictionary value, whose sum is the acceptance probability. Approximate the vector on one finite physical partition by rational vectors in the subprobability simplex. The assumed effective approximation makes the sum of its $L^1(\nu)$ coordinate errors arbitrarily small, say at most γ. Domination of the actual report law bounds its actual conditional decision discrepancy by $\bar{u}\gamma$ on every history, and its unnormalized loss discrepancy by $L\bar{u}\gamma$. Approximate the actual finite exported values inside H as well.

Choose γ so small that $\bar{u}\gamma < 1/(2b)$ and $bL\bar{u}\gamma \leq F(b,m)$, and choose export error at most $\sqrt{F(b,m)}$. The conditional acceptance still exceeds $1/b$, so the expected cost is at most $\kappa_+ b$. Summing loss and applying the stopped L^2 readout estimate gives a fixed multiple of $F(b,m)$. For $1 \leq b < 4/l_+$, accept the first attempt and use one finitely described value in H. Bounded loss is a fixed multiple of the strictly positive lower bound for $b^{-2/\alpha}$ on this interval, uniformly in m.

The existence argument can be effected by finite enumeration using the supplied integration certificates and effective dictionary nets: the above choices leave strict mass and loss margins, so a candidate with certified acceptable intervals is eventually found. This need not be an efficient search. Current partition indices and rational randomization are charged within-call work; they do not create new intercall roles. All chosen descriptions and precision parameters are finite. Exact zero risk on an atom requires an exactly implementable atom classifier and readout; otherwise Proposition S42.1, not an unqualified finite-code zero claim, governs that component. □

Proposition S42.3 (Resource-certified causal morphisms)**.** *A causal simulator must specify its parameter-independent strategy lift, legal command map, report transformation, internal state, clock and terminal task readout. Suppose a target program has ℓ live and t terminal roles. A compatible simulator has at most K continuing internal states, including every persistent phase, coin and clock, and at most E terminal export classes sufficient for its final decoder. Then a source implementation uses at most $\ell K + tE$ state-stopped labels or $\max(\ell K, tE)$ event-stopped labels. Without a terminal sufficient-class certificate, retaining the whole simulator state gives the safe product bound.*

If the complete stopped scored transcript, including mapped stopping time and premature commitment, is within total variation ε of the target law uniformly over the declared models and policies, then the bounded common-task risk changes by at most $L\varepsilon$. Two compatible simulators multiply their continuing and terminal resource factors and add their deficiencies. Raw-call count and physical-time transfer require separate actual cost or killed-potential certificates. Reverse risk inequalities require a reverse simulator or an actual lower witness; terminal-law proximity alone does not provide them.

Proof. Before commitment retain the pair of target live role and continuing simulator state. After commitment retain the pair of target output role and sufficient export class. Compatibility supplies legal transitions and commands without an uncharged port or phase. Lemma 17.4 gives the two packing counts. Serial composition retains triples; rebracketing yields the products K_1K_2 and E_1E_2. For a loss in $[0, L]$, integration of its superlevel probabilities gives $|\mathbb{E}_P \ell - \mathbb{E}_Q \ell| \leq L\|P - Q\|_{\mathrm{TV}}$. Total

variation contracts under a common Markov kernel, so the triangle inequality adds complete-law deficiencies. Costs are generally unbounded functions of stopped paths and are not controlled by this bounded-loss argument; sum their certified increments or use Lemma S40.6 instead. □

Combining Propositions S42.1 and S42.3 adds $L\varepsilon$ to (S42.1) while charging the simulator's states and actual cost map. This composes approximation and simulation in one task metric. A bound on an allowed error is not, without attained alternatives, an unavoidable lower bound. The following construction supplies such alternatives on a single task.

Theorem S42.4 (A same-task resource–calibration–deficiency law)**.** *Take any core class satisfying Theorem 16.1 with identity exports and B common live roles. Add one compulsory paid observation of an independent fair sign E_0, erased with probability 2ε, $0 \le \varepsilon \le 1/2$, and a final paid audit of*

$$(Z, E_0, s\delta_{\mathrm{cal}}), \qquad s \in \{-1, 1\}.$$

The fixed unknown sign s affects no pre-audit law. The core, including its costs, has no access to E_0 beyond the initial visible tag. Use squared Euclidean loss on this product task, and charge these two additional calls. The core lower from (16.2)*, evaluated with budget $N-2$ and safe projected dictionary width M, plus $2\varepsilon + \delta_{\mathrm{cal}}^2$, is a lower bound for either stopping interface.*

Let $U(n, m)$ denote the upper expression in (16.4) *for core budget n and output width m, or its first-attempt alternative. For state stopping choose any $m \ge 1$ with $1 + 3B + 3m \le M$; for event stopping choose any $m \ge 1$ with $\max(1 + 3B, 3m) \le M$. Then an upper is*

$$U(N-2, m) + 2\varepsilon + \delta_{\mathrm{cal}}^2. \tag{S42.2}$$

In the relative-order regular class, for M large enough that these sufficient widths are a fixed fraction of M, this yields

$$R_{\mathrm{joint}}(N, M, \delta_{\mathrm{cal}}, \varepsilon) \asymp F((N-2)/\kappa_-, M) + \delta_{\mathrm{cal}}^2 + \varepsilon. \tag{S42.3}$$

The early sign channel has causal deficiency exactly ε relative to the perfect-sign channel when the virtual sign must be supplied before any core call. The state counts are sufficient, not minimal for the enlarged experiment.

Proof. For the upper use one initial role, then three tagged copies of the core live roles for observed $+$, observed $-$ and erasure. Use three copies of its output dictionary. Decode the known sign or zero on erasure, and decode zero for the inaccessible last coordinate. This uses $1 + 3B$ live and $3m$ terminal roles, giving the stated widths by Lemma 17.4. The common core upper is uniform after each tag. The other losses are exactly 2ε and δ_{cal}^2. The two compulsory calls give core budget $N-2$.

For the lower use the same actual core witness as in the central theorem. Conditioning on the initial tag does not change its full-report certificates; the initial sign and observer coins are additional independent randomization. The projected core decoder has at most M values, so the actual-history proof gives its core lower. On erasure, induction over the core kernels and the stopping decisions leaves E_0 fair conditional on all available pre-audit information; its conditional squared prediction loss is at least one. Average the two indistinguishable fixed choices $s = \pm 1$. For any last coordinate r, their mean loss is $\delta_{\mathrm{cal}}^2 + r^2 \ge \delta_{\mathrm{cal}}^2$. The core and erased-sign lower bounds hold under both of these identical pre-audit laws. Thus the three contributions hold simultaneously in one worst-case total task, rather than under incompatible separately optimized models. Regular geometry and fixed width ratios prove (S42.3).

On an erased branch every immediate virtual sign is independent of the true fair sign. The event that they agree has probability at most $1/2$, giving complete scored-law deficiency at least ε. Reporting the known sign otherwise and an independent fair sign on erasure gives a coupling with mismatch probability exactly ε. Run the same future core and audit in this coupling. The early-delivery requirement prevents waiting for the final audit to recover the sign, and proves the opposite inequality. □

A finite calibration acquisition with guarded stopping. The reference and task-order certificates above are not inferred automatically from finite data. Here is one precise finite-sample certificate and its cost when the unknown quantities are the fixed probabilities of a physical command sampler. It also explains why high confidence alone cannot certify almost-sure stopping.

Proposition S42.5 (Finite command calibration and a counted guard)**.** *Suppose a physical sampler has a fixed unknown distribution $(\lambda_j)_{j=1}^J$ throughout calibration and deployment, its draws are independent, and deployment draws are independent of the current hidden input. The known target probabilities are $1/J$. Observe n independent calibration draws at cost c_{cal} each, and let $\widehat{\lambda}_j$ be their frequencies. Fix $0 < \zeta < 1$, and choose a certified rational bound*

$$r_n \geq \sqrt{\frac{\log(2J/\zeta)}{2n}}.$$

Declare a relative certificate with parameter $\eta < 1$ only when all intervals $[\widehat{\lambda}_j - r_n, \widehat{\lambda}_j + r_n]$ lie in $[(1-\eta)/J, (1+\eta)/J]$. The probability of a falsely declared certificate is at most ζ.

On certificate failure, commit by a bounded-loss first-attempt fallback. On certificate success, deploy the corresponding upper of Theorem 16.1, with design hazard $1/b$, but force commitment with a fixed decoder after at most $T \geq 1$ productive attempts. The whole-attempt cost bound κ_+ is required whether or not the certificate is true. If the true sampler lies in a tighter relative band $\eta_0 < \eta$ and $2r_n \leq (\eta - \eta_0)/J$, then the entire guarded procedure commits almost surely and has

$$\mathbb{E}C_{\mathrm{total}} \leq nc_{\mathrm{cal}} + \kappa_+(b + \zeta T), \tag{S42.4}$$

$$R_{\mathrm{total}} \leq U + L\big(2\zeta + e^{-T/b}\big), \tag{S42.5}$$

where U is the certified core upper. The calibration phase can be implemented with at most $(n+1)^J$ count-vector labels. Deployment uses a counted attempt index, multiplying its live-role bound by at most T; fallback and terminal roles are additional explicit roles. These resources may not be omitted from an autonomous budget.

Proof. For a Bernoulli variable X, the log moment-generating function of $X - \mathbb{E}X$ has second derivative at most $1/4$ under every exponential tilt, and value and derivative zero at the origin. Twice integrating gives $\log \mathbb{E}e^{t(X-\mathbb{E}X)} \leq t^2/8$. Independence, the exponential Markov inequality and optimization in t imply $\mathbb{P}(|\widehat{\lambda}_j - \lambda_j| > r) \leq 2e^{-2nr^2}$. A union bound proves that all confidence intervals contain the true probabilities except on an event of probability at most ζ. Hence a false certificate has probability at most ζ. If the tighter-band condition holds, this same simultaneous event puts each empirical interval within the declared wider band, so rejection of a true certificate also has probability at most ζ.

On a true certificate, the conditional hazard proof bounds survival through T attempts by $(1 - 1/b)^T \leq e^{-T/b}$. The loss contributed by earlier acceptances is at most U; the forced last commitment adds at most $Le^{-T/b}$. Expected attempts on this branch are at most b. On a certificate failure the first-attempt fallback costs at most κ_+ and loses at most L. On a falsely accepted certificate the deterministic guard uses at most T attempts and has loss at most L, even if the unguarded acceptance set has zero actual mass. Every attempt terminates almost surely by its finite whole-attempt cost bound. Combining these cases gives (S42.4) and (S42.5), with the non-false cases bounded by $b \geq 1$ attempts and the false-acceptance case by ζT additional attempts.

A count vector stores every calibration total, including the number of draws already taken, and has at most $(n+1)^J$ possible values. At total n, the threshold comparison may be performed in charged scratch using a fixed certified finite arithmetic implementation; the resulting branch choice is encoded in the selected live role. The counts can then be discarded because the deployed dictionary is fixed for the nominal law and declared band, not fitted to the realized frequencies. The deployment guard requires its own retained index with T values. This gives the asserted conservative finite-state

accounts. Without the tighter-band assumption one still has the false-certificate probability bound and guarded termination, but no small rejection-risk bound; we do not omit that distinction. □

This result controls one genuinely acquired calibration quantity, not arbitrary unknown-kernel learning. It counts pilot observations, a physical fallback, the stopping guard and its memory. It prevents a rare incorrect calibration certificate from being hidden inside an unjustified robust almost-sure stopping assertion.

S43. A predictive quotient constructed from positive-evidence histories

The squared-task coordinates in (2.1) need not separate all future experiments. We therefore give the predictive realization independently of those coordinates, for a fixed prepared law with its declared legal test class. The theorem is a sufficient realization result and a detailed application of compact selection; predictive minimality itself has classical antecedents [4, 5].

Let H be the standard Borel space of finite visible histories on which specified conditional versions are defined, embedded in a metrizable history carrier. Reports and resource records are included in the history. Actions belong to a finite set, with Borel legal domains. Histories with different legal continuations must be separated. Let $q_j(h) \in [0,1]$ be a countable determining list of conditional response coordinates for all finite open-loop future report words and bounded tests. Include legal-domain indicators and the coordinates necessary to determine report and cost-record laws. The list is closed in the following determining sense: the one-call prefixes of these future tests, localized by a countable measure-determining family of current-report functions, are determined by the list. This closure is a property of the declared executable tests, not a finite-moment assertion. Set $q(h) = (q_j(h))_{j\ge1}$ in $[0,1]^{\mathbb{N}}$.

The realization theorem in this appendix concerns one fixed prepared law, not an unknown-model quotient oracle. Its compact-fiber selection step is a concrete instance of classical measurable selection [32]; the explicit construction below checks its needed measurable domains instead of assuming a selector.

Theorem S43.1 (Compact-exhausted executable predictive realization)**.** *Suppose $\mathsf{H} = \bigcup_{n\ge1} C_n$ with C_n compact in its metrizable carrier and every q_j continuous on each C_n. Assume the specified response coordinates are the consistent conditional laws of the raw causal kernels and satisfy the determining and continuation conditions above. Then the actual image $S = q(\mathsf{H})$ is Borel and has a Borel section $s : S \to \mathsf{H}$. Equality of q is exactly equality of all declared finite executable future laws, including legal domains. The report and next-quotient law factor through q as a Borel kernel. This factorization is modulo the common current-report null sets of each compared pair of histories; it is not a pointwise claim on impossible reports.*

If another Borel statistic T determines the response coordinates by Borel maps $q_j = g_j \circ T$, then $q = g \circ T$ measurably, where $g = (g_j)_{j\ge1}$ on the realized range. Thus q is minimal for these executable tests. Equality extends to bounded stopped tests which terminate almost surely under both compared histories, by finite truncation. Unbounded costs and infinite-horizon limits require their separate integrability or convergence hypotheses.

Proof. The countable product is compact metrizable, and coordinate continuity makes q continuous on every C_n. Thus $q(C_n)$ is compact and $S = \bigcup_n q(C_n)$ is Borel. A section can be constructed rather than assumed. Given $x \in S$, first choose the least n for which $x \in q(C_n)$, a Borel decision. For this compact C_n, choose finite closed covers at each scale, refining each previously chosen compact cell, with diameters tending to zero. Empty cells are discarded. At each level choose the first child compact cell D for which $x \in q(D)$. Such a child exists because its parent covers the nonempty fiber; the test is Borel because $q(D)$ is compact. The chosen cells are nested compact sets with diameters tending to zero and each intersects the fiber. Choose a fixed representative in each possible cell. Their sequence is Borel in x and converges to a unique point of C_n. Continuity of q on C_n shows

that this point maps to x. The countable piecewise construction in the least n gives the required Borel section.

Suppose $q(h) = q(h')$. The legal-domain coordinates give the same permitted next commands. The determining current-report coordinates give the same next-report law R_a for a common legal command a. For a future response coordinate q_j and a member g of the determining current-report function family, continuation consistency gives a conditional joint-test value

$$\int g(y)q_j(hay)R_a(dy).$$

The prefix closure and equality of $q(h), q(h')$ imply equality of this integral with the expression using $h'ay$. Uniqueness of finite measures on a determining family implies $q_j(hay) = q_j(h'ay)$ for R_a-almost every y. Intersecting these sets over the countable coordinates gives $q(hay) = q(h'ay)$ almost surely. Concatenations outside the positive-evidence history domain may be completed arbitrarily, because they have zero report mass from the histories in question.

Define the quotient report/update kernel by running the raw row from $s(x)$ and pushing the continued history through q. The section, the raw kernel and concatenation map are Borel, so this is a Borel kernel. The preceding almost-sure equality shows that it agrees with the pushforward from every representative of x, not just from $s(x)$. Induction gives finite-prefix equality under a common quotient policy. More generally finite adaptive policies depending on the visible future history are handled by the same raw conditional continuation laws: with finitely many actions, sum over action words and multiply their open-loop report laws by the measurable feedback probabilities of the policy. Equality extends from bounded determining functions to Borel sets by the monotone-class argument. The converse follows since each listed response is itself a declared test coordinate.

For another statistic with the stated Borel response maps, the product map g is Borel and satisfies $q = g \circ T$; the asserted minimality is precisely this factorization, not a finite-state claim. For a stopped test truncate at time n with a distinguished surviving outcome. Its finite law is determined. Almost-sure termination under both histories makes both surviving probabilities tend to zero, so the bounded stopped law agrees. This limit does not control expected stopping cost without an additional bound. □

Proposition S43.2 (Primitive positive-evidence patches)**.** *The continuity and exhaustion conditions of Theorem S43.1 hold for the following raw class. Hidden-state and report coordinates lie in compact metric spaces, the action set is finite, and primitive transition–report kernels have bounded continuous densities with respect to fixed finite reference measures; the preparation is fixed. A finite-dimensional positive instrument whose matrix entries depend continuously on compact report coordinates is another such class. Finite physical phases are included as discrete coordinates, with continuous legal-domain indicators on the corresponding patches. Only histories of positive prepared likelihood are included.*

Proof. Fix a finite action word and history length. Iterated integration of products of bounded continuous primitive densities gives a continuous nonnegative history likelihood $p(h)$. The same statement holds for the numerator of a conditional finite future test whose report test function is continuous; the integration domains are fixed compact spaces with finite measures. Uniform boundedness on each finite product and dominated convergence justify continuity. In the matrix case these quantities are traces of products of continuous instrument maps followed by finite integration, with the same conclusion.

The sets where $p(h) \geq 1/n$ are compact and exhaust the positive-likelihood histories for this action word. On each, division by $p(h)$ is continuous and all conditional response coordinates are continuous. Countably many lengths, finitely many action words per length, and countably many n give the compact exhaustion. On a compact metric report space there is a countable uniformly dense family of continuous functions; include their finite products and all finite action words. Their integrals determine finite report laws, and adjoining finite prefixes gives the required determining

closure. Costs and legal phases must be included among the stipulated continuous or discrete observable coordinates. The probability of a zero-likelihood report history is zero under the prepared law. The construction does not claim that the likelihood has a uniform positive lower bound over the whole experiment. □

The proposition is about continuity of raw finite-word responses. It does not assume a pre-existing posterior manifold or a covariance geometry, and it does not give a finite algorithm for discovering all quotient classes. An unknown-model family of separate posterior laws is not automatically one executable posterior. Applying this appendix uniformly to a model family would require a common determining response construction and compatible null-set/regularity conditions; the finite-state upper in the main theorem instead obtains its uniformity from the explicitly checked task-order program and stores no such family.

S44. Sharp boundaries of the proved comparisons

The following witnesses isolate distinct hypotheses. They are mathematical counterexamples to specific uncharged inferences, not a replacement of the main theorem by an impossibility statement.

Proposition S44.1 (Absolute task error can be amplified by selection)**.** *For every $b \geq 1$ and $0 \leq e \leq 1/b$, there are positive prepared and actual joint laws with identical report marginals and one-call absolute squared-task defect at most e, whose optimal stopped risks at expected call budget b differ by be, even with one output value. The difference occurs in either direction. A fixed relative task order is therefore strictly stronger than a small absolute task defect near a zero-loss component.*

Proof. Draw an independent uniform report $x \in [0,1]$ at every call and let D have mass $1/b$. All audits have conditional mean zero. Off D, both laws have a fresh fair-sign audit of magnitude one. On D, the comparison has a fair sign of magnitude $\sqrt{be}$ and the actual law has audit zero. For every $q \in [0,1]$ and finite-image readout $r(x)$, their unnormalized squared losses differ by $be \int_D q(x)dx \leq e$. The comparison residual variance is everywhere at least be, so every committing strategy has risk at least be; selecting D and outputting zero attains it with expected cost b. The actual law attains zero by the same procedure. Reversing the laws proves the other direction. A single event-stopped label can reject at itself and commit with a final event; a state-stopped realization uses an additional live label. On D a task zero under one law is positive under the other, so no two-sided relative order with a positive lower factor can hold. □

A report-only comparison also fails: a constant mark can have the same marginal under two laws while an additional visible bit reveals a fair audit sign. A two-output decoder then has zero loss, although a profile formed from the constant mark alone has residual variance one. The full-report conditional moments in (2.3) exclude this omission.

There is also an all-program witness for the necessity of more than report balance. Take the three admitted models $\theta \in \{-1, 0, 1\}$, let every precommit report be the same constant, and let the unseen audit at commitment be $Z = \theta$. There is one legal productive command, costing one, and no command changes θ. The actual comparison member $\theta = 0$ has $\mathcal{Q}_m(b) = 0$. Nevertheless the distribution of every committed readout R is identical under $\theta = \pm 1$, because all precommit laws, including stopping and coins, are identical. The average of their risks is $1 + \mathbb{E}R^2 \geq 1$, and output zero attains worst-case risk one. This holds for every almost-surely terminating policy, at every feasible budget and every dictionary size. Report masses match exactly; the missing first task moment is the obstruction. It is not repaired by choosing a different causal rejection rule.

A live-state count alone cannot justify a terminal-width subtraction. In the two-port interface, use labels $0, 1$ with live commands gate and productive respectively, and decoder values $-1, +1$. Gate failure continues at 0, success at 1; a rejected productive report returns to 0. A productive commitment emits a final event with label 0 or 1 according to its chosen output. Two live and two

output roles have used only two labels. The state-stopped model cannot share those roles because its halt map is intrinsic. This is the exact semantic repair retained from the earlier work.

Finally, bounded-law discrepancy is not a cost certificate. A source procedure can agree with a target terminal output with probability one while performing an independent preliminary waiting time of arbitrarily large finite mean, or of infinite mean. Terminal total variation is then zero and expected cost is uncontrolled. Our simulator definition includes the scored stopping information when its law is compared, and still requires a separate expected-cost certificate because time is unbounded. Likewise, neither fixed-time equivalence nor finite-dimensional positive matrix identities imply large-deviation equivalence, unrestricted optimal-control equivalence, or closure of unbounded operators.

What is and is not generalized. The central theorem generalizes exact and absolute-calibrated causal profile transfer to a common affine task order and derives its all-scale relative and stratified consequences. The positive-matrix characterization concerns localized task order; the quotient construction concerns compact-exhausted continuous-response experiments; the pilot concerns a fixed independent command sampler. These results do not establish necessity of a common acquisition program for every causal experiment, nor the minimal possible simulator state under all implementations, nor matching lower bounds for program length, scratch and arithmetic time. Those stronger mathematical questions are distinct from the fully stated sufficient-class theorem proved here. The general theorem does not depend on a conclusion taken from either realization. The retained nonintervening sensing, occupation-flow, exact role and absolute-defect results remain substantive independent parts of the complete companion.

S45. The scope of the strengthened theory

The boundaries in Appendix S44 concern the inherited causal transfer theorem. The present synthesis theorems add necessary-and-sufficient conclusions on explicitly stated classes; they do not turn that sufficient transfer theorem into a classification of all standard Borel causal experiments.

The task-defect theorem is exact for a fixed selected loss family, its declared decision carrier, and one global finite decoder. Its occupation version holds for an actual occupation measure supplied by a competitor. Neither assertion manufactures that measure or an acquisition program. The scalar squared-task order is strictly weaker than the affine-square matrix order; the latter remains an exact characterization of its own larger test family. The Bregman comparison preserves squared-loss exponents only under the stated curvature bounds.

The finite-obstruction hierarchy characterizes one common full-history program for finitely branching instruments, bounded terminal scores and a strictly positive call-cost floor. It permits arbitrarily many operating models, but finite testing of an arbitrary uncountable family is not a decision procedure for its infinite synthesis problem. Compactness produces a policy, not a uniformly bounded number of live labels. Its finite dual concerns one specified horizon and finite model set. Fallback completion requires a genuinely legal common continuation and charges its cost and memory. The current supplement treats deadlocks, restricted preparation faces and the distinction between small survival mass and small expected cost tails.

The finite-memory certificate is intrinsic to a fixed finite-label observer on a declared time-homogeneous interface. Its projections and potential matrices are proof certificates, not stored observer variables. The finite algebraic decision conclusion requires finitely many models, algebraic primitive data and a semialgebraic decision carrier. It does not assert efficient synthesis, rational optimal probabilities, an exact finite-bit implementation of arbitrary algebraic randomization, or optimal program, scratch and arithmetic lower bounds. Exact initial-state uniformity does not replace a missing certificate at later roots of a history-dependent experiment. Returning from a balancing subroutine must be compiled at the current productive report before its prospective audit, as specified in the supplement.

The two-detector theorem has no balancing group and no free reset. It characterizes all integer classical live-memory budgets for two projectors, unit call costs, first-success commitment and no other physical command. The maximum principal-angle cosine is the worst spectral block of that model, not a universal rank or memory exponent. The initial-state saddle at unrestricted memory is proved from the physical instrument. The exact two-label minimax gap shows that a single preparation need not supply a sharp robust converse after imposing a nonconvex controller restriction. It does not imply such a gap for every memory-constrained experiment. Randomized initial labels are a separately stated interface, not an uncharged repair of a fixed-start observer.

All inherited valid results remain either in the native main inputs, the complete preceding companion, or their pinned independent realization lineages. In particular, the ordered-measurement local geometry and allocation theory, exact two-stage reset and message-discrimination theory, rank constraints and spectral variational values are not asserted to follow from the detector example. Conversely the current article does not abandon them because its general structural results have become central. The accompanying preservation and dependency maps identify their locations and prevent a change of research identity from being hidden in a reorganization.

Open mathematical goals remain: an intrinsic classification beyond finite branching; minimal memory under arbitrary resource-certified simulators; a broad nonintervening controlled-sensing application beyond the retained directional model; matching computational-resource lower bounds; and the independent unbounded-operator, large-deviation and wider A1/A2 pipeline targets. These are not hypotheses standing in for a missing proof of a theorem stated here. The present proofs establish their displayed conclusions with their displayed quantifiers. Finite diagnostics test particular algebraic identities, boundary cases and implementations; they are not formal proofs, independent expert assessment or an editorial decision on novelty and significance.

The report-resolved continuation theorem is exact for linear marked task equalities of a supplied time-homogeneous finite-return routine. It does not identify the reachable positive cone with every positive matrix on its support, nor replace a one-sided reachable cost bound by an unsupported operator inequality. The reset-instrument classification applies to failure branches which physically prepare known states and to first-success commitment. It includes arbitrary finite rank-one families, but not arbitrary quantum instruments or an arbitrary-rank projection without input-erasing failure. The zero-risk spectrum is exact for the acquired profile; its physical threshold has the same converse and implementation hypotheses as the exact transfer theorem. These distinctions separate the new structural results from the independent wider research goals.

The primitive finite-model route in Section S26 removes the reset and reference assumptions on its own declared interface. It supplies a common fallback, monotone two-sided approximations to the least feasible cost, and bounded-risk approximations at every strictly interior feasible budget with counted finite implementations. It does not replace the preparation-free all-initial-state quantifier by an arbitrary restricted preparation set, and it assumes finite models, finite globally legal commands/reports, time homogeneity and bounded positive call costs. It is not unknown-kernel statistical learning, a universal closed formula for the comparison profile, or optimal computational complexity. The six-regime diagram solves one explicit mixed-strata comparator with its original asymptotic qualification intact.

The compact-family route extends the primitive finite-model risk approximation to a continuously parameterized compact uncertainty class on a common finite globally legal physical interface. It controls model error and restores the same budget at every $N > N_{\min}$. Its algebraic specialization is a terminating approximate synthesis result for supplied exact primitive data, not learning from finite data, polynomial-time synthesis or a solution of the boundary finite-memory problem. The model stays fixed during a run. Exact geometric reference-profile realization, arbitrary unknown-model quotient construction and minimal computational resources remain separate mathematical questions. The examples showing boundary discontinuity and failure of vertex-only certification delimit the positive theorem rather than replace it.

The current optimality layer distinguishes three assertions. Boundary attainment and complete performed-law compactness use the shared finite-branching tree, positive cost floor and bounded terminal loss, with a fixed dictionary for total variation. Finite spectral witnesses additionally require the compact continuous primitive family and strict prefix feasibility. Same-budget finite-label approximation uses a real common fallback with strict cost slack. Hard optimized output width is not convexified, and several expected resources require a jointly safe fallback. The timing application assumes a stateless single-command device with no free clock; its exact attainment count is not a universal lower for devices that report time. None of these theorems supplies an arbitrary unknown-model quotient or a general polynomial-time synthesis method.

S46. Preparation faces and reachable supports

Let $\mathcal{A} = \bigoplus_{s=1}^{\ell} M_d(\mathbb{C})$ be the live controller–physical algebra. A fixed controller has a completely positive trace-nonincreasing continuation map $\mathcal{T}$, a cost observable $g \succeq c_* I$ and terminal score observables f_j satisfying

$$\alpha_j(I - \mathcal{T}^* I) \preccurlyeq f_j \preccurlyeq \beta_j(I - \mathcal{T}^* I).$$

The initial live label is s_0. A preparation face is specified by a nonzero physical projection Q, and all initial density matrices supported on Q are admitted. Let ι insert a matrix into block s_0, and put

$$X_0 = \iota(Q), \quad D = \ell d, \quad R = \sum_{n=0}^{D-1} \mathcal{T}^n X_0, \quad P = \operatorname{supp} R.$$

Theorem S46.1 (Exact face-restricted potential certificate)**.** *The projection P is the least invariant block support containing X_0. A fixed controller is proper at expected cost at most N for every admitted preparation, with prescribed terminal operators A_j on Q, if and only if there exist an invariant block projection $\widetilde{P}$, with $\widetilde{P} X_0 = X_0$, a positive $H = \widetilde{P} H \widetilde{P}$, and Hermitian $K_j = \widetilde{P} K_j \widetilde{P}$ such that*

$$\widetilde{P}(H - \mathcal{T}^* H - g)\widetilde{P} \succeq 0, \qquad Q H_{s_0} Q \preccurlyeq N Q,$$
$$\widetilde{P}(K_j - \mathcal{T}^* K_j - f_j)\widetilde{P} = 0, \qquad Q (K_j)_{s_0} Q = A_j,$$
$$\alpha_j \widetilde{P} \preccurlyeq K_j \preccurlyeq \beta_j \widetilde{P}.$$

One may choose $\widetilde{P} = P$. On that least support, finite-cost proper operation from the preparation face is equivalent to $\operatorname{spr}(\mathcal{T}_P) < 1$, and the exact minimal potentials are the supported Neumann resolvents. No transience is required on an unreachable block.

Proof. The supports of $R_n = \sum_{k=0}^{n} \mathcal{T}^k X_0$ increase. When two successive supports agree, R_n dominates a positive multiple of its support projection. Positivity then implies that the support is invariant under $\mathcal{T}$. Until that event the rank increases, and its ambient bound is D; hence stabilization occurs by $D - 1$. Every invariant support containing X_0 contains every iterate, proving leastness.

For sufficiency, iterate the cost superpotential inequality on $\widetilde{P}$ and pair with an initial density matrix supported on Q in block s_0. Dropping the positive remainder bounds every partial expected cost by N. Monotone convergence bounds total cost. The positive floor bounds $c_* \sum_{k \ge 0} \Pr(\tau > k)$ by N, proving properness. Iterate the score equation. Its terminal remainder is bounded in absolute trace by $\max(|\alpha_j|, |\beta_j|) \Pr(\tau > n)$, which vanishes. Its partial sum contains the actual disjoint terminal-score contributions. Thus the initial compressed block is exactly A_j.

Conversely the cost bound on every initial density in the face gives $\sum_{k \ge 0} \langle g, \mathcal{T}^k X_0 \rangle \le N \operatorname{rank} Q$. The same series starting at $\mathcal{T}^n X_0$ is a nonnegative tail of that series. Summing for $0 \le n < D$ yields a finite bound $DN \operatorname{rank} Q$ on the series starting at R. Since R is strictly positive on its support, every positive supported matrix is bounded by a finite multiple of R. Thus the increasing

partial cost potentials on that support have bounded norm and converge. Their limit $H^{\min}$ solves $H^{\min} - \mathcal{T}_P^* H^{\min} = PgP$ and has the required compressed initial budget.

The floor implies norm convergence of $\sum_k (\mathcal{T}_P^*)^k P$. Positivity bounds the iterates of any supported Hermitian matrix by its norm times the iterates of P, so every such iterate tends to zero. The same holds for all complex matrices by decomposing into Hermitian parts. In finite dimension this forces every eigenvalue of $\mathcal{T}_P^*$ to have modulus less than one, proving transience and the Neumann formula. The bounded-score series also converges. Its normalization bounds follow from telescoping the terminal effect $P - \mathcal{T}_P^* P$, whose series sums to P. The initial face identity follows from the assumed terminal scores. Finally iteration shows that every positive superpotential dominates every partial exact cost potential, hence their limit. □

An arbitrary preparation set is not automatically its whole support face. For equality of task expectations, its linear span, not its support face, controls what is required. The main manuscript's report-resolved continuation theorem treats that distinction. Extending a preparation family to its full face strengthens the quantifier and must be declared explicitly.

S47. Legal physical domains and deadlocks

A physical command can have a restricted domain. A time-homogeneous face-valued domain is represented by projections D_a on the physical space: command a is permitted only on states supported on D_a. The controller is not given that support or a free physical-port tag. Let P_s be the controller's least reached physical support at label s, computed from the allowed preparations and the actual continuing instrument.

Proposition S47.1 (Fixed-row legality on reached supports)**.** *A command row $p_{s,a}$ is legal at every reached history if and only if*

$$p_{s,a} > 0 \quad \Longrightarrow \quad P_s \preccurlyeq D_a$$

for every label and command. Equivalently $p_{s,a}(I - D_a)P_s = 0$. For a finite algebraic model family with face domains, these are polynomial conditions on the shared row probabilities and modelwise support projections. They may be added to the finite-memory potential system without giving the controller an extra domain input.

Proof. If the support inclusion holds, every reached positive state at label s is supported on D_a, so the command is legal. Conversely, legality at every reached positive state annihilates all of them by $I - D_a$ whenever $p_{s,a} > 0$. Their positive sum over a spanning finite set of reached supports has support P_s; hence $I - D_a$ also annihilates P_s. This gives the inclusion. The scalar multiplication formula handles zero row probabilities without introducing a division or a nonclosed strict-positivity condition. It is polynomial in real and imaginary matrix coordinates. Apply it separately in each model, retaining the same row probabilities. □

This proposition concerns declared support-face domains. A legal domain specified by a different predicate on states need not admit this polynomial encoding. If a protocol port is already a deterministic function of the visible history, a full-history policy can recover it, but a finite-label controller must store enough information to choose a legal row without a free port input.

For the full-history finite-obstruction hierarchy, the main manuscript assumes a nonempty legal command set at every open decision node. Deadlocks can instead be admitted explicitly as forbidden nonproductive leaves. A node with no legal decision must carry zero reached mass. For a finite set of models and finite horizon, impose the corresponding positive-operator equality for each such leaf on the admitted preparation face. In the common realization plan it is the equality

$$Q_\theta \, \gamma(h) \mathcal{I}_{\theta,h}^*(I) \, Q_\theta = 0.$$

Here $\gamma(h)$ excludes physical report probabilities, as in the main manuscript.

Proposition S47.2 (Deadlock-aware finite hierarchy)**.** *With finitely branching legal histories, a common initial preparation face in each finite-dimensional model, a positive call-cost floor and bounded terminal scores, the finite-intersection proof of the causal synthesis hierarchy remains valid after adding the deadlock zero-mass constraints. Constraints at depth T include every deadlocked leaf reached by that depth. An open nondeadlocked leaf still owes at least one further paid call. A deadlocked leaf cannot be used as a fictitious terminal output or a free fallback.*

Proof. Give an empty decision node a singleton dummy coordinate solely to make the product of nodewise controller coordinate spaces compact. This coordinate causes no physical action and is constrained to have zero reached mass in each admitted model. It is not a legal executed command. All finite-prefix mass, cost and score expressions remain continuous in the product coordinates. Deadlock constraints are closed and remain true under extension to later horizons. Restricting a later feasible prefix therefore preserves them and the usual cost reserve and score intervals. Conversely, every proper legal policy satisfies them because it never reaches a deadlock with positive probability.

For any finite collection of model, horizon and deadlock constraints, choose its largest horizon and union of model indices. Feasibility at that level gives one common prefix satisfying the collection. Compactness gives an infinite common coordinate assignment satisfying all levels. No dummy decision is executed with positive probability. The same positive-cost survival bound tends to zero, so the resulting process commits almost surely without reaching a deadlock, and monotone convergence supplies its budget. Bounded terminal-score tails recover its targets. This repeats the hierarchy argument with the missing legal-boundary event explicitly accounted for. □

S48. Prospective audits and post-report return compilation

A routine's logical return and the experiment's irrevocable halt are different events. Let its final productive call produce a visible report y. At this report the routine selects a return mark x and a finite exported decoder choice, using current coins. Its prospective audit instrument defines the score that would be obtained if the outer caller accepted. Those scores do not require executing the audit on a rejected branch.

Proposition S48.1 (Compilation without a free audit or hidden mode)**.** *Suppose the caller's acceptance probability and decoder are functions only of the current finite return mark and the existing finite decoder dictionary. Suppose the physical rejection operation is legal and either executes within the current productive call or has an explicitly counted finite-state return subroutine. Then the routine can be compiled into an event-stopped observer by replacing each logical return transition by its acceptance and rejection branches. On acceptance it emits the outer halt event and retains the decoder label; on rejection it executes the declared return and reenters the existing start label. Prospective audit statistics on accepted branches, complete physical paths and charged costs agree with this construction. The state-stopped implementation uses disjoint live and final-output types instead.*

Proof. At the current productive report, first sample exactly the routine's original return choice. Sample the caller's independent acceptance coin next. On acceptance, emit the outer commitment event, store its dictionary value and only then perform the declared scored audit. On rejection do not perform that audit; follow the declared rejection instrument and reenter the start. This reproduces each branch's raw physical instrument and its unnormalized accepted audit measure. Iteration gives equality of every finite stopped prefix and every raw path cost. Properness is supplied by the positive conditional-hazard argument, not by the compilation alone.

All information used to decide the branch is available during the current call. No value of the future audit is read. If the return needs another physical call, its phase persists and is stored in the added return-subroutine labels; its cost is included. Otherwise the existing start label suffices for continuation, and the event-stopped live/output packing introduces no continuing mode bit. A

state-stopped implementation sends accepted branches to disjoint halt-typed labels. This proves the assertions while keeping the different observer models distinct. □

A synthesized terminal score list alone is not enough to prove this proposition's hypotheses. In particular, a score obtained only by actually measuring and revealing the audit cannot be treated as an unobserved prospective score on rejection. Likewise a kernel identity at the first routine root cannot substitute for its report-resolved continuation certificate. The current main manuscript supplies that additional finite test for time-homogeneous finite-return routines.

S49. Survival mass is not expected cost tail

Proposition S49.1 (The exact limitation of a mean-cost bound)**.** *For a positive-integer stopping time with $\mathbb{E}\tau \le b$, one has*

$$\Pr(\tau > T) \le \min\{1, (b-1)/T\} \quad (T \ge 1).$$

This does not imply a uniform vanishing bound on $\mathbb{E}[(\tau - T)_+]$ over the same budget class. In particular the main manuscript's score-tail convergence, which uses bounded scores, cannot be repurposed as an expected cost-tail convergence statement.

Proof. Pointwise $\tau \ge 1 + T\mathbf{1}_{\{\tau > T\}}$, which gives the probability bound. Fix $b > 1$. For an integer $K \ge b$, set $\tau = K$ with probability $(b-1)/(K-1)$ and $\tau = 1$ otherwise. This is a finite-branching, unit-call-cost terminating experiment and $\mathbb{E}\tau = b$. At a fixed cutoff $T < K$,

$$\mathbb{E}[(\tau - T)_+] = (b-1)\frac{K-T}{K-1} \longrightarrow b-1 \quad (K \longrightarrow \infty).$$

Thus its supremum over the budget class never tends to zero with T. The survival probability at that same cutoff can tend to zero. Bounded terminal scores are controlled by the latter probability; unbounded remaining call cost is not. □

The finite-prefix fallback theorem uses a conditional cost bound on an actual fallback, not an unproved uniform tail-cost estimate. For a fixed finite-state transient instrument its own supported resolvent gives a finite tail-cost bound, but its constants need not be uniform over a family approaching the boundary of transience.

S50. Correction boundary and preserved companion

The single-atom proposition inherited by the current manuscript is corrected only in its necessary-threshold quantifier. The statement that its specified atom yields zero loss when $b\lambda \ge 1$ remains valid, as do its lower and upper comparisons with the remainder profile. Necessity based on that atom alone requires positive residual variance almost everywhere on the remainder, or another condition excluding all other finite-dictionary zero-loss mass. The current main theorem gives the complete criterion through the largest atomic masses of the zero-residual predictive measure, and an explicit two-atom counterexample shows why the qualification matters.

The byte-preserved R58 native companion has its own older statements and provenance. Retention is not a claim that its historical wording supersedes a correction in the current manuscript. The preservation map identifies the current controlling result for every corrected claim. The original acquired-law geometry, absolute-calibration proofs, nonintervening Gaussian sensing and exact stopping constructions remain available in that companion, and the independent ordered-measurement, reset-communication and rank-constrained discrimination developments remain at their pinned source identities.

S51. The primitive frontier: budget mixing and limiting scopes

This section supports the current primitive-instrument theorem. Its baseline is not a hidden-state reset model: a reported state may retain all dependence on the unknown input. The fixed fallback must be valid from every initial state in every one of the finitely many declared models.

Lemma S51.1 (A finite mixture without an uncounted persistent seed)**.** *Let two finite-label observers share the same physical command/report interface and fixed terminal dictionary. If their live-label sets have sizes n_1, n_2, a mixture chosen with one fixed probability at initialization is realizable with at most $1 + n_1 + n_2$ live labels and the same terminal dictionary. Its expected cost and terminal-loss operators are the corresponding convex combinations in every model. Event stopping permits the usual live/output reuse; state stopping uses disjoint output labels.*

Proof. Use an initial label whose command row mixes the two initial command rows with the prescribed probability. At this first call sample the branch and command jointly using fresh private randomness. On the current report, either commit according to the sampled branch or enter its tagged continuing label. Subsequently all transitions stay inside that branch's counted live-label set. If a transition is specified only by a kernel conditioned on the executed command and report, integrate out the branch coin: for a fixed executed command its posterior branch weights depend on the two command probabilities, but not on the physical state, since the report instrument for that command is identical in both branches. Zero command probabilities are assigned arbitrary unused conditional rows. This is a legal post-report kernel, not a persistent external random tape. Coupling with the explicit within-call coin proves equality of every path law and its nonnegative costs. The same retained decoder is used in both branches, and the halt convention gives the two total-label counts. □

Proposition S51.2 (Why the uniform qualifications cannot be omitted)**.** *Modelwise properness does not imply a uniform finite-cost fallback in an infinite uncertainty class, even in physical dimension one. A robustly minimax initial command row also need not be pointwise optimal for each known initial state.*

Proof. For the first claim take one command and models indexed by integers $n \geq 1$, with productive probability $1/n$ and unit call cost. Every individual model terminates almost surely at finite mean n, but every legal observer must wait for the first productive event or longer. Its worst-model mean is infinite. The finite-model theorem instead takes a finite maximum of strictly positive termination margins.

For the second claim take the two orthogonal rank-one failure projections $P = |0\rangle\langle 0|$ and $Q = |1\rangle\langle 1|$ with unit calls and immediate first-success commitment. Fair first choice followed by switching after failure has robust value $3/2$. At the known initial state $|0\rangle$, calling the detector with failure projection Q succeeds at cost one, while the fair-first policy has mean $3/2$. Thus one uniform minimax row is not a simultaneous pointwise optimizer. The current reset-control statement makes this distinction explicit; the complete R61 wording remains in the retained original. □

The risk approximation does not bound the expected cost remaining in an arbitrary original policy after a cutoff. It executes a different, uniformly cost-bounded fallback, and hence bounds the new remaining cost by κ times survival mass. For budget $N > \kappa$, mixing with this fallback restores the original budget exactly. At $N = \kappa$ that mixing argument has no positive cost slack; only the stated enlarged-budget conclusion follows unless the fallback itself costs the one-call floor. The absence of a same-budget conclusion there is a limit of this proof, not a no-go assertion about the optimal boundary policy.

S52. Finite diagnostic experiments accompanying the proof

The retained R62 diagnostics include two separate classical finite-prefix programs. One is binary costly observation with a hidden fixed bit and two possible noise levels, using a common decision plan for both models and both initial bits. The other is a heralded experiment with nonproductive reports and two possible productive-report laws. Their linear programs implement the common sequence weights, terminal loss and open-path cost reserve directly. An intentional removal of the reserve at one unit of budget produces a false zero-loss open prefix; the correct constraint excludes it. Exact finite checks separately test the sharp nilpotent detection depth, a noncommuting two-projection fallback resolvent and the budget-restoring probability. Neither the numerical linear programs nor the finite examples prove the semidefinite statement for all dimensions.

The six-regime phase calculation is the exact solution of the explicitly displayed comparator, which estimates the acquired risk only up to the inherited geometric constants. Numerical checks compare it with an independently optimized scalar convex objective. The two nonsmooth intervals are not replaced by a differentiable surrogate. The plotted curve is not fitted to data or presented as an exact universal physical risk curve.

S53. Certified finite approximations for compact model uncertainty

The current compact-family theorem compares raw report-resolved instruments, not only final-output distributions. In this section use the same common finite interface, bounds c_*, c^*, L, model-cover errors $\epsilon_I, \epsilon_g, \epsilon_F$, horizon T and uniform fallback cost $\kappa < N$ as in the main manuscript. The fallback must work from every current physical state of every actual model. A returned numeric optimizer value, by itself, is neither a valid all-model upper nor a proved lower value.

Proposition S53.1 (A posteriori certificate with rigorously bounded solver errors)**.** *Suppose a normalized common finite realization plan has certified bounds*

$$C_{\phi,T} + c_* S_{\phi,T} \preccurlyeq (N+s)I, \qquad V_{\phi,T} \preccurlyeq (q+r)I \quad (\phi \in F), \qquad s, r \geq 0.$$

Its normalization and positive weights are exact; s, r may be proved interval bounds on constraint residuals. Put

$$\begin{aligned} D &= \min\{1, ((N+s)/c_* - 1)/T\}, \\ e &= s + (\kappa - c_*)D + \frac{N+s}{c_*}[\epsilon_g + \epsilon_I(c^*(T-1) + \kappa)], \\ z &= r + LD + L\frac{N+s}{c_*}\epsilon_I + \epsilon_F. \end{aligned}$$

Completion by the actual fallback, followed by the counted initial mixture of weight $(N-\kappa)/(N-\kappa+e)$, gives a proper common observer of uniform expected cost at most N and risk at most

$$\min\{L, q + z + Le/(N - \kappa + e)\}.$$

If a separate exact dual certificate gives a lower value $\underline{q}$ for the finite-model problem at budget N, then $\underline{q}$ is also a lower bound for the fixed-dictionary full-family risk. For a decision net subtract its uniform rounding modulus from that lower. A feasible approximate primal value q is not automatically such a lower certificate.

Proof. The proved source budget $N + s$ bounds source prefix occupation by $(N+s)/c_*$ and open mass by D. Completing the same plan on the source adds at most $(\kappa - c_*)D$ to its reserved cost. The occupation-sensitive comparison in the main manuscript bounds the discrepancy to any covered target by the displayed instrument and cost term, so the actual completed cost is at most $N + e$. Its loss is at most $q + z$, using the source upper $q + r$ and at most L on the open terminal mass. Both bounds are uniform in the initial density matrix. The fallback alone costs at most κ and has loss at

most L. Its stated initial mixture therefore costs at most N; the same convex calculation gives the risk upper. All calls, return phases and branch memory are those of the compiled observer.

For the lower, every full-family budget-N policy restricts to a feasible finite-model prefix with no greater nonnegative loss. Weak duality for an exactly feasible finite dual therefore gives $\underline{q}$ no greater than every such policy's worst loss. The rounding argument for one common decoder net adds only the modulus at its actual single terminal event. Nothing in this proof converts numerical feasibility tolerances into exact probability normalization or a numerical objective into a rigorous dual certificate. □

S53.1. **Size of one certified construction.** Write $b = ay$ for the maximum command–report branching factor and assume $b \geq 2$. At horizon T, the number of pre-call node roles is at most

$$L_T = \sum_{t=0}^{T-1} b^t.$$

For a fixed dictionary of at most m values, a common realization program has $O((a+m+1)\sum_{t=0}^{T} b^t)$ scalar weights. Apart from normalization and nonnegativity, the finite cover problem uses two d-dimensional Hermitian inequalities per cover model: its reserved cost and terminal loss bound. Constructing their coefficients still involves the raw finite paths; these counts do not include the cost of evaluating those coefficients, exact arithmetic or finding a certified optimizer. The terminal dictionary is common across the entire model family.

For an explicit upper count, suppose a supplied model parameterization in a bounded subset of $\mathbb{R}^p$ admits certified covers of size at most $C\eta^{-p}$ with all primitive errors $O(\eta)$, and suppose the decision loss modulus is Lipschitz. At fixed positive slack $N - \kappa$, target error ξ is achieved by $T = O(\xi^{-1})$, model resolution $\eta = O(\xi^2)$, and decision resolution $O(\xi)$. The displayed gap formulas prove these choices by direct substitution. Thus $O(\xi^{-2p})$ model representatives and $b^{O(\xi^{-1})}$ live node roles are sufficient; a decision net contributes at most its cardinality to the power m possible decoder tuples in exhaustive enumeration. If $b = 1$, use the exact sum $L_T = T$ instead. These are conservative *construction-size upper bounds*, not matching memory lower bounds or polynomial-time complexity claims. All constants depend on the declared physical bounds, moduli and slack, not on a data-fitted unknown uniformity claim.

For the compact algebraic class, the main manuscript instead supplies a fully quantified polynomial procedure. Common finite-prefix probabilities and one decoder tuple are variables; model parameters are universally quantified in the cost and loss inequalities. Real-algebraic elimination can produce the values and witnesses, and the explicit cutoff error tells when to stop. This does not turn a quantified problem into an ordinary finite-grid semidefinite problem. The endpoint $N = N_{\min}$ need not be decided by the strict-feasibility search. Algebraic probabilities are exact descriptions in the stated ideal sampler model, not proofs of a bounded-bit physical randomization implementation or optimal arithmetic cost.

S53.2. **Independent finite diagnostics for the compact theorem.** The new diagnostic program builds four common finite-prefix linear programs for a noisy observation of an unknown fixed bit, with three noise models and both initial bits. After numerical optimization, it recovers rational primal and dual vectors and checks their feasibility and equality of objective values using exact rational arithmetic. The same recovered plan is completed and budget-mixed at finitely many off-grid rational noise parameters. The analytic cover radius and the main theorem, not the sampled off-grid points, supply the continuum implication.

A separate check constructs normalized noncommuting qubit instruments and compares independently computed forward path sums with backward cost, survival and loss operators. The occupation-sensitive inequalities are tested in operator order on these finite examples, not only at one pure input. Exact negative controls cover boundary risk discontinuity, the required remaining-horizon

cost term, failure of vertex-only certification even for affine primitive data, and the distinction between a fixed unknown model and a switching adversary. These diagnostics cannot prove the theorems in all dimensions, but their failures would expose implementation or formula inconsistencies. Their actual counts and environment are emitted by the executable, rather than inserted into the mathematical proof.

S54. Explicit common recursion for the spectral dual

We give the finite recursion used by the main article's primal–dual theorem. Its cost atoms are $(\phi_i, \rho_i, \alpha_i)$ and its task atoms are $(\theta_j, \sigma_j, \beta_j)$, where preparations are density matrices, weights are nonnegative, $\sum_j \beta_j = 1$, and $\lambda = \sum_i \alpha_i$. The theorem permits pure preparations and sparse lists; the recursion itself works for any such finite lists. A pre-call history h records actions and reports of its t completed calls. Set

$$\rho_{i,h} = \mathcal{I}_{\phi_i,h}(\rho_i), \qquad \sigma_{j,h} = \mathcal{I}_{\theta_j,h}(\sigma_j).$$

No controller probability occurs in these unnormalized matrices. Define

$$a(h,a) = \sum_i \alpha_i \operatorname{tr}(G_{\phi_i,a}\rho_{i,h}),$$

$$b(h,a,y,r) = \sum_j \beta_j \operatorname{tr}(F_{\theta_j,a,y,r}\sigma_{j,hay}),$$

$$U(h) = c_* \sum_i \alpha_i \operatorname{tr}\rho_{i,h} \quad (|h| = T).$$

For $|h| < T$, compute backwards

$$W(h,a,y) = \begin{cases} \min\{\min_{r\in C} b(h,a,y,r), U(hay)\}, & y \text{ productive}, \\ U(hay), & y \text{ nonproductive}, \end{cases}$$

$$U(h) = \min_{a \text{ legal at } h} \{a(h,a) + \sum_y W(h,a,y)\}.$$

At a compulsory terminal report omit the continuation alternative. At an impossible word all relevant matrices may vanish; the arbitrary minimizing choice there has no physical consequence. The dual value is $U(\varnothing) - \lambda N$.

Proposition S54.1 (Exact slack accounting for a common finite tree)**.** *Let $\gamma(h,a)$ be a plan's sequence weight before the physical report and let $\gamma(h,a,y,r)$ and $\gamma(h,a,y,c)$ be its commitment and continuation weights. The difference between its scalarized reserved objective and $U(\varnothing) - \lambda N$ equals the sum of its nonnegative local Bellman slacks, weighted by these sequence weights. In particular every positive-weight alternative must have zero slack at equality. The matrices in the recursion are analytic data used to choose a single common tree, not an observer-owned model register.*

Proof. For a command choice the slack is $a(h,a) + \sum_y W(h,a,y) - U(h) \geq 0$. For a productive commitment it is $b(h,a,y,r) - W(h,a,y) \geq 0$, and for continuation it is $U(hay) - W(h,a,y) \geq 0$; the latter is zero when continuation is forced. Multiply by the corresponding sequence weights and sum over the tree. The common flow equalities cancel every internal occurrence of U and W. At the root its incoming weight is one. At a depth-T open leaf the remaining term is exactly the reserved cost; at a terminal leaf it is exactly its weighted task loss. Thus the uncancelled terms are the claimed objective difference. Finiteness makes all sums literal finite identities. A finite sum of nonnegative terms vanishes exactly when every term with positive coefficient vanishes. □

If several physical atoms assign zero probability to a reached word, no division by those probabilities is taken. This is why a common unnormalized recursion remains valid with changing supports. A dual-minimizing deterministic tree may have large cost or risk in an individual model. It is not a

primal optimal strategy unless the original robust constraints and complementary slackness are also verified.

S55. Conic compression and what a certificate proves

The sparse bound uses affine coordinates on the plan polytope, not the number of tree variables before flow reduction. At an optimal feasible x_*, each active risk gradient is augmented by 1 and each active cost gradient by 0. Their conic combination is $(g, 1)$, with g in the negative normal cone. If the positive support has more than $k+1$ generators, choose a nonzero linear dependence $\sum_i t_i a_i = 0$. Reverse its sign if necessary so some $t_i > 0$, and replace positive coefficients w_i by

$$w_i' = w_i - \left(\min_{t_j > 0} \frac{w_j}{t_j} \right) t_i.$$

All new coefficients are nonnegative and at least one vanishes. The augmented vector is unchanged, so the risk mass remains one. There is no requirement to preserve the old total cost-multiplier mass: all cost atoms are active at x_*, and the strict feasible point bounds the new mass by $L/(N-\kappa)$. Iteration proves finite support. This is the classical conic Carathéodory mechanism, applied to the specified common-plan constraints.

The certificate therefore has four logically separate components: a robustly feasible plan, active physical risk and cost preparations, a common-tree minimum of the scalarized expression, and a paid completion if the object being certified is an infinite stopped experiment. Exact arithmetic may verify each finite identity. A numerical optimizer which only reports convergence verifies none of the universal model inequalities by that fact alone. For a semialgebraic compact model set, the universal inequalities have the real-algebraic description given in the main article; a numerical grid is not substituted for them.

S56. Boundary compactness is not compactness of occupation measures

The common infinite-plan space is compact in its countable product coordinates. The bounded-cost subset is closed because every finite cost reserve is continuous modelwise. The terminal loss is bounded, so its tail is uniformly at most $L(N/c_* - 1)/T$. These facts prove boundary attainment and increasing finite lower convergence. They do not prove continuity of total cost in that topology. The earlier example with a small probability of a very long run has bounded mean while retaining a nonvanishing expected cost tail. Thus the map from a strategic plan to its full occupation measure need not be continuous. Occupation-measure compactness and uniform absorption have their own hypotheses in the existing theory; the argument here neither assumes them silently nor claims to improve that theory in arbitrary Borel control spaces.

For the boundary theorem, each finite tree has finitely many actions and reports. Its countable product is compact metrizable, which is why a subsequence suffices in that proof. This is not a replacement of the older profile's weak-star compactness argument by an unjustified sequential argument. Optimized output tuples lie in compact metric D^m and use a common decision modulus. A hard output-width class is not convex in general; randomizing two dictionaries does not preserve its width.

S57. Weighted boundary realization: explicit primitive diagnostics

The new weighted theorem applies beyond model-independent report words. For example let a two-dimensional protected state have failure Kraus matrix $A = \operatorname{diag}(\sqrt{u}, \sqrt{v})$ and herald Kraus matrix $B = \operatorname{diag}(\sqrt{1-u}, \sqrt{1-v})$, with (u, v) in a compact rectangle inside $(0,1)^2$. One call costs one and a rejected herald retains $B\rho B / \operatorname{tr}(B\rho B)$. The mean herald probability depends on the current state unless $u = v$, and neither branch resets that state. The first-herald potential is exactly

$$H = \operatorname{diag}((1-u)^{-1}, (1-v)^{-1}), \qquad H - A^* H A = I.$$

For a fixed prepared state the maximum cost occurs at the upper-right corner, where every finite unary word has positive mass. Thus the cost-support coverage holds. This example verifies the enlarged interface directly and does not replace its theorem by a simulation.

For a prescribed stochastic observer with transitions Q_y, its block killed map retains the same total physical failure mass. Indeed summing the output memory blocks removes Q_y by row normalization. Iteration proves the uniform survival bound on positive block inputs. Its score potential is therefore the unique bounded solution of the main article's equation (S28.7); the first block evaluated on the prepared density is the actual performed risk. In contrast, a finite Hankel factorization with negative transition entries has no such probability interpretation.

S58. Scope of exact algebraic memory certification

The finite test in Proposition S28.7 fixes a desired probability response, not merely finitely many empirical samples. Its linear representation is an input identity valid for all words. Equality at words through length $M + r - 1$ implies equality everywhere by the invariant-span argument; finite tests on arbitrary fitted data do not carry that conclusion. The number of tested words is at most $\sum_{j=0}^{M+r-1} |Y|^j$, each coordinate equality has polynomial degree at most $M + r$, and the unknown controller has $|Y|M^2 + M|C|$ entries before row-sum elimination. These are finite syntactic bounds, not an efficient bound for quantifier elimination.

For the damped qubit the polygon construction supplies a finite search bound, with algebraic vertices and algebraic barycentric weights for algebraic α. Real-algebraic feasibility can therefore return an algebraic minimal controller. Its transition-table description, the encodings of those algebraic numbers, and the resources of exact sampling are additional to $\lceil \log_2 M \rceil$ retained-state bits. In the protected classical chain example the displayed rational probabilities already give an exact k-label controller, and the probability-valued Hankel rank proves that this label count is minimal. Neither proof gives a matching universal program or working-storage lower bound.

The classical chain and the quantum calibration problem score replication of a terminal probability, not recovery of the realized hidden state. The observer's private random chain is independent of the protected chain; only their terminal output laws agree. This distinction makes clear why no extra observer-owned quantum memory or hidden-state oracle has entered the construction.

S59. A nonreset, noncommuting exact instrument realization

Here is an explicit small realization of the branch-interiority theorem. Let the core vertices be the six Pauli eigenstates

$$\tau_{j,\pm} = (I \pm \sigma_j)/2, \qquad j = x, y, z.$$

Their convex hull is the Bloch octahedron. For any positive matrix B with trace t and unnormalized Bloch coordinates $b_j = \operatorname{tr}(\sigma_j B)$, if $\sum_j |b_j| \le t$, then

$$B = \sum_{j,\pm} w_{j,\pm}\tau_{j,\pm}, \qquad w_{j,\pm} = \frac{t - \sum_l |b_l|}{6} + \max\{\pm b_j, 0\}. \tag{S59.1}$$

All coefficients are nonnegative, their sum is t, and their signed coordinate differences are b_j, proving the identity directly. If $B = 0$ every coefficient is zero. Thus this formula uses no undefined zero-report posterior.

Take any qubit instrument and, after each reported branch, physically apply $\mathcal{D}(B) = B/4 + 3\operatorname{tr}(B)I/8$. Its normalized output has Bloch length at most $1/4$; by Cauchy–Schwarz its Bloch coordinate sum is at most $\sqrt{3}/4 < 1$. Formula (S59.1) gives exact branch rows with six core states for any finite command family. A separately specified initial state can be retained as one extra anchor. This seven-state upper is better for this example than the general dimension-only bound; it is not asserted minimal.

The instruments used in the exact diagnostics include amplitude damping with Kraus matrices

$$A_0 = \mathrm{diag}(1, 3/5), \qquad A_1 = \begin{pmatrix} 0 & 4/5 \\ 0 & 0 \end{pmatrix},$$

and its conjugate by the Hadamard unitary. Their report probabilities depend on the input state and may vanish. After the declared depolarization their unconditional channels have distinct fixed Bloch vectors $(0, 0, 16/91)$ and $(16/91, 0, 0)$, so there is no common stationary density matrix. Their report branches are not all replacement maps: in particular the invertible branch retains a nonzero dependence on the input. The main theorem therefore has content beyond a shared fixed point or a reset model.

Two additional weak-measurement commands, with Kraus matrices $\mathrm{diag}(3/5, 4/5)$ and $\mathrm{diag}(4/5, 3/5)$ and their Hadamard conjugates, have both report effects bounded away from zero. A two-label observer alternates these commands after report zero and stops at report one. Its physical block cost and score potentials are computed independently of the seven-state simulator's fourteen-state joint killed matrix. Exact rational inversion gives the same initial cost and squared-audit risk. This is a diagnostic of the all-policy interpolation proof, not its substitute.

S60. Interiority can first occur at a genuine second call

Let one command have the single reported channel $\Phi(B) = (HBH^* + SBS^*)/2$, where H is Hadamard and $S = \mathrm{diag}(1, i)$. The Kraus choice is not a recorded report. On Bloch vectors its matrix is

$$A = \begin{pmatrix} 0 & -1/2 & 1/2 \\ 1/2 & -1/2 & 0 \\ 1/2 & 0 & 1/2 \end{pmatrix}.$$

The largest singular value of A is one, so no positive uniform one-call floor holds. But $\|A^2\|_{\mathrm{F}}^2 = 5/8 < 49/64$, hence every output of Φ^2 has Bloch length at most $7/8$ and spectral floor at least $I/16$. Thus the block theorem applies with $L = 2$ and $\eta = 1/8$. The realization still exposes both actual calls and allows stopping after the first. Treating the two unrecorded Kraus choices as if they were available reports would be a different instrument.

The average-interiority counterexample in the main article goes in the opposite direction: the random unitary choice is actually recorded, so its branch states remain pure and its record cannot be averaged away. These two examples fix exactly which information is erased and which remains in the causal interface.

S61. Positive coordinates, arbitrary effects and a forbidden inference

A state-positive simulator has a linear reconstruction map $p \mapsto \sum_i p_i \tau_i$ from its simplex. It does not have a positive affine inverse defined on every quantum state. If such an inverse $E : \mathcal{D}_d \to \Delta_M$ reproduced every density matrix exactly, then every pure state would be a convex combination of the fixed τ_i. Extremality would force that pure state to be one of the finitely many τ_i, an impossibility for $d \geq 2$. Our construction instead fixes finitely many initial preparations and uses the strict branch margin after the first call. It is not an entanglement-preserving channel simulation.

Theorems S23.2 and S23.4 use positive density coordinates. The lower count for uniform noisy states therefore cannot be imported into the general optimal-face realization problem without checking that its generator decoders have this structure. Conversely the fixed-angle spectral exclusion in Corollary S23.8 requires only a finite stochastic matrix and a probability decoder, so it does not have that extra restriction. Its asymptotic lower rate is much weaker than the state-positive noise exponent and is not claimed sharp.

Fanizza, Lumbreras and Winter's Theorem 2 [50] bounds the maximal positive-realization dimension across a varying phase family in a different noisy quantum process. It includes an inverse-square-root noise dependence in that worst-phase statement. The present fixed-angle rational

exclusion should not be presented as an improvement to it. The new uniform upper instead supplies a constructive finite realization from a condition on every actual report branch, and its same-policy corollary keeps the simulator hidden under control and stopping.

S62. Leakage boundaries and exact bounded-decoder conventions

The transient-core lemma in the main article uses a nonzero response tending to zero, a prescribed initial label and a decoder in $[0,1]$. The assumptions ensure that the initial label belongs to the output-visible core even when its immediate output is zero. The positive path argument tests every active entry, not only the norm of one arbitrarily chosen physical state. The appended zero-output label remains live until the productive herald. Removing it by declaring a hidden leakage event terminal would change the experiment.

Proposition S62.1 (Degenerate leakage parameters)**.** *For $u_n = \lambda^n[1+\gamma\alpha^n\cos(n\vartheta+\varphi)]/2$, interpret the initial term separately at zero parameters. With $0<\lambda<1$, zero contrast gives least live memory two. At $\alpha=0$ the least size is two if $\gamma\cos\varphi=0$, and three otherwise. At $\lambda=0$ it is one when $u_0=0$ and two otherwise. At $\lambda=1$ there is no mandatory leakage label: the inherited unattenuated theorem applies in its stated nonreal-mode case. Background $\eta=1$ gives the constant response $1/2$ and needs one label.*

Proof. For zero contrast, the response is $\lambda^n/2$. One live label cannot change its output; a transient label of retention λ and output $1/2$, followed by an absorbing zero-output label, realizes it with two.

At $\alpha=0$, let $a=u_0=(1+\gamma\cos\varphi)/2$. The subsequent terms are $\lambda^n/2$, $n\ge1$. A three-label construction has a starting label of output a, leading deterministically on its first failure to a transient label with retention λ and output $\lambda/2$, followed by a zero-output label. Its response is exactly the required one. A two-label realization has, by Lemma S21.1, a one-dimensional active core, so its entire response must be ar^n. Matching the positive terms at $n=1,2$ forces $r=\lambda$ and $a=1/2$. If $a=0$, nonzero later output already excludes a one-dimensional core. Thus two are possible precisely in the stated case.

At $\lambda=0$ only the initial output can be nonzero. A starting label of that output followed by a zero-output label suffices; one label is possible precisely when all terms are zero. At $\lambda=1$ the output no longer has the decay hypothesis used to force a zero-output label. The remaining claims follow directly from the constant response or the inherited theorem. □

The theorem's nonreal-mode assumption also matters. If ϑ is a multiple of π, exponential modes can coincide or cancel. One must minimize the resulting actual response rather than apply a nonreal spectral count. No claim about those coincidences is hidden in the four-mode Hankel statement.

S62.1. **The physical audit is unchanged by the realization argument.** For the six-dimensional squared embedding, the herald probability is p in every model; it is not the leakage probability. Its first compulsory call and every failed call cost one. Hence the least cost is $1/p$ and every boundary-feasible policy accepts the first herald. The flag parameter t changes an unobserved flag channel, not the herald hazard. The two signs are physical audit-model parameters with the same fixed loss $(r-z)^2$, not a choice of loss after observing the answer. The three effects in the inherited construction have positive mass and sum to the identity. Their computed moments give

$$\mathbb{E}_{t,\sigma}(r-Z)^2 = 2 + 2\varepsilon\sigma p\sum_{n\ge0}(qt)^n(b_n-u_n).$$

Averaging the two signs gives two for every rule. Thus the target response attains two even on the finite witness family, while exact finite-memory equality requires both opposed inequalities at every test parameter. The hidden sink is never supplied as an observer report.

S62.2. **What the finite test certifies.** The generating function of an M-label stochastic decoder has denominator $\det(I - zQ)$ and numerator degree at most $M - 1$. The determinant is nonzero throughout $[0, q\bar{t}]$ because $q\bar{t} < 1$. The target's denominator is likewise nonzero there. Consequently the $M + r$-point theorem is a polynomial identity argument with a known degree bound. It does not certify an arbitrary fitted function from finitely many samples, nor identify an unrestricted full-history rule from those finitely many equations.

For the leaking irrational-phase example, the exact fixed-M optimization can be encoded with variables Q, d, v_j, w and the inequalities

$$Q \geq 0,\ Q\mathbf{1} = \mathbf{1}, \quad 0 \leq d \leq \mathbf{1}, \quad (I - qt_jQ)v_j = d, \quad -w \leq (v_j)_1 - U(qt_j) \leq w.$$

The minimizing w exists and is strictly positive. Quantifier elimination over this compact graph computes it as an algebraic number and returns a minimizing controller. The proof supplies this terminating procedure for every prescribed size; the finite diagnostics do not claim to have run a global elimination for all sizes. The number of model pairs grows with M, which is essential to the scope of the finite witness assertion.

Proposition S62.2 (An exact finite-witness one-label value)**.** *In Theorem S21.6, take $M = 1$, the four parameters $t = 0, 1/6, 1/3, 1/2$, and both audit signs. The least one-live-label risk is*

$$2 + \frac{331}{13573},$$

attained by the constant probability $d = 12249/13573$ of decoder $+1$.

Proof. The generating target is

$$U(z) = \frac{-8z^2 + 28z - 40}{5z^3 - 22z^2 + 44z - 40}, \qquad U(0) = 1, \quad U(1/4) = 2144/1939.$$

Every one-label rule is a constant $d \in [0, 1]$. Its uniform resolvent error is at least $\max\{1 - d,\ 4d/3 - 2144/1939\}$ from the two endpoint tests. Their intersection gives $d_* = 12249/13573$ and lower error $1324/13573$. Direct rational substitution at $z = 1/12$ and $z = 1/6$ gives errors respectively $-465602268/9412509029$ and $6704202/487474295$, whose absolute values are strictly below this lower error. Thus all four tests are bounded by the endpoint value at d_*. Finally $2\varepsilon p = 1/4$ converts resolvent error to the stated excess squared risk. This proves optimality over all stochastic one-label decoders; no local numerical optimizer is used. □

S62.3. **Background, initial mixing and precision.** The background effect $E_\eta = \eta I/2 + (1 - \eta)E$ is a valid actual effect on the qutrit, with eigenvalues in $[\eta/2, 1 - \eta/2]$. Its positive floor changes which terminal responses must vanish on the stationary face. It does not add a report, a reset, an observed physical state or a free random seed.

The construction stores the T exceptional prefix values. Only its final failure row chooses one of two disjoint suffix machines, and that choice is subsequently represented by the retained label. Their average realizes the complete suffix from its first step. This is why the stated count is $T + 2 + m(\lambda\alpha, \vartheta)$, with no additional initialization label and no uncounted intercall mixture memory.

For algebraic input the background target has an algebraic linear representation of dimension at most four. A rationally certified polygon or the existing spectral size theorem supplies a finite size upper. The prescribed-size algebraic equality test then terminates a minimal-size search. Exact algebraic encodings and their sampling cost are separate from the live-label count. The upper logarithmic dependence on $1/\eta$ is not a matched precision or program-bit lower bound.

S63. Arithmetic synthesis and diagnostic boundaries

The main arithmetic-envelope theorem constructs a stochastic matrix, not merely a rational angle close to a target. If a/q is the chosen approximation, pass to the explicitly counted multiple l of q. The target eigenvalue belongs to the triangle with vertices zero and the two adjacent lth roots of

unity. If its radial boundary is r, the two chord coefficients give that boundary point; scaling them by α/r and assigning the remaining weight to $J = \mathbf{1}\mathbf{1}^{\mathsf{T}}/l$ gives a stochastic circulant matrix. Its row from the prescribed initial vertex produces the complete response from time zero. This explains why passing to a multiple costs labels but no physical invocation.

The arithmetic envelope itself also admits finite certified bounds. For any integer $B \geq 1$, write

$$A_B = \min_{1 \leq q \leq B} \max\{3, q, e_x(q)/\epsilon\}.$$

Then

$$\min\{A_B, B+1\} \leq \mathfrak{A}_x(\epsilon) \leq A_B. \tag{S63.1}$$

Indeed every omitted integer denominator is at least $B+1$. In particular, once $B+1 \geq A_B$, enumeration gives the exact envelope if its finitely many arithmetic comparisons are exact. For a computable irrational phase, rational interval enclosures instead give certified upper and lower values by replacing each error with its interval bounds. Precision is increased whenever a comparison or a polygon margin remains unresolved. A supplied strictly feasible polygon upper is a finite exact construction; this does not assert decidability of arbitrary equality cases from finite-precision phase samples.

The two noise sequences used in the main proof have different functions. For a convergent denominator Q with error $e_x(Q)$, noise $e_x(Q)/Q$ is a low-memory scale. For the next denominator Q', noise $1/[Q(Q')^2]$ is a crossover scale at which every candidate denominator below $Q'/2$ has excessive error. Both statements use all candidate rationals in their lower arguments. Restricting a numerical search to convergents alone is not the proof of the universal envelope or of the crossover exclusion.

S63.1. **Finite computations supplied with this revision.** The R71 diagnostic program separately records exact rational checks and floating matrix probes. The exact part tests finite continued-fraction identities, all denominators below the tested convergent bound, the rational separation used at crossover scales, complete finite envelope enumeration and its certified stopping condition. Rational finite tails are deliberately identified as rational test data. Their finite check results are not used to infer an irrationality exponent of a limiting number.

The floating part independently constructs the full circulant transition matrix and compares its forward probabilities with raw qubit iteration and with the target scalar formula. It checks bounded decoders, row normalization, nonnegative mixture weights, spectral eigenvectors, the envelope constants and the power-sector obstruction on finite examples. A separate sequence of large partial quotients probes the two distinct noise scales in the fluctuation proof; it is not labeled a numerical proof of the limiting exponents. Exact special cases and deliberately invalid shortcuts provide negative controls.

S63.2. **Resources and unresolved sharp constants.** The upper construction may use more states than a minimal stochastic realization by the universal factor in Theorem S19.1. Its main purpose is a uniform matching law and exact realization, not a claim that the polygon is optimal at every phase. At algebraic eigenvalues, the inherited finite spectral-membership test can search the actual minimal size below a supplied upper. For general computable phases, finite interval arithmetic can certify strict feasible constructions without deciding all exact spectral boundary equalities.

The growth exponents concern $\log M/\log(1/\epsilon)$ for the least live-label count. In a finite mixture, taking a maximum before a lower limit is essential; no simultaneous Diophantine property is supplied by separate one-dimensional irrationality exponents. Classical rational approximation, stochastic spectral geometry, positive realization and the physical squared embedding retain their distinct roles in this proof chain. No formal proof-assistant check or independent human priority assessment is claimed by the finite diagnostics or the native build.

S64. Verification coordinates for common reversible responses

In the compact-group argument the map from hidden rows to response functions is defined only on the reachable row span R. A whole realizing simplex need not lie in that span. Thus the admissible polytope is the image of $R \cap \Delta_M$, not an unspecified positive cone on the minimal physical support. The support-count estimate uses the fact that two distinct extreme points of the section cannot have exactly the same positive coordinates. It is only a size certificate for a supplied realization, not a constructive upper on minimal stochastic memory.

The quantum specialization removes the common kernel of all commanded terminal tests. Each unitary conjugation restricts to an isometry on the finite-dimensional reachable Hermitian span. Forward invariance and finite dimension imply backward invariance, both for that span and the invisible kernel. The quotient therefore has a common invariant inner product. No controllability or informational completeness of the terminal measurement is assumed; a constant effect has a trivial observable group even if the full physical group is infinite.

For the two reflections the physical chronological convention is $\Phi_{a_n} \cdots \Phi_{a_1} \rho$, while the observer row is $e_1^{\mathsf{T}} Q_{a_1} \cdots Q_{a_n}$. The intertwining equations use row vertices and the transpose of the physical Bloch map. This is why the word $(01)^n$ corresponds to powers of $Q_0 Q_1$ and to the rotation produced by the physical map $S_1 S_0$. The explicit diagnostic uses both equations and a separate raw density-matrix evolution. Reversing only one convention would break the mixed-word identity.

S65. Partition synthesis, anchors and physical interpretation

The initialization row in the periodic construction chooses its recurrent block only after the first actual transition. At time zero it uses a separately assigned decoder b_0. Entering phase one, rather than phase zero, ensures the first observed response is correct. The lower's extra label is not an arbitrary implementation preference: two reachable closed recurrent classes cannot contain the same prescribed initial state. A model permitting an initially mixed state is a different resource convention and can remove that label.

In the decay-normalized lower, the full nonnegative core need not be irreducible, row stochastic or similar by one strictly positive global Perron vector to a stochastic matrix. Its powers are bounded because every active entry is bounded by a fixed shifted target-output path. Perron scaling is performed only on a critical irreducible diagonal block. The no-critical-chain lemma forbids recovering an initialization label by sending a critical class into another required critical class. The proof counts nonnegative paths, so cancellation cannot conceal the resulting quadratic cumulative growth.

The periodic response has a prepared block-qubit realization and a fixed terminal effect. The established flag embedding realizes the associated exact-response requirement using one fixed squared loss and three actual audit outcomes at the original first-herald budget. Internal damping, physical leakage and an observed terminal event are different operations. The new finite-memory laws concern exact replication of the terminal probability, not predicting the realized hidden bit or the distinct squared-sign decision task. They do not give the protected system to the observer.

S66. Independent diagnostics for the present revision

The new program compares subset dynamic programming with an independent enumeration of all set partitions on finite input lists. Rational and algebraic examples check row normalization, bounded decoders, the initialization phase, exact damped and leaked coefficients and random-horizon resolvents. A positive substochastic two-block example shows exactly how linking critical classes yields linearly unbounded rescaled powers. These checks do not replace the class-decomposition lower proofs.

For the controlled theorem, a shared three-vertex algebraic construction verifies both reflection matrices and their intertwining identities exactly. Every tested mixed word is independently evolved

through the raw qubit channels. Additional numeric polygon constructions check longer binary words, both physical maps, their pairwise product and phased periodic mixtures. They establish finite implementation consistency only. A finite angular or continued-fraction sample is never used to infer an irrationality exponent or uniform asymptotic class.

S67. Verification status

The accompanying scripts compare finite algebraic examples, solve common finite-prefix control problems and small finite reset-control instances independently of the manuscript's closed formulas, check report-resolved span counterexamples, test boundary resource values, and compile the complete native manuscript, this supplement and the retained companion. Exact rational checks and floating-point residual checks are separately counted. Neither kind proves an unbounded theorem by sampling. Native LaTeX compilation verifies source completeness and references, not mathematical correctness. No formal proof assistant, newly commissioned independent human referee or editorial acceptance is claimed.

S68. Response degeneracies and the stochastic spectral certificate

The exact spectral criterion in the main article deliberately assumes a nonreal mode and nonzero contrast. The following closes its real and zero-mode boundaries, without applying a three-mode rank argument where the modes coalesce.

Proposition S68.1 (Complete scalar boundary cases)**.** *Let* $b_n = \frac{1}{2}(1 + a\lambda^n)$*, with* $|a| \le 1$*,* $-1 \le \lambda \le 1$*, and* $\lambda^0 = 1$*. Its least stochastic live memory with a prescribed initial label is one if* $a = 0$ *or* $\lambda = 1$*, and otherwise two. In particular this covers the original sinusoidal family at phases* 0 *and* π *with* $a = \gamma\cos\varphi$*, and at* $\alpha = 0$ *with* $\lambda = 0$*.*

Proof. In the constant cases a single label with the constant output probability suffices. In every other case $b_0 \neq b_1$, whereas a one-label stochastic machine has a constant output probability. For an upper construction use

$$Q = \frac{1}{2}\begin{pmatrix} 1+\lambda & 1-\lambda \\ 1-\lambda & 1+\lambda \end{pmatrix}, \qquad d = \frac{1}{2}\begin{pmatrix} 1+a \\ 1-a \end{pmatrix}.$$

The constant vector is fixed, and $(1,-1)^{\mathsf{T}}$ is an eigenvector at λ. Starting in the first label gives the required response. This includes negative real eigenvalues and both unit-circle endpoints. At $\alpha = 0$ the formula gives its one exceptional initial output followed by a constant half output. □

S68.1. **Polynomial verification without an initial randomization.** For $z = x + iy$, $y \neq 0$, the real variables of (S20.3) encode a complex eigenvector $v = a + ic$. The equations $Qa = xa - yc$, $Qc = ya + xc$ imply $Qv = zv$, while $v_1 = 1$ prevents a zero witness. The inequalities $a_j^2 + c_j^2 \le 1$ ensure legal decoder probabilities for every $0 < \gamma \le 1$ and every fixed phase. Conversely a maximal-modulus eigenvector coordinate can always be normalized this way. Thus the certificate does not assume a special circulant form, a full-support stationary distribution, a reversible chain or a hidden preparation mixture.

With algebraic $x, y, \gamma, \cos\varphi, \sin\varphi$, every nonempty fixed-size certificate has an algebraic sample by real-closed-field elimination. A dense transition table has M^2 entries and a binary decoder has M probabilities, in addition to the retained $\lceil \log_2 M \rceil$ state bits. Algebraic-number representations and exact sampling have their own costs. The cubic-root theorem does not silently grant a polynomial bit implementation of an arbitrary real Diophantine rotation.

S68.2. **Constructing a directional circulant.** Let $q \geq 3$ and place ϑ between adjacent root angles $2\pi j/q$ and $2\pi(j+1)/q$, choosing a representative on the circle. Write $\beta = 2\pi/q$ and $\psi = \vartheta - 2\pi j/q \in [0, \beta]$. If $\alpha \leq R_q(\vartheta)$, set

$$a = \alpha \frac{\sin(\beta - \psi)}{\sin \beta}, \qquad b = \alpha \frac{\sin \psi}{\sin \beta}, \qquad c = 1 - a - b.$$

All three are nonnegative. The matrix $Q = aC_q^j + bC_q^{j+1} + cJ$ is stochastic and has eigenvector $v_l = e^{2\pi i l/q}$ at $z = \alpha e^{i\vartheta}$. Its decoder is $d_l = (1 + \gamma \cos(\varphi + 2\pi l/q))/2$. This is a directly specified program at a fixed phase, not an abstract existence proof requiring the physical state to be observed. A root-angle endpoint is permitted and may reduce the actual minimal size. The construction is sufficient; it is not asserted to describe every part of the Karpelevič boundary.

The cyclic eigenvector gives the all-time certificate $Qv = zv$. Checking a finite list of output probabilities without that identity would not certify exact realization. The new diagnostic program tests both the intertwining identity and independent raw qubit evolution, and labels the latter tests as finite probes.

S68.3. **Constants and quantifiers in the arithmetic theorem.** For $h \leq \pi/3$ and $0 \leq \delta \leq h$, the inequalities used in the lower proof are

$$\sin \delta \geq \frac{3}{4}\delta, \qquad 1 - \cos \delta \leq \frac{1}{2}\delta^2, \qquad \tan h \geq h, \qquad \cos \delta + \sin \delta \tan h \leq \sec h \leq 2.$$

The first follows, for example, from concavity of sine and $\sin(\pi/3)/(\pi/3) = 3\sqrt{3}/(2\pi) > 3/4$. Thus there is no numerical fitting in the constant 16 of the cubic-root lower. The exact first-sector law proves the worst-phase lower, not a relaxation to a larger convex hull of all low-order roots. In contrast, the fixed-phase upper uses a polygon whose size is a Diophantine denominator; the two constructions have different quantifiers.

For the golden quadratic phase $x = (3-\sqrt{5})/2$, the integer-polynomial proof in the main text gives a global constant $c = 1/3$, valid at every positive denominator. Inspecting a finite continued-fraction prefix would not give that global conclusion. For $(3+4i)/5$ the Gaussian-integer argument gives the displayed logarithmic lower but not a badly-approximable certificate. These facts prevent transferring the cubic-root exponent to a different phase by numerical resemblance.

The exact-memory count is discontinuous at vanishing contrast: all nonzero contrasts have the same spectral obstruction, but zero contrast is constant. This statement concerns equality of every response coefficient. Approximate risk at a fixed tolerance can of course become insensitive to small contrast; that is a different optimization problem and is not contradicted by the exact criterion.

S68.4. **Relation to classical results and to other resource curves.** The stochastic sector inequality, Karpelevič regions and first-sector lazy-cycle classification are classical; their pertinent statements are credited in the main text and proved there in the form used. The noise-dependent classical-memory bounds in Fanizza–Lumbreras–Winter concern processes and quantifiers distinct from the current one-response squared task. The present argument neither claims priority for the unit-circle obstruction nor uses it as a substitute for the matched arithmetic rate.

The protected system in the squared embedding has dimension four, independent of the observer labels. The instrument-level state-positive simulator of Section S23 is a separate construction: it preserves every declared terminal effect and adaptive experiment, whereas the current stochastic response realization preserves only the fixed optimal task kernel. The old two-label squared-sign frontier and delayed-release timing frontier likewise have different objective functions. Their proofs remain current and no general-M risk formula for either is inferred from the new exact-response law.

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